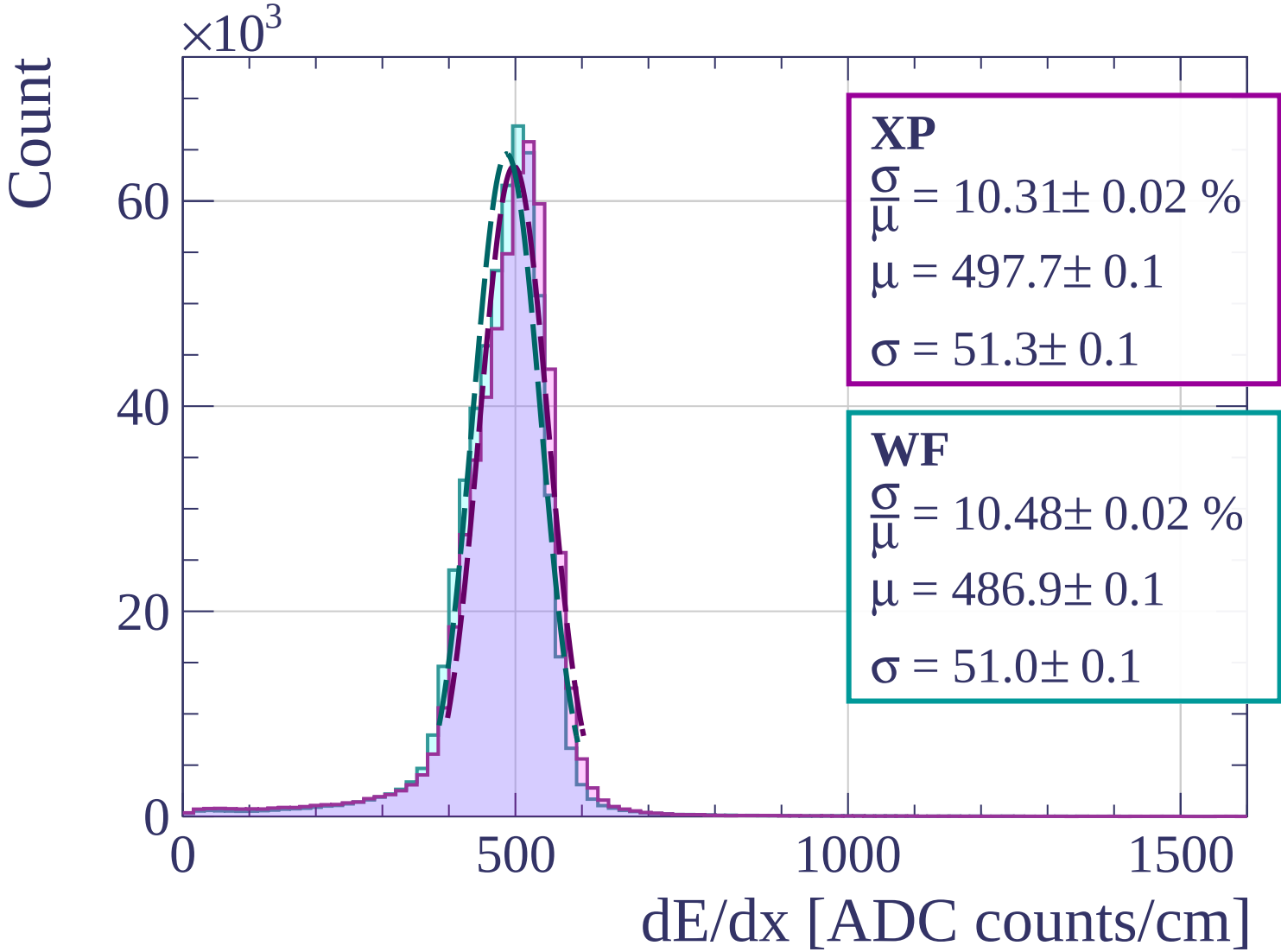
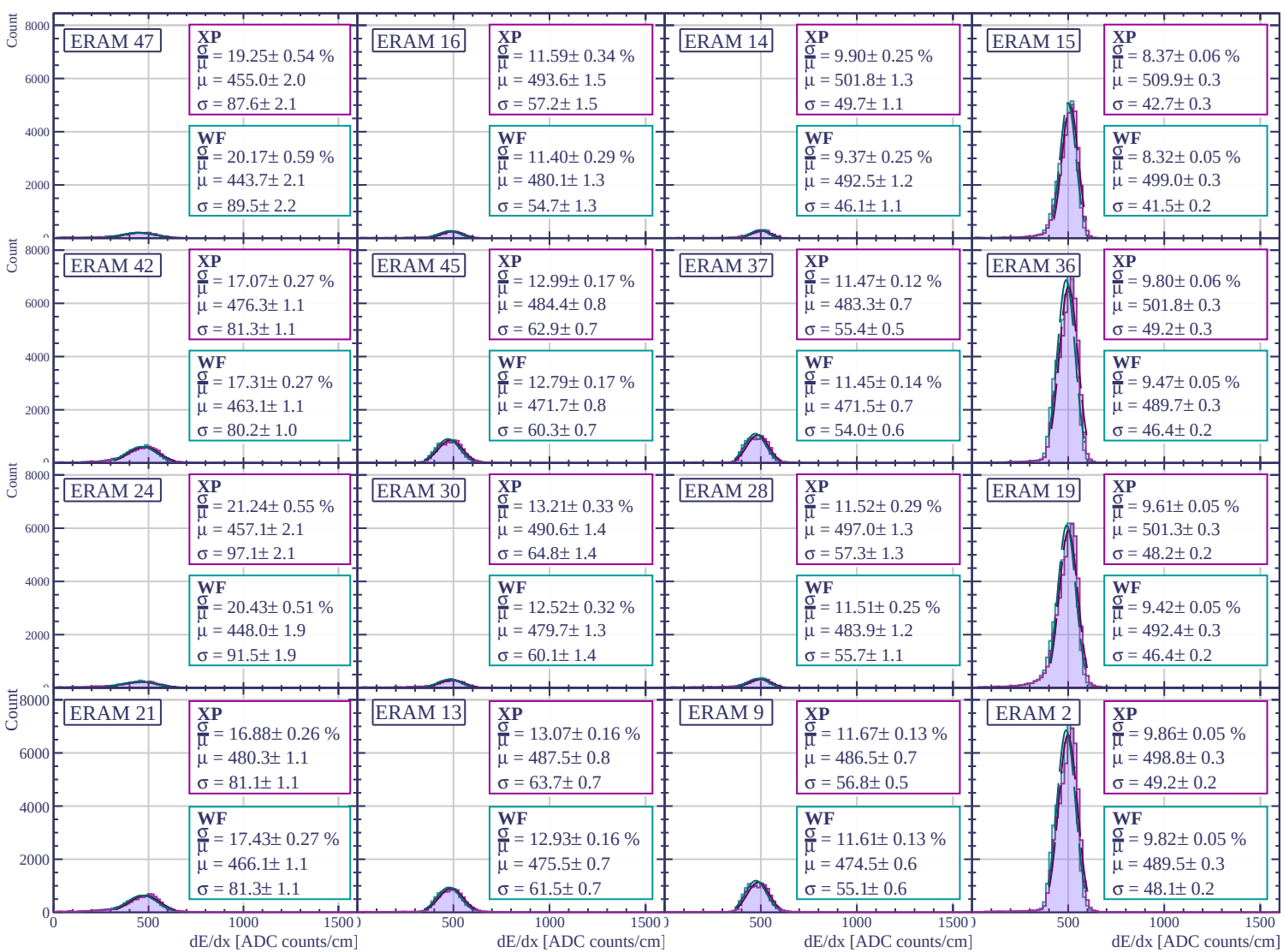
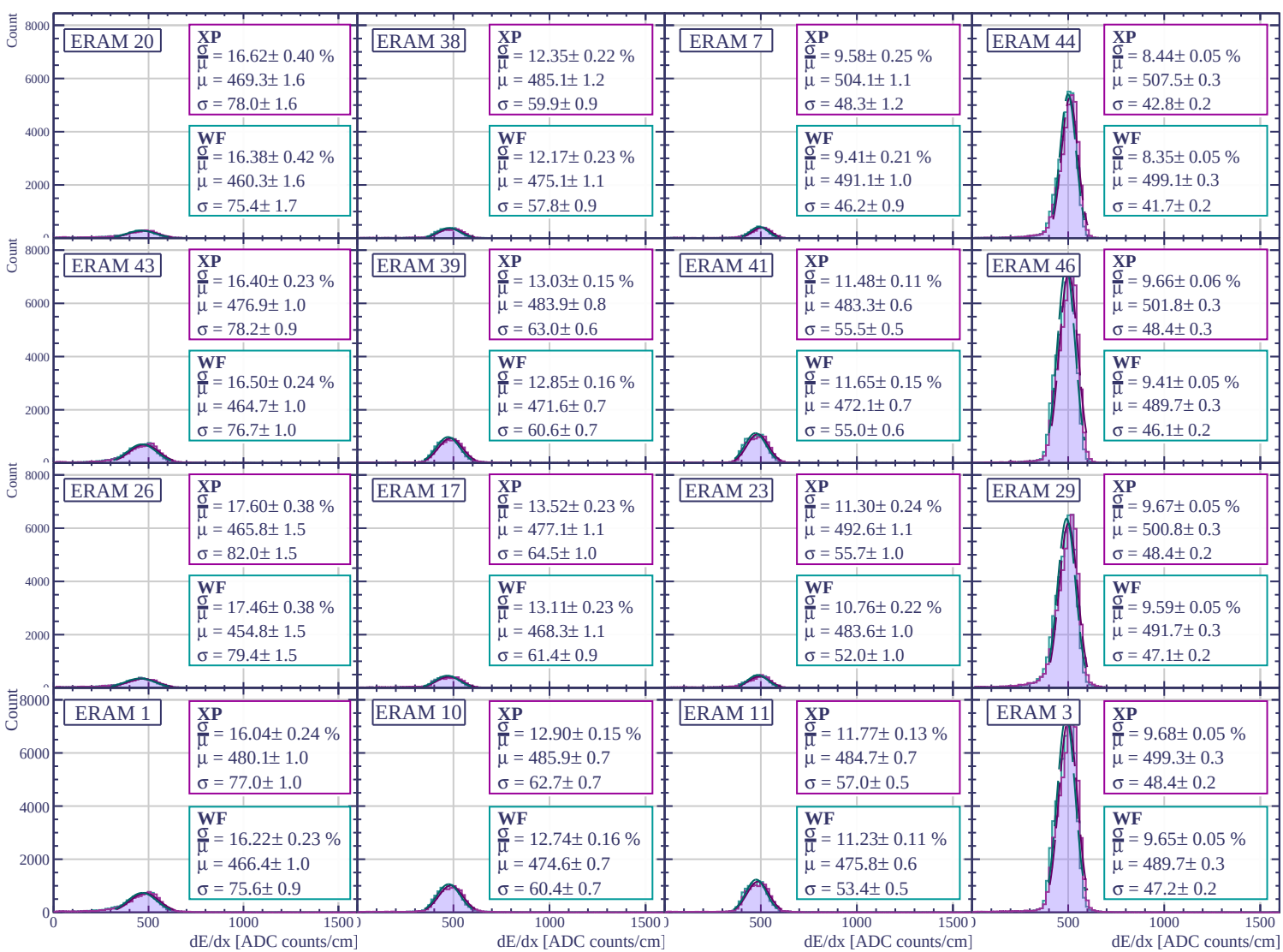
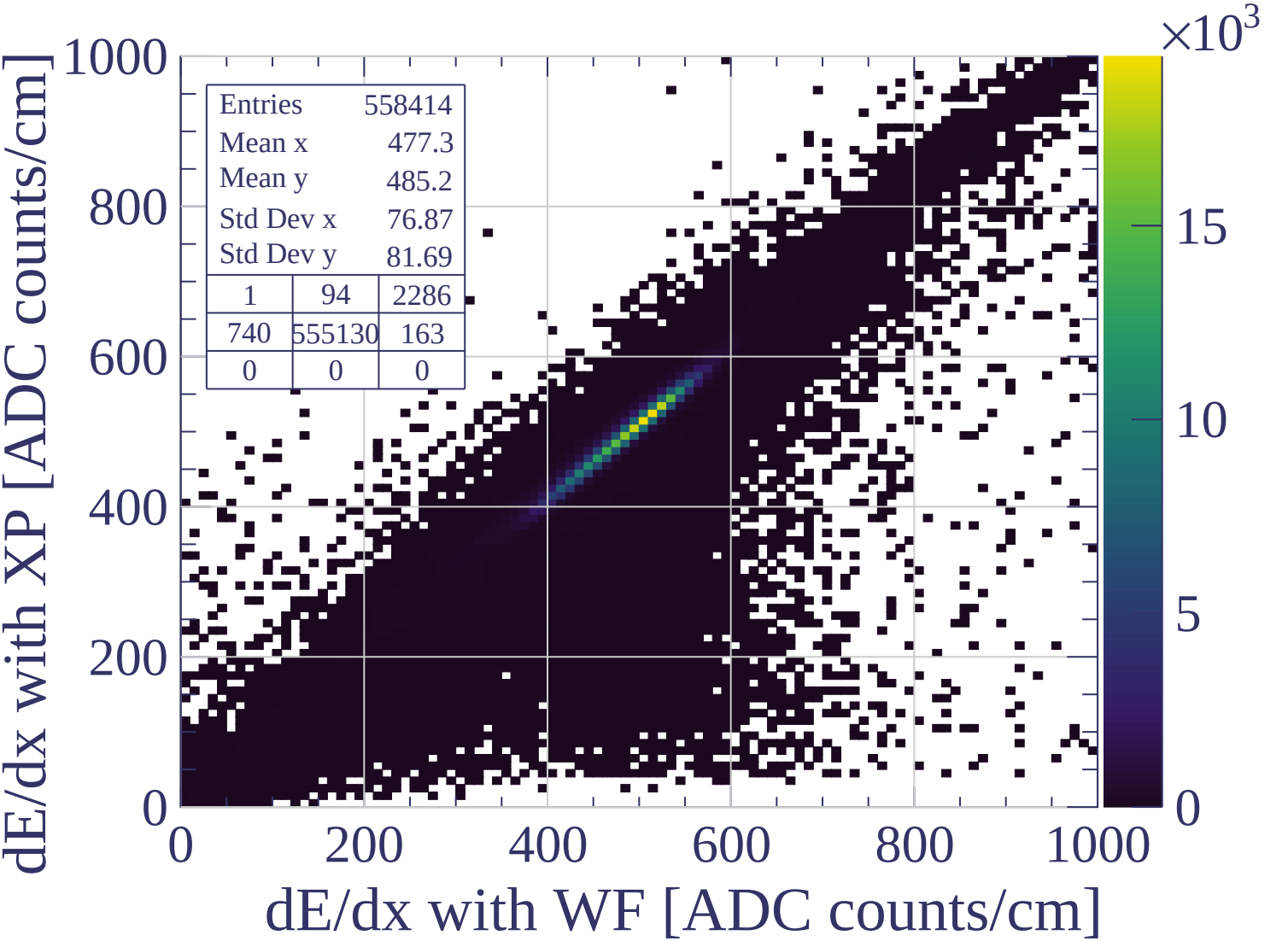
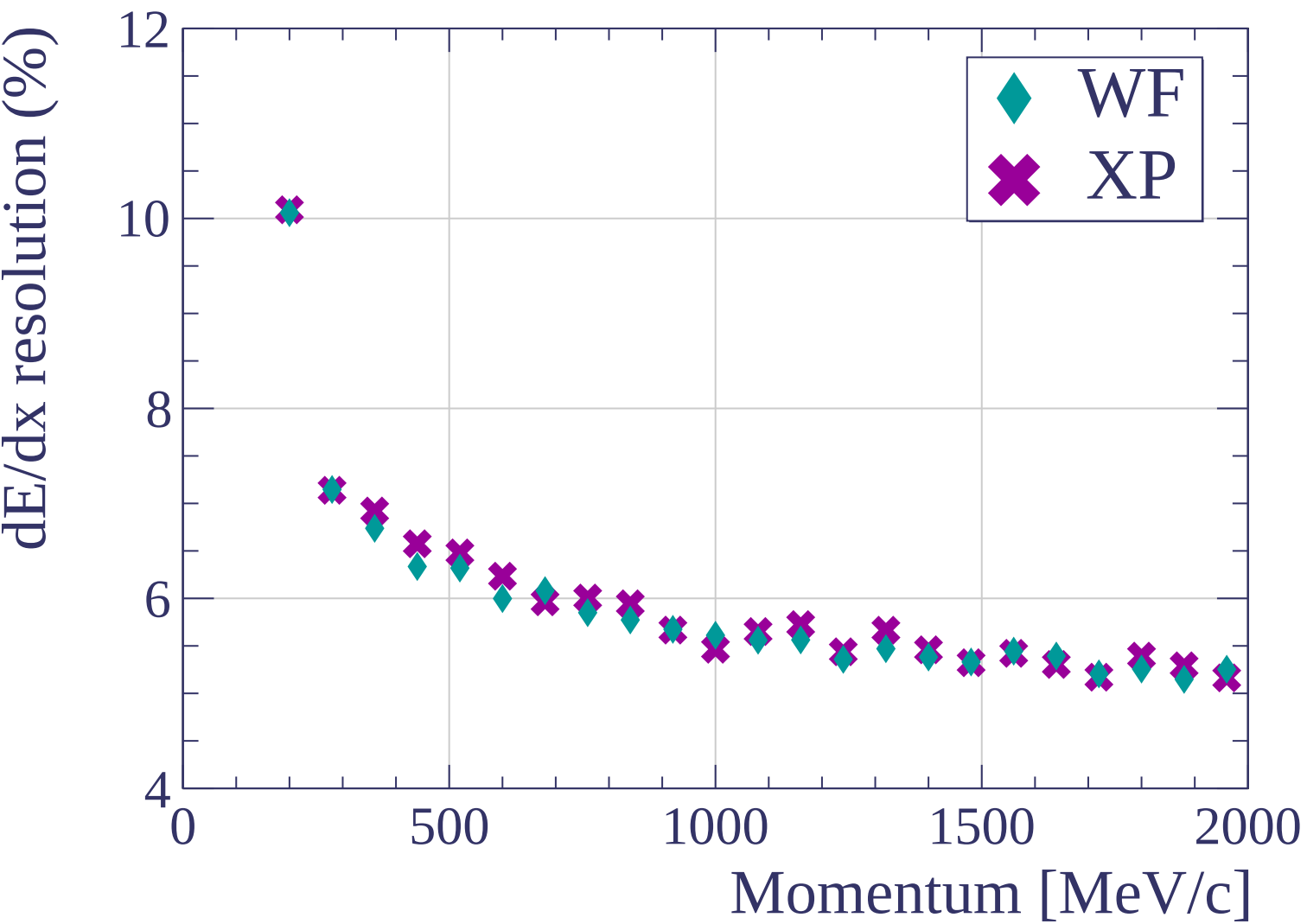


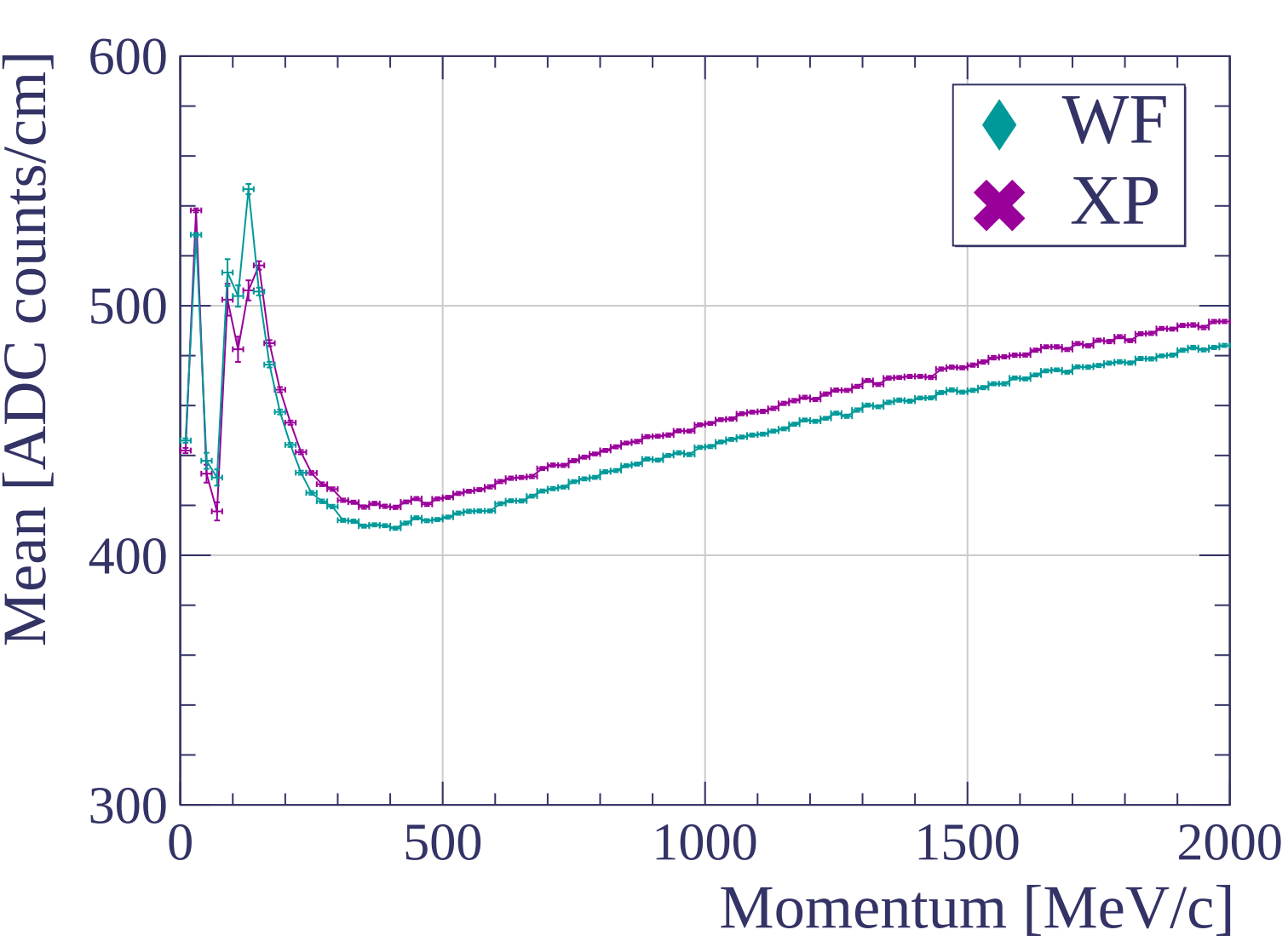


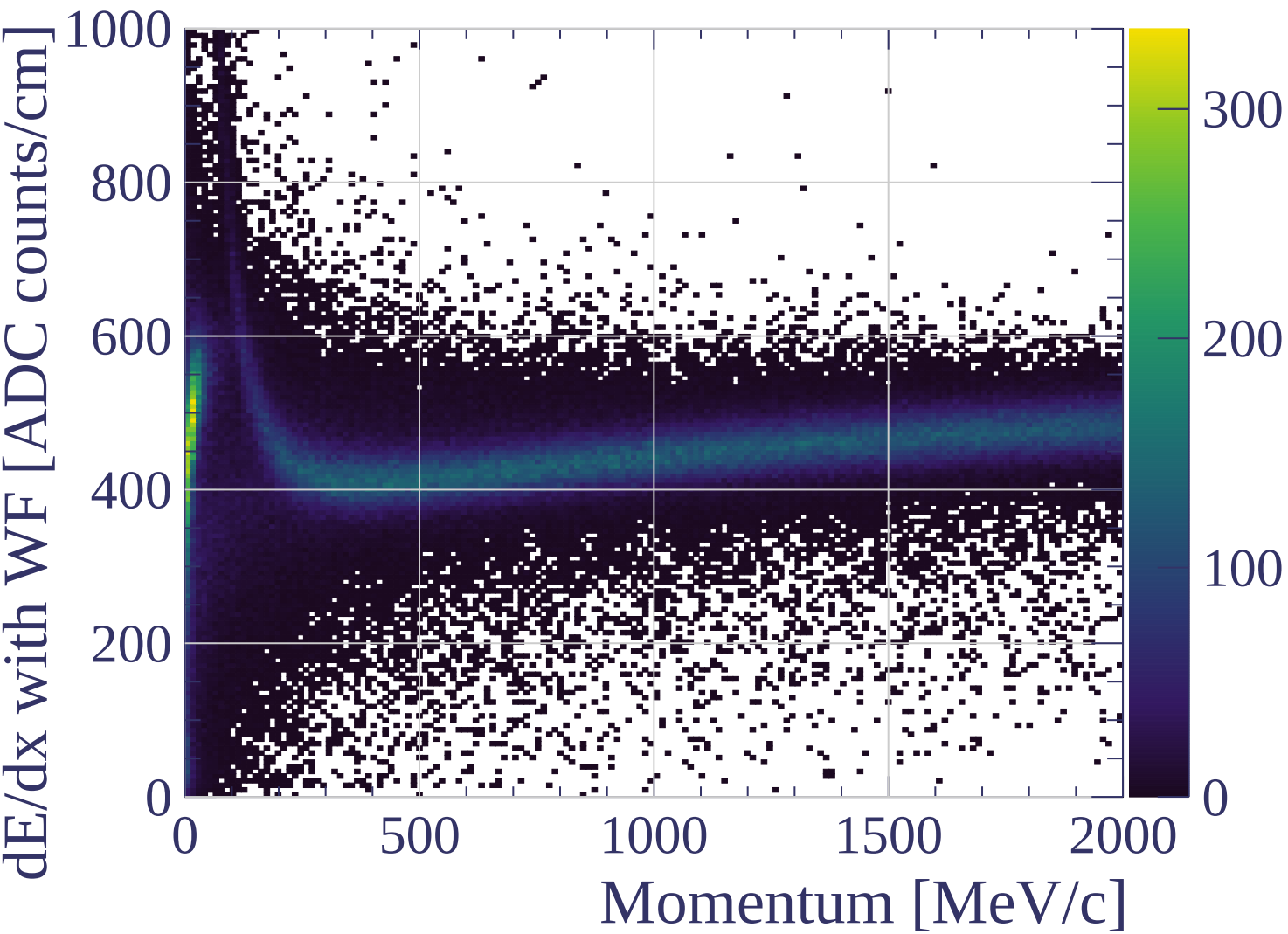


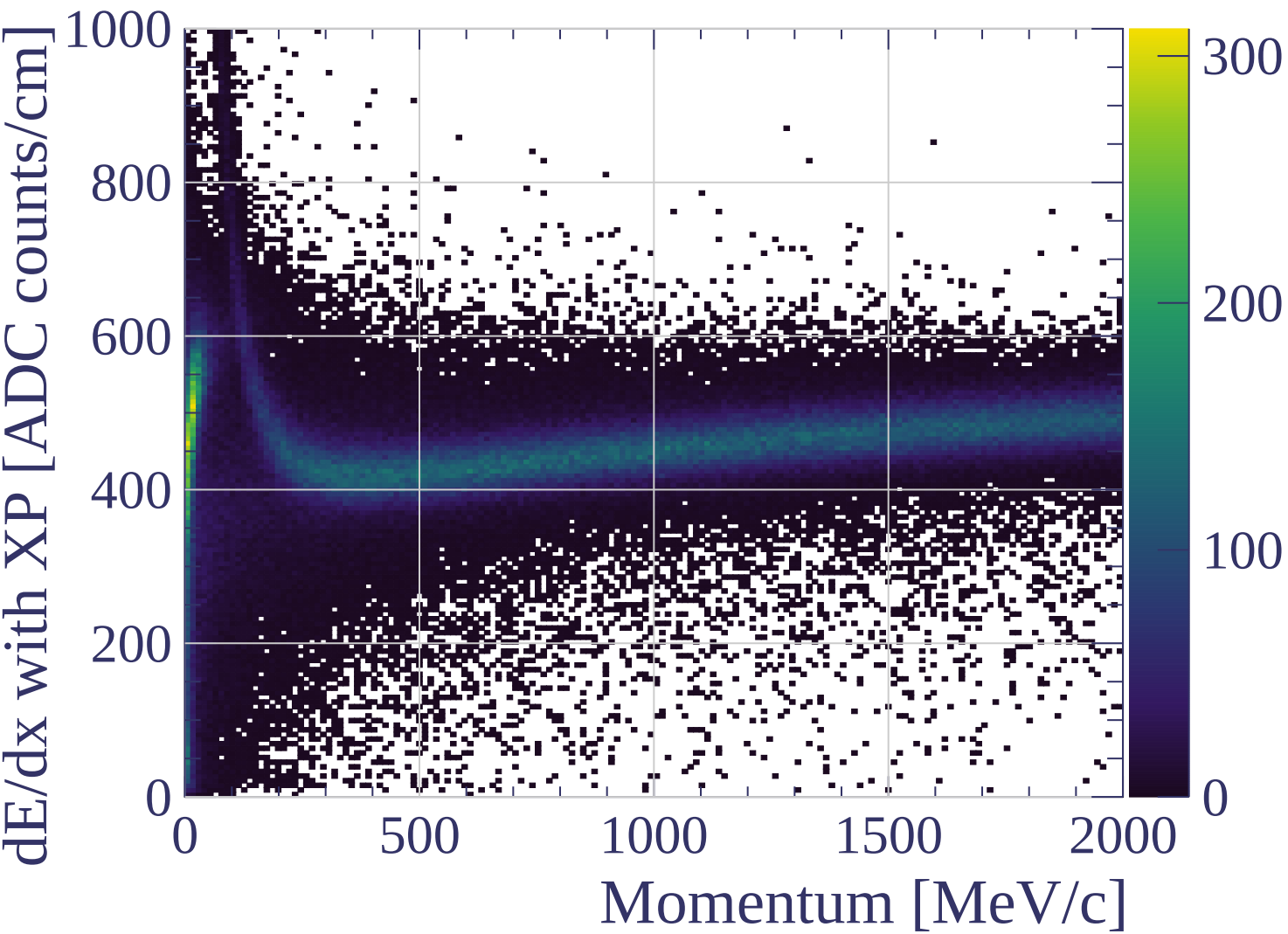


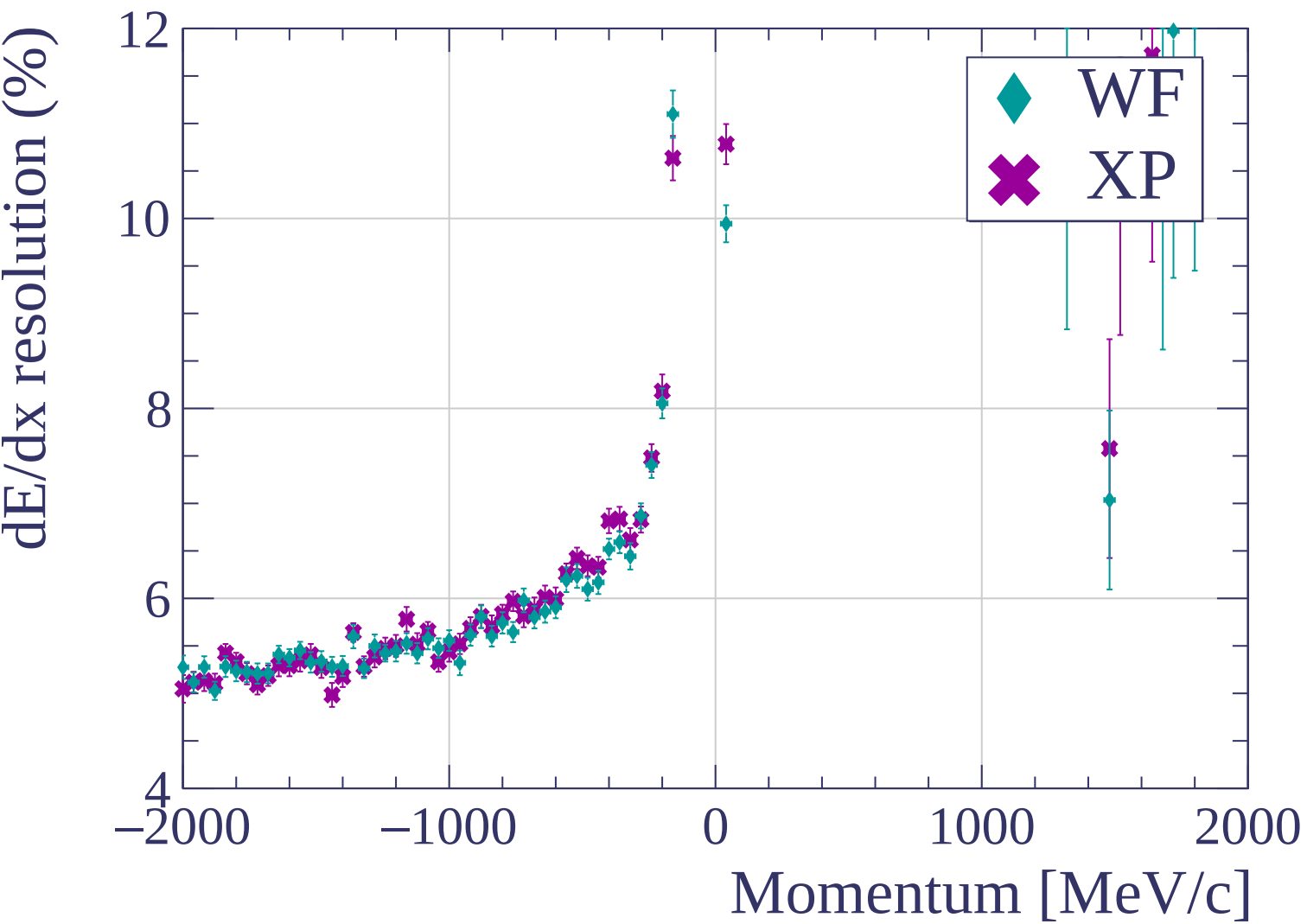


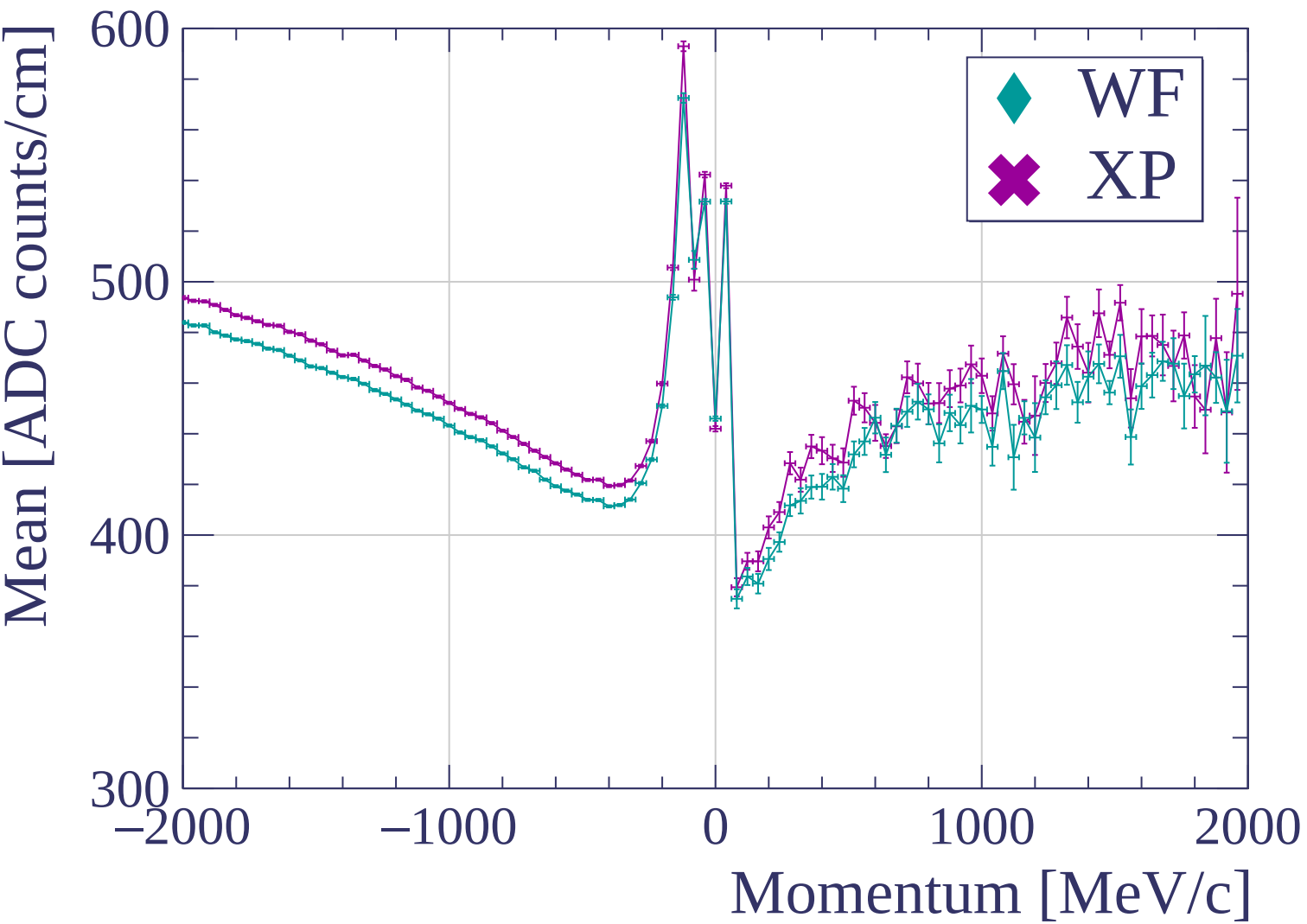


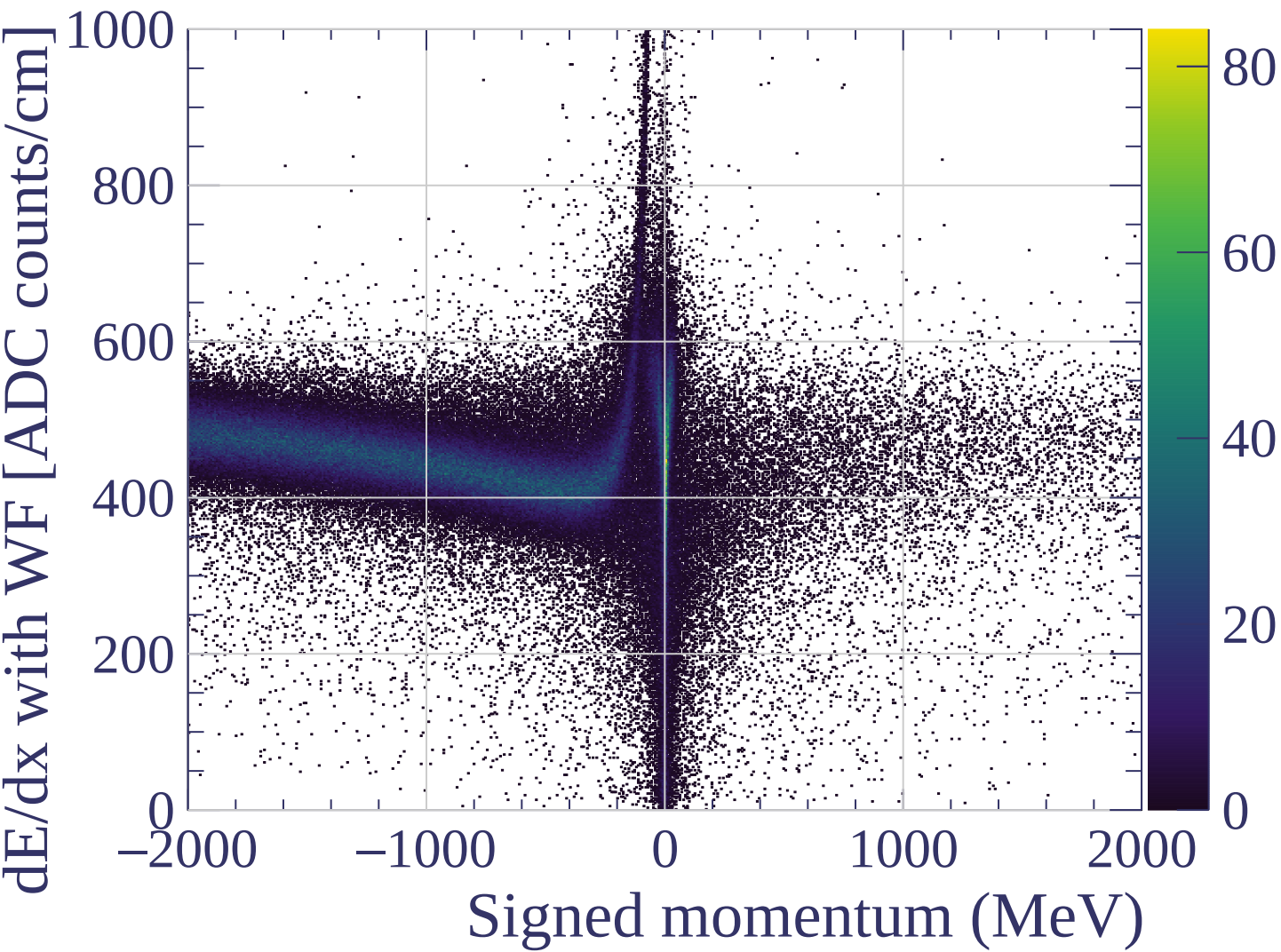


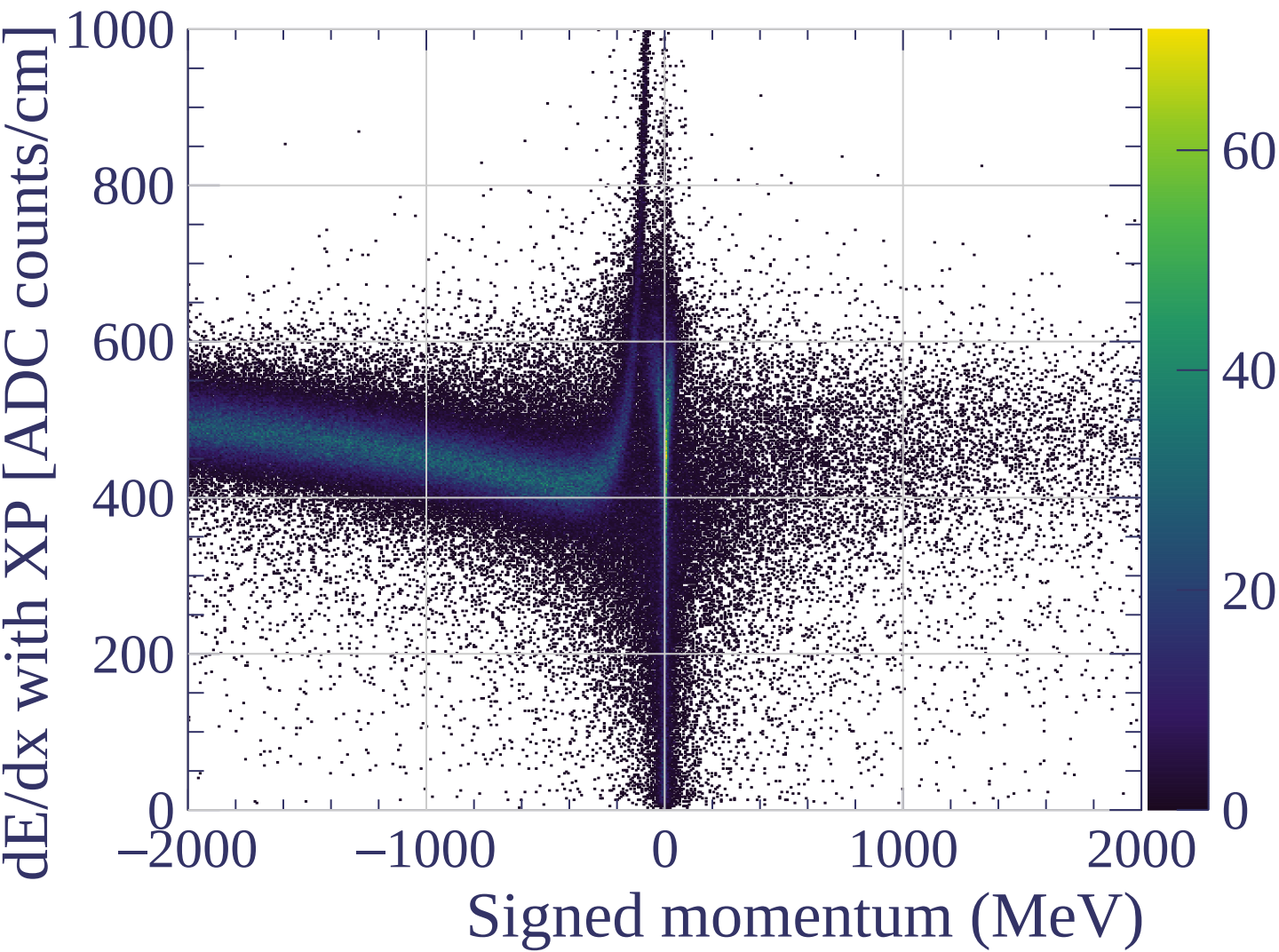


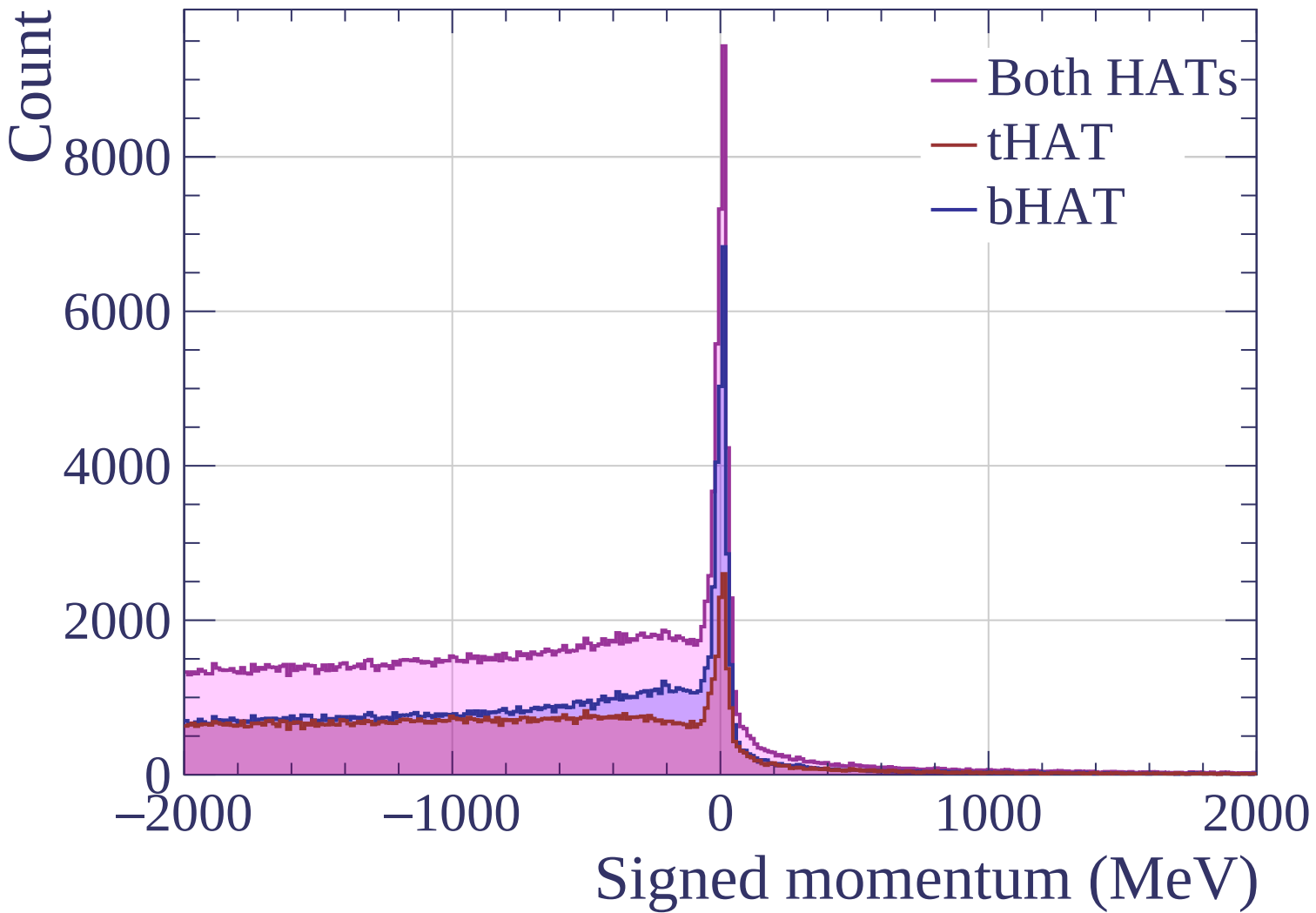


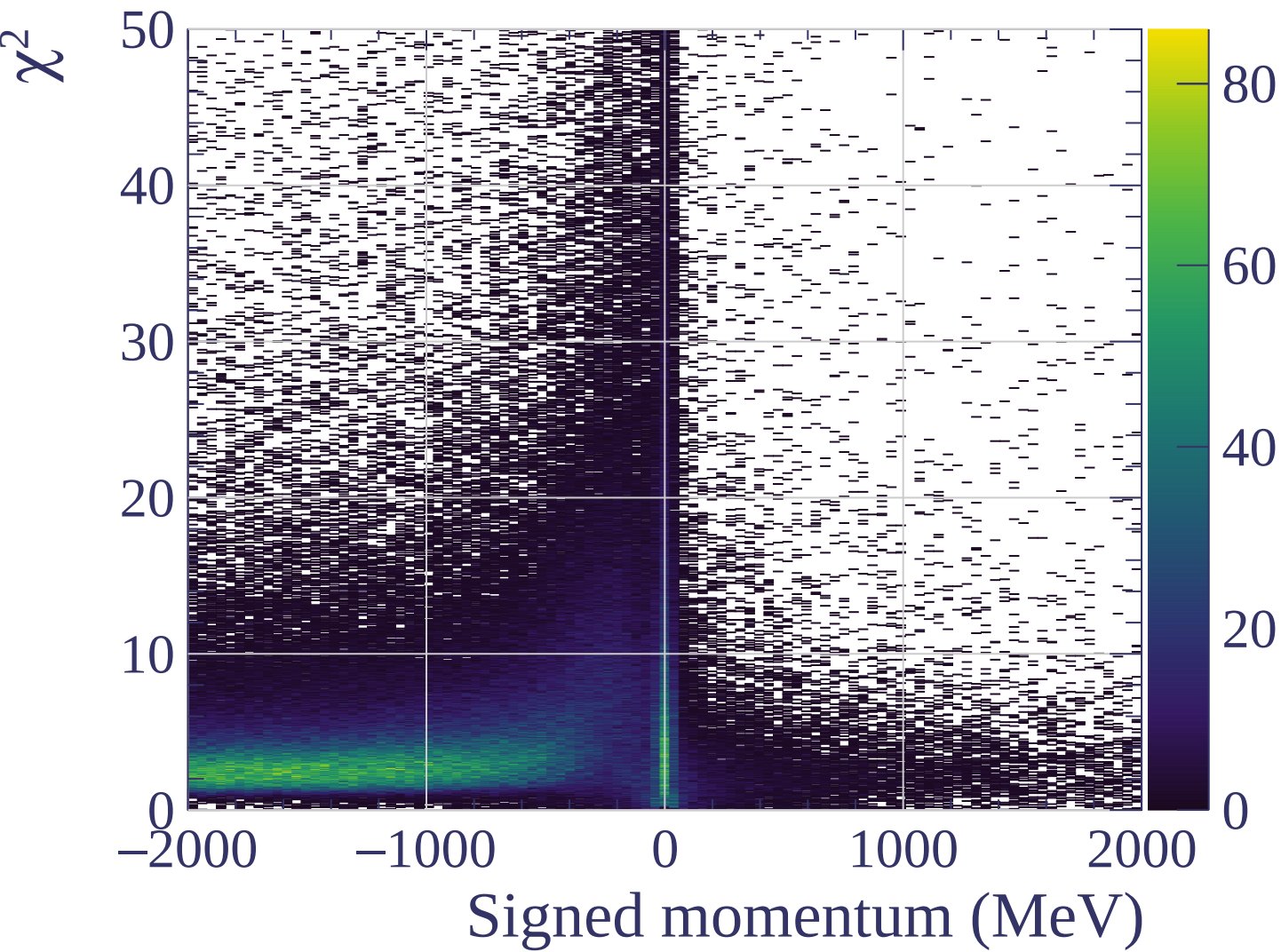


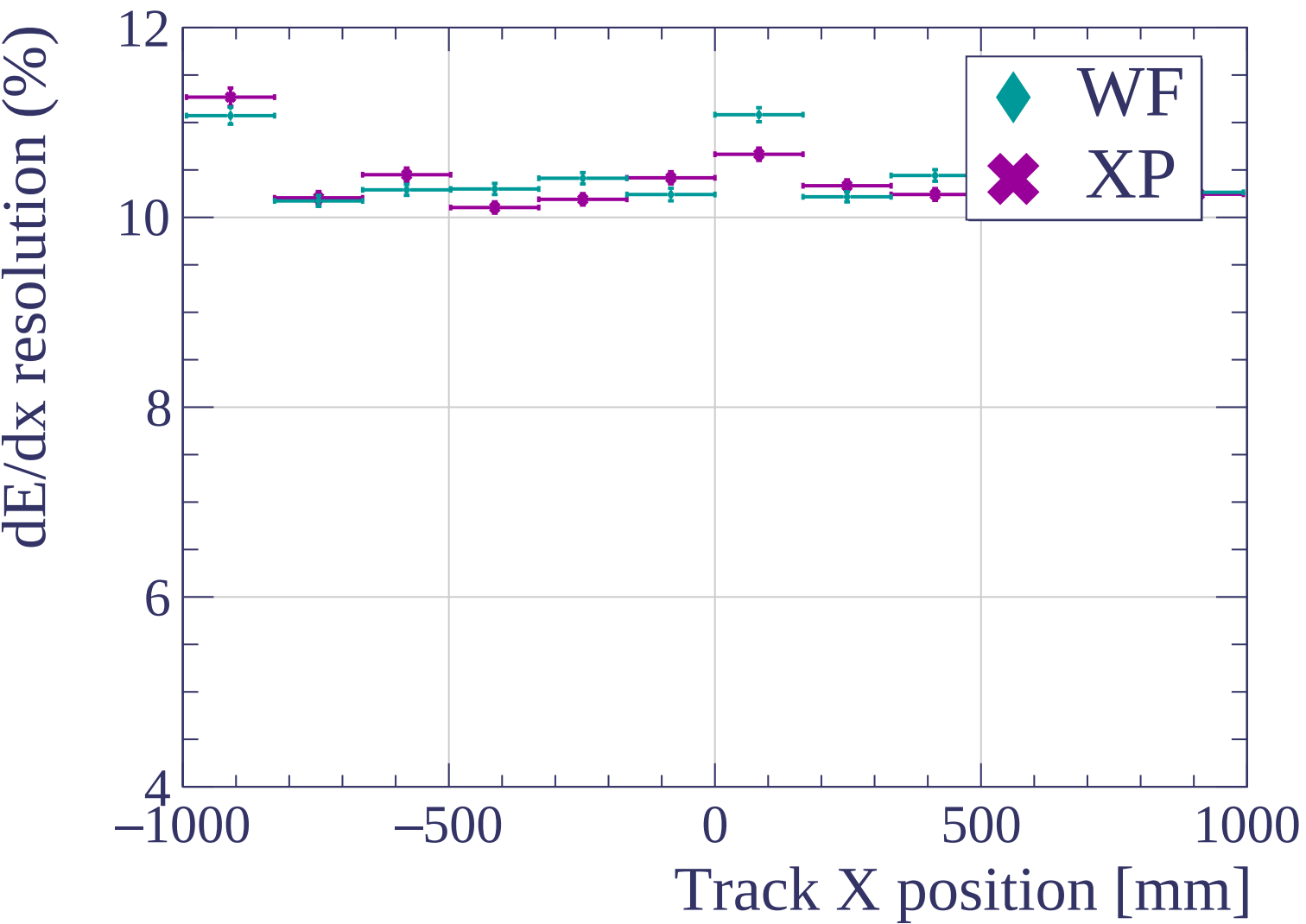


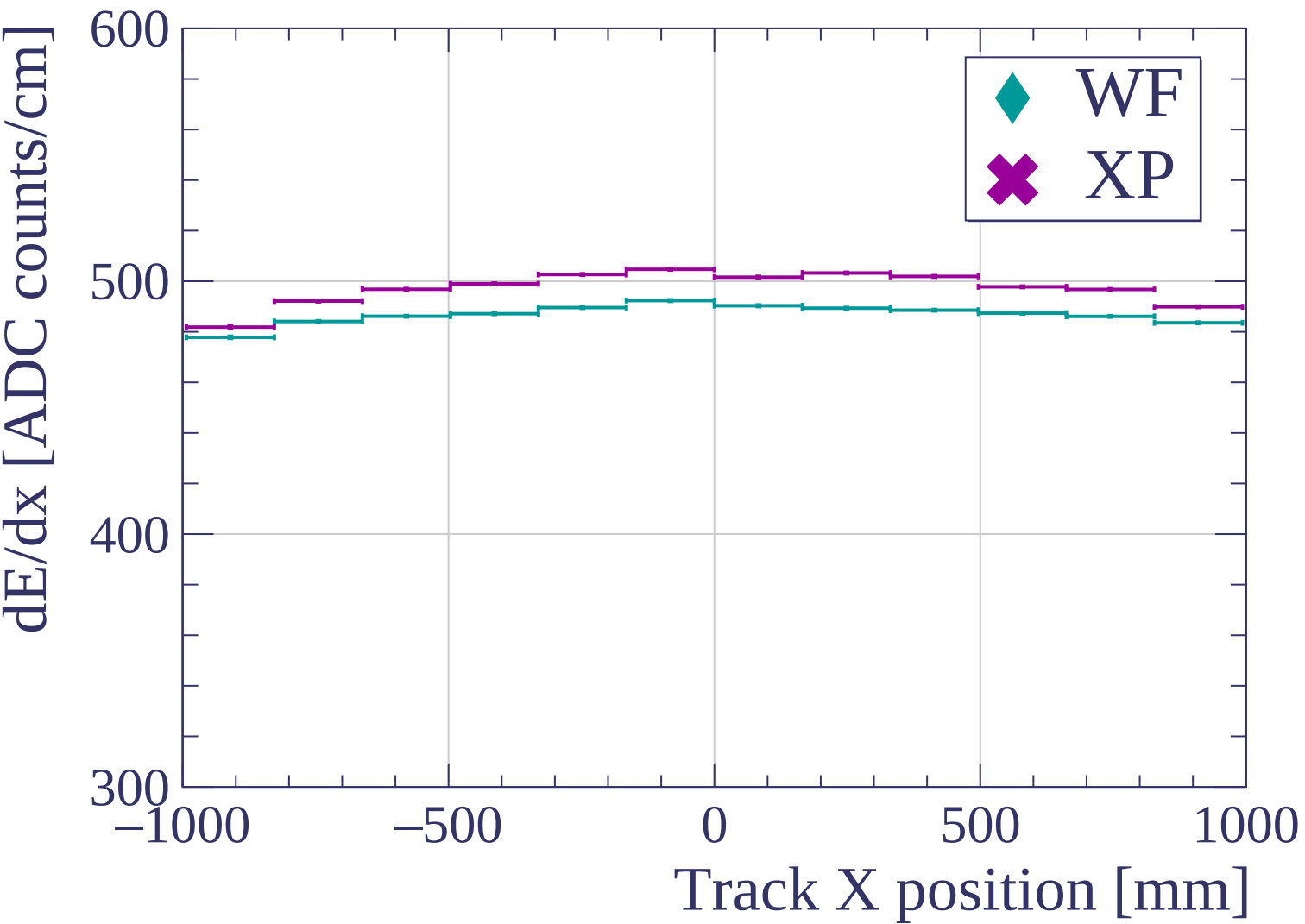


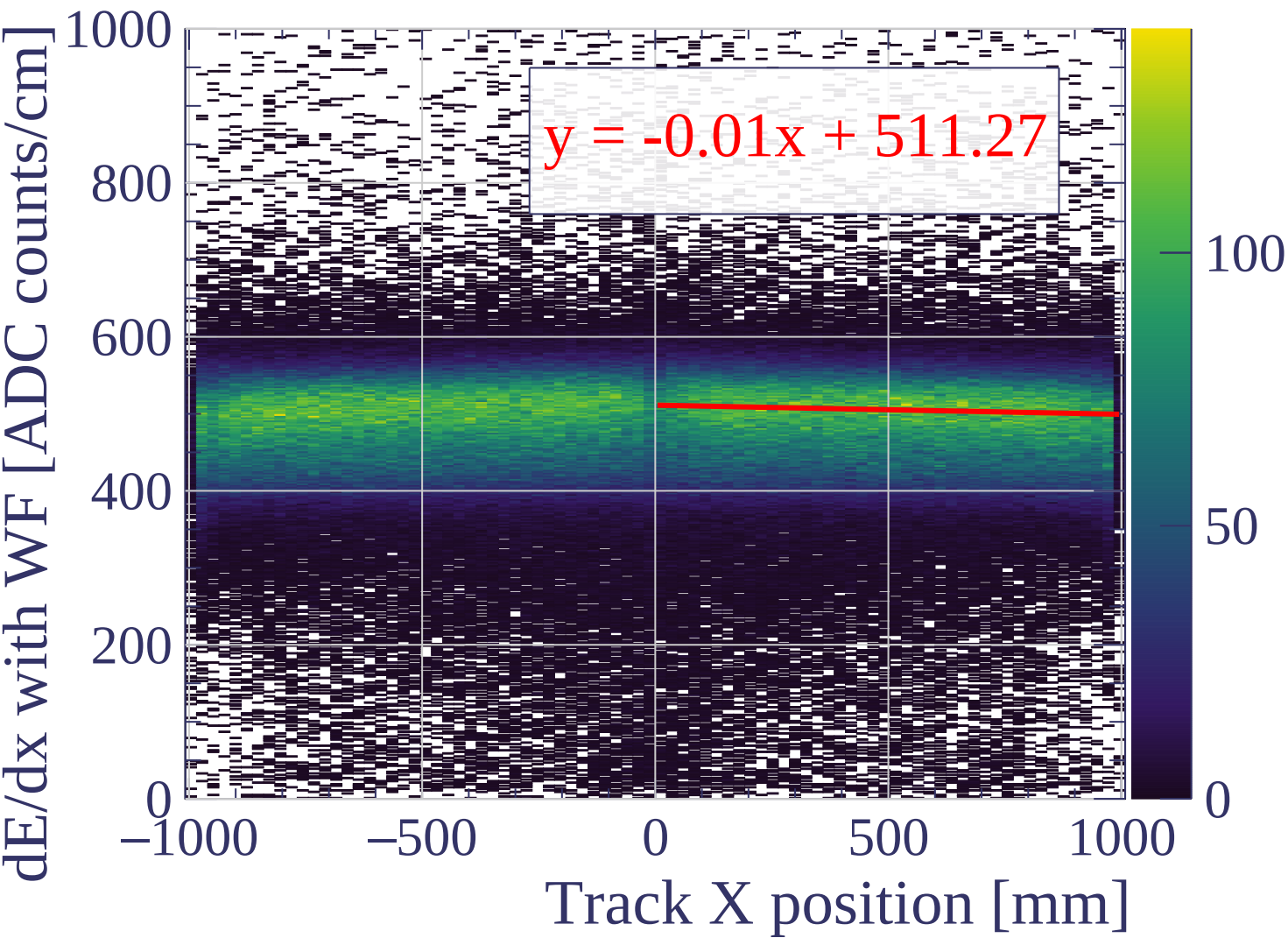


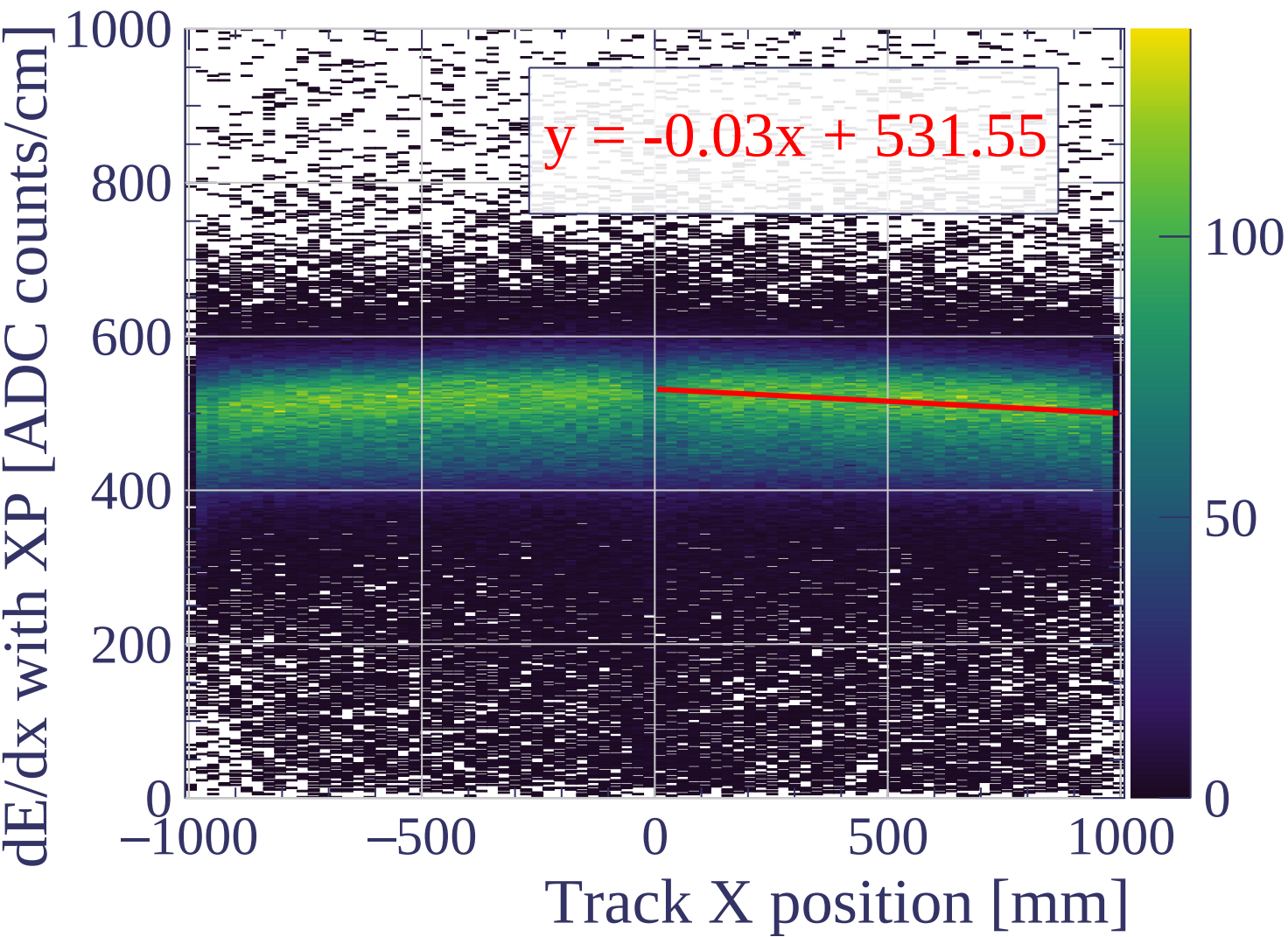


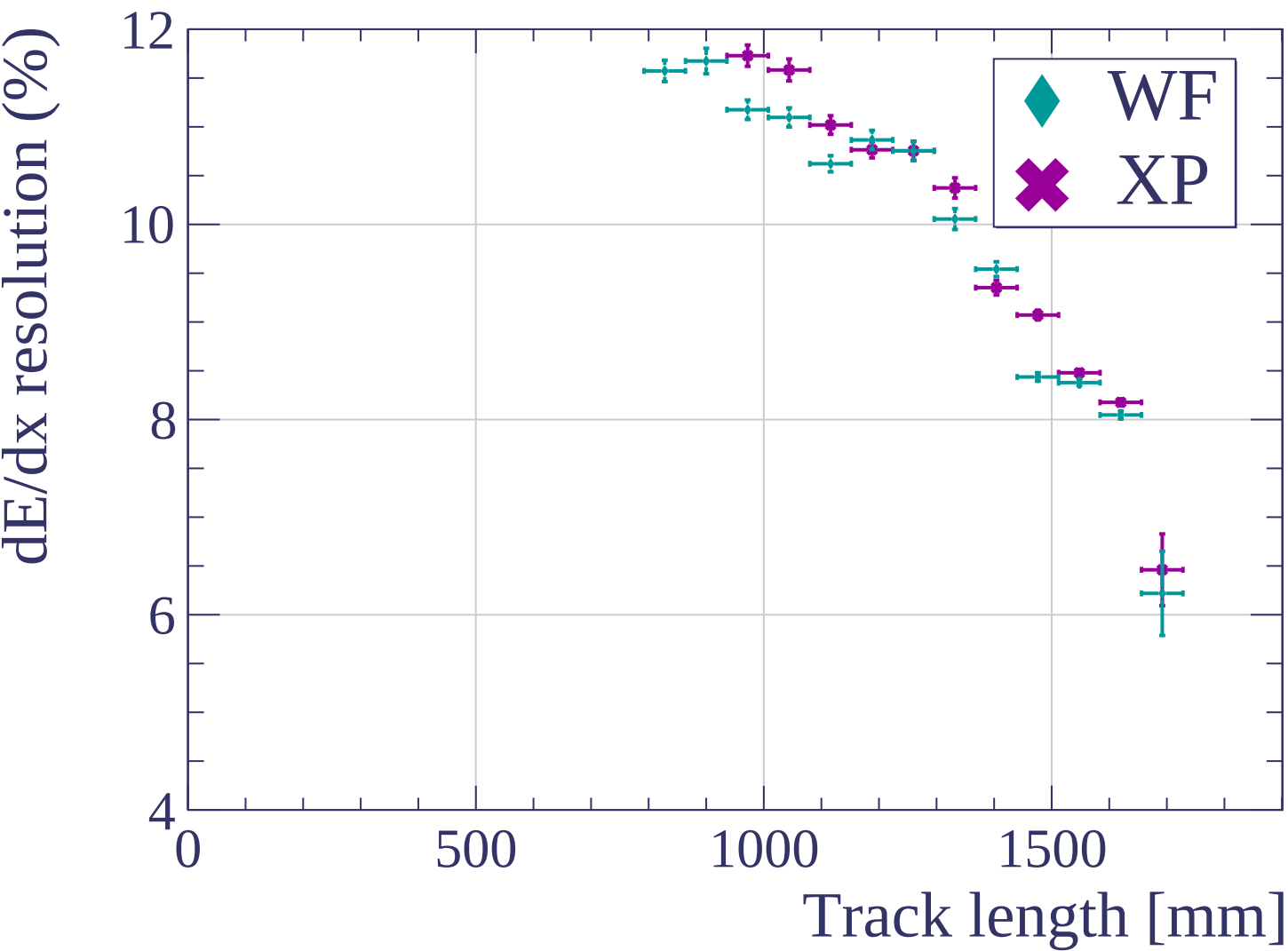


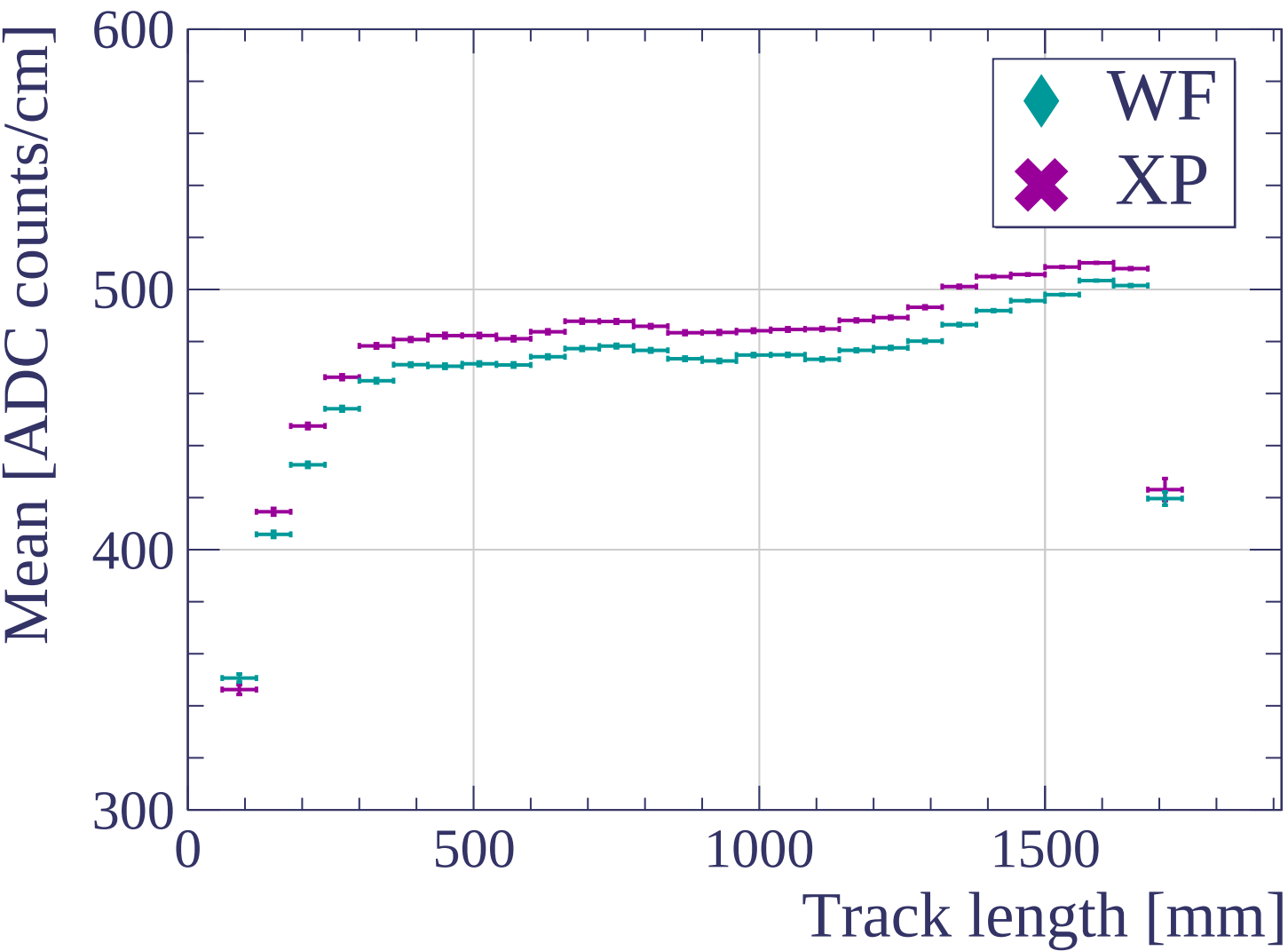


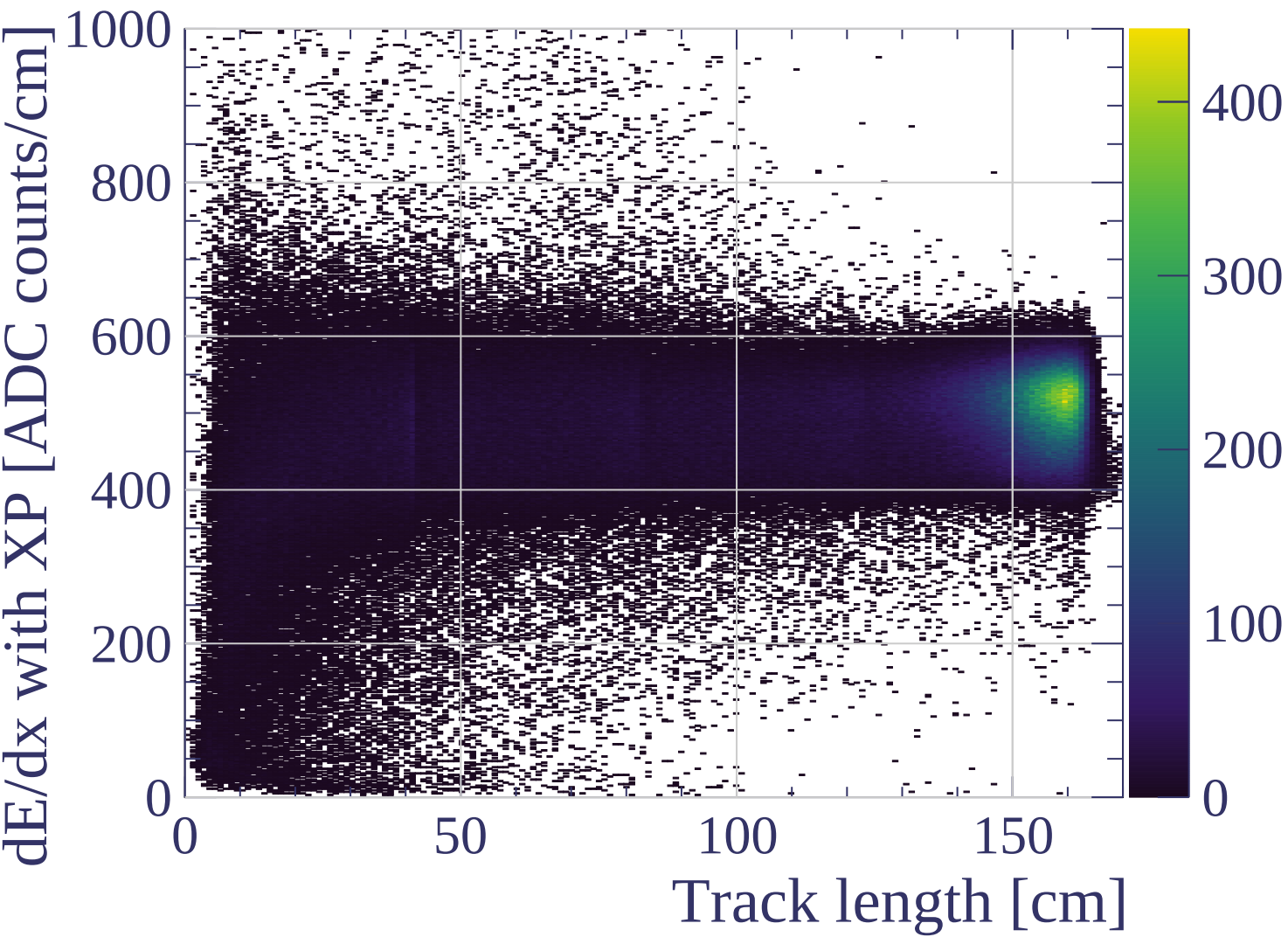


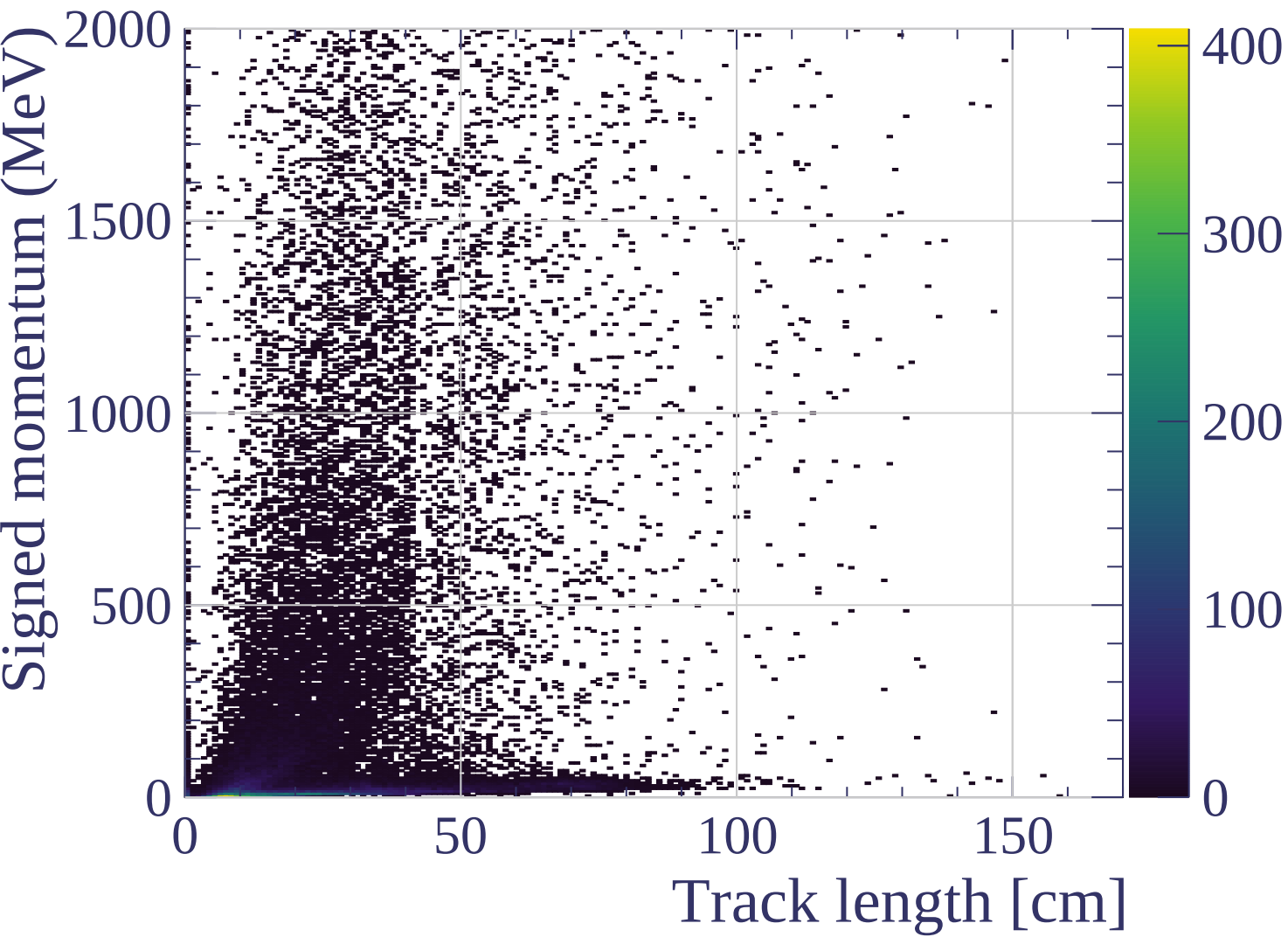


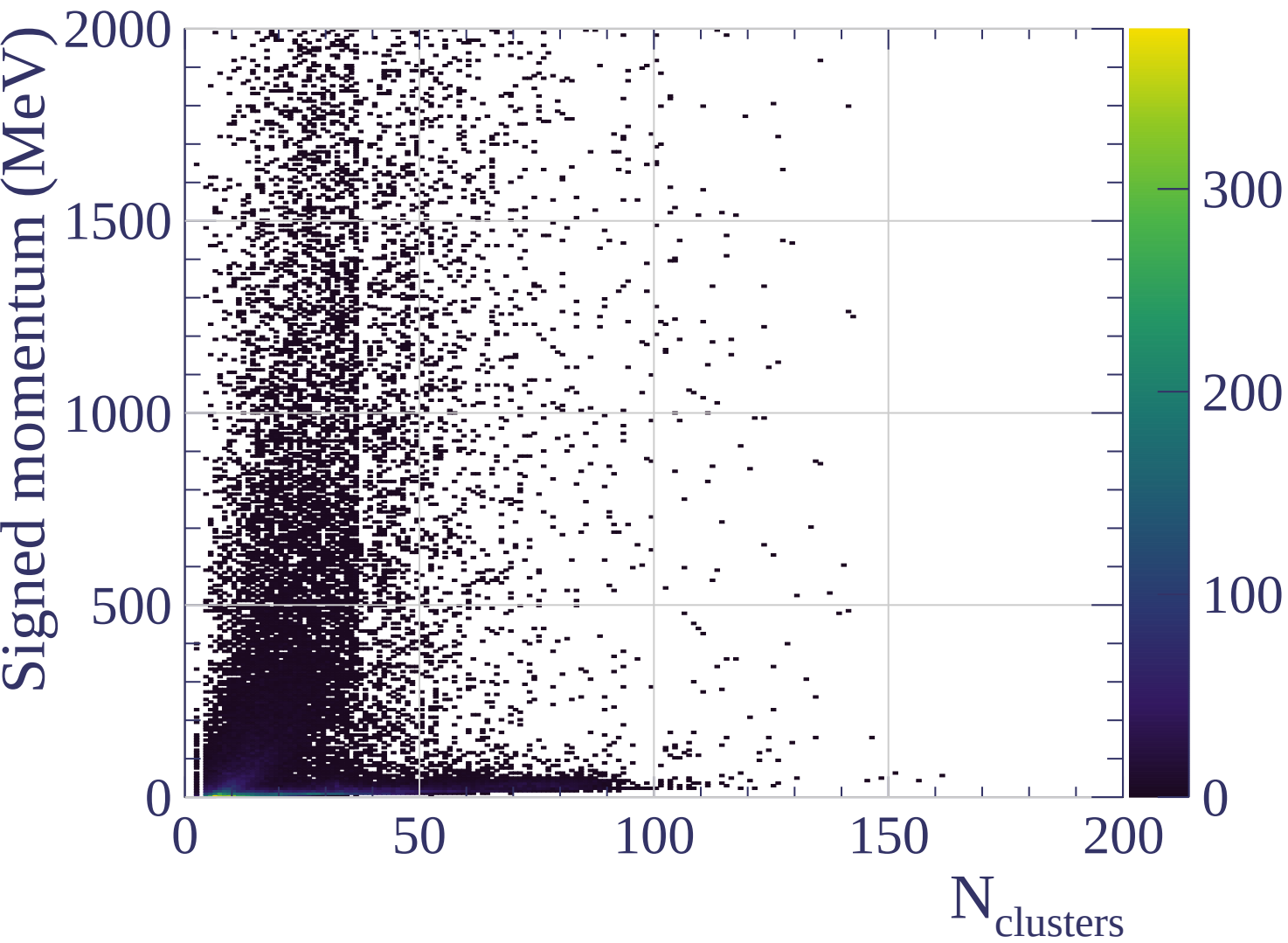




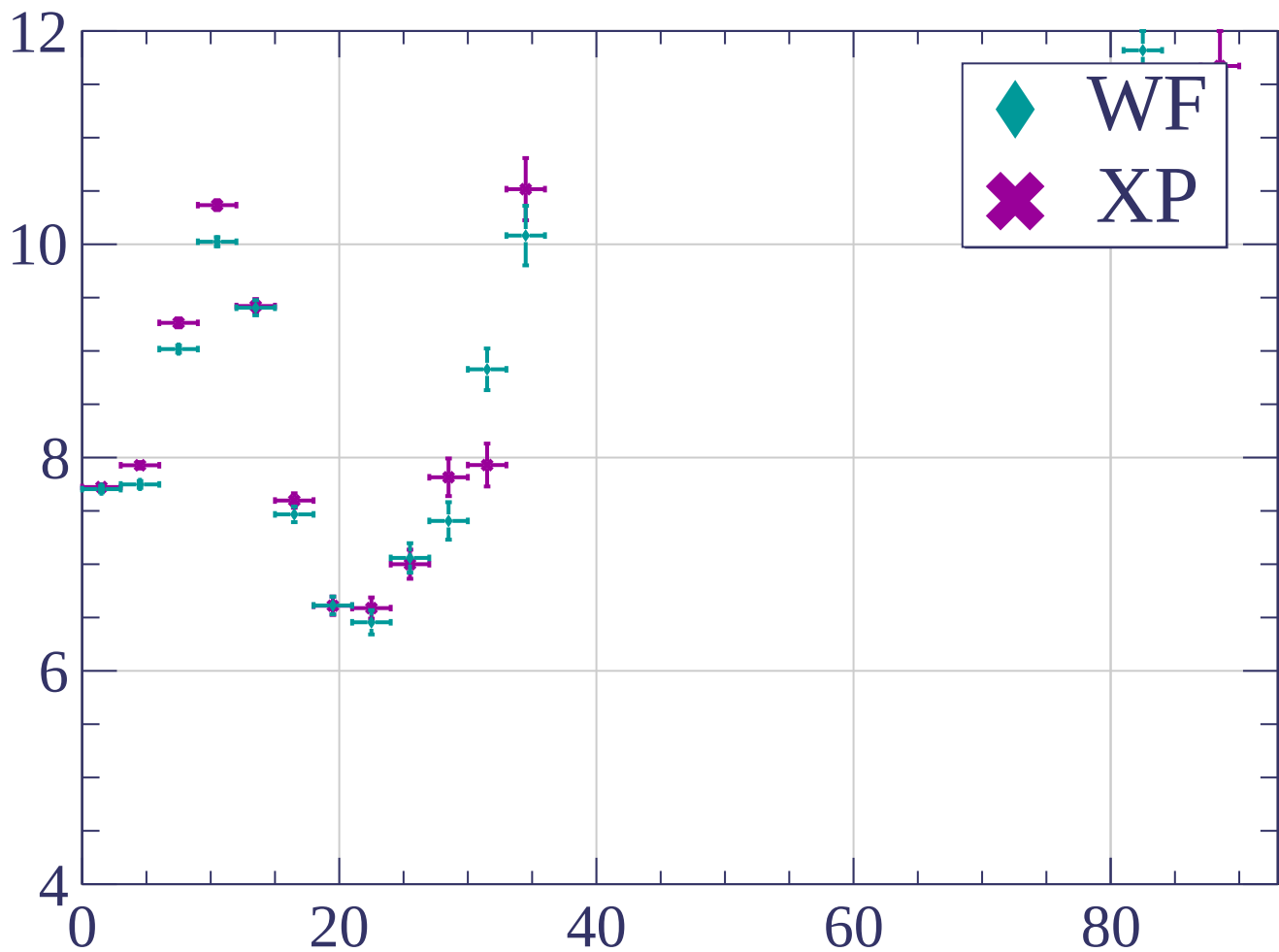




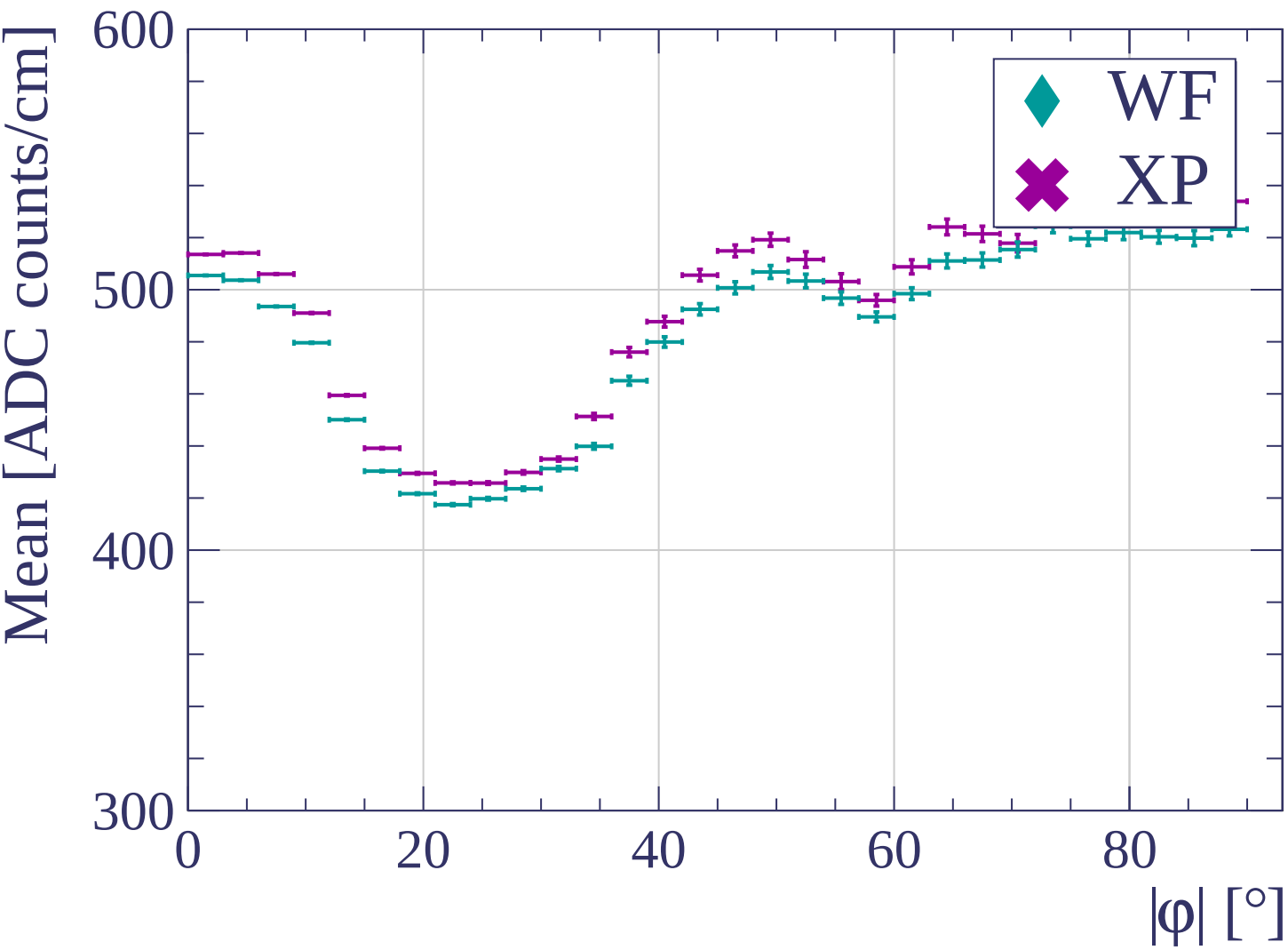


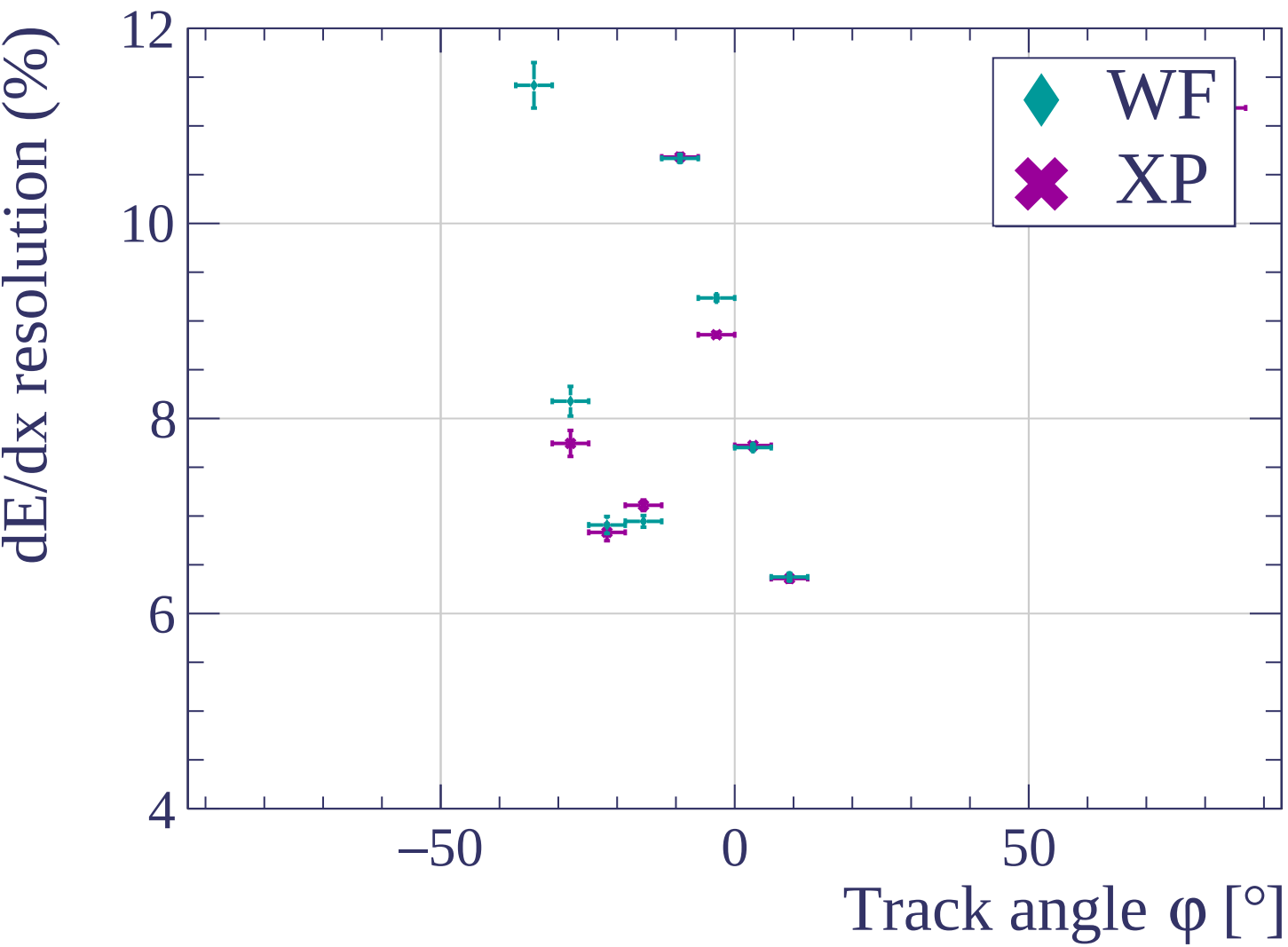


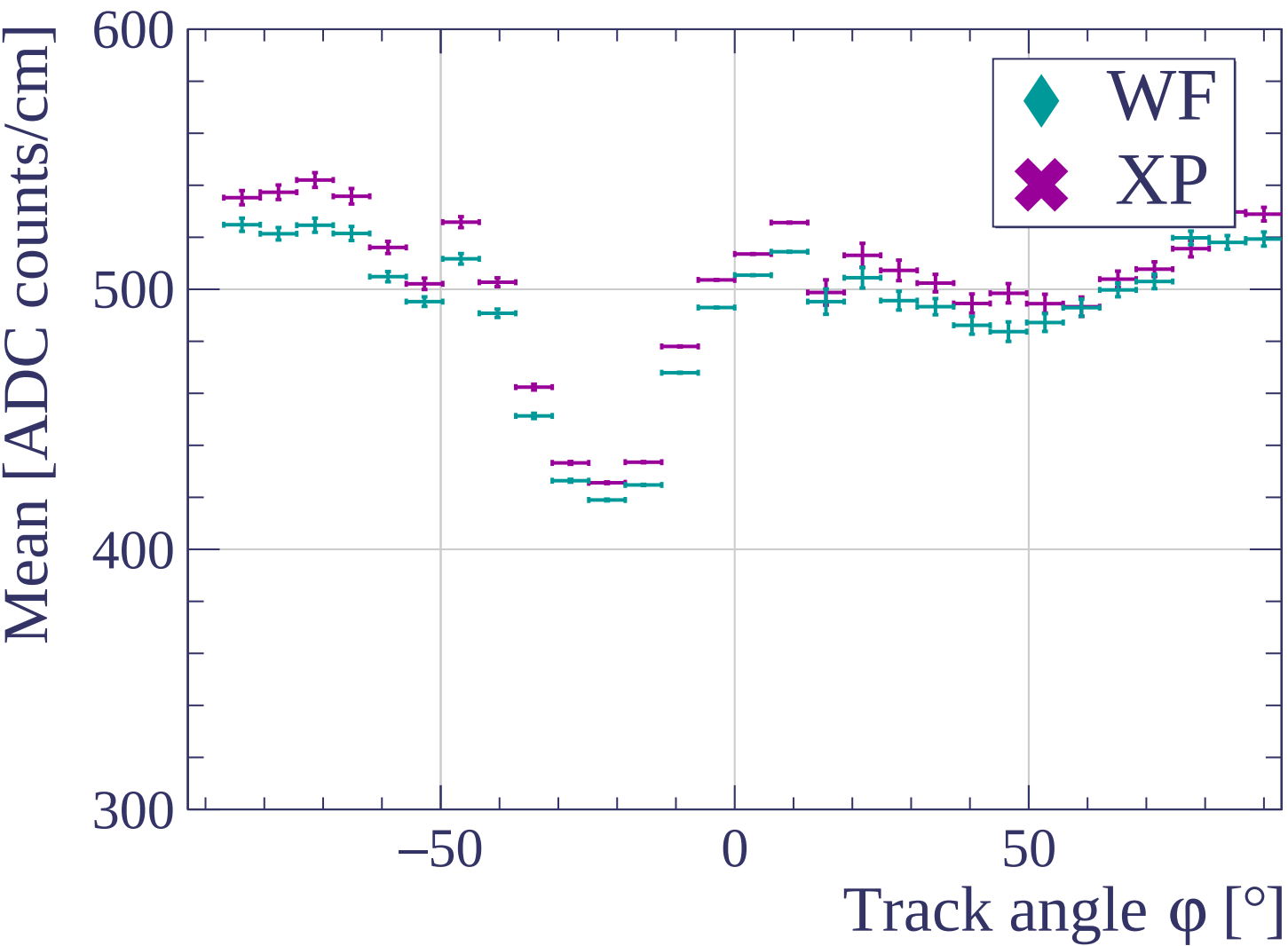
$dE/dx$  resolution (%)

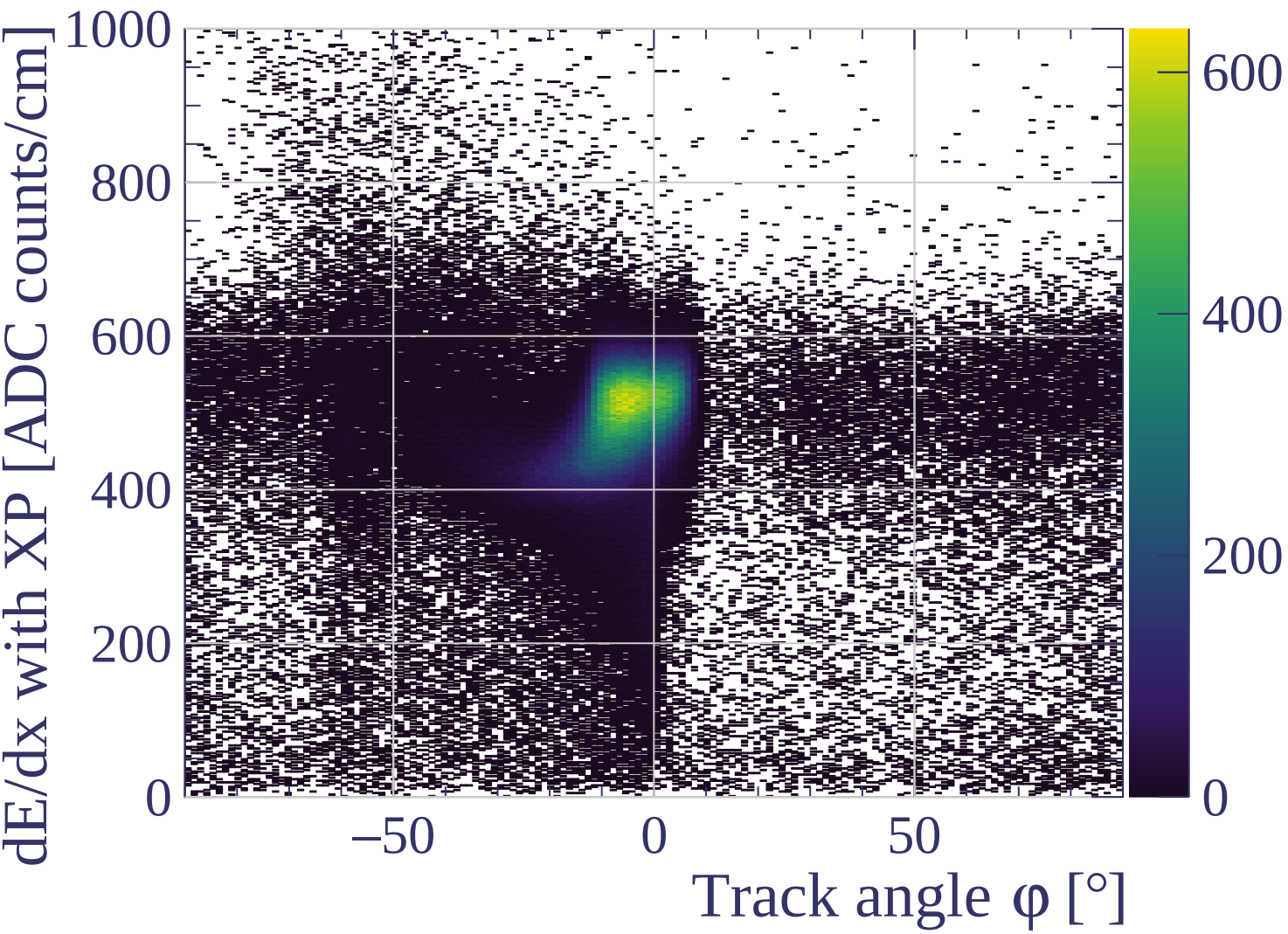


Track angle  $\phi$  [°]

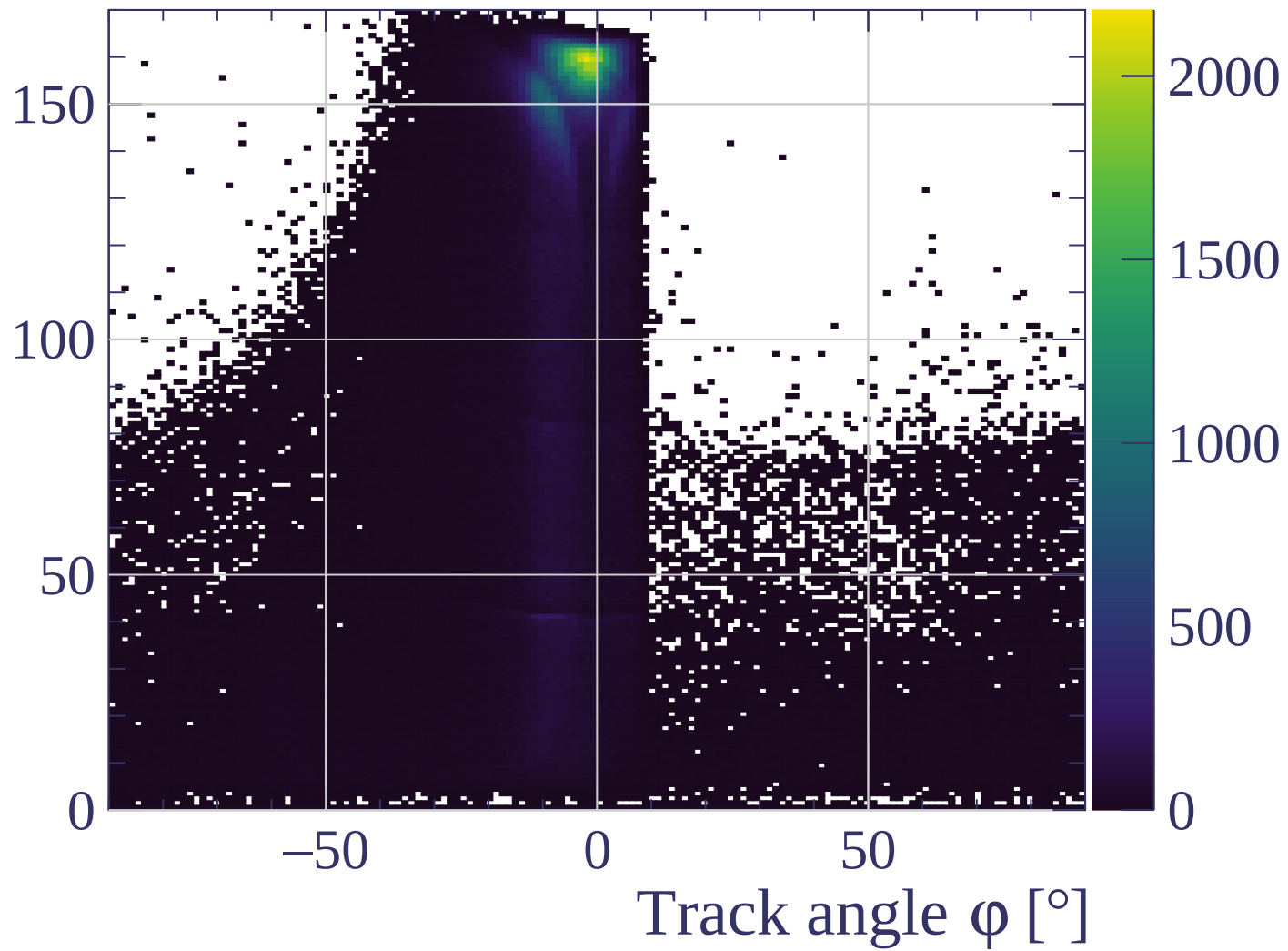


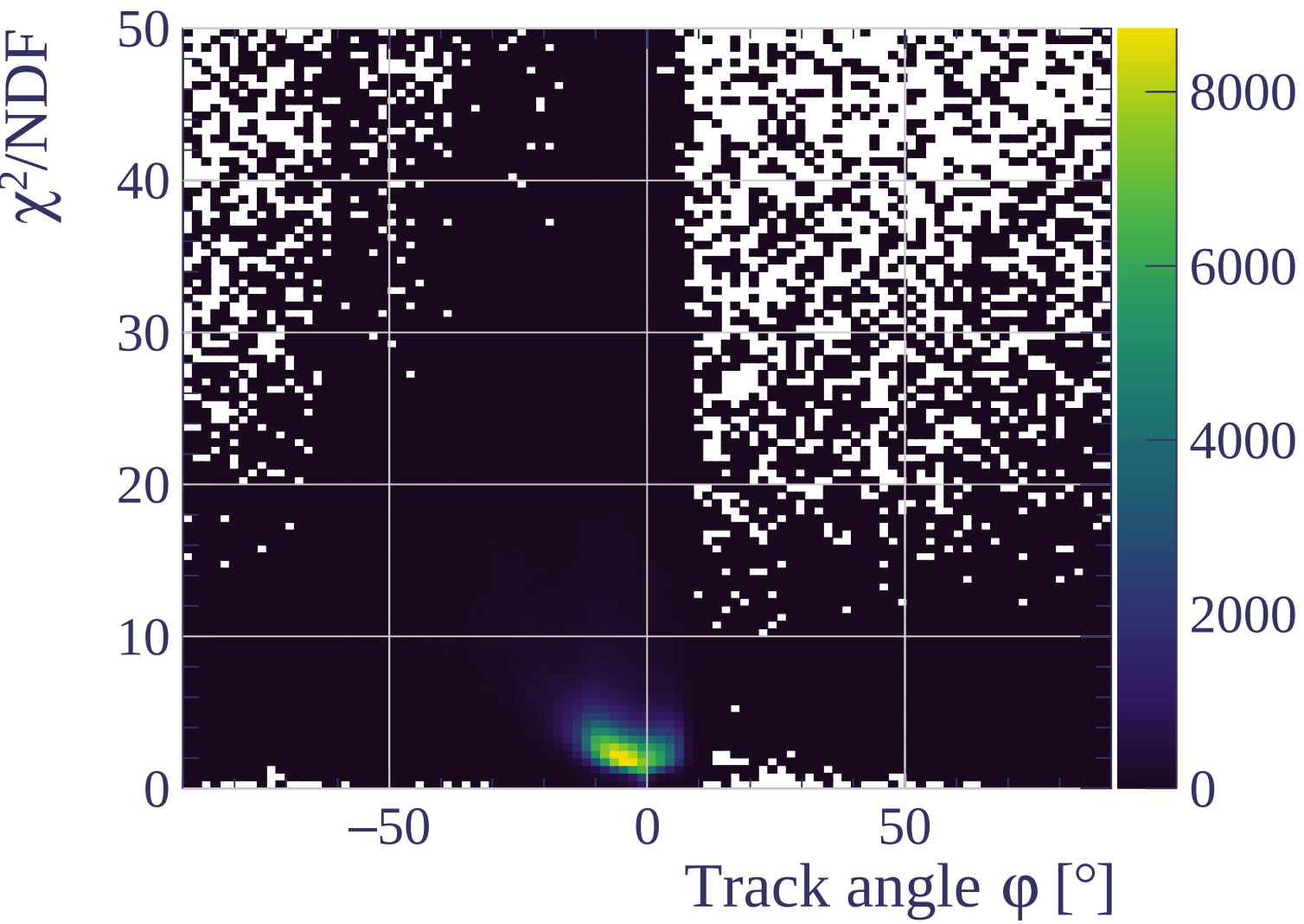


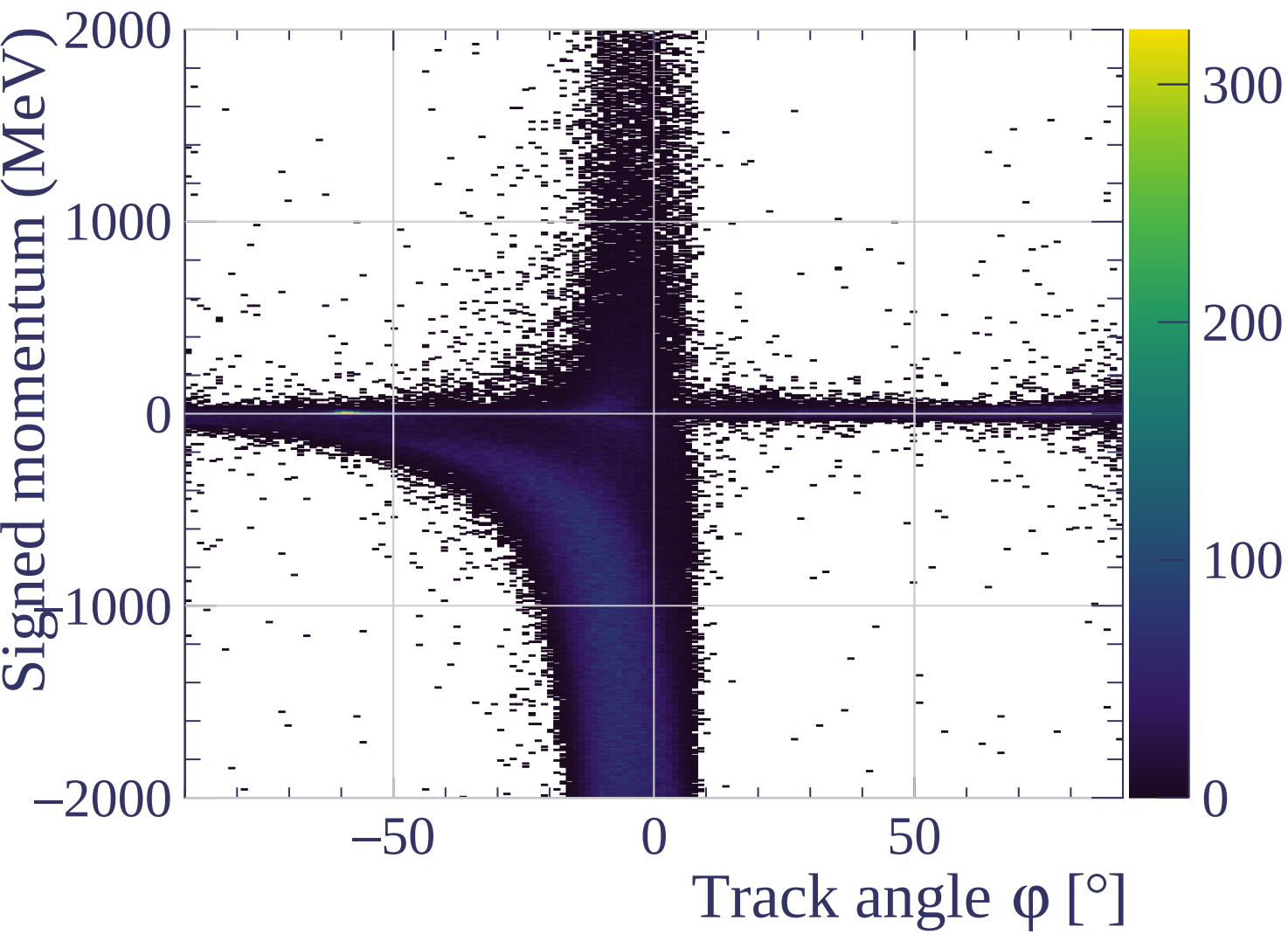


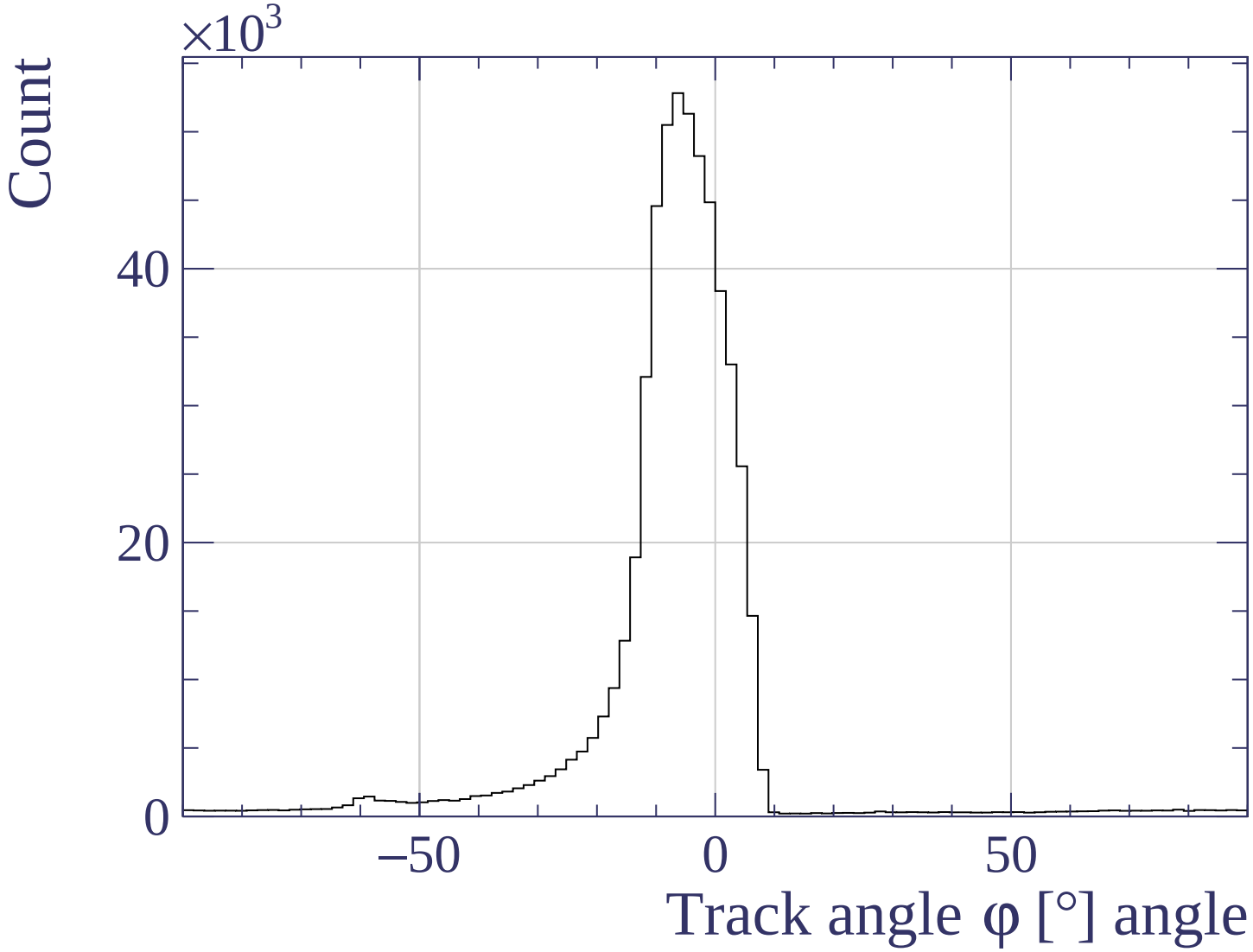


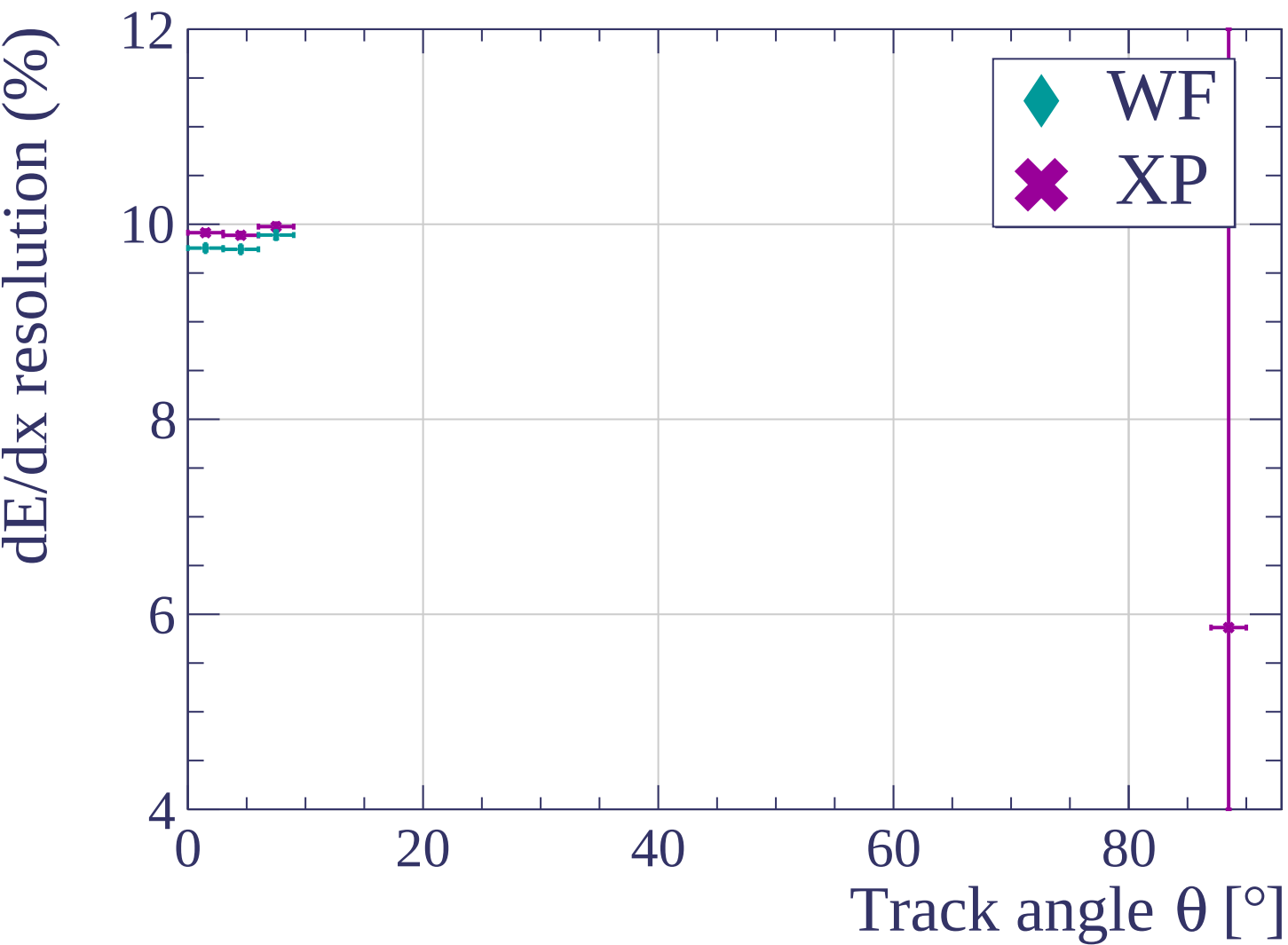
Track length [cm]

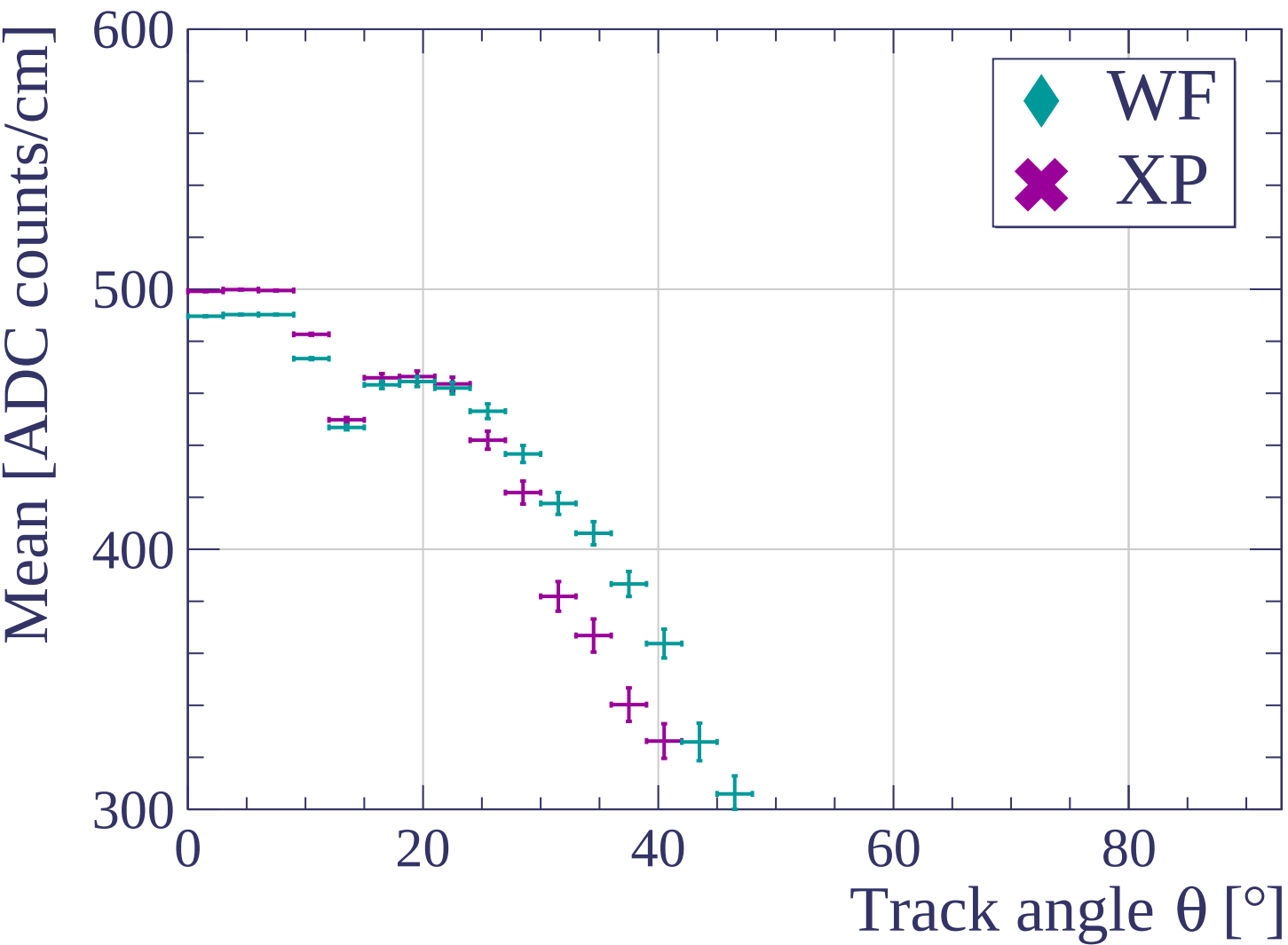


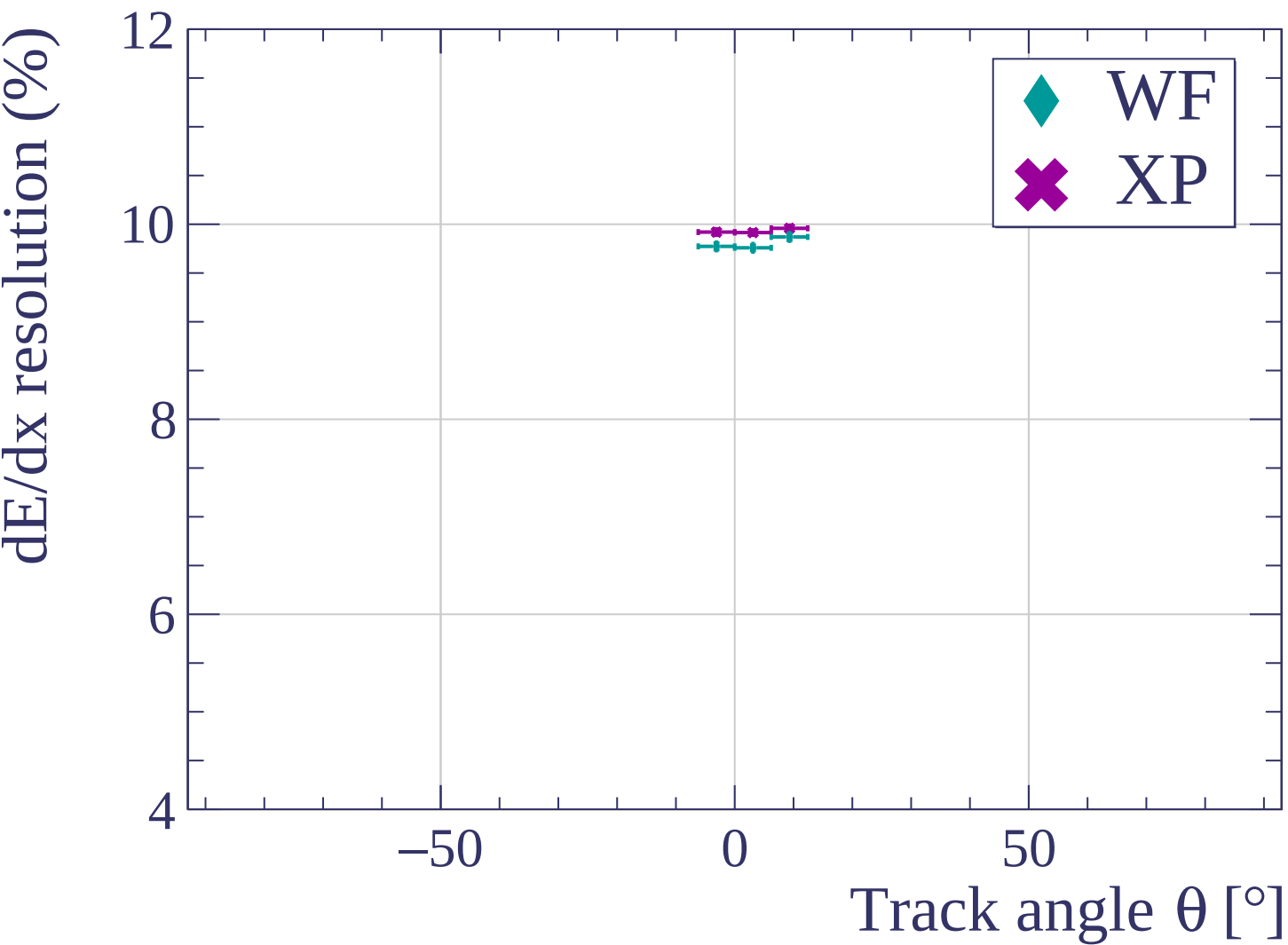


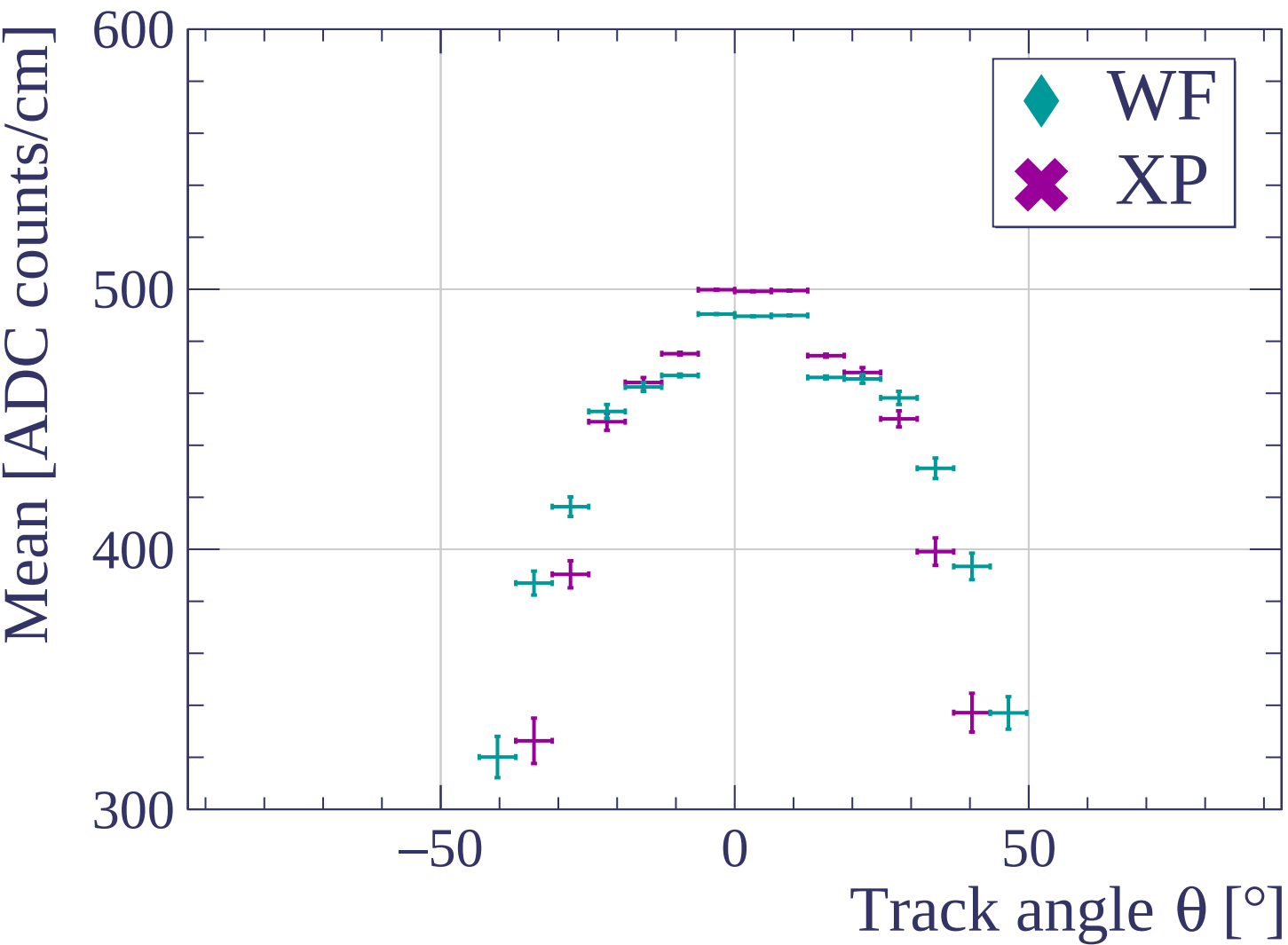


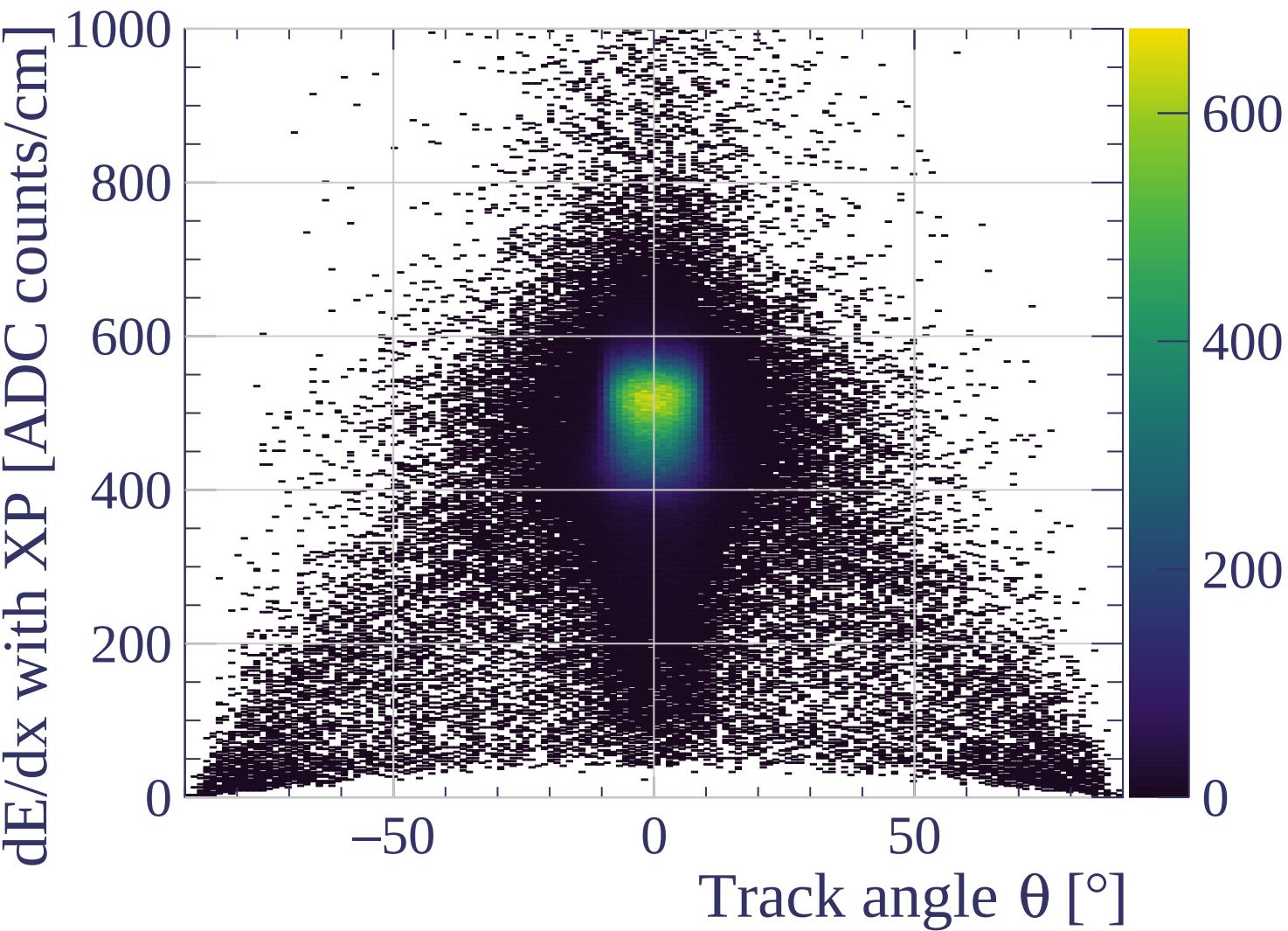




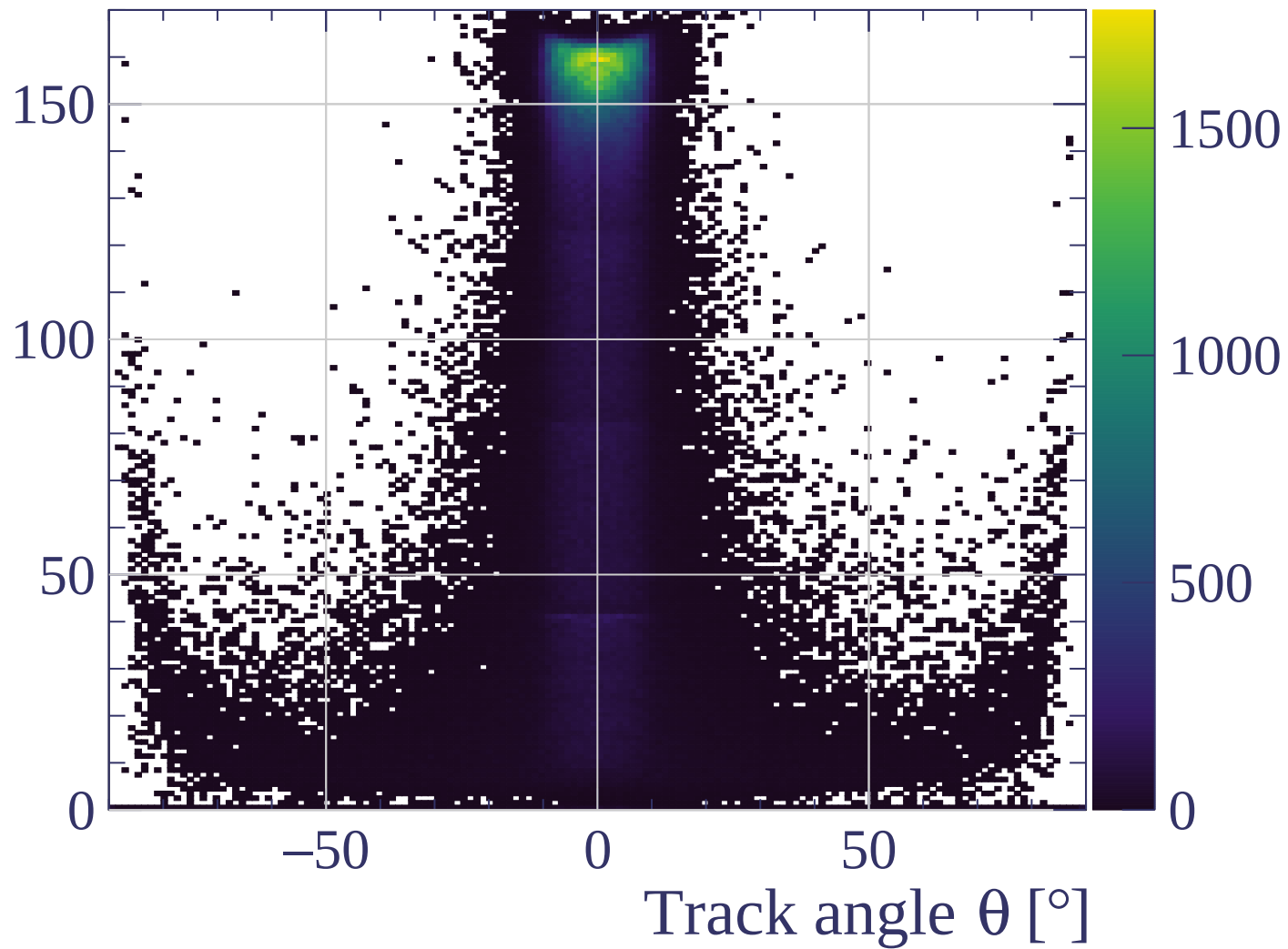


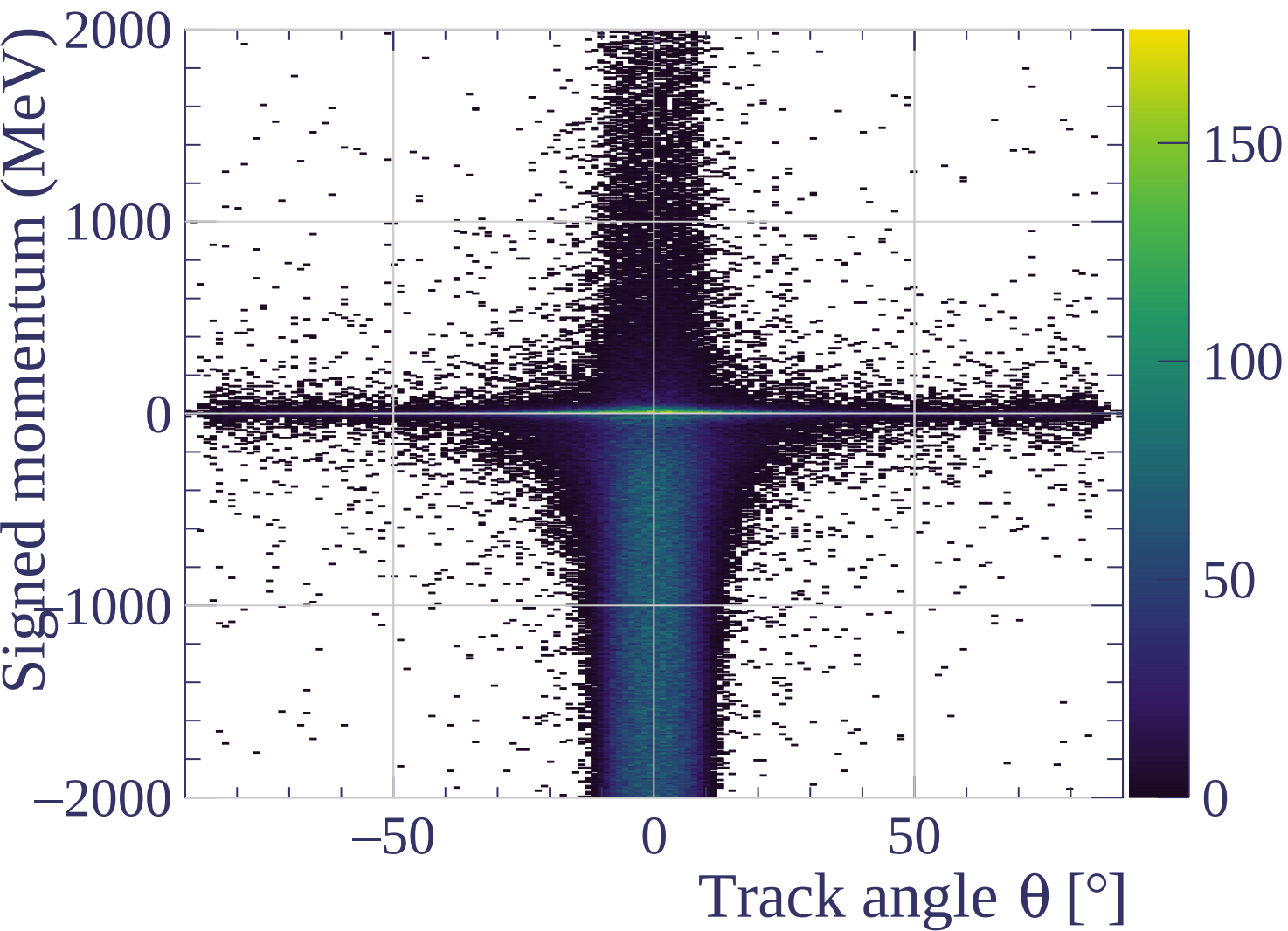


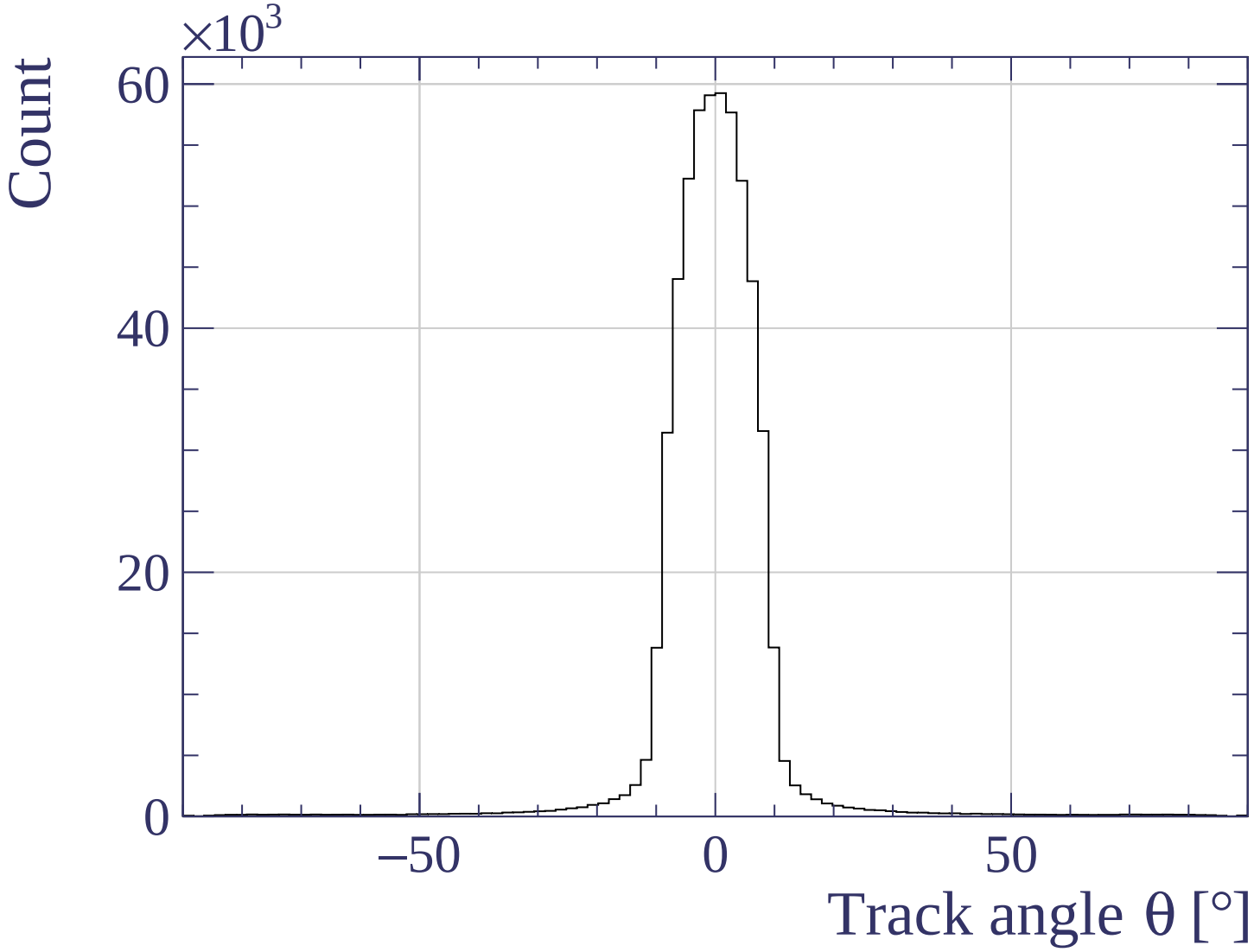




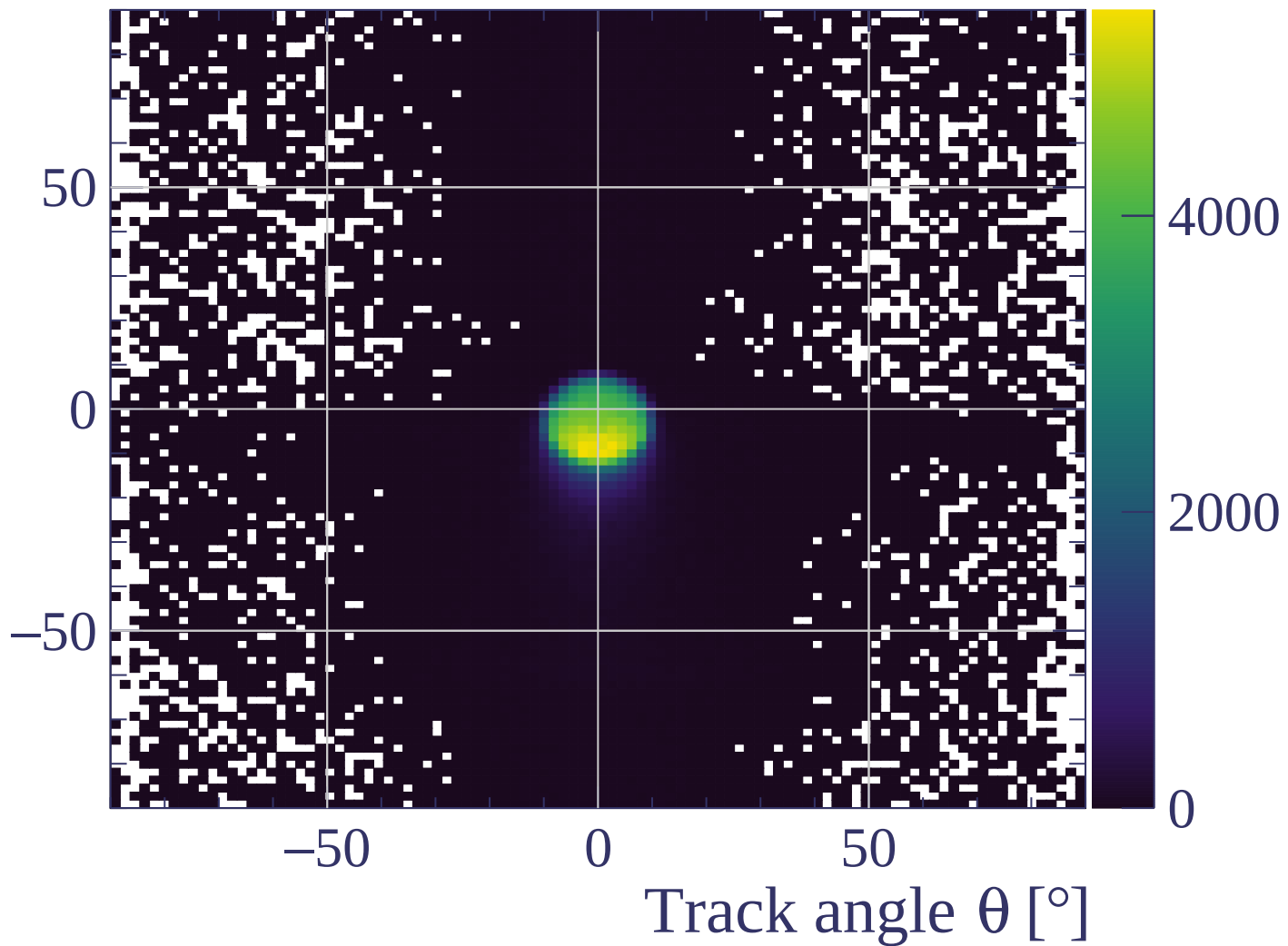
Track length [cm]



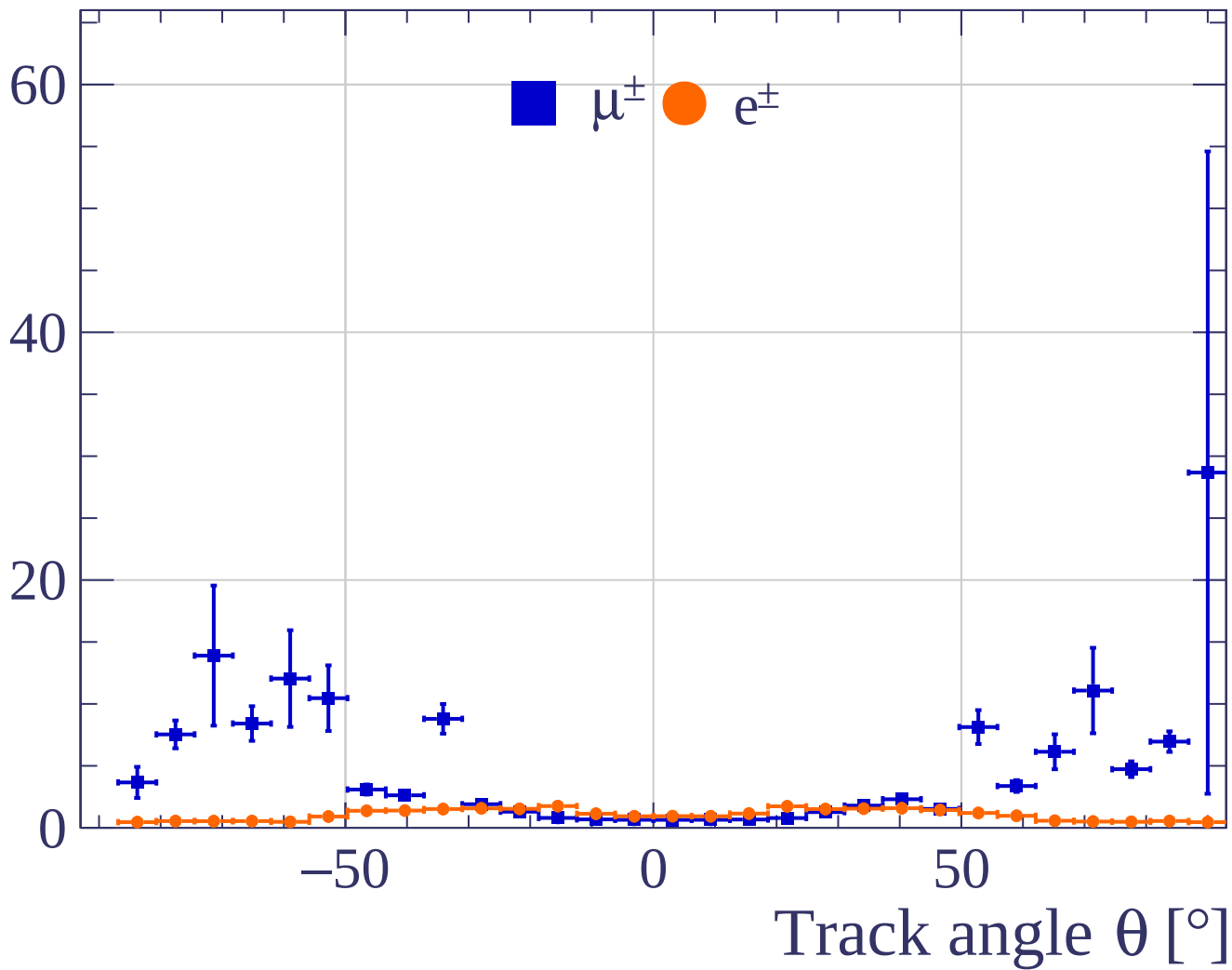




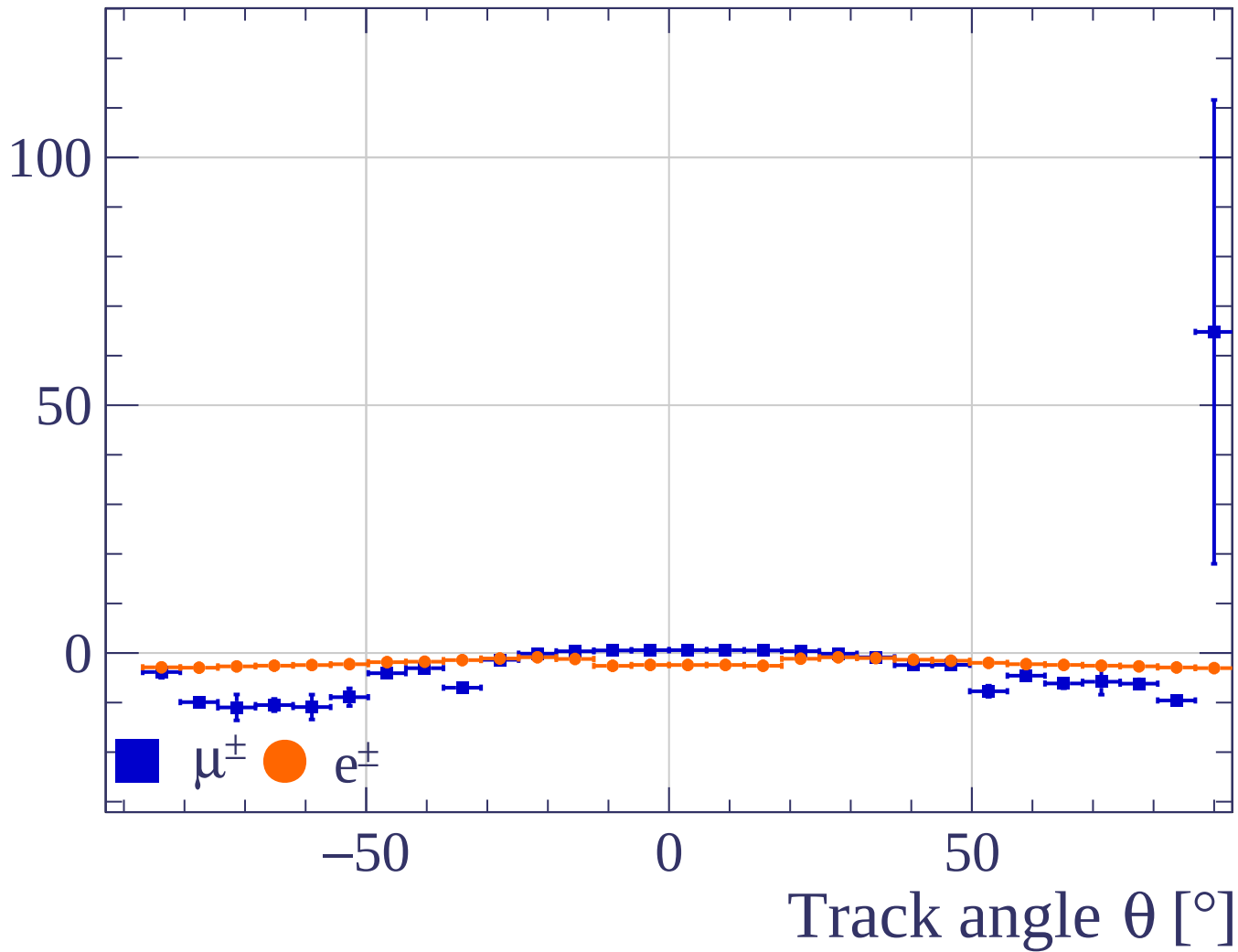
Track angle  $\phi$  [°]



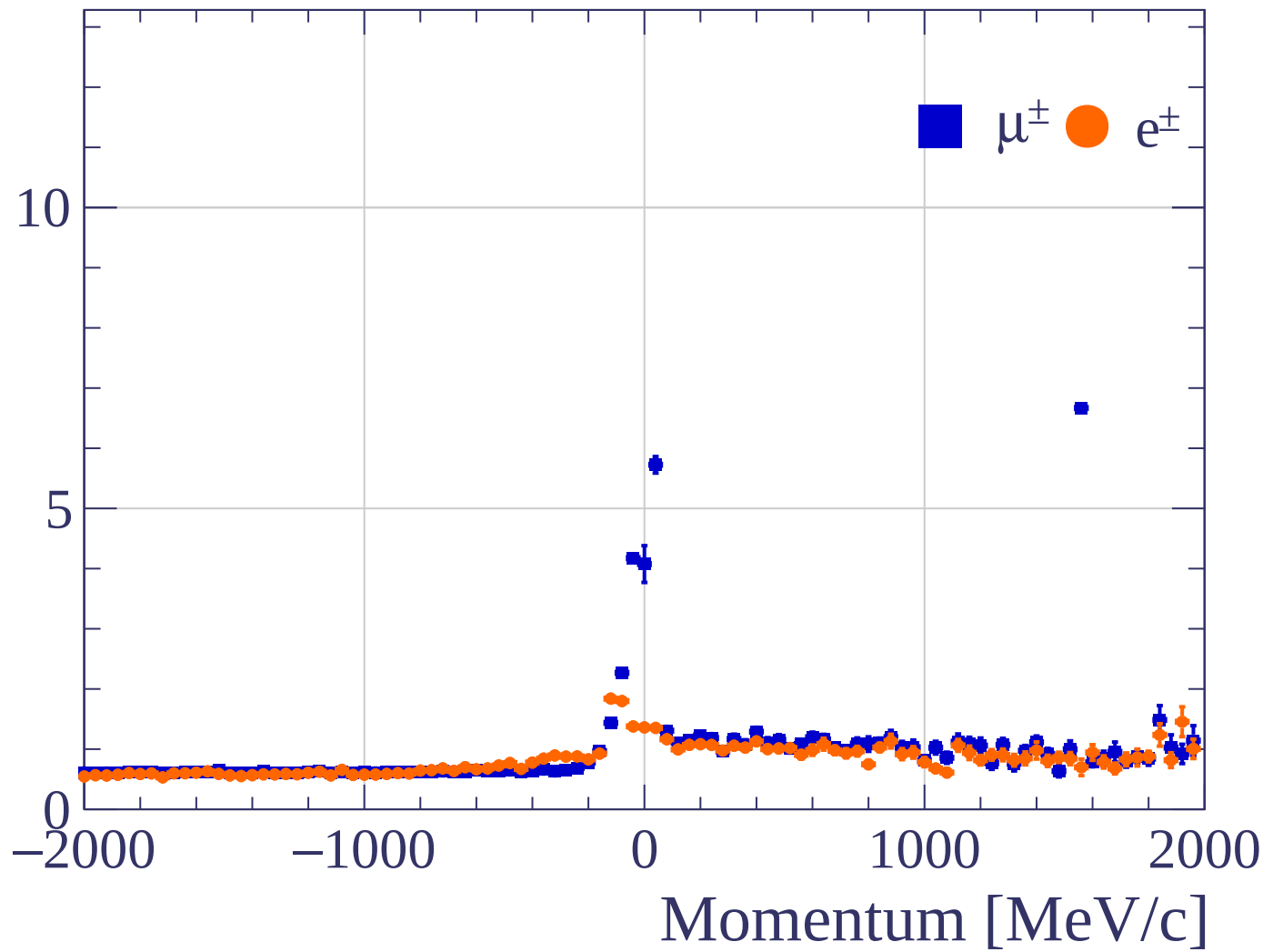
Pulls standard deviation



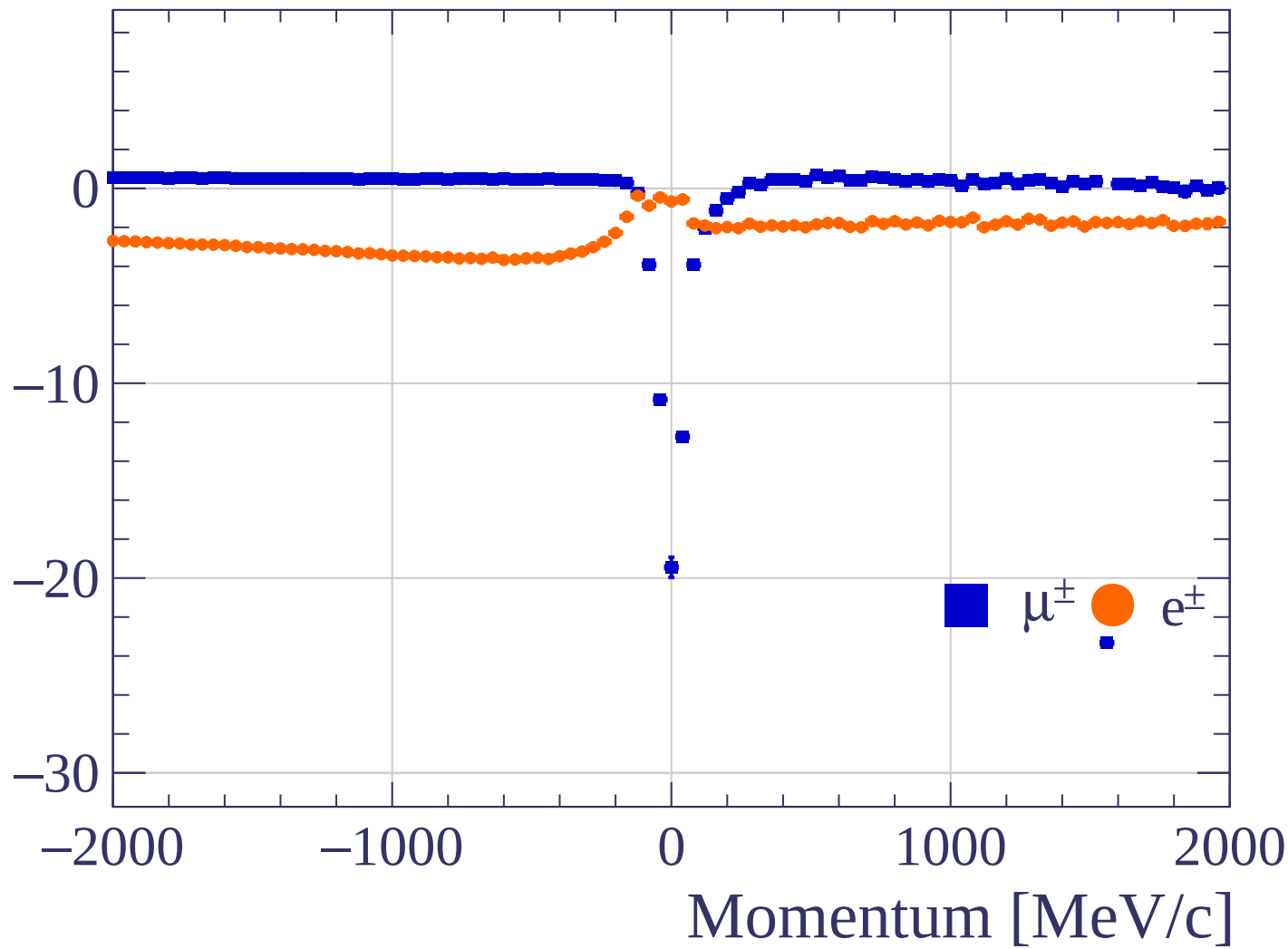
Pulls mean

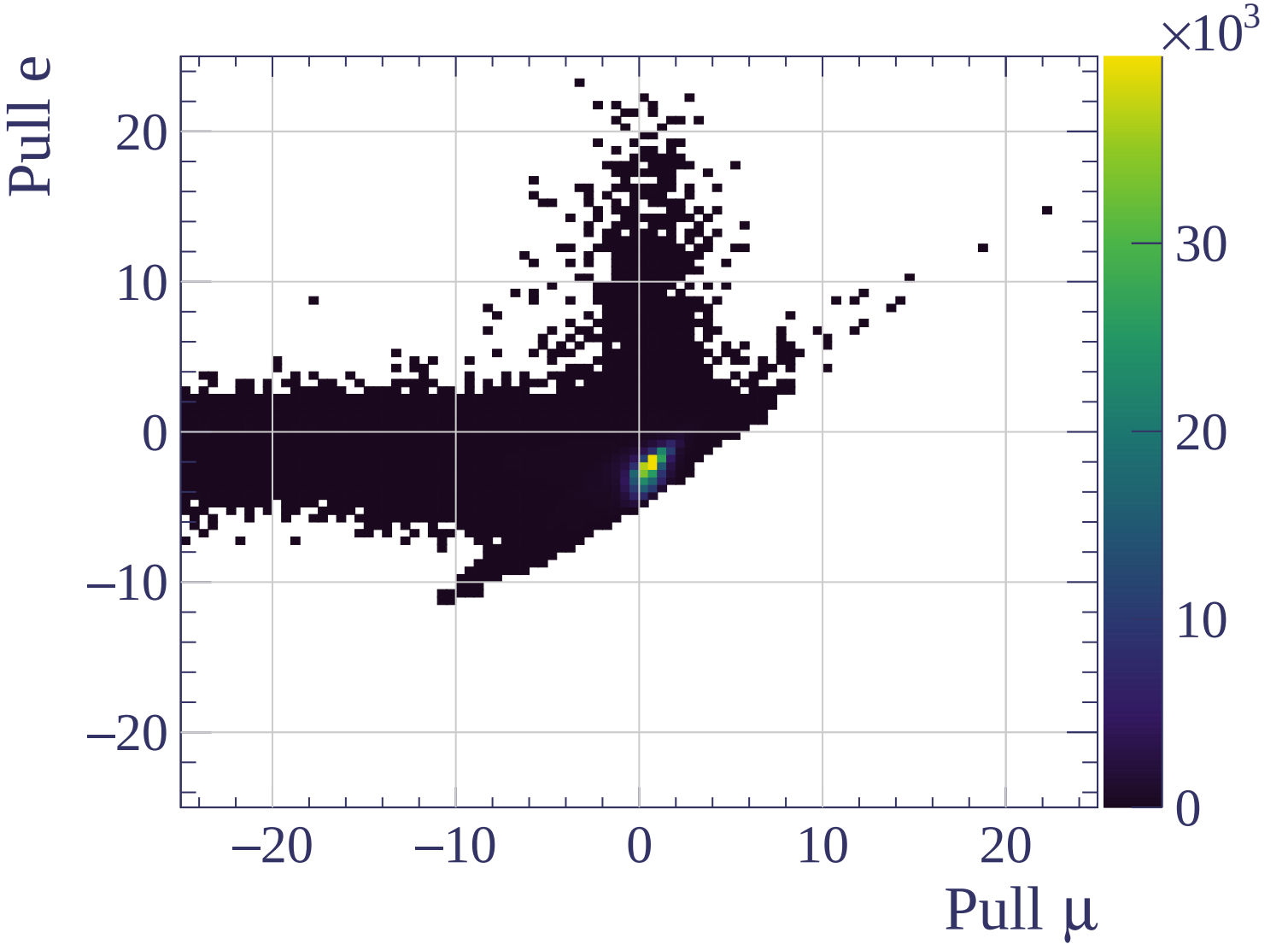


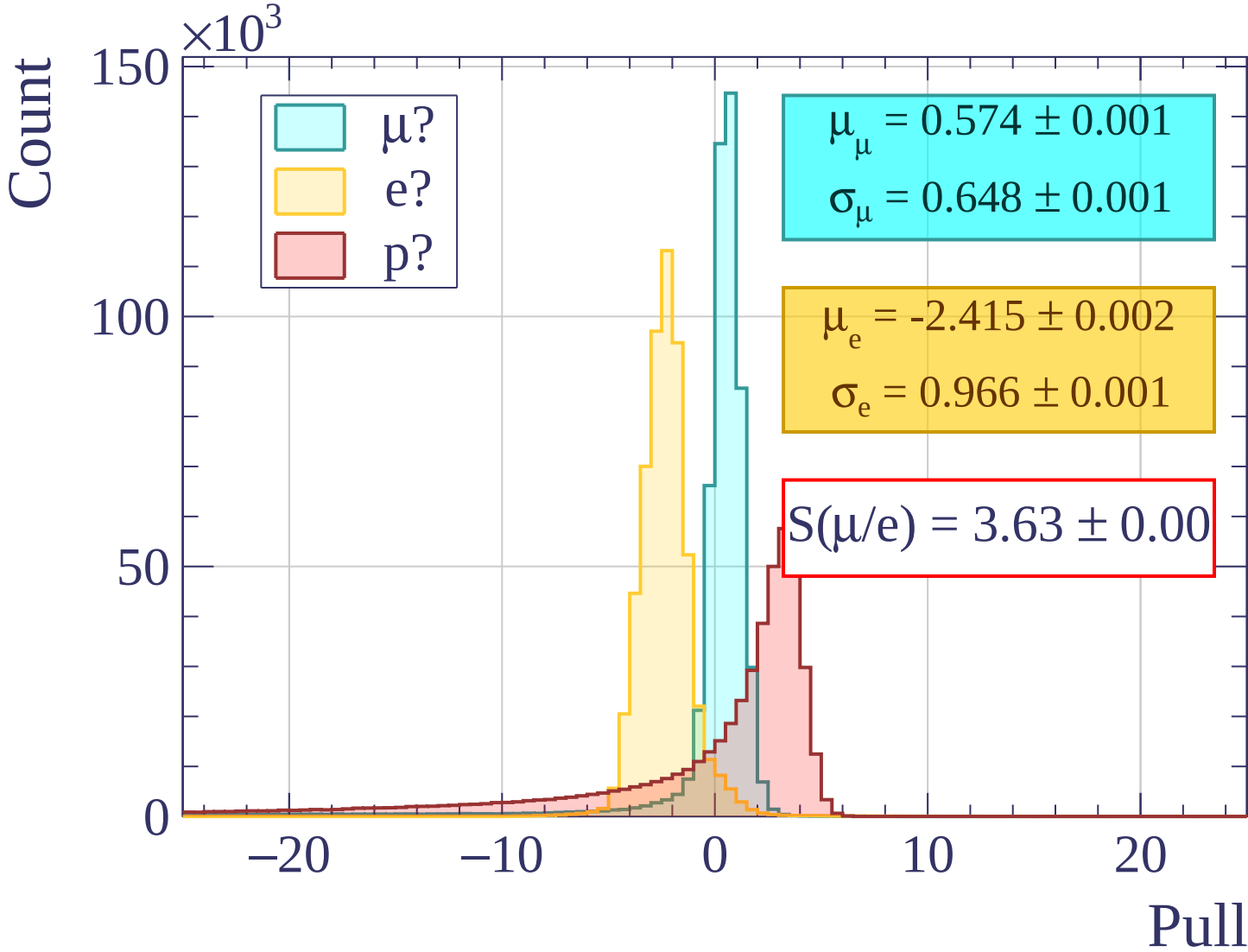
Pulls standard deviation

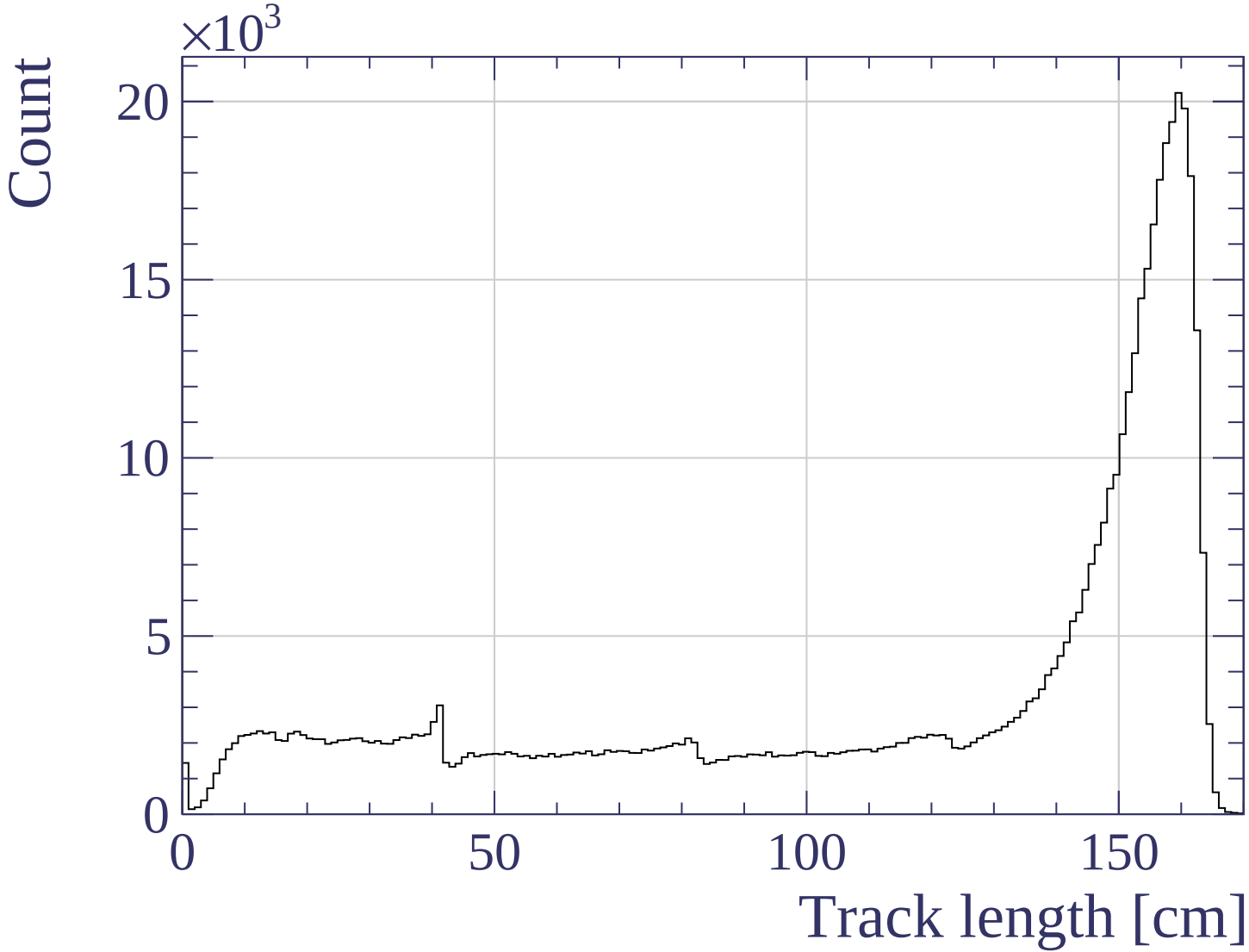


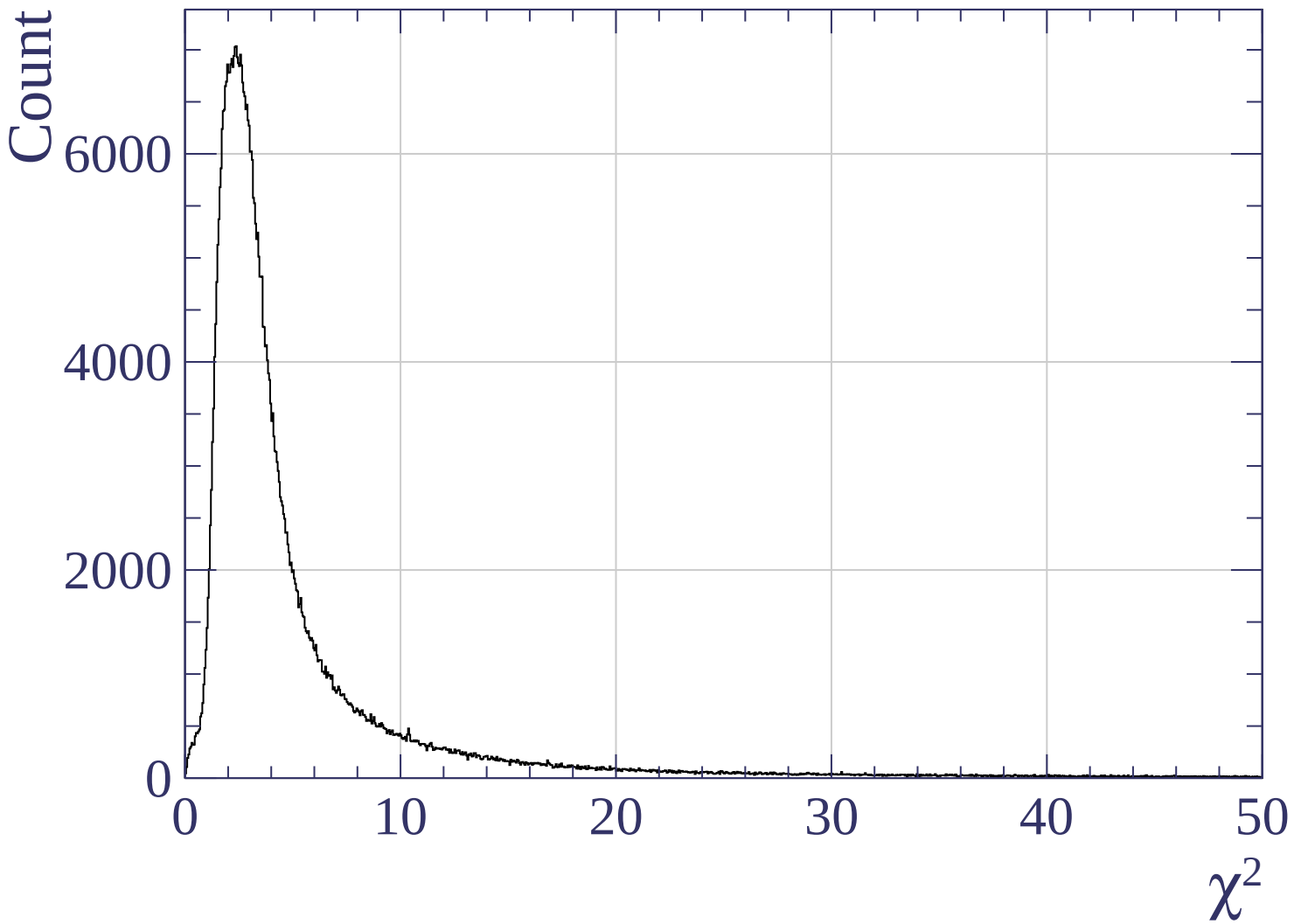
Pulls mean

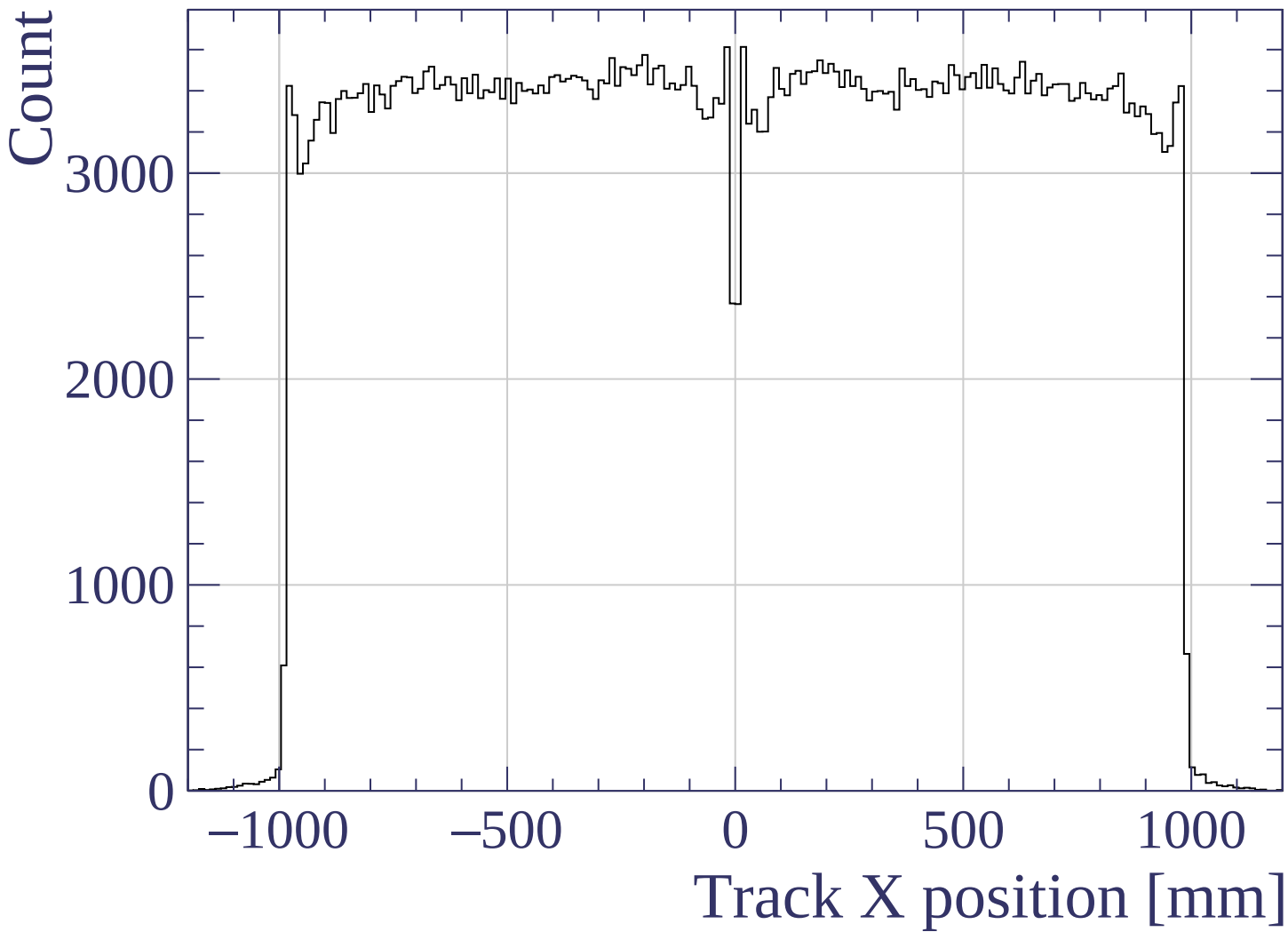


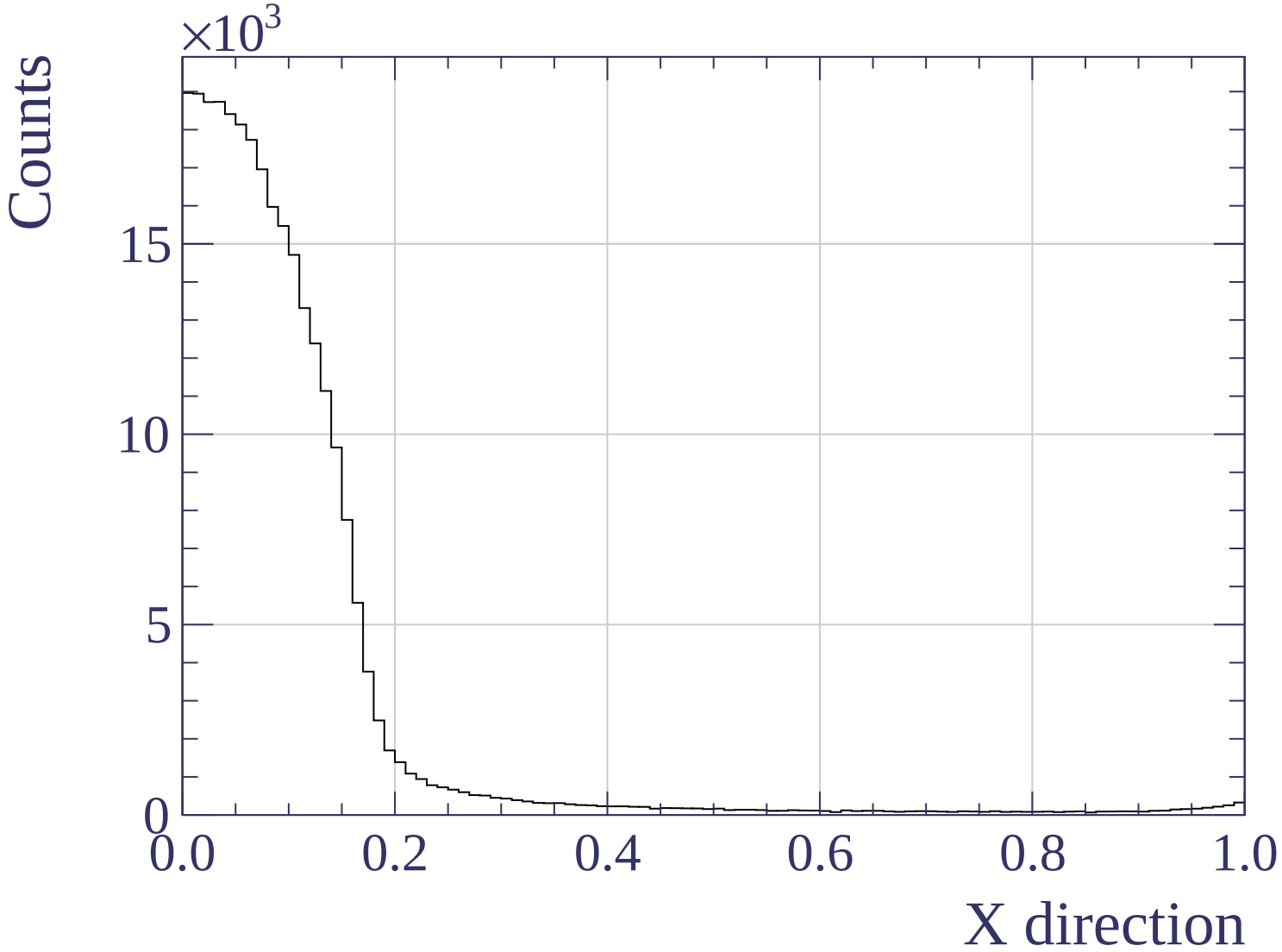


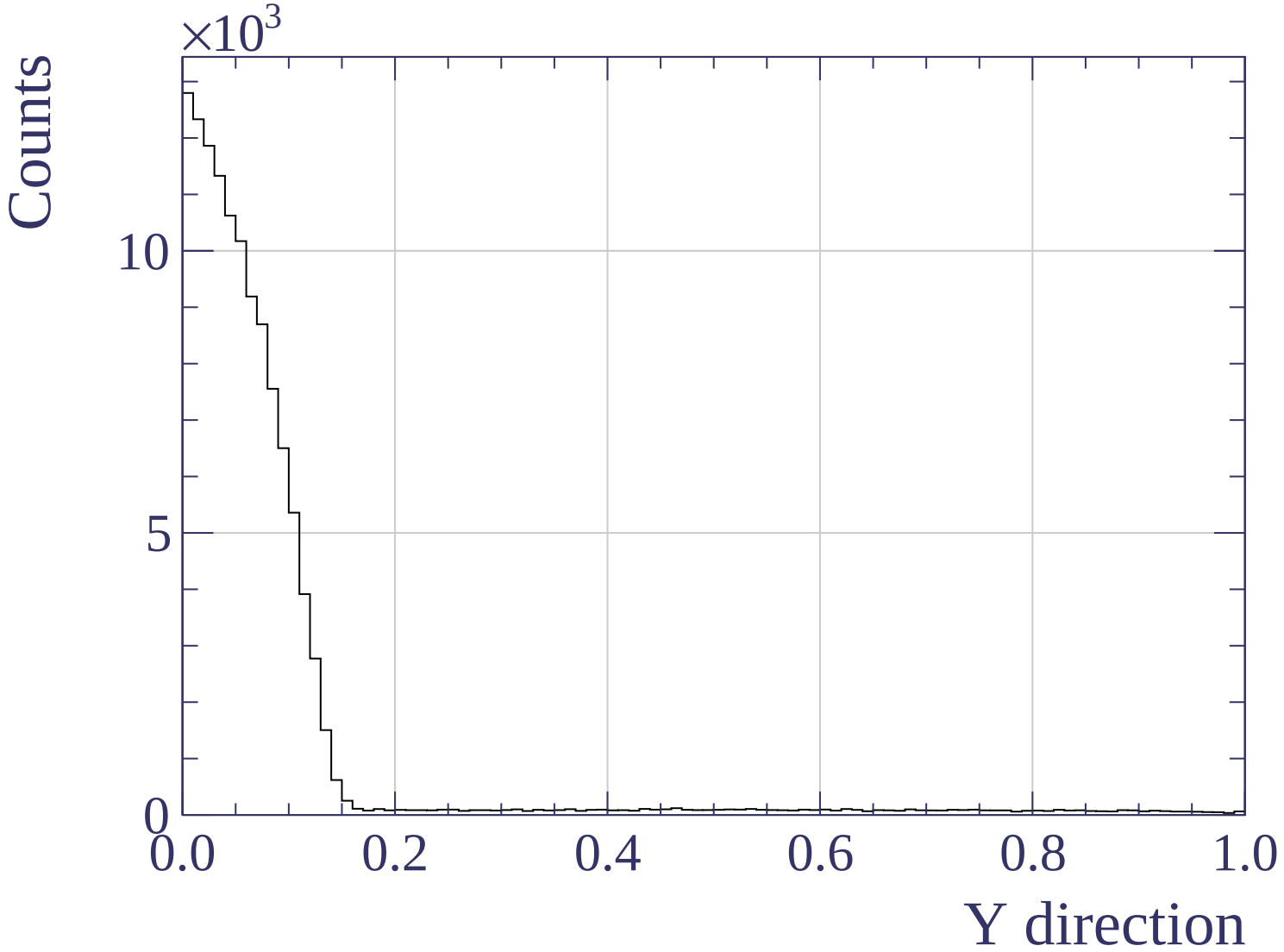


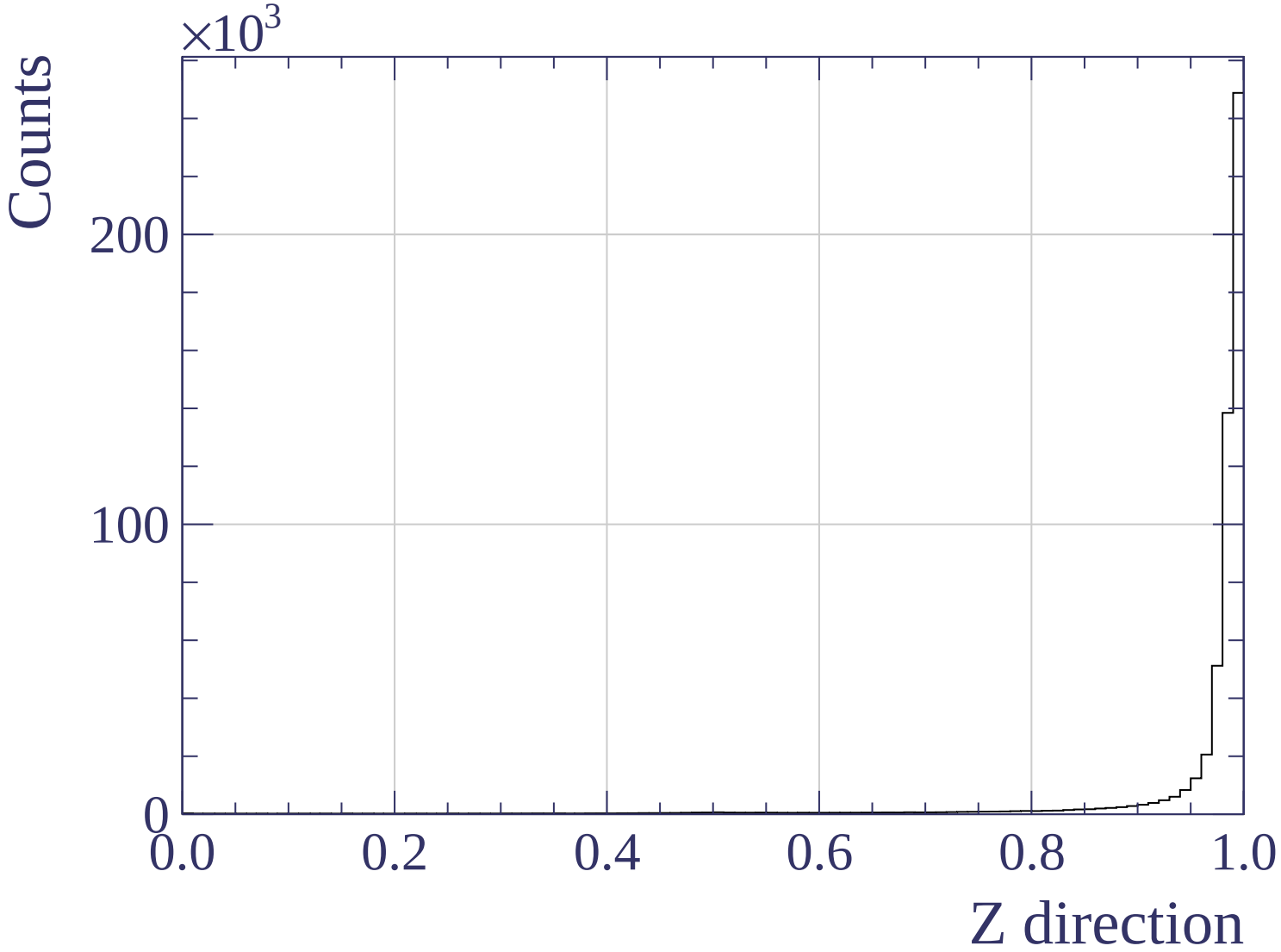


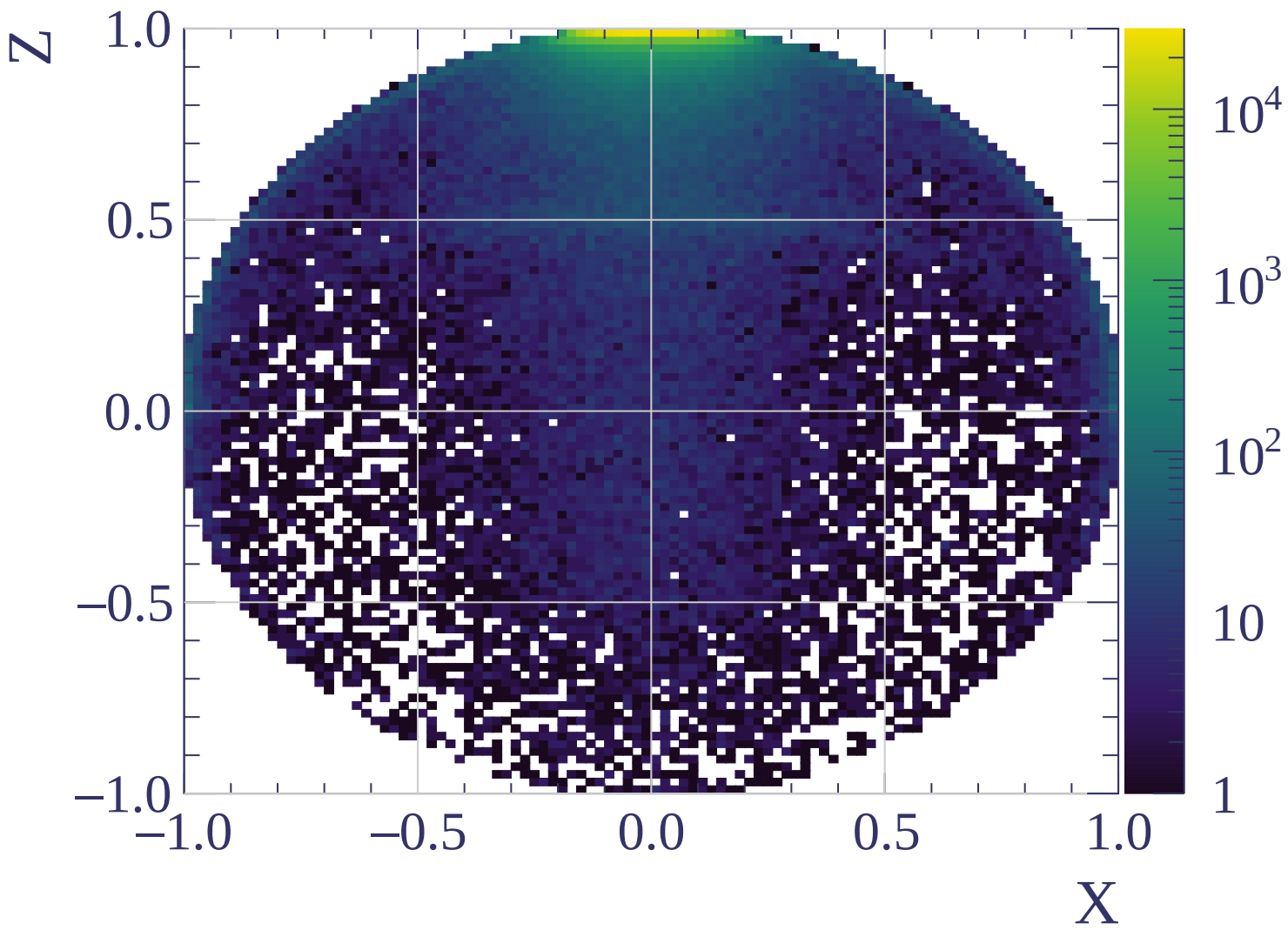


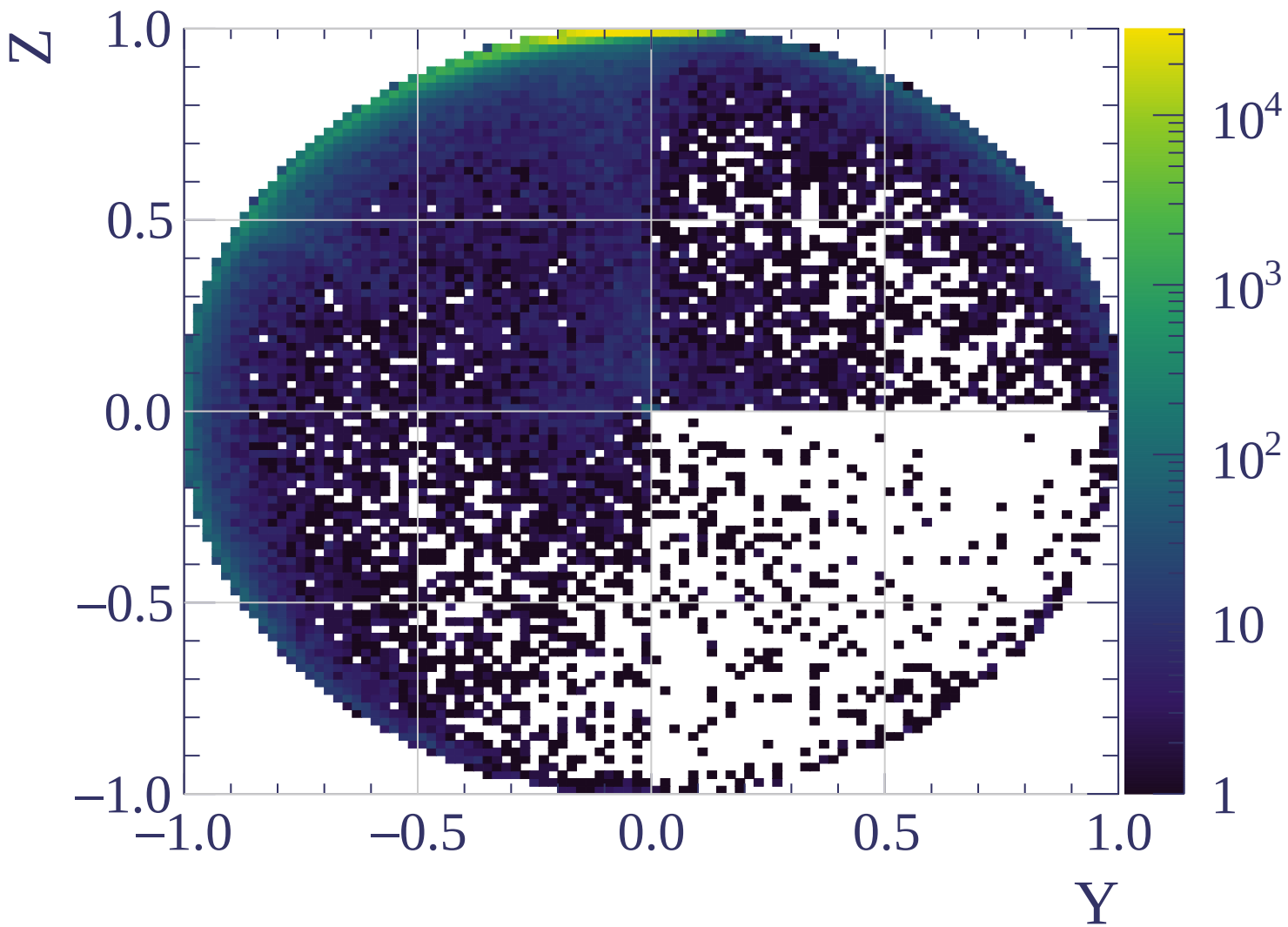


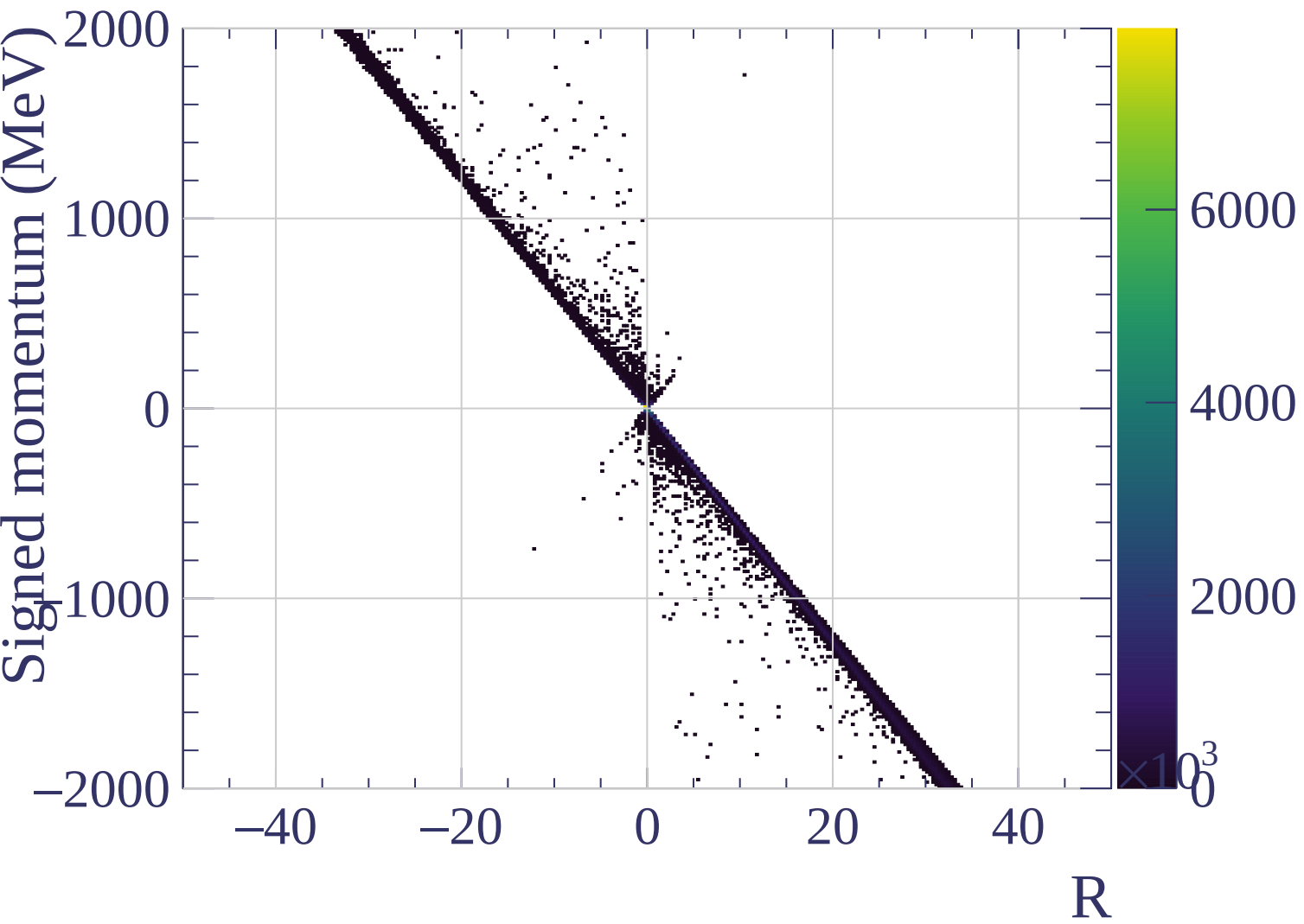


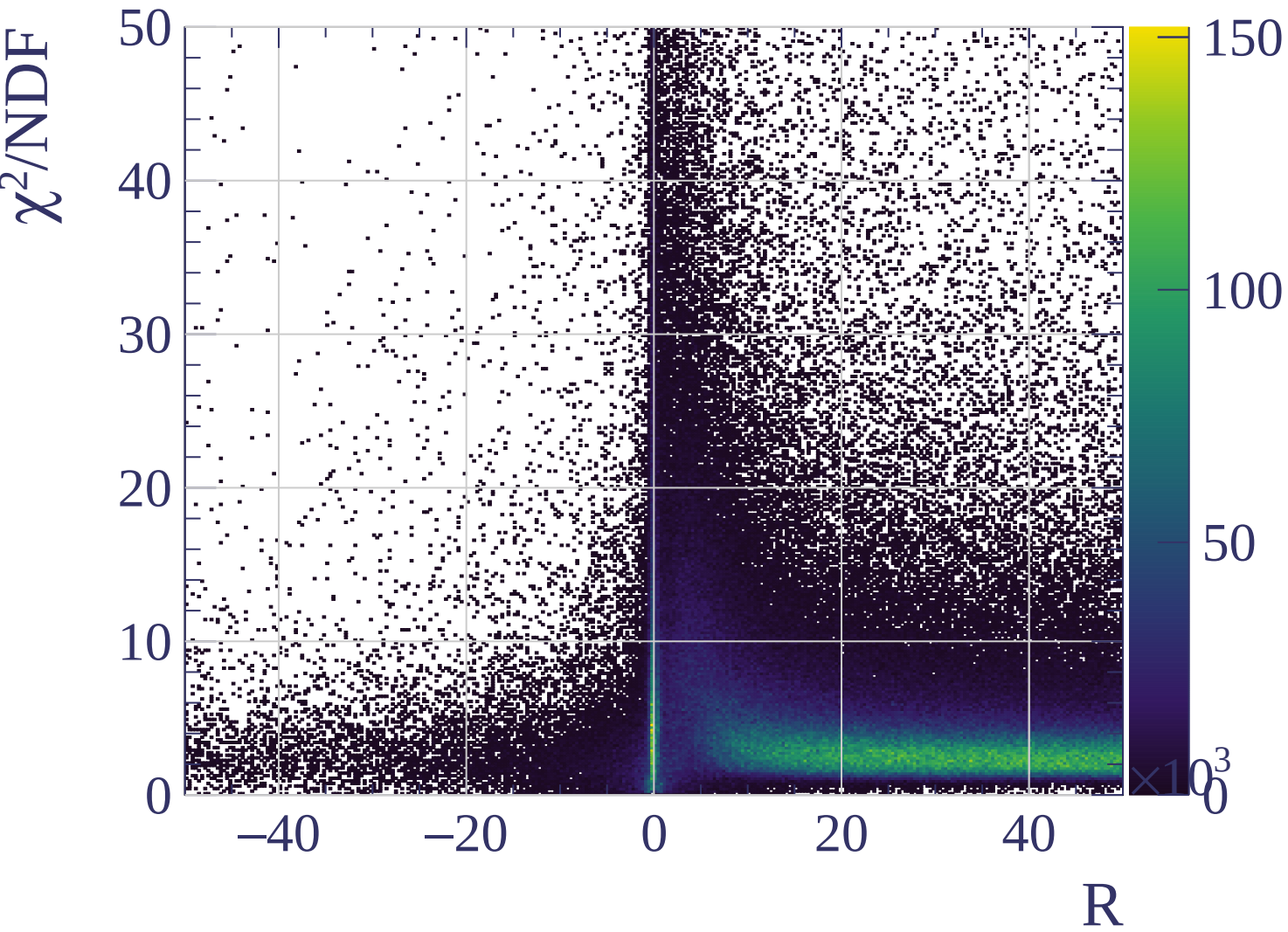




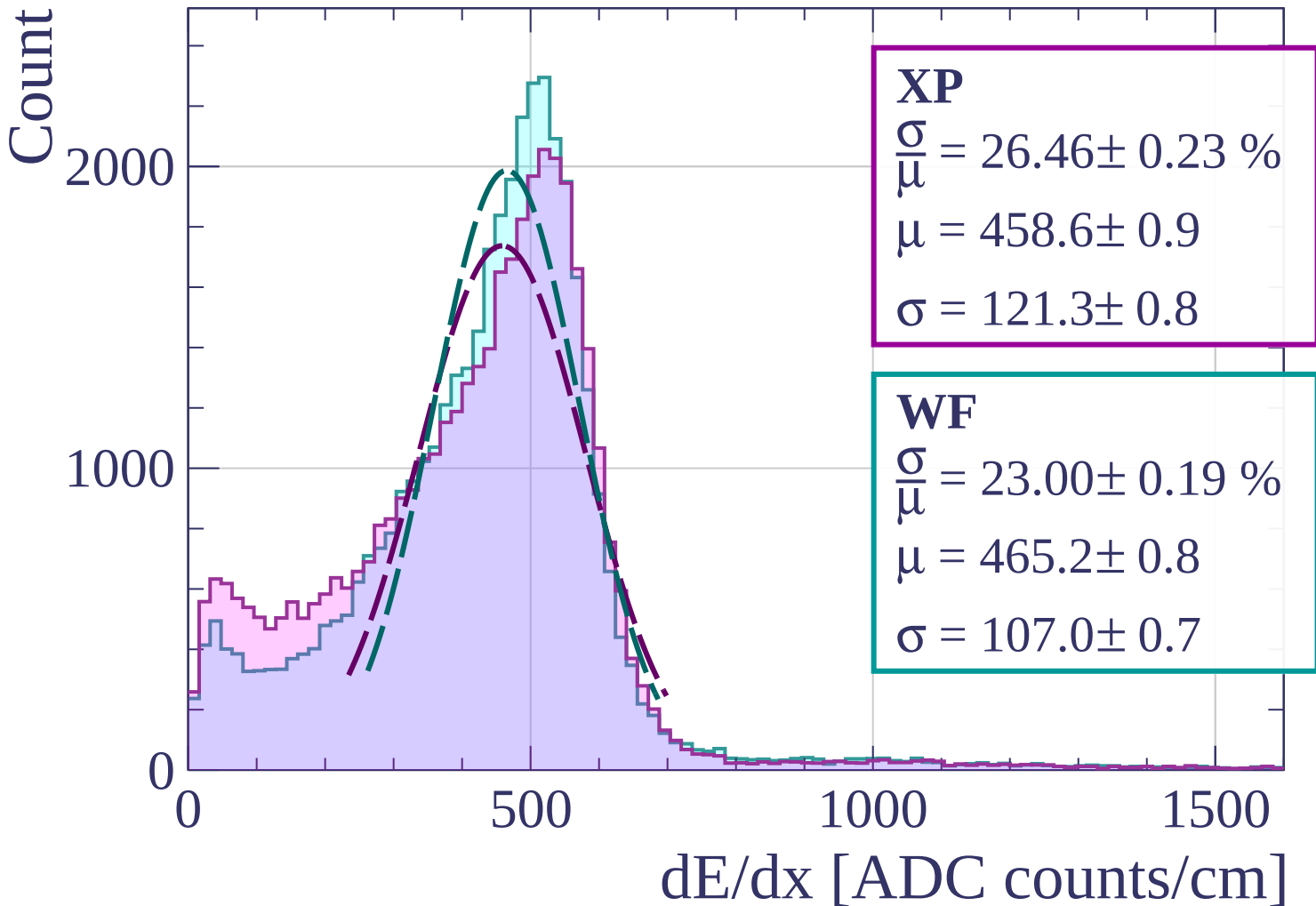






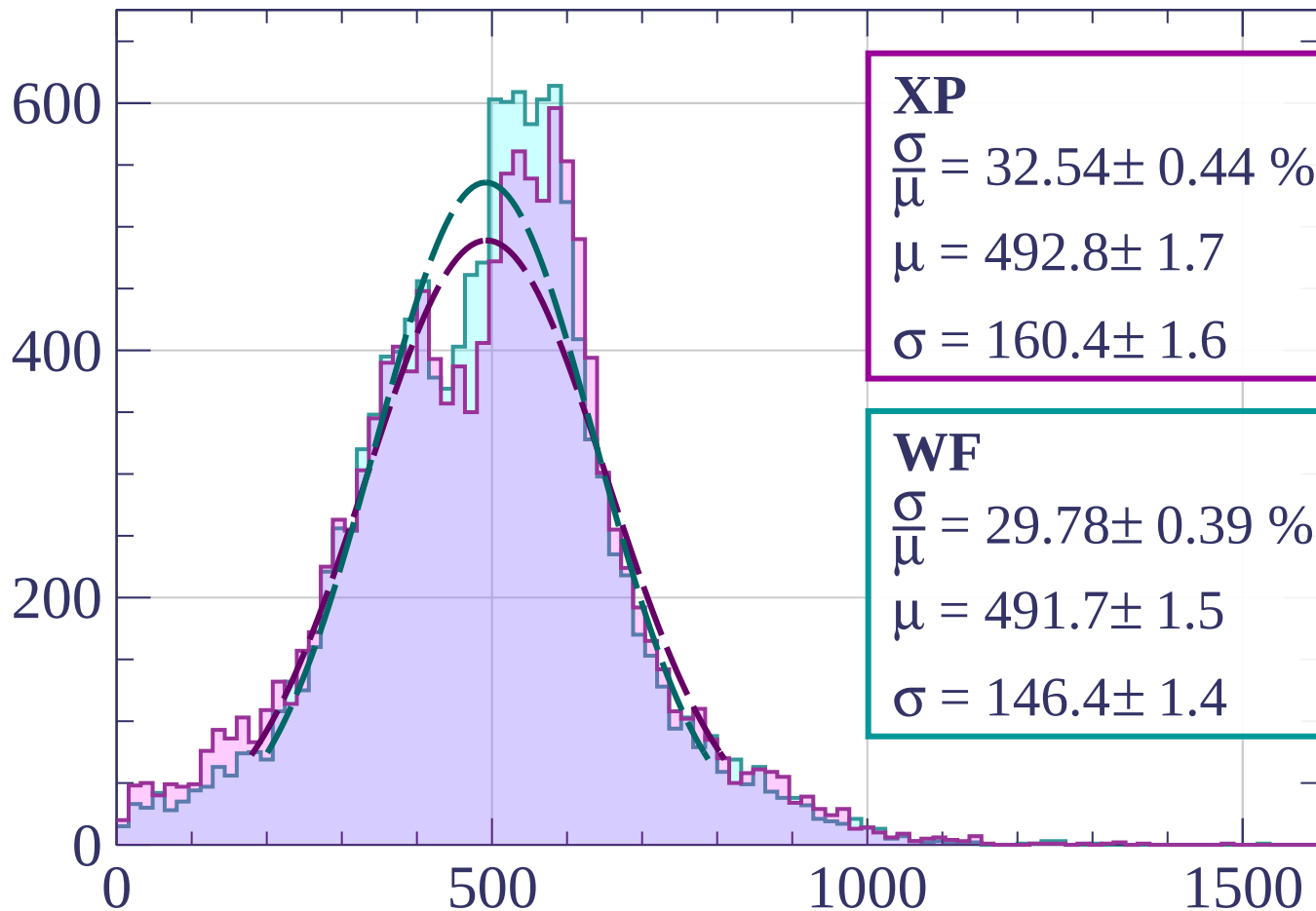


Energy loss |  $0 < p < 80$



Energy loss |  $80 < p < 160$

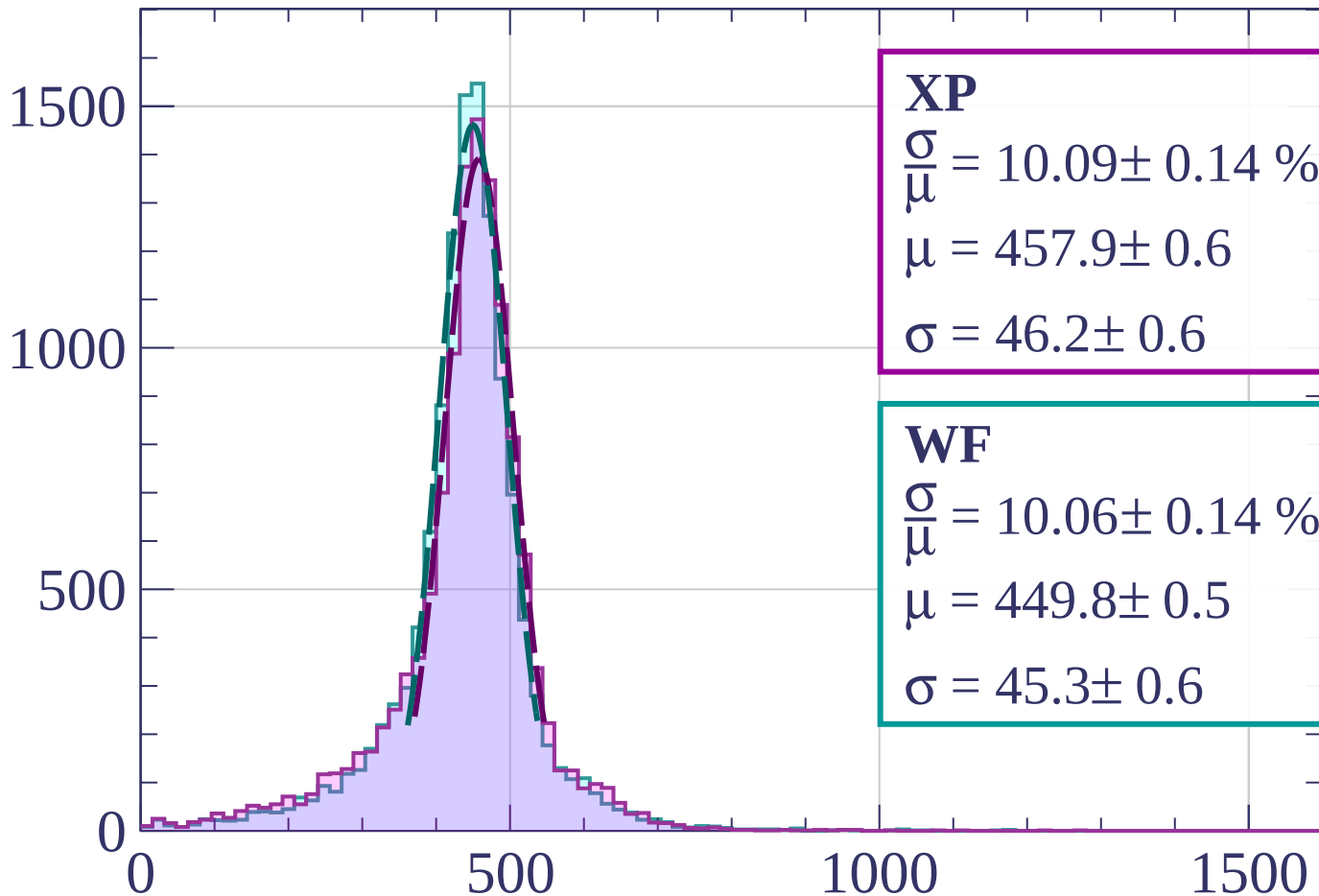
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $160 < p < 240$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.09 \pm 0.14 \%$$

$$\mu = 457.9 \pm 0.6$$

$$\sigma = 46.2 \pm 0.6$$

**WF**

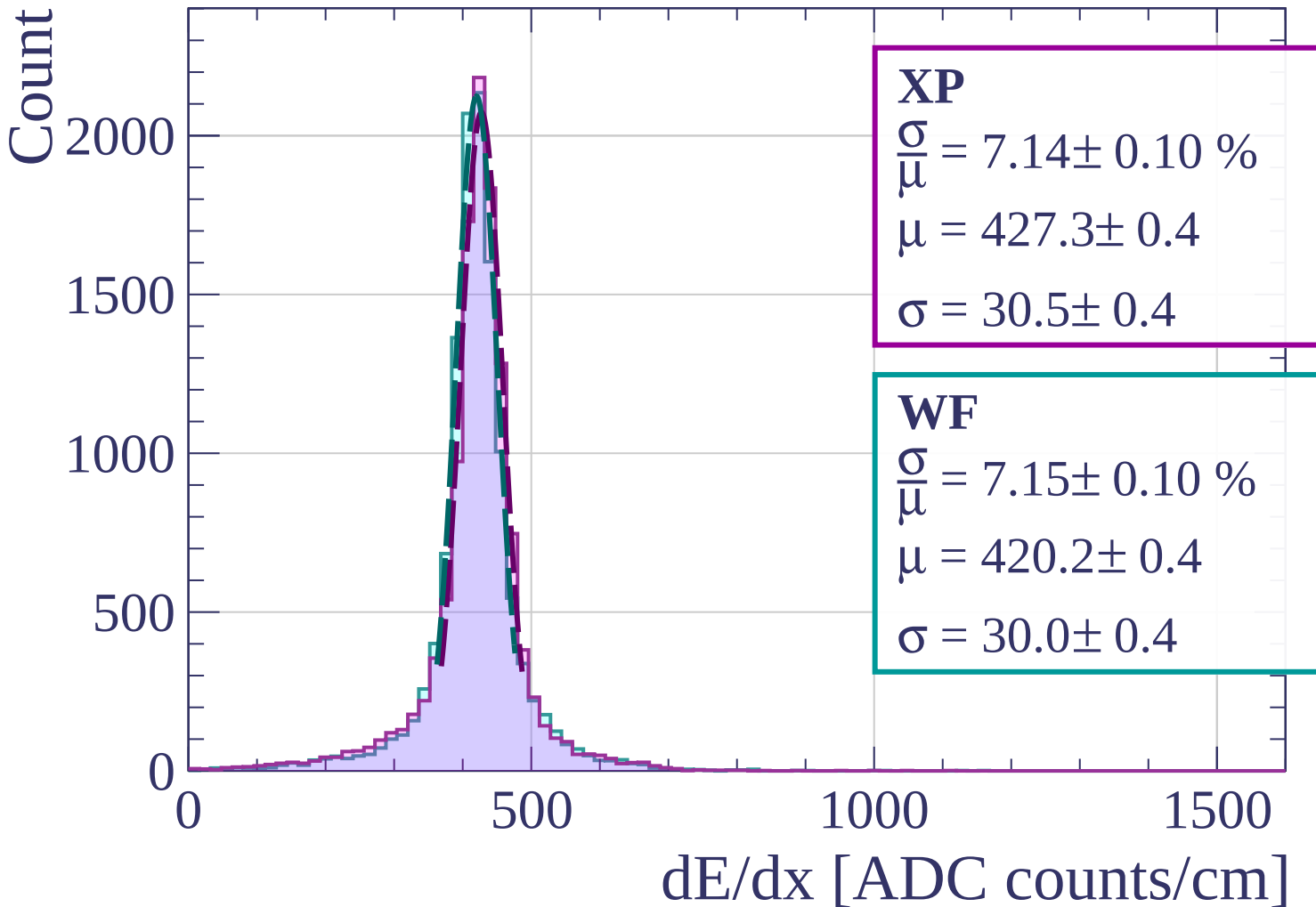
$$\frac{\sigma}{\mu} = 10.06 \pm 0.14 \%$$

$$\mu = 449.8 \pm 0.5$$

$$\sigma = 45.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss |  $240 < p < 320$



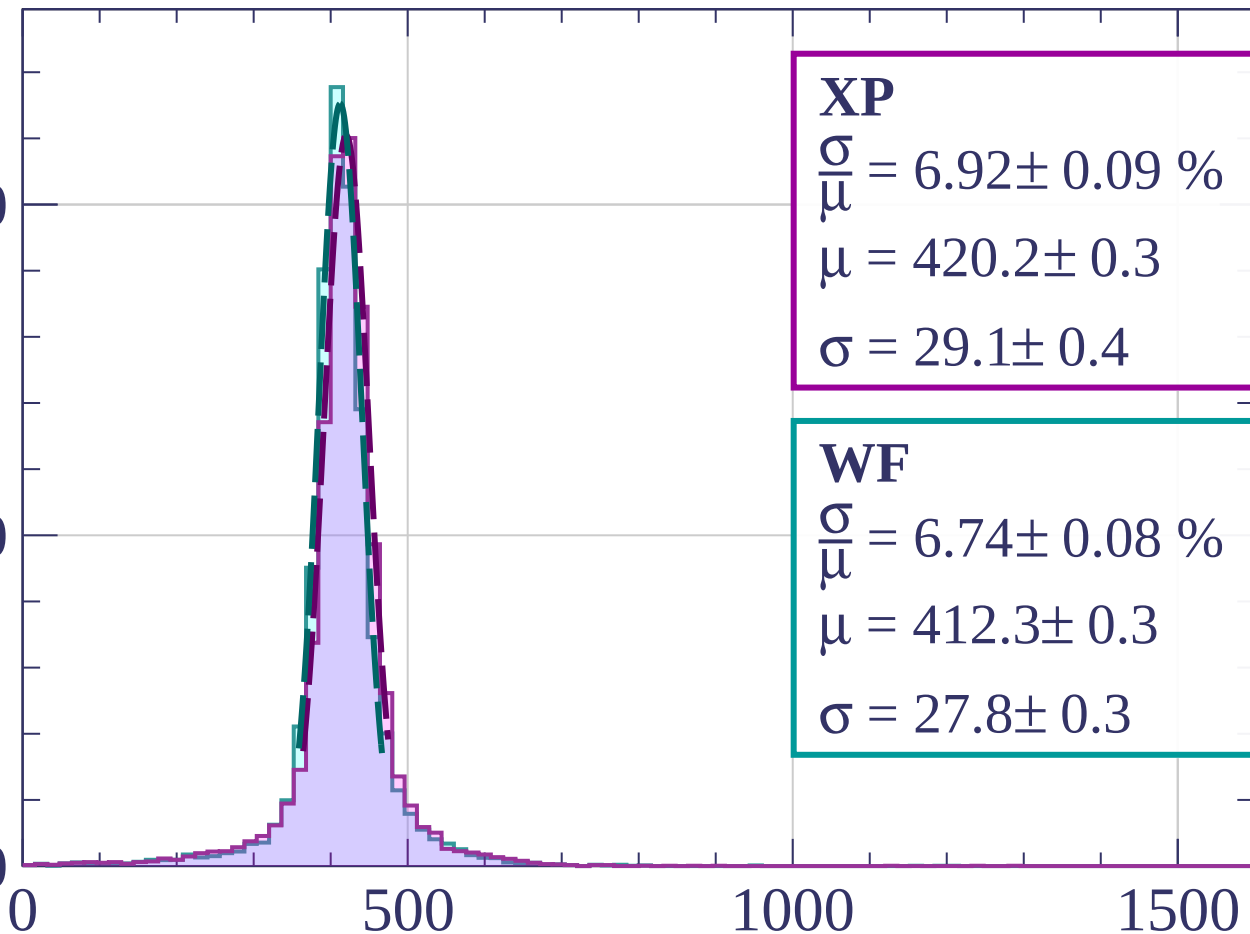
Energy loss |  $320 < p < 400$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 6.92 \pm 0.09 \%$$

$$\mu = 420.2 \pm 0.3$$

$$\sigma = 29.1 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 6.74 \pm 0.08 \%$$

$$\mu = 412.3 \pm 0.3$$

$$\sigma = 27.8 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

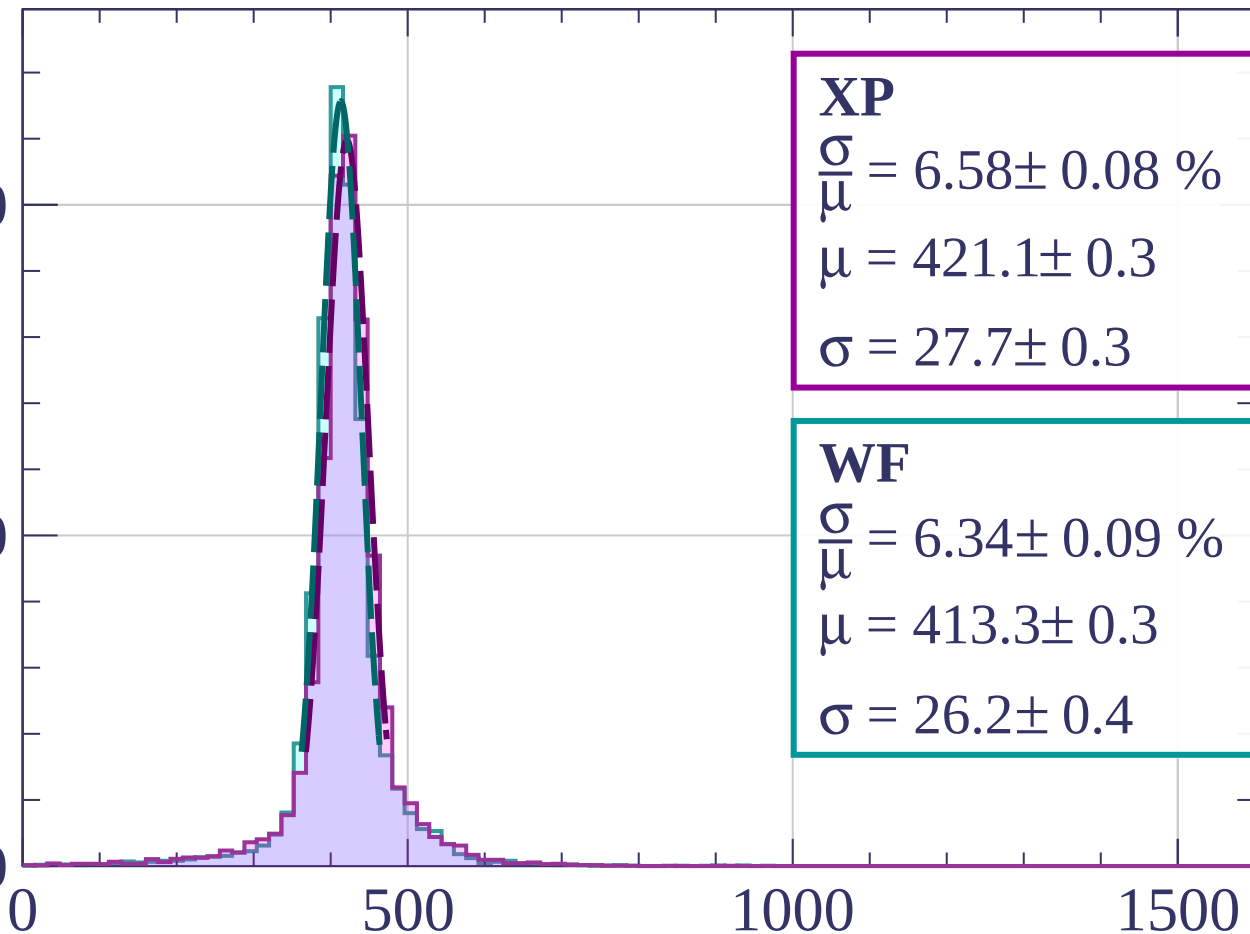
Energy loss |  $400 < p < 480$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 6.58 \pm 0.08 \%$$

$$\mu = 421.1 \pm 0.3$$

$$\sigma = 27.7 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.34 \pm 0.09 \%$$

$$\mu = 413.3 \pm 0.3$$

$$\sigma = 26.2 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

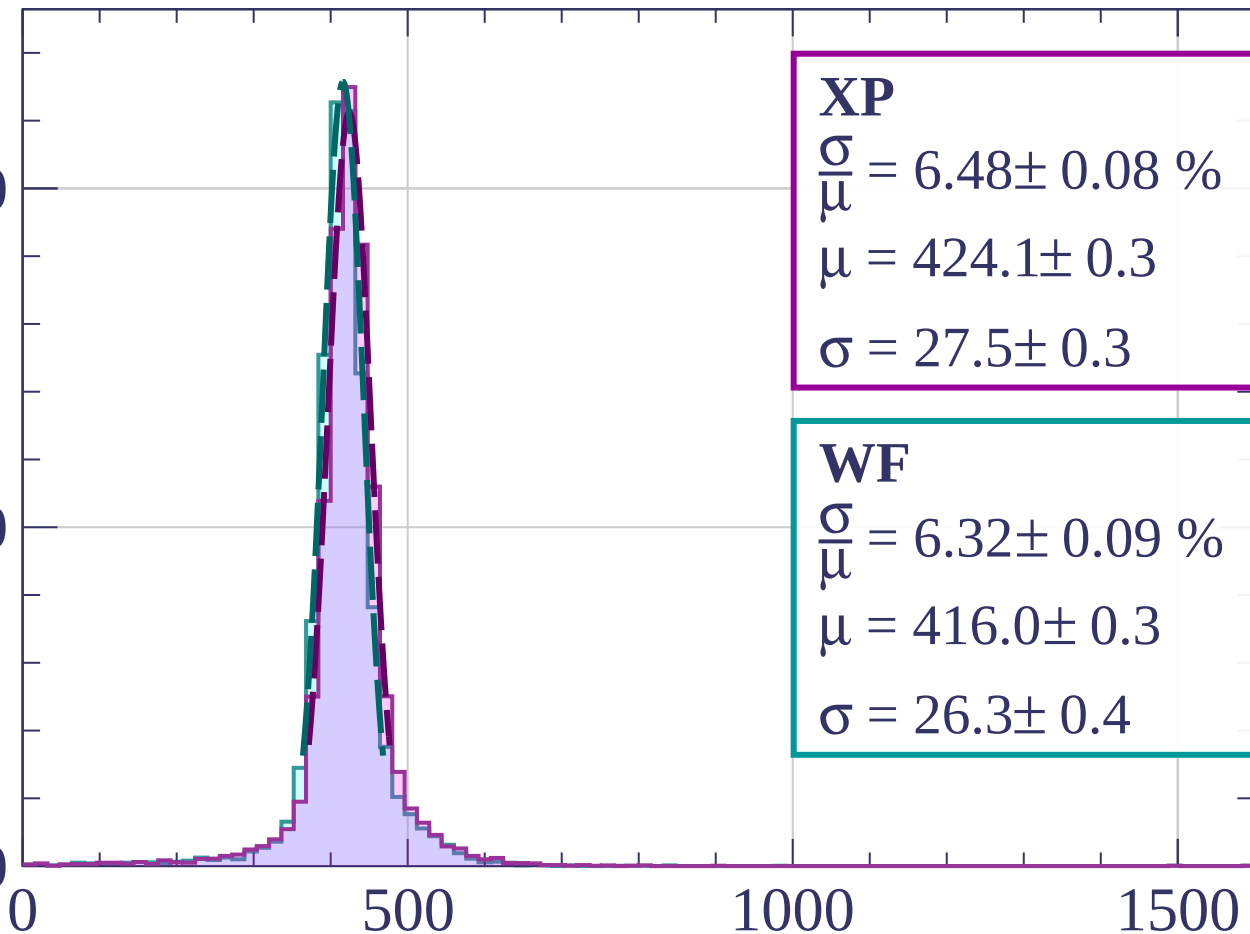
Energy loss |  $480 < p < 560$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 6.48 \pm 0.08 \%$$

$$\mu = 424.1 \pm 0.3$$

$$\sigma = 27.5 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.32 \pm 0.09 \%$$

$$\mu = 416.0 \pm 0.3$$

$$\sigma = 26.3 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $560 < p < 640$

Count

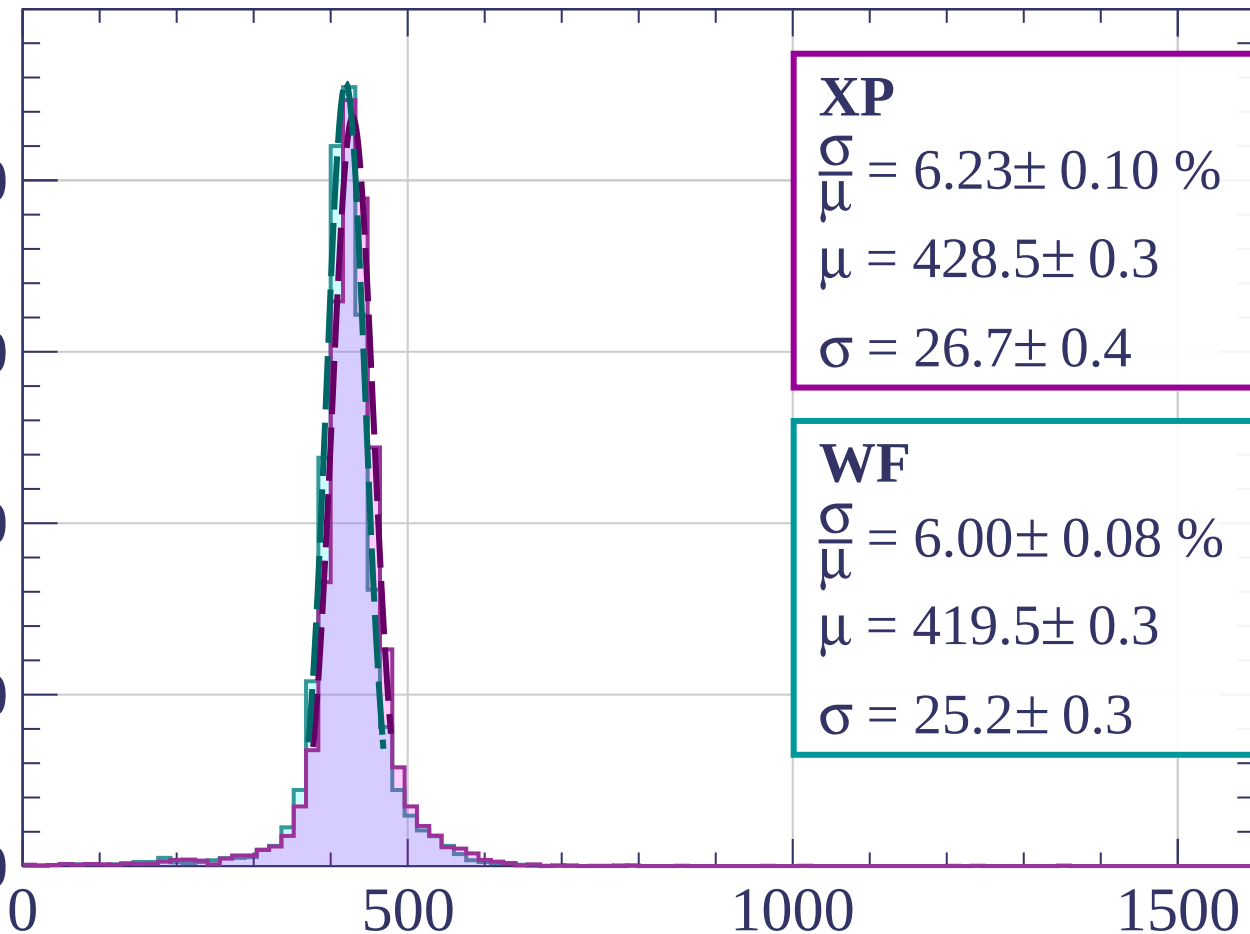
2000

1500

1000

500

0



1000

1500

$dE/dx$  [ADC counts/cm]

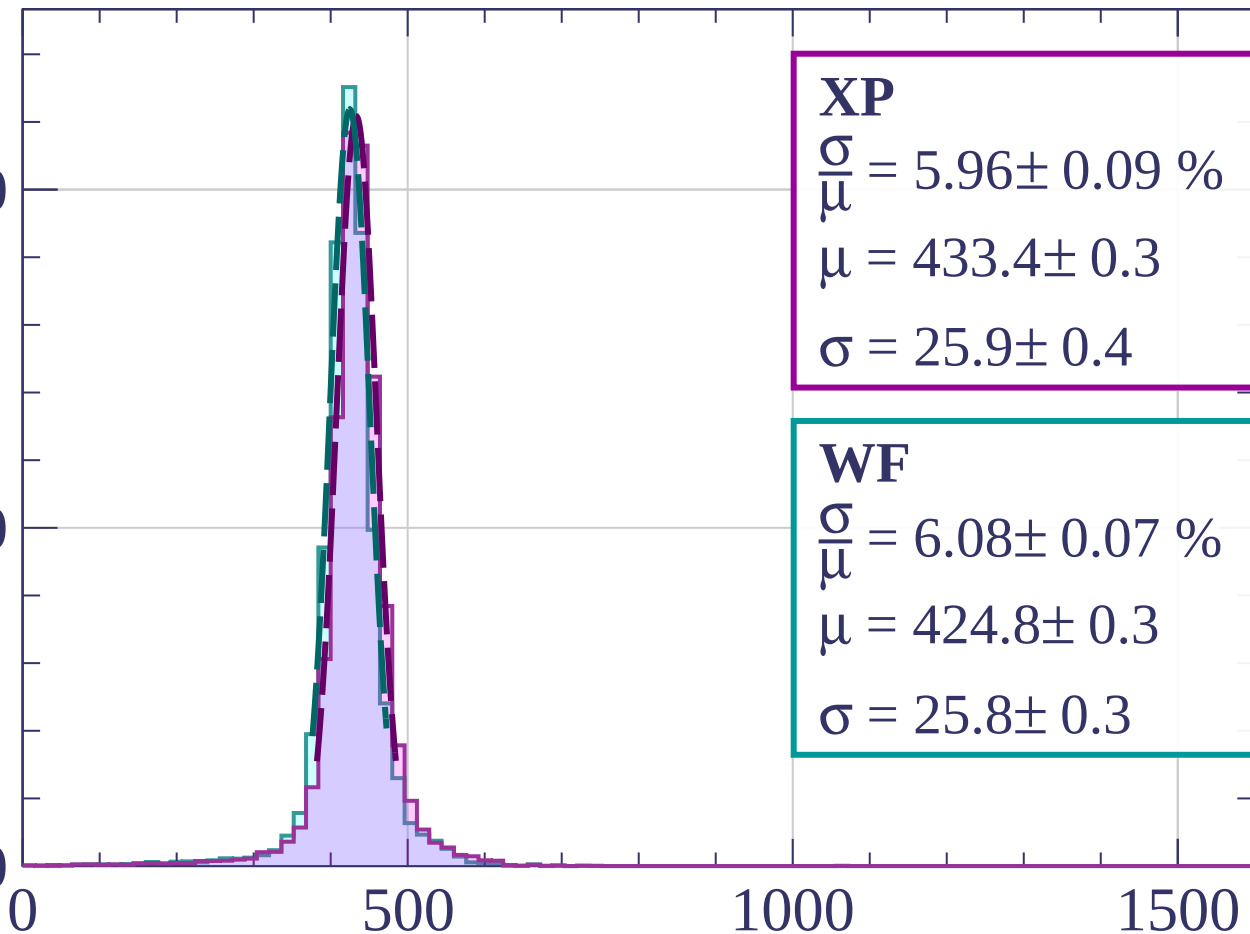
Energy loss |  $640 < p < 720$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 5.96 \pm 0.09 \%$$

$$\mu = 433.4 \pm 0.3$$

$$\sigma = 25.9 \pm 0.4$$

**WF**

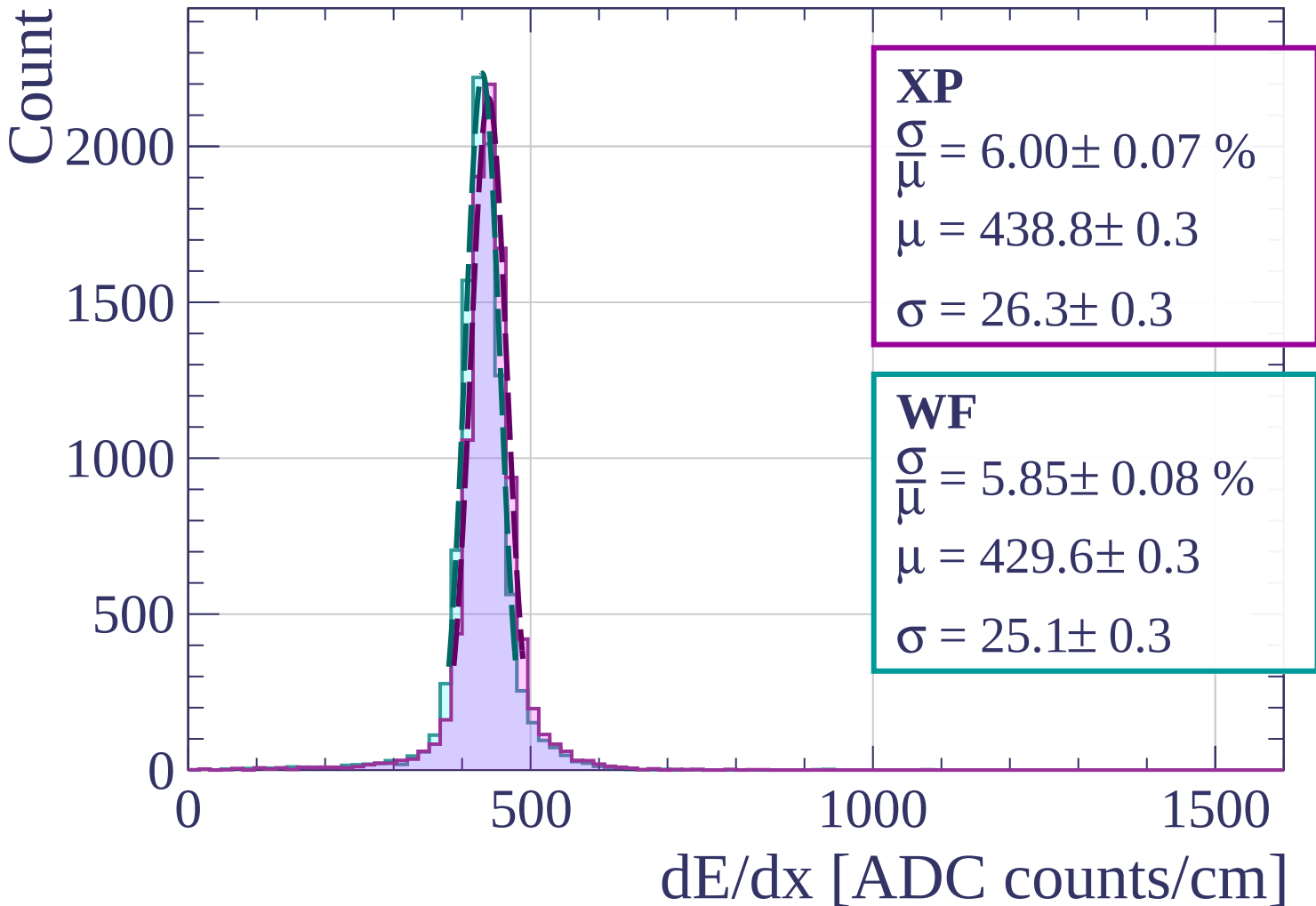
$$\frac{\sigma}{\mu} = 6.08 \pm 0.07 \%$$

$$\mu = 424.8 \pm 0.3$$

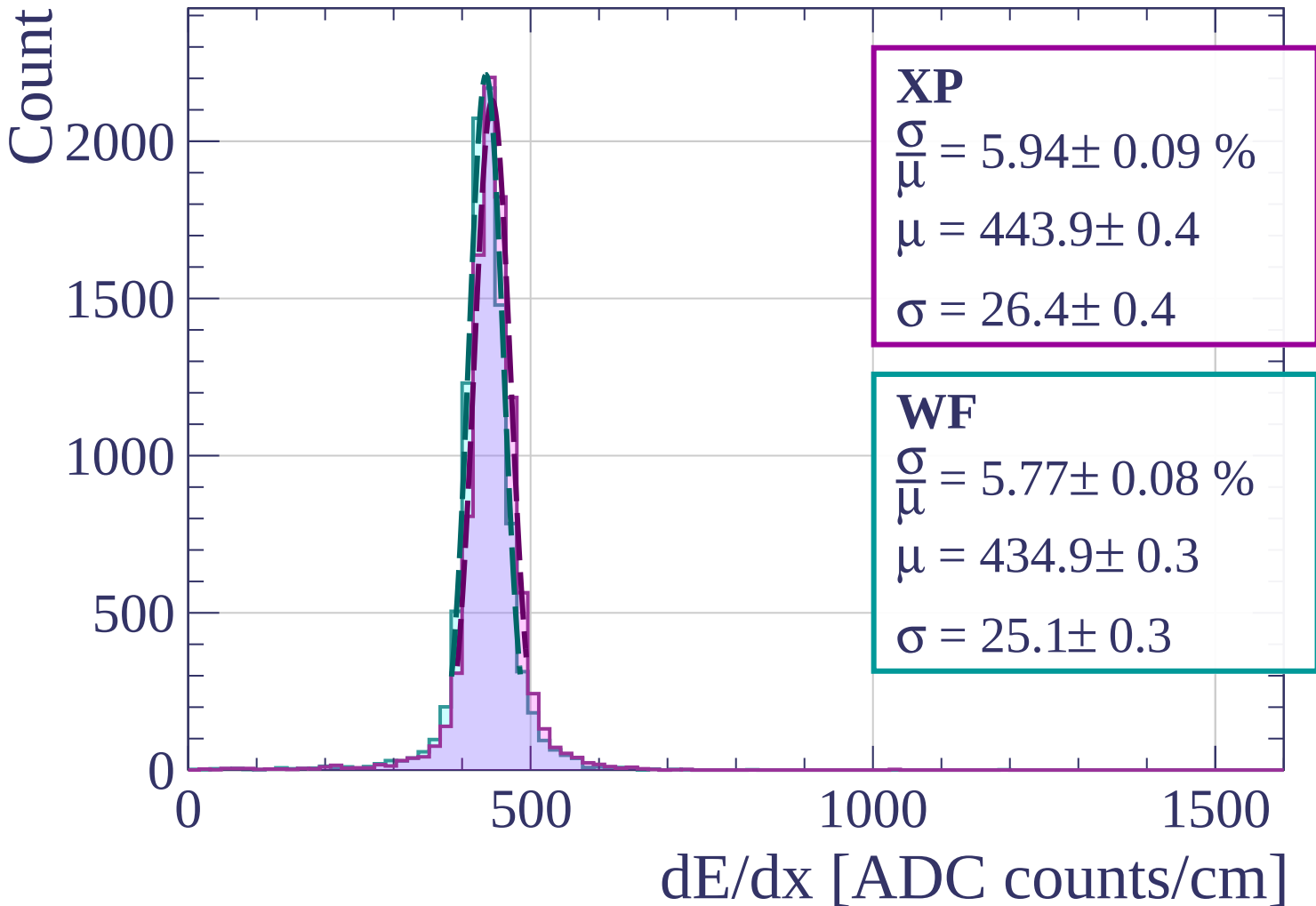
$$\sigma = 25.8 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

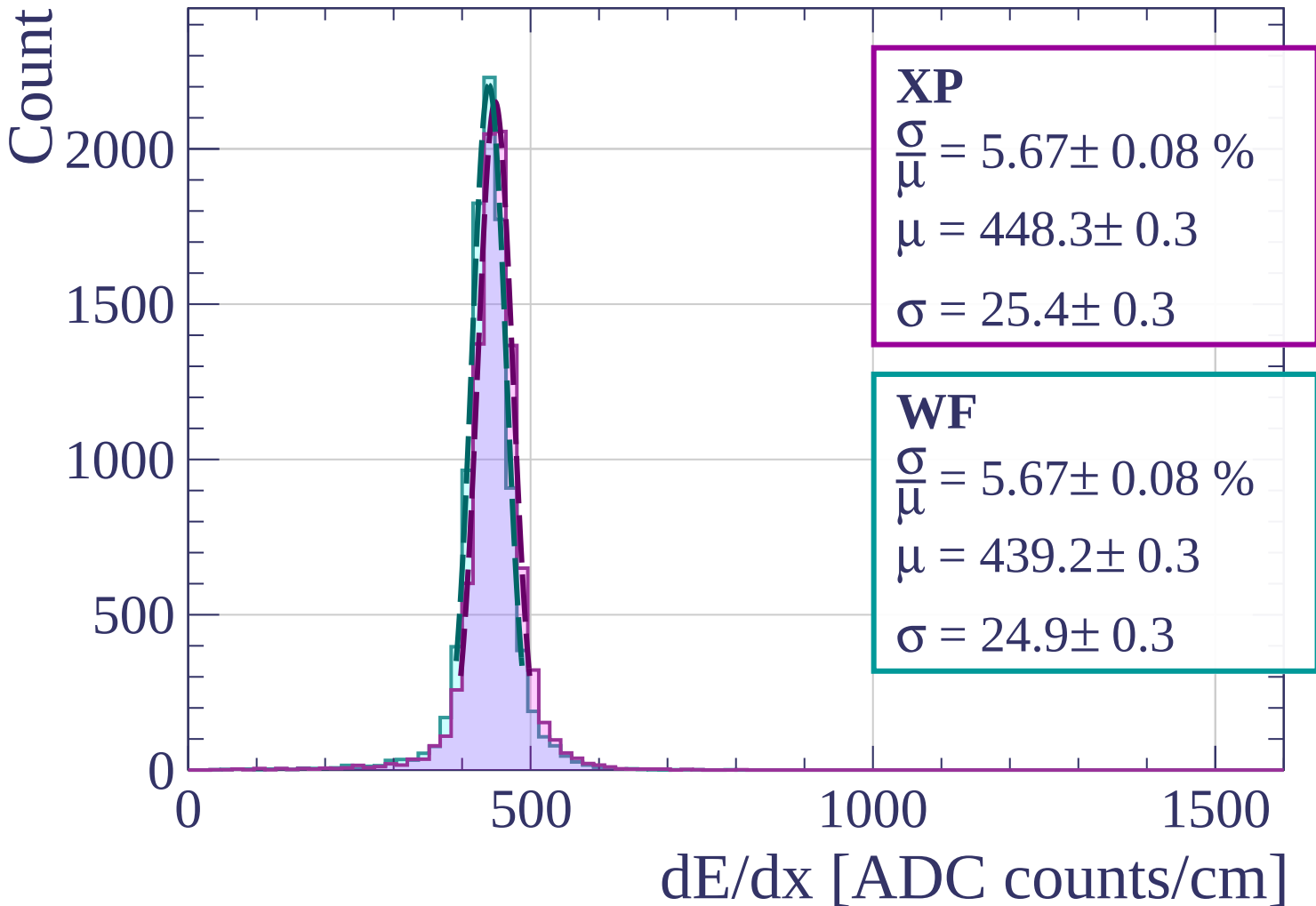
Energy loss |  $720 < p < 800$



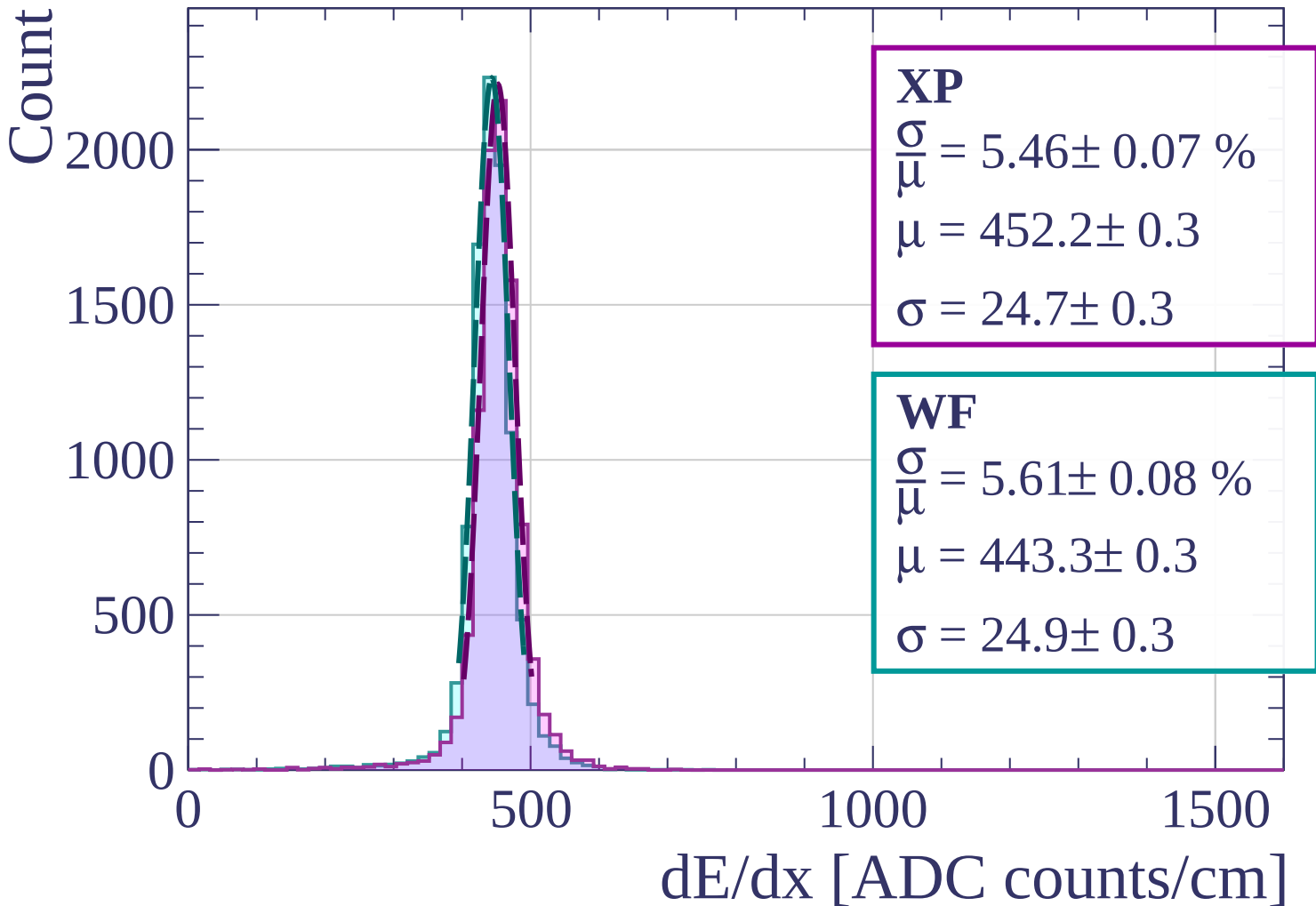
Energy loss |  $800 < p < 880$



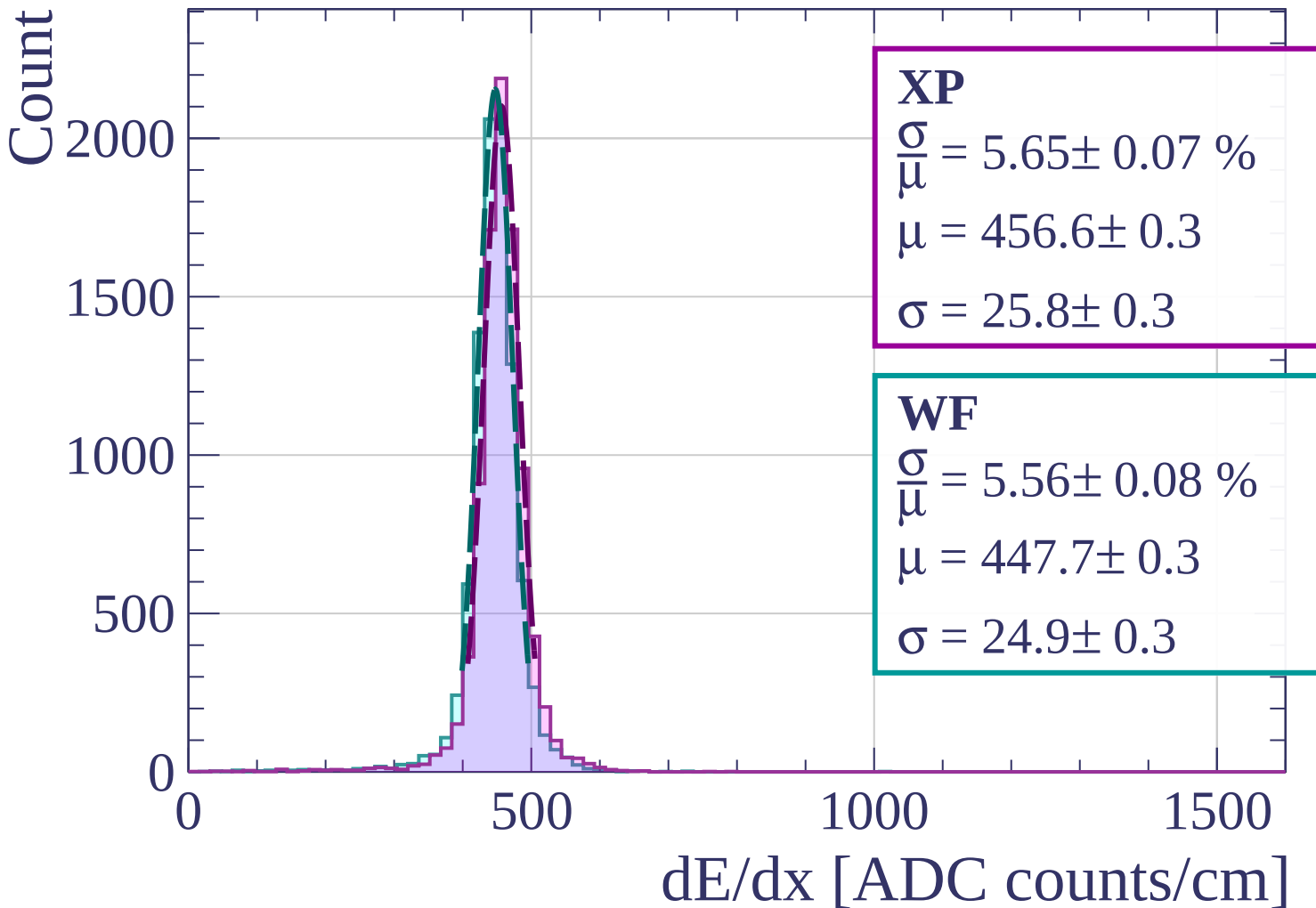
Energy loss |  $880 < p < 960$



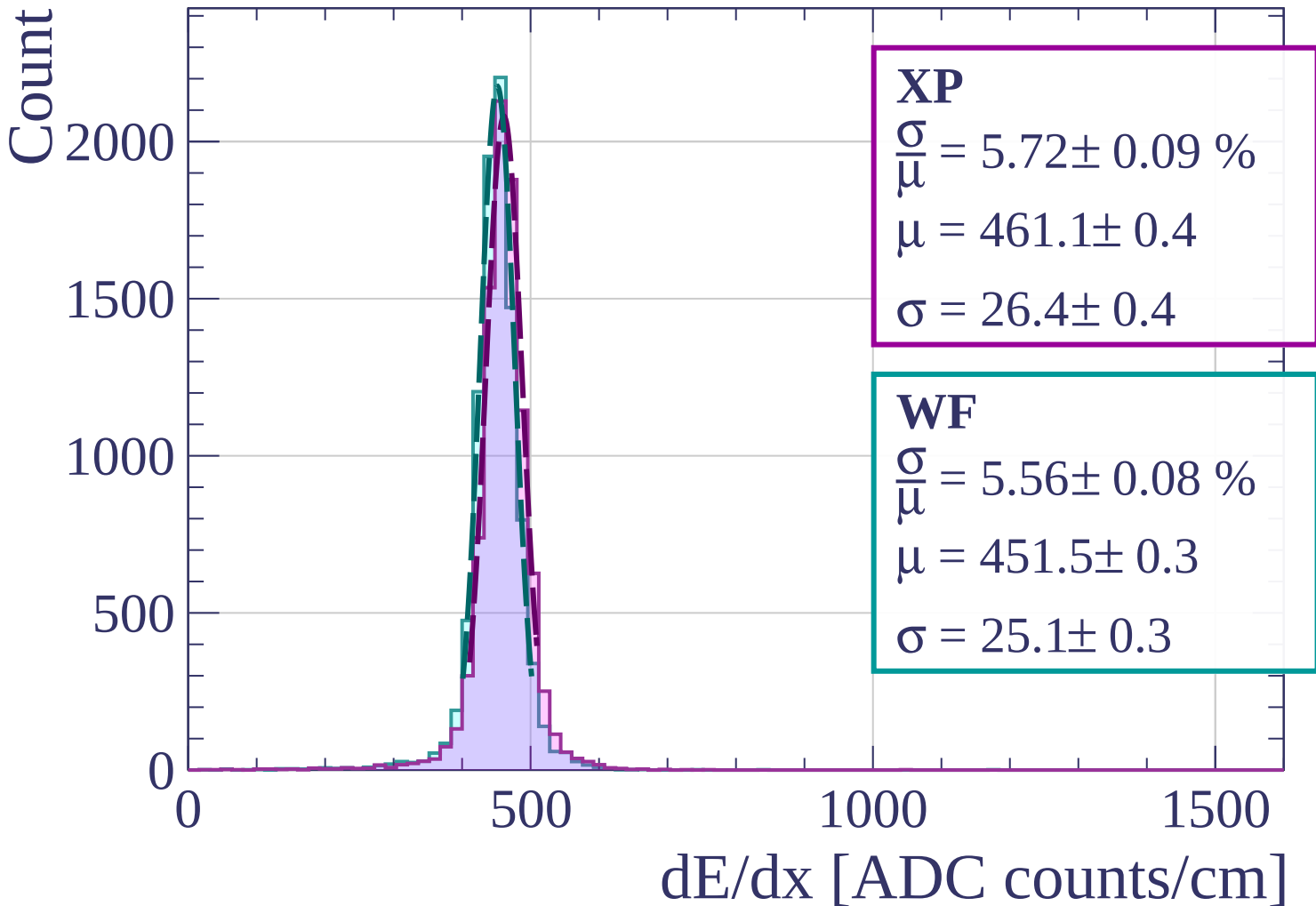
Energy loss |  $960 < p < 1040$



Energy loss |  $1040 < p < 1120$



Energy loss |  $1120 < p < 1200$



Energy loss |  $1200 < p < 1280$

Count

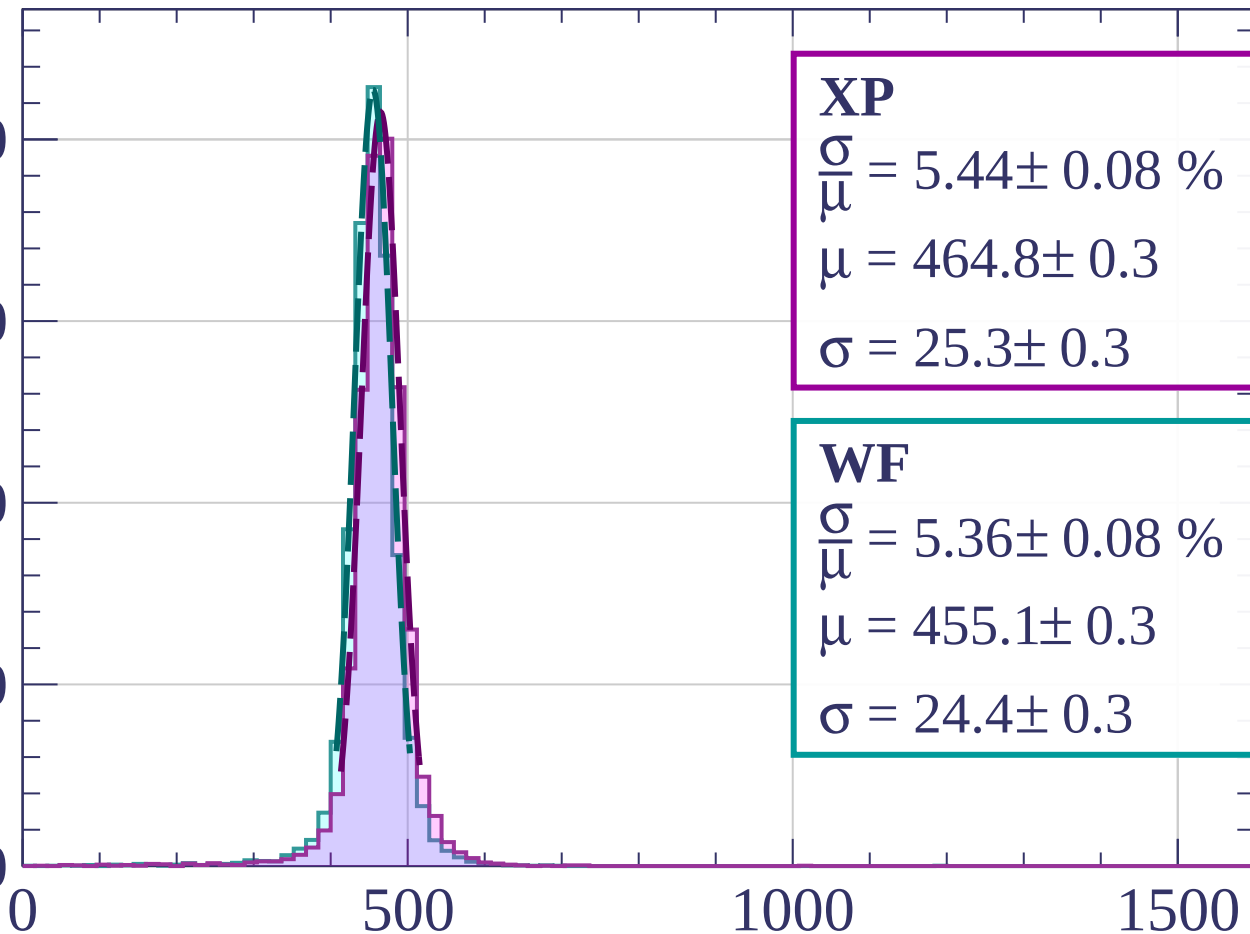
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.44 \pm 0.08 \%$$

$$\mu = 464.8 \pm 0.3$$

$$\sigma = 25.3 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.36 \pm 0.08 \%$$

$$\mu = 455.1 \pm 0.3$$

$$\sigma = 24.4 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1280 < p < 1360$

Count

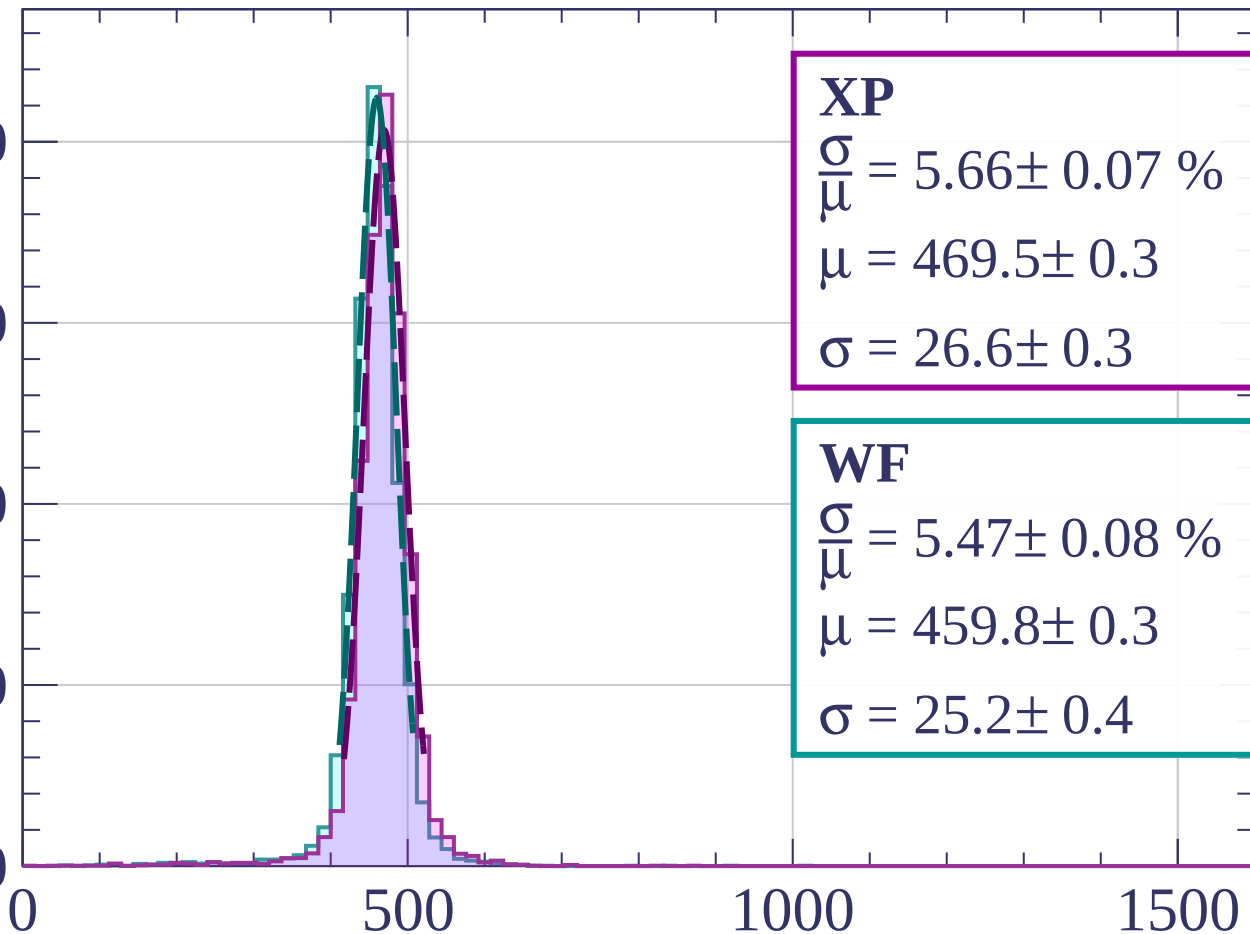
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.66 \pm 0.07 \%$$

$$\mu = 469.5 \pm 0.3$$

$$\sigma = 26.6 \pm 0.3$$

**WF**

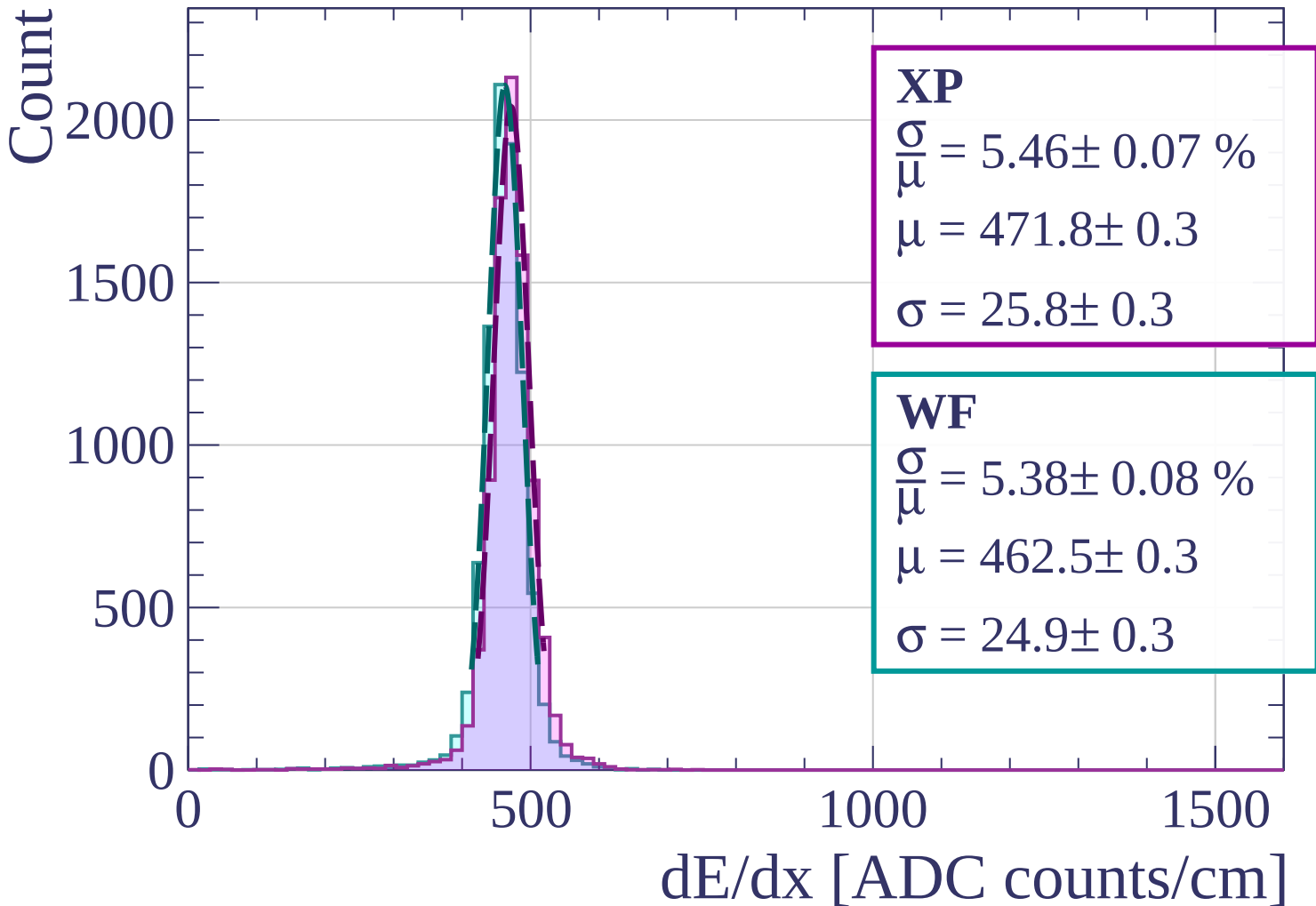
$$\frac{\sigma}{\mu} = 5.47 \pm 0.08 \%$$

$$\mu = 459.8 \pm 0.3$$

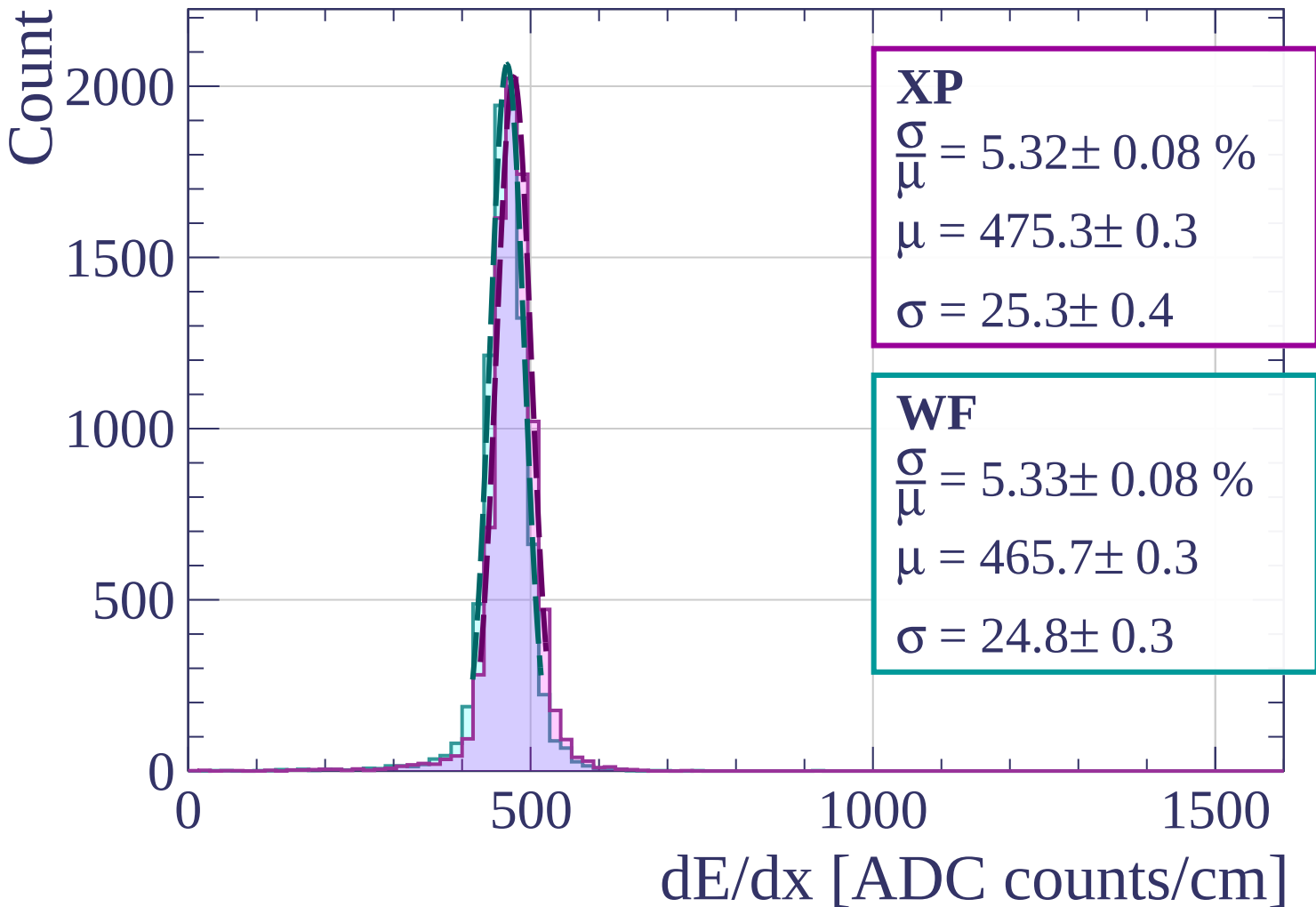
$$\sigma = 25.2 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1360 < p < 1440$



Energy loss |  $1440 < p < 1520$



Energy loss |  $1520 < p < 1600$

Count

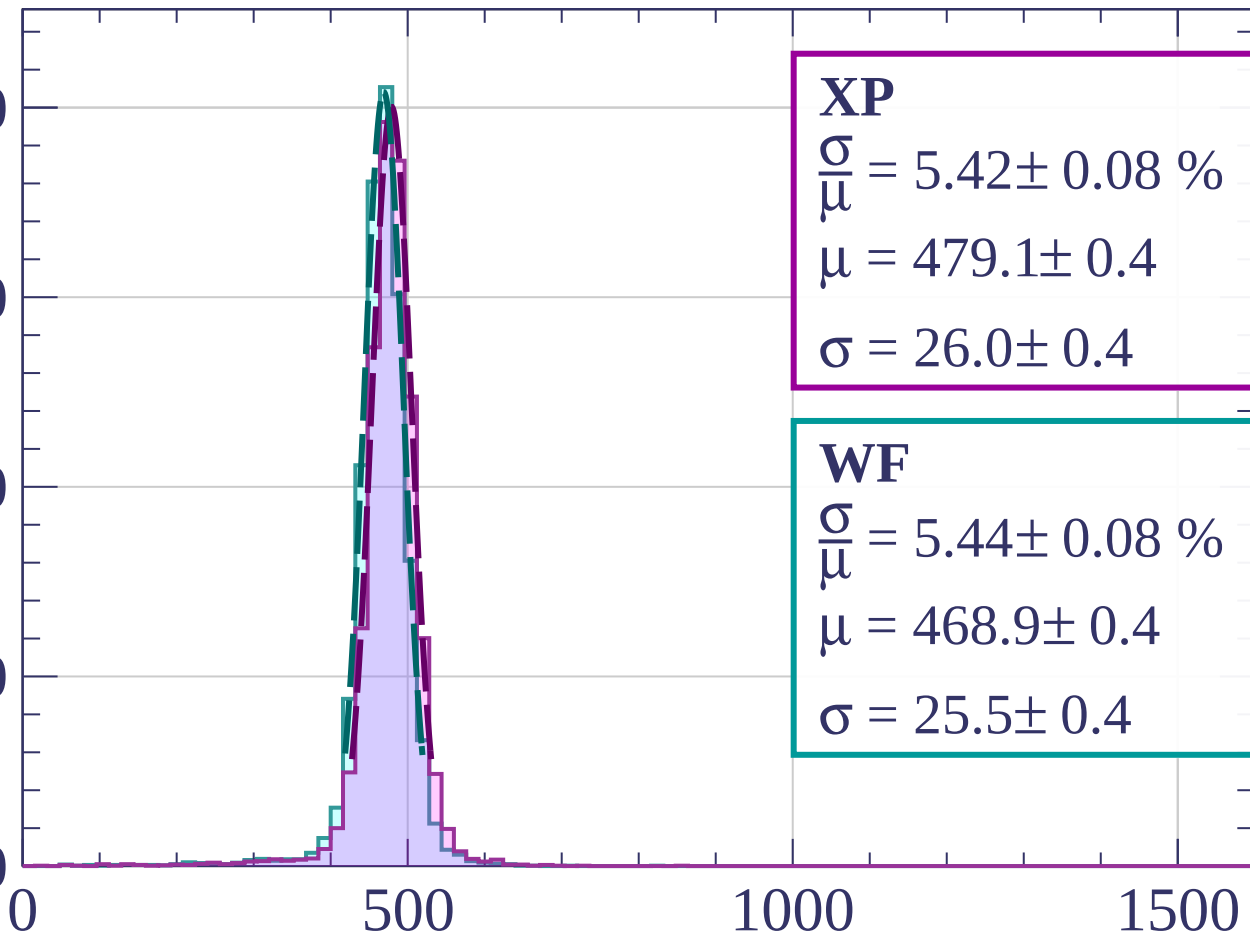
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.42 \pm 0.08 \%$$

$$\mu = 479.1 \pm 0.4$$

$$\sigma = 26.0 \pm 0.4$$

**WF**

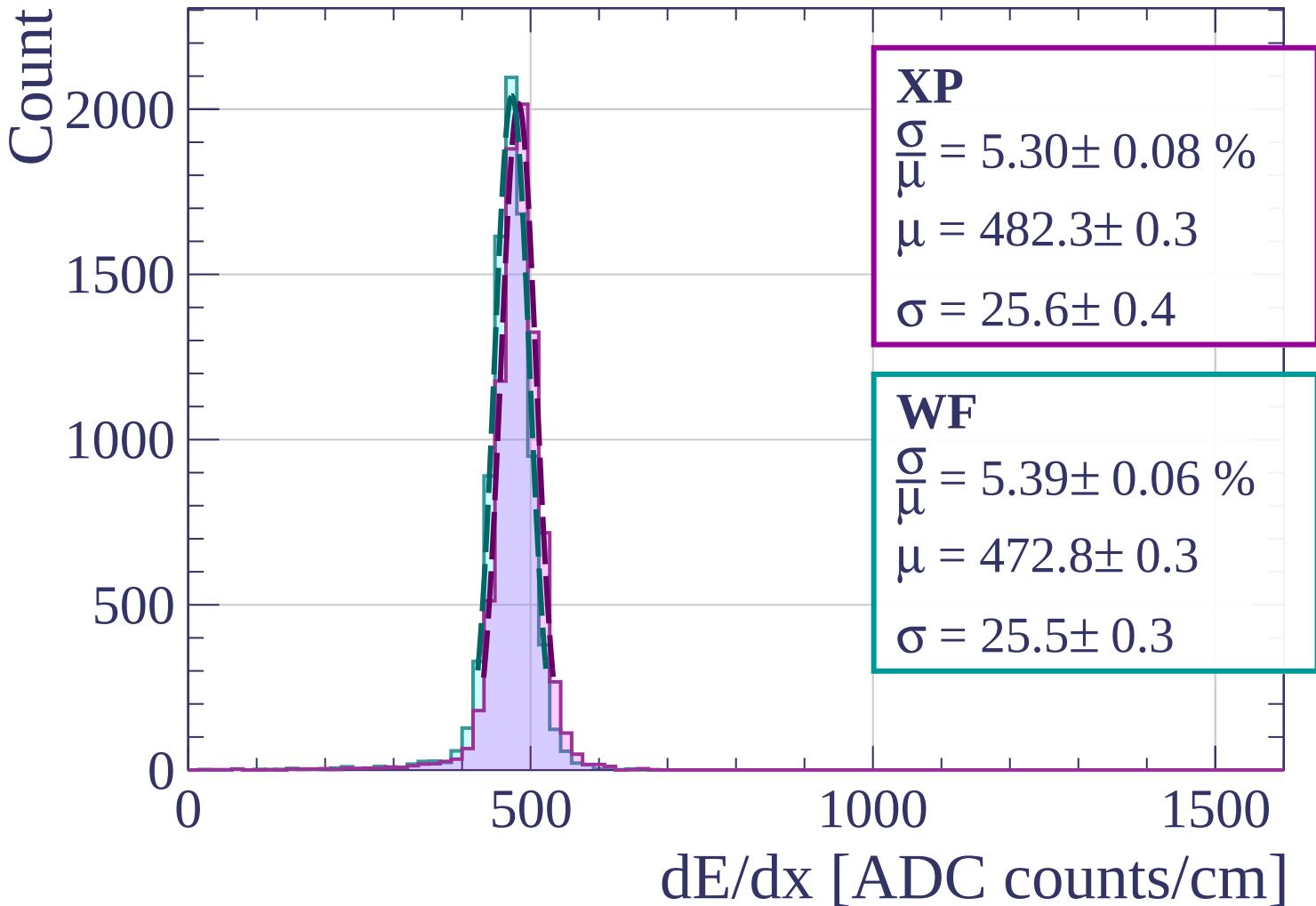
$$\frac{\sigma}{\mu} = 5.44 \pm 0.08 \%$$

$$\mu = 468.9 \pm 0.4$$

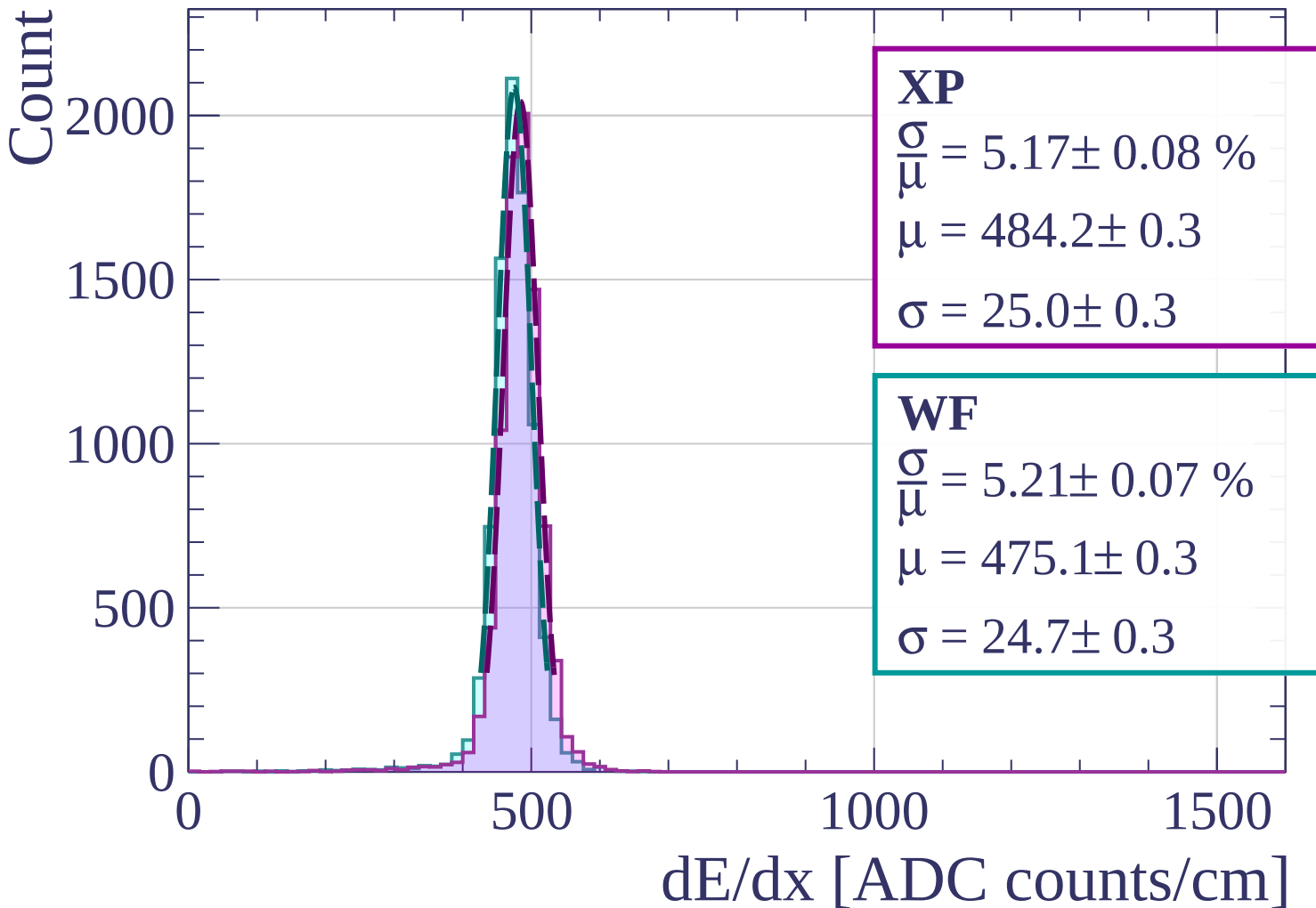
$$\sigma = 25.5 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1600 < p < 1680$



Energy loss |  $1680 < p < 1760$



Energy loss |  $1760 < p < 1840$

Count

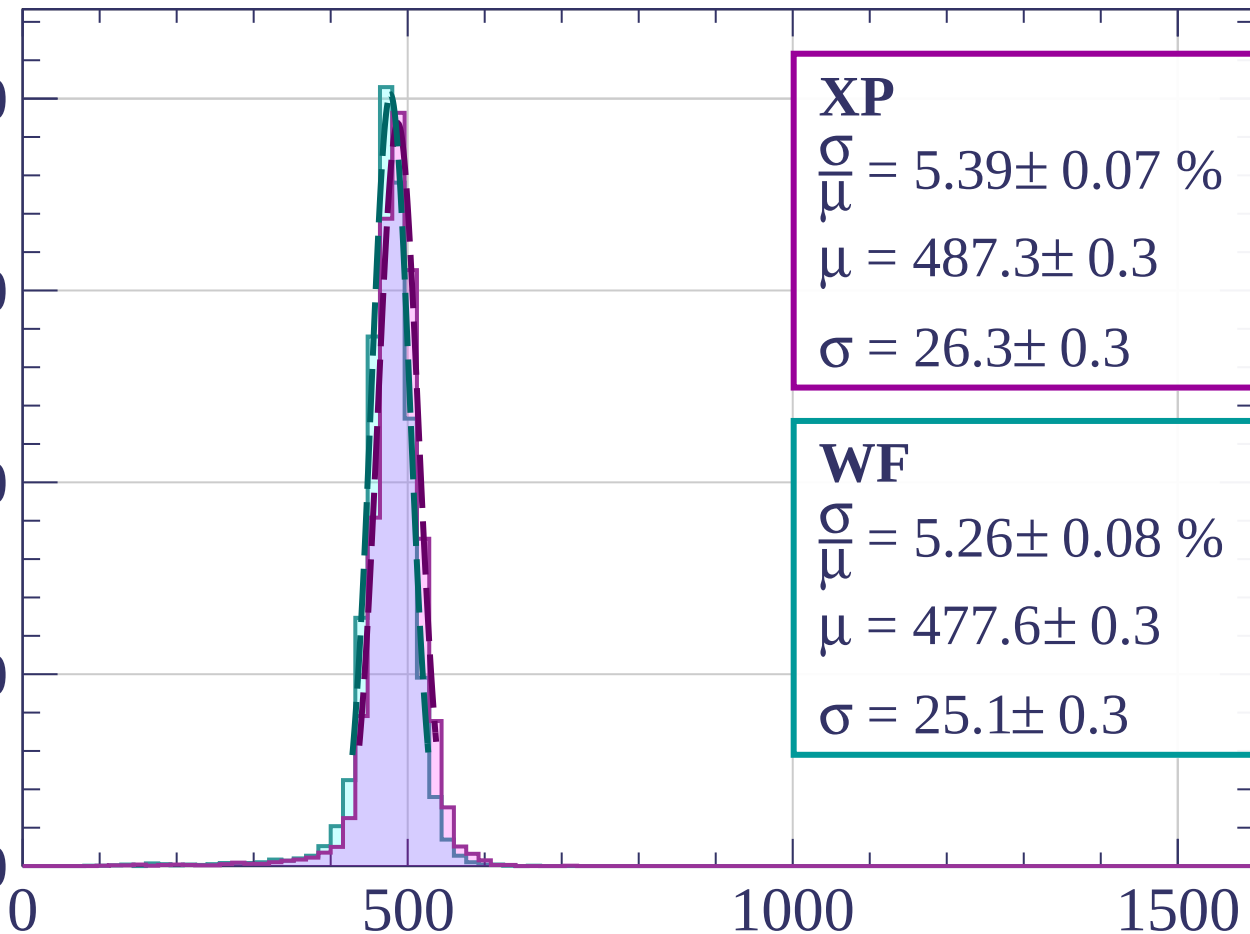
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.39 \pm 0.07 \%$$

$$\mu = 487.3 \pm 0.3$$

$$\sigma = 26.3 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.26 \pm 0.08 \%$$

$$\mu = 477.6 \pm 0.3$$

$$\sigma = 25.1 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1840 < p < 1920$

Count

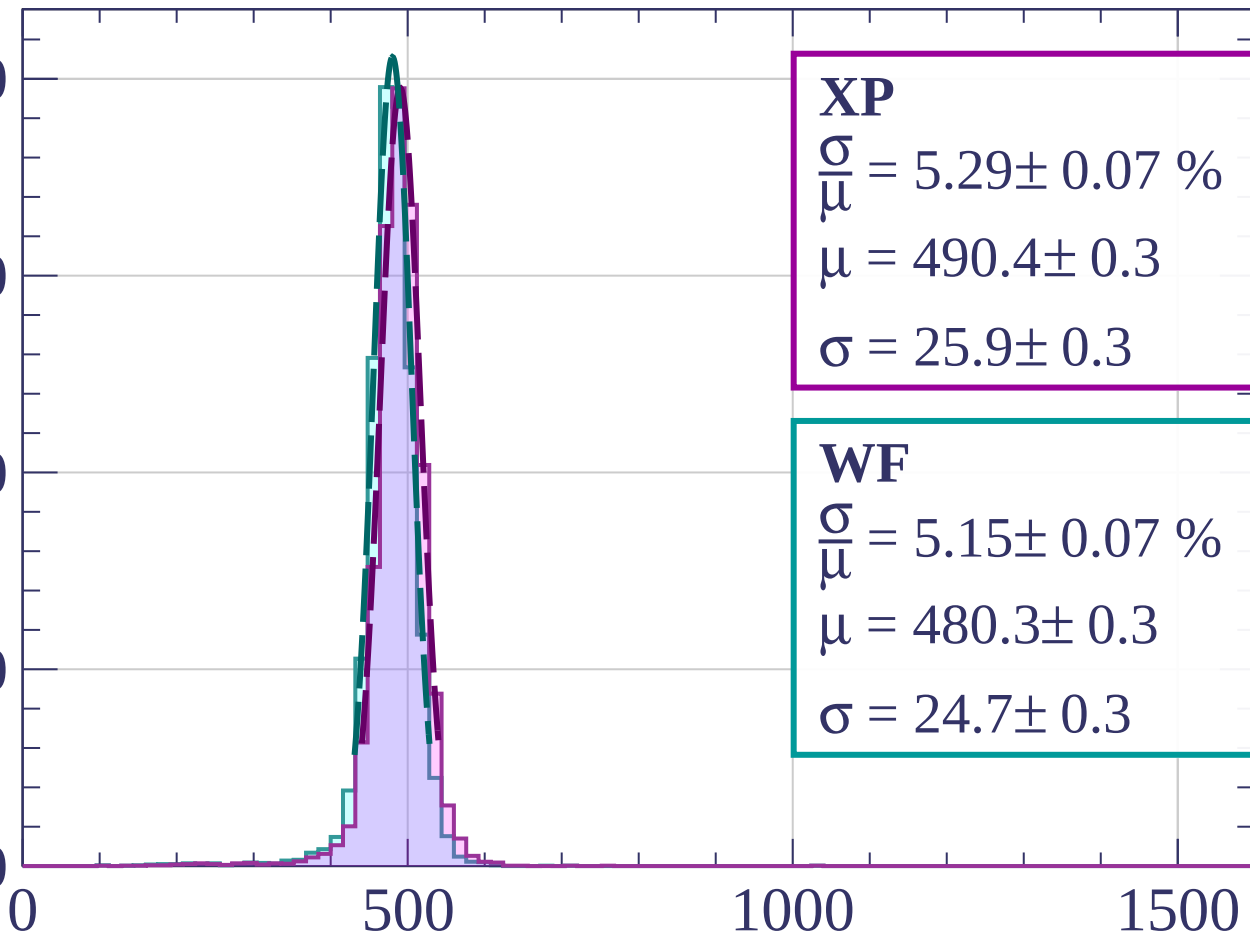
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.29 \pm 0.07 \%$$

$$\mu = 490.4 \pm 0.3$$

$$\sigma = 25.9 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.15 \pm 0.07 \%$$

$$\mu = 480.3 \pm 0.3$$

$$\sigma = 24.7 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1920 < p < 2000$

Count

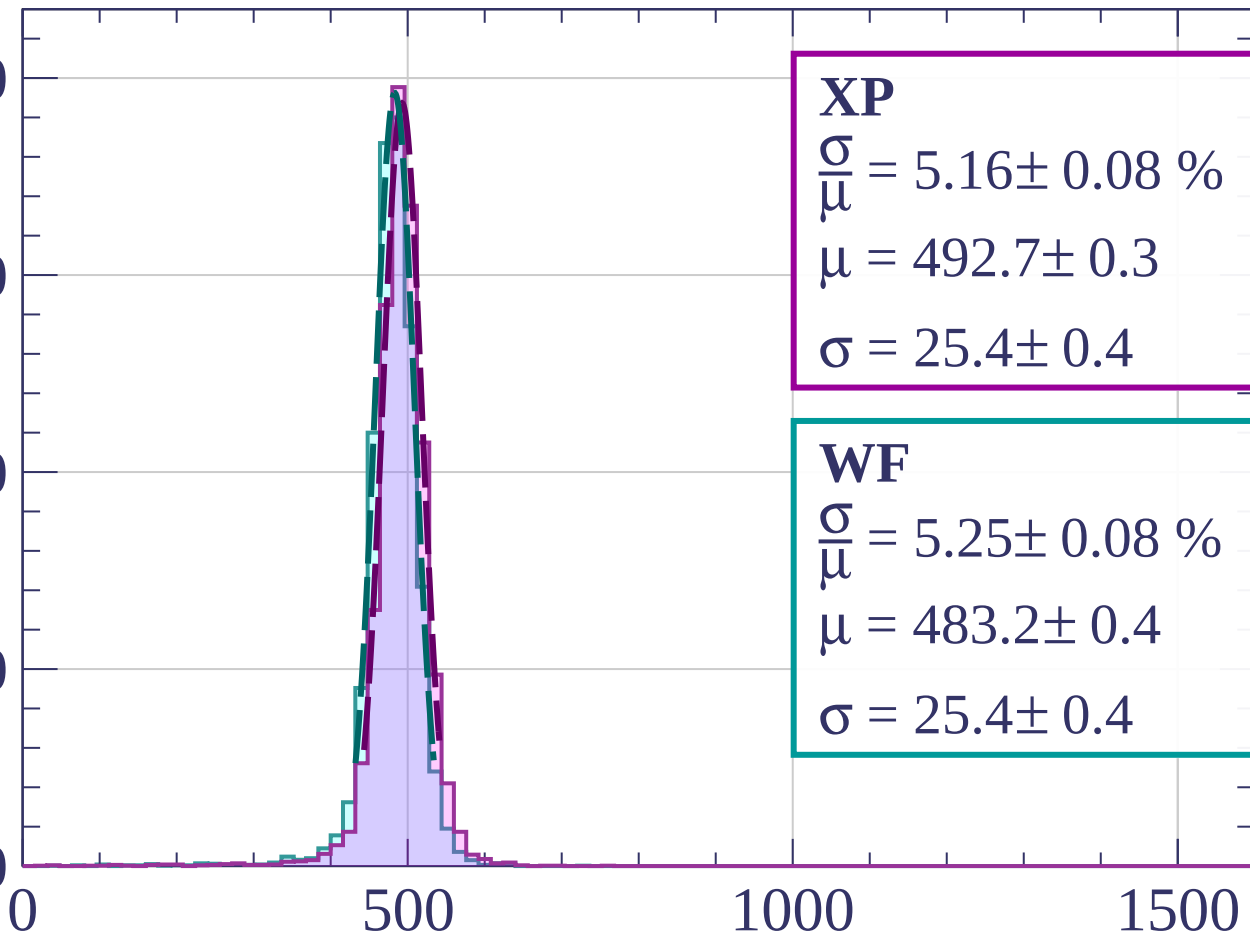
2000

1500

1000

500

0



**XP**

$$\frac{\sigma}{\mu} = 5.16 \pm 0.08 \%$$

$$\mu = 492.7 \pm 0.3$$

$$\sigma = 25.4 \pm 0.4$$

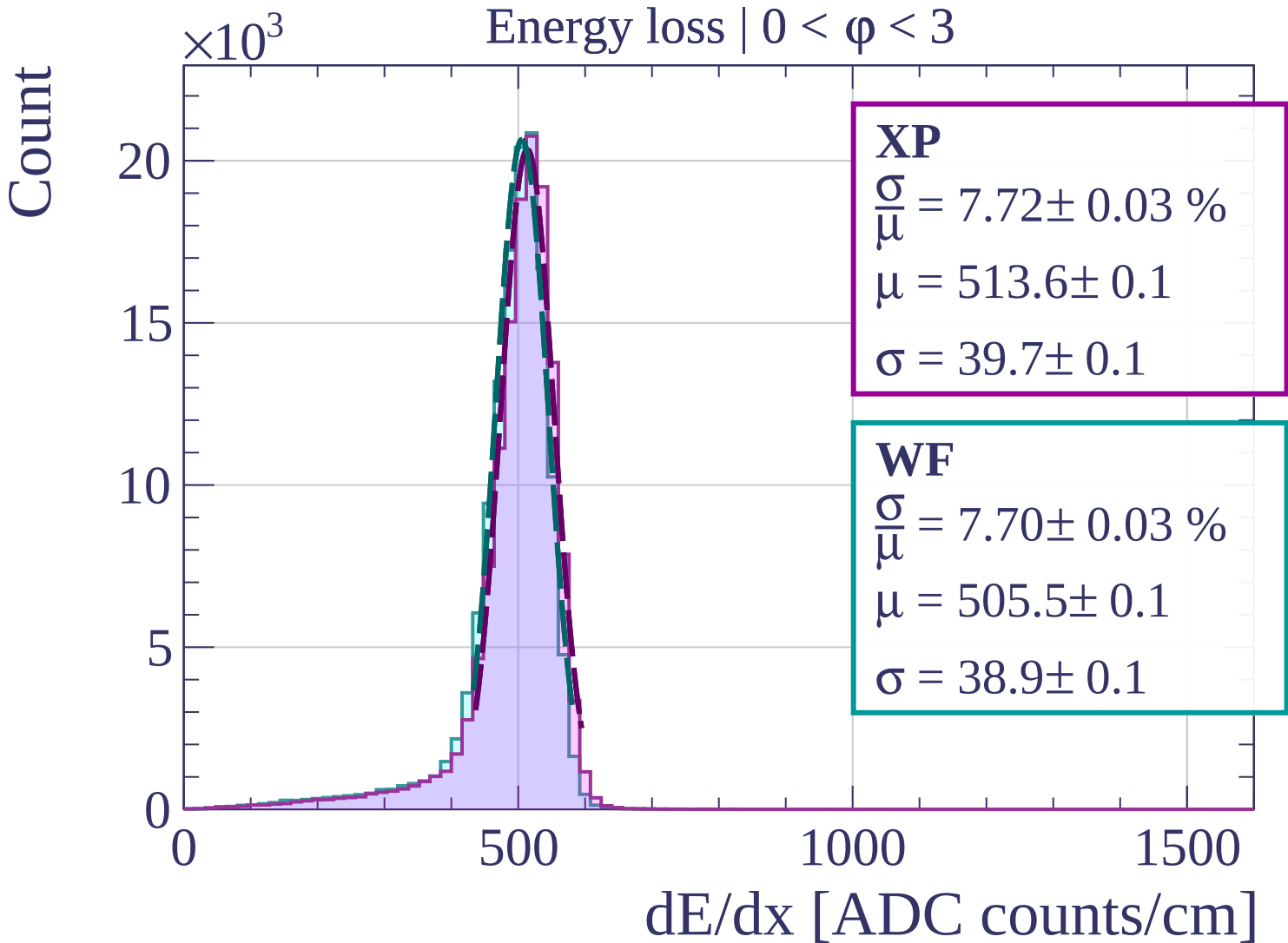
**WF**

$$\frac{\sigma}{\mu} = 5.25 \pm 0.08 \%$$

$$\mu = 483.2 \pm 0.4$$

$$\sigma = 25.4 \pm 0.4$$

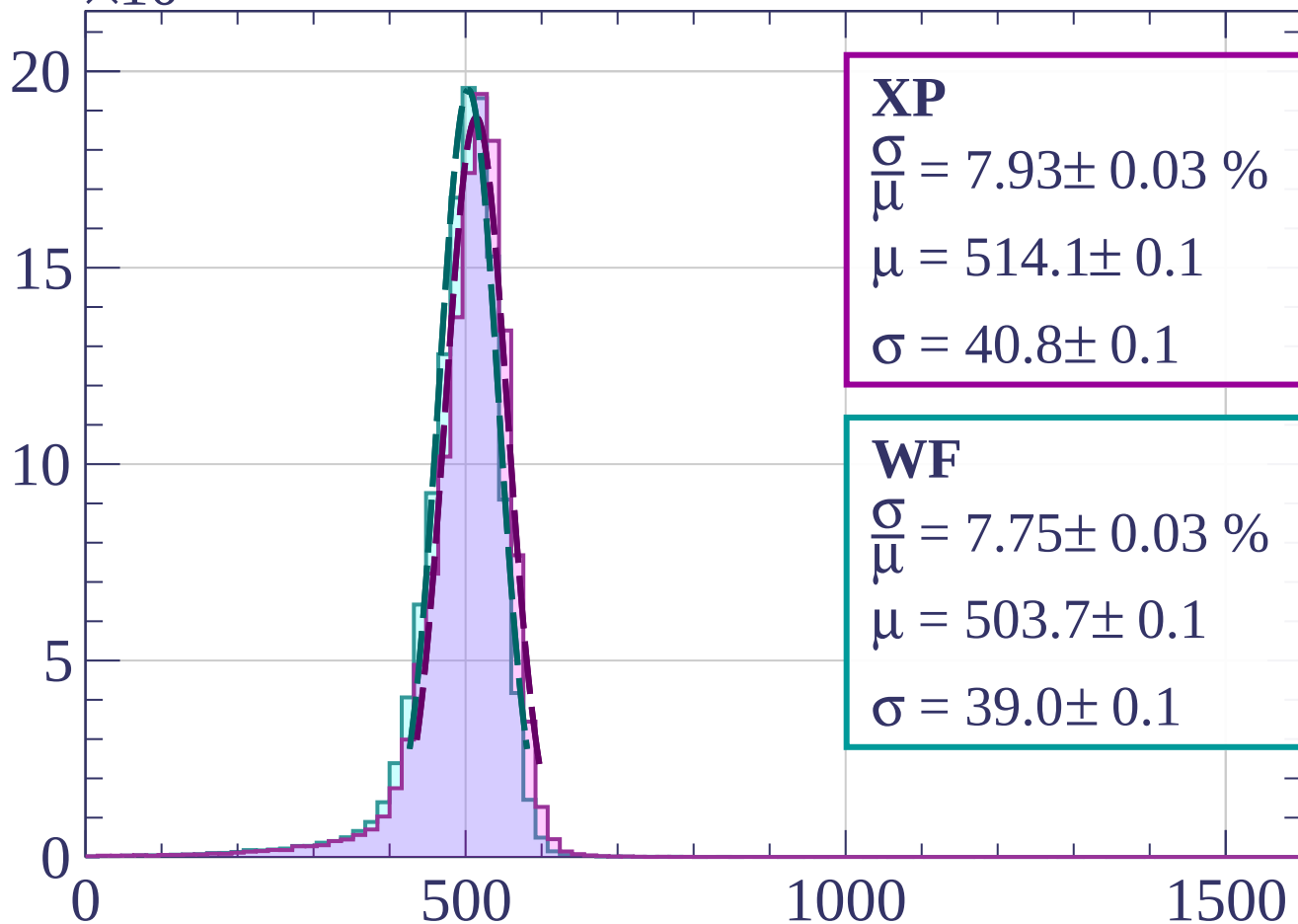
$dE/dx$  [ADC counts/cm]



Count

$\times 10^3$

Energy loss |  $3 < \phi < 6$



**XP**

$$\frac{\sigma}{\mu} = 7.93 \pm 0.03 \%$$

$$\mu = 514.1 \pm 0.1$$

$$\sigma = 40.8 \pm 0.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.75 \pm 0.03 \%$$

$$\mu = 503.7 \pm 0.1$$

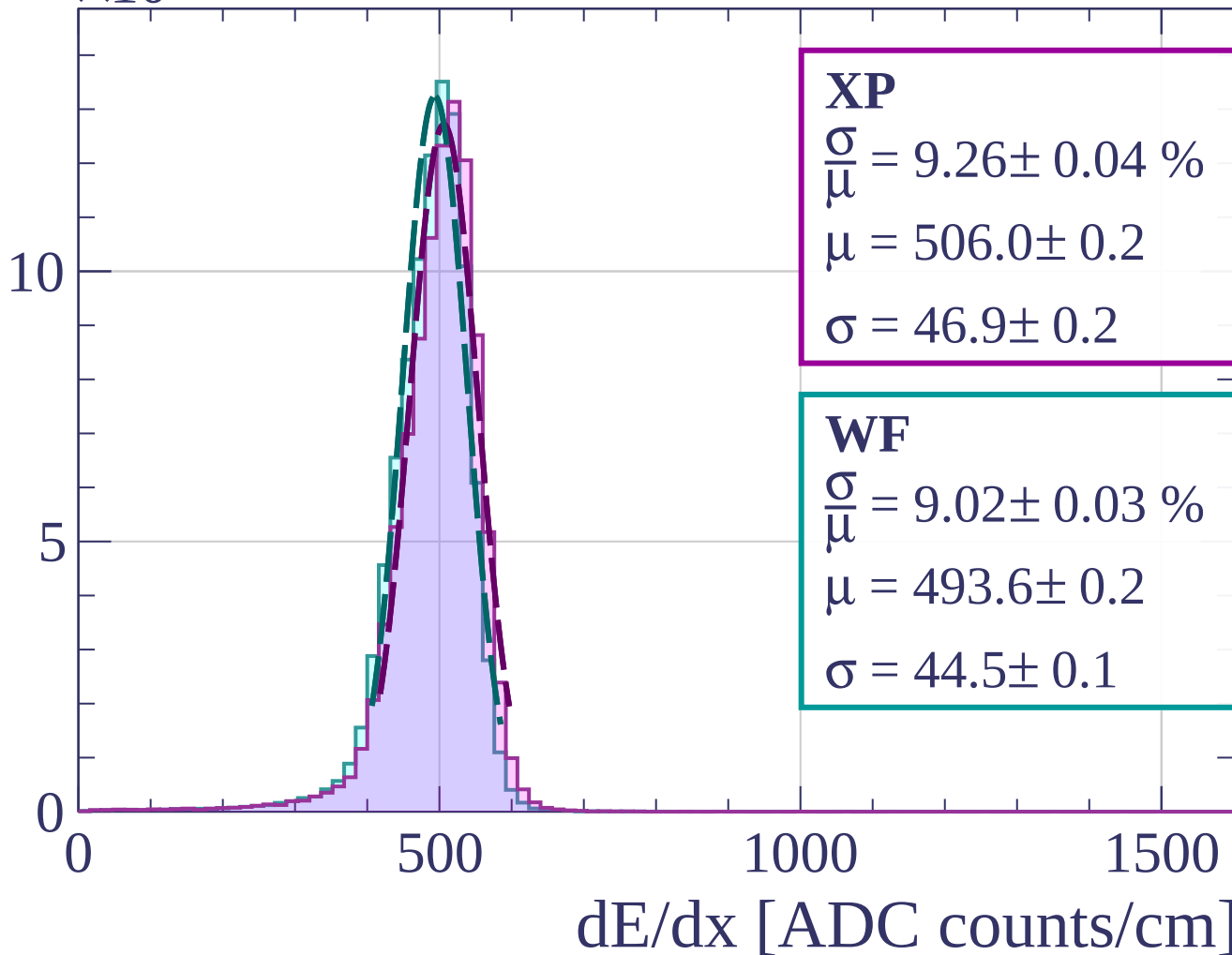
$$\sigma = 39.0 \pm 0.1$$

dE/dx [ADC counts/cm]

Count

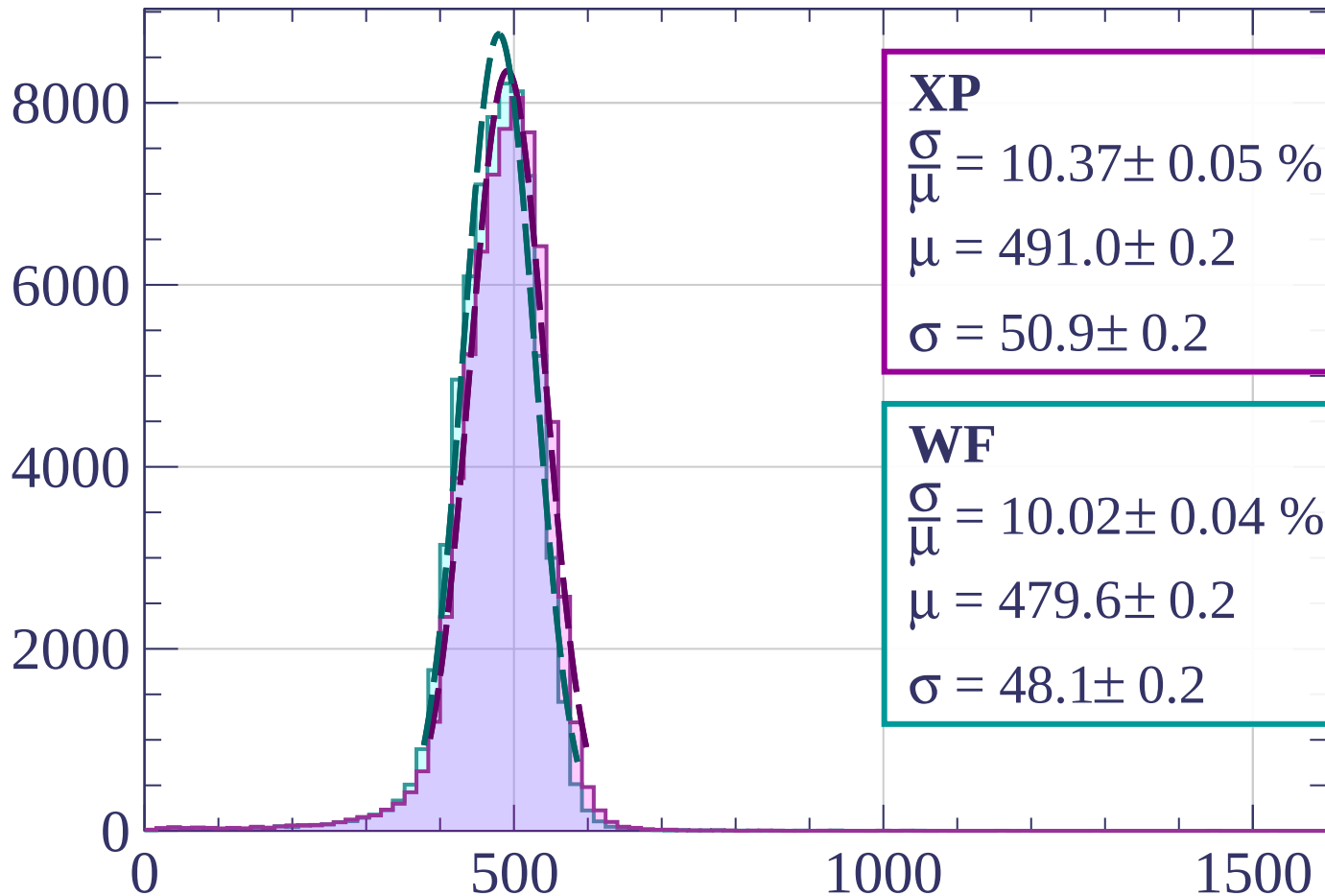
$\times 10^3$

Energy loss |  $6 < \phi < 9$



Energy loss |  $9 < \phi < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.37 \pm 0.05 \%$$

$$\mu = 491.0 \pm 0.2$$

$$\sigma = 50.9 \pm 0.2$$

**WF**

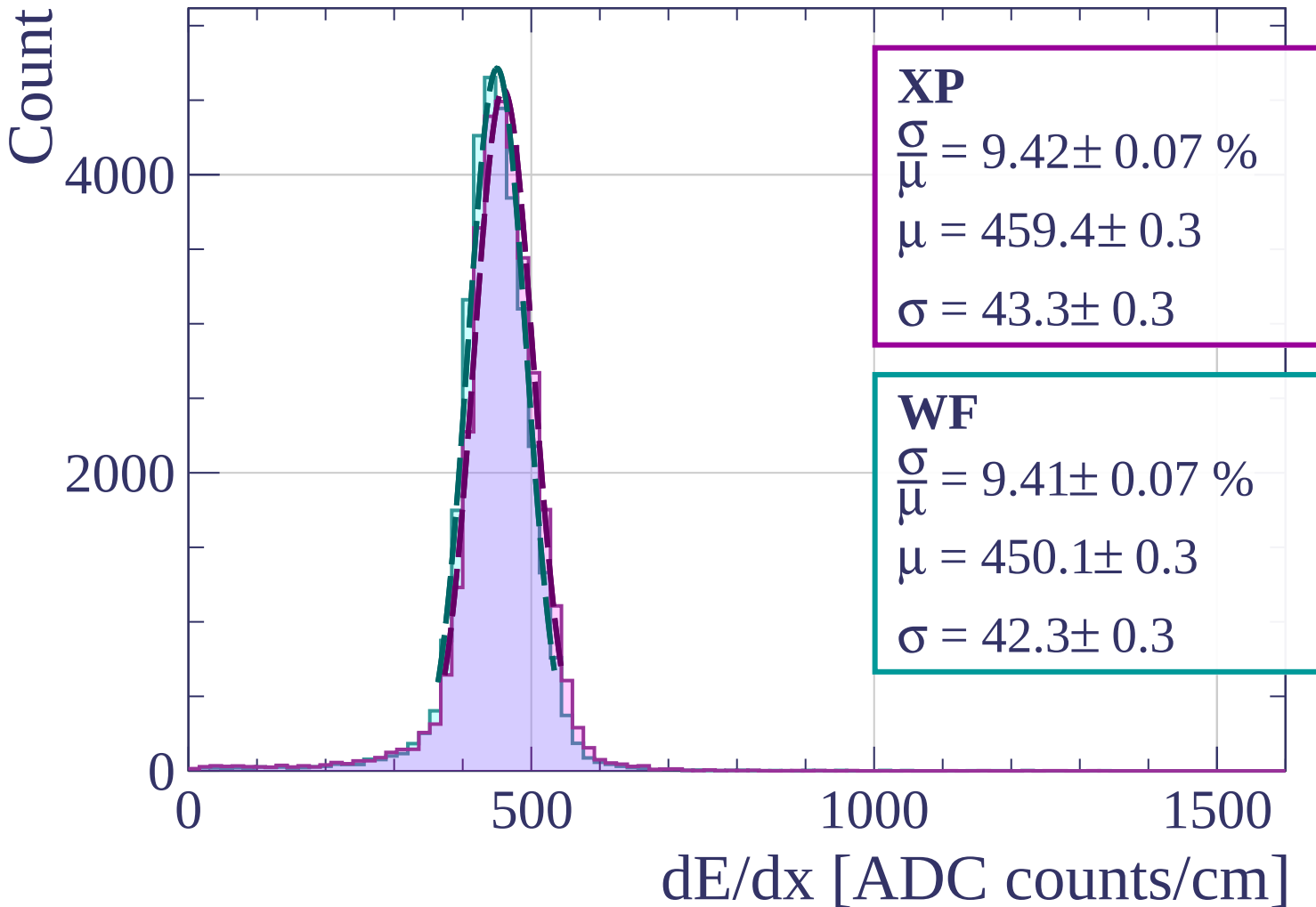
$$\frac{\sigma}{\mu} = 10.02 \pm 0.04 \%$$

$$\mu = 479.6 \pm 0.2$$

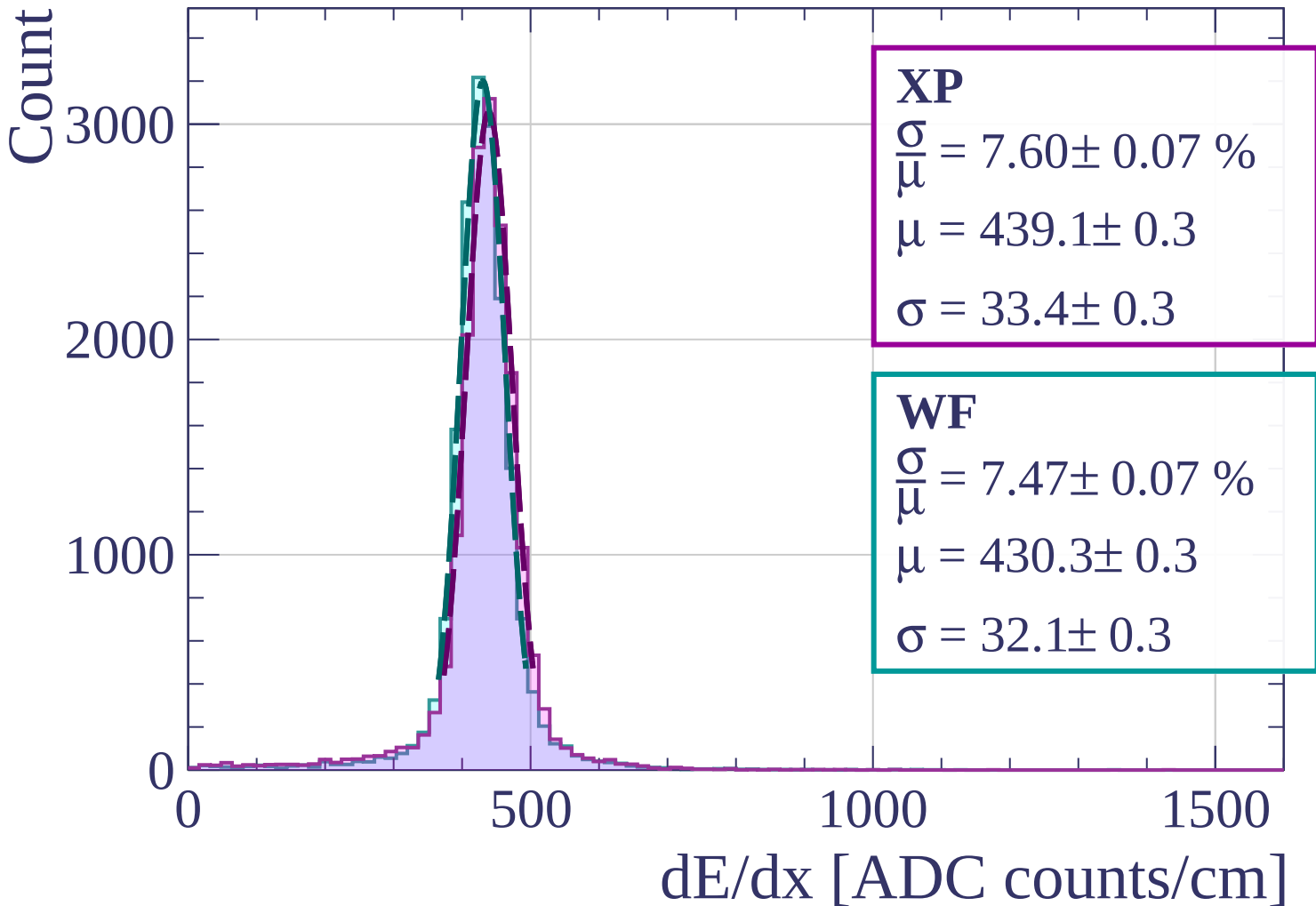
$$\sigma = 48.1 \pm 0.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $12 < \phi < 15$



Energy loss |  $15 < \phi < 18$



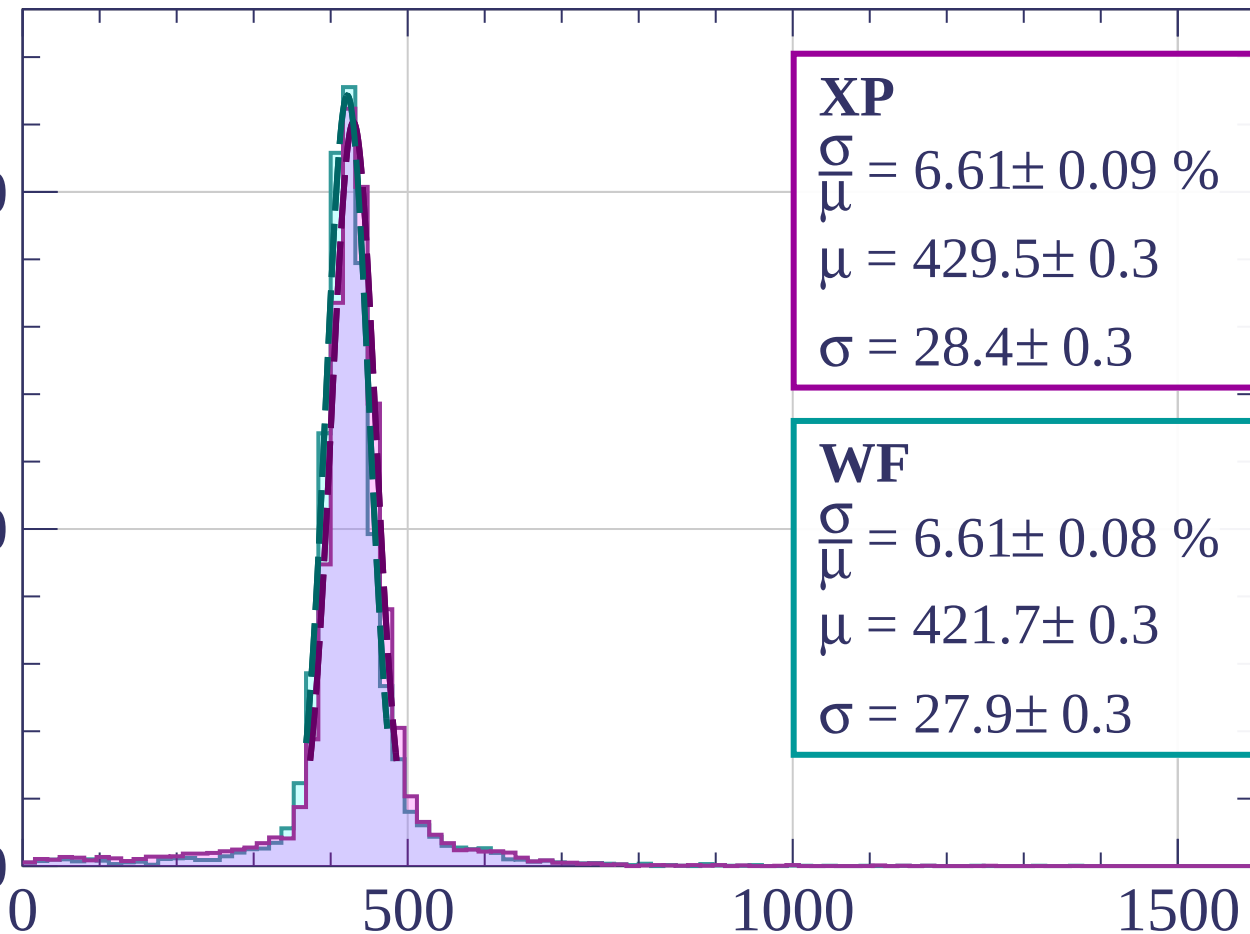
Energy loss |  $18 < \phi < 21$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 6.61 \pm 0.09 \%$$

$$\mu = 429.5 \pm 0.3$$

$$\sigma = 28.4 \pm 0.3$$

**WF**

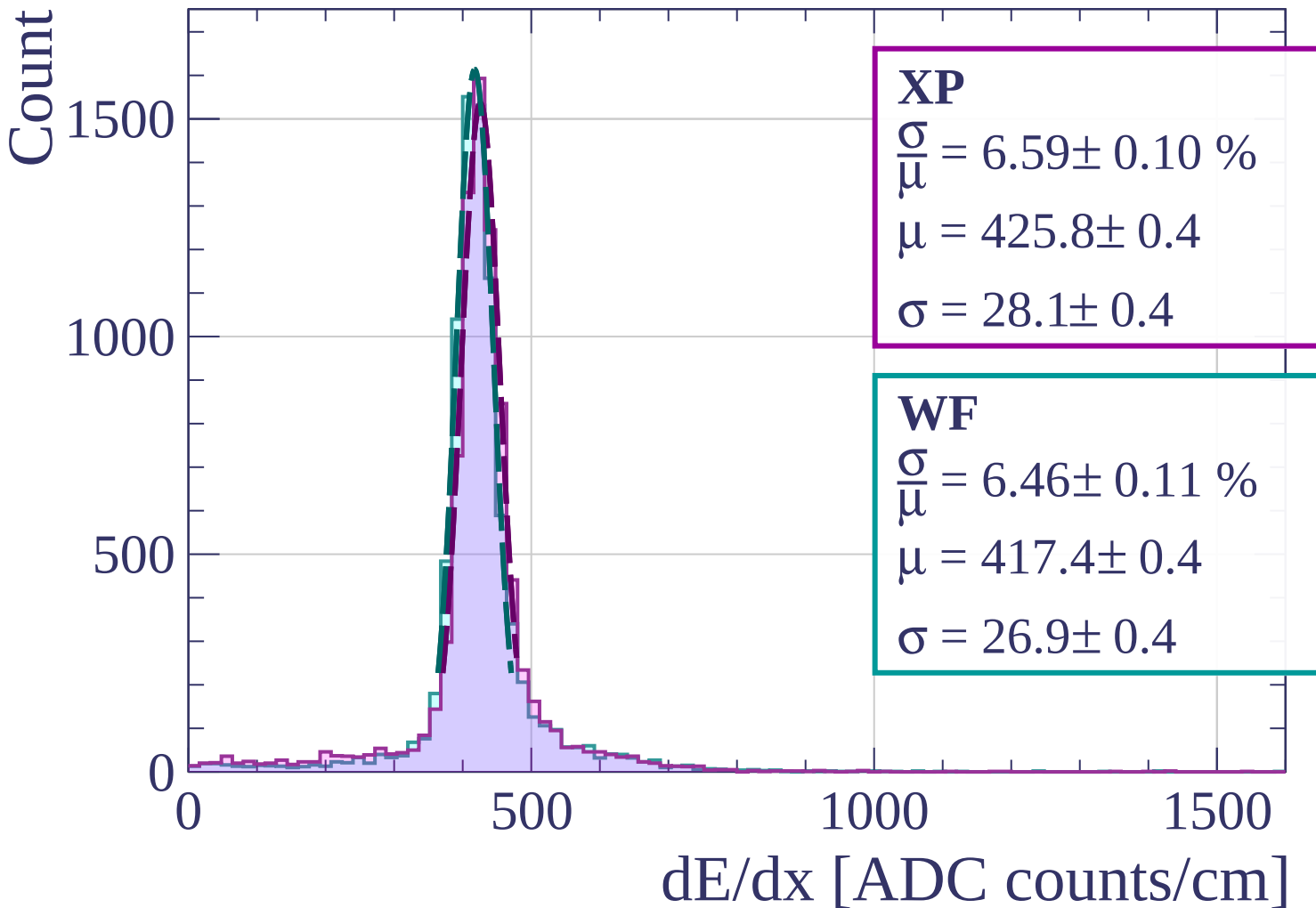
$$\frac{\sigma}{\mu} = 6.61 \pm 0.08 \%$$

$$\mu = 421.7 \pm 0.3$$

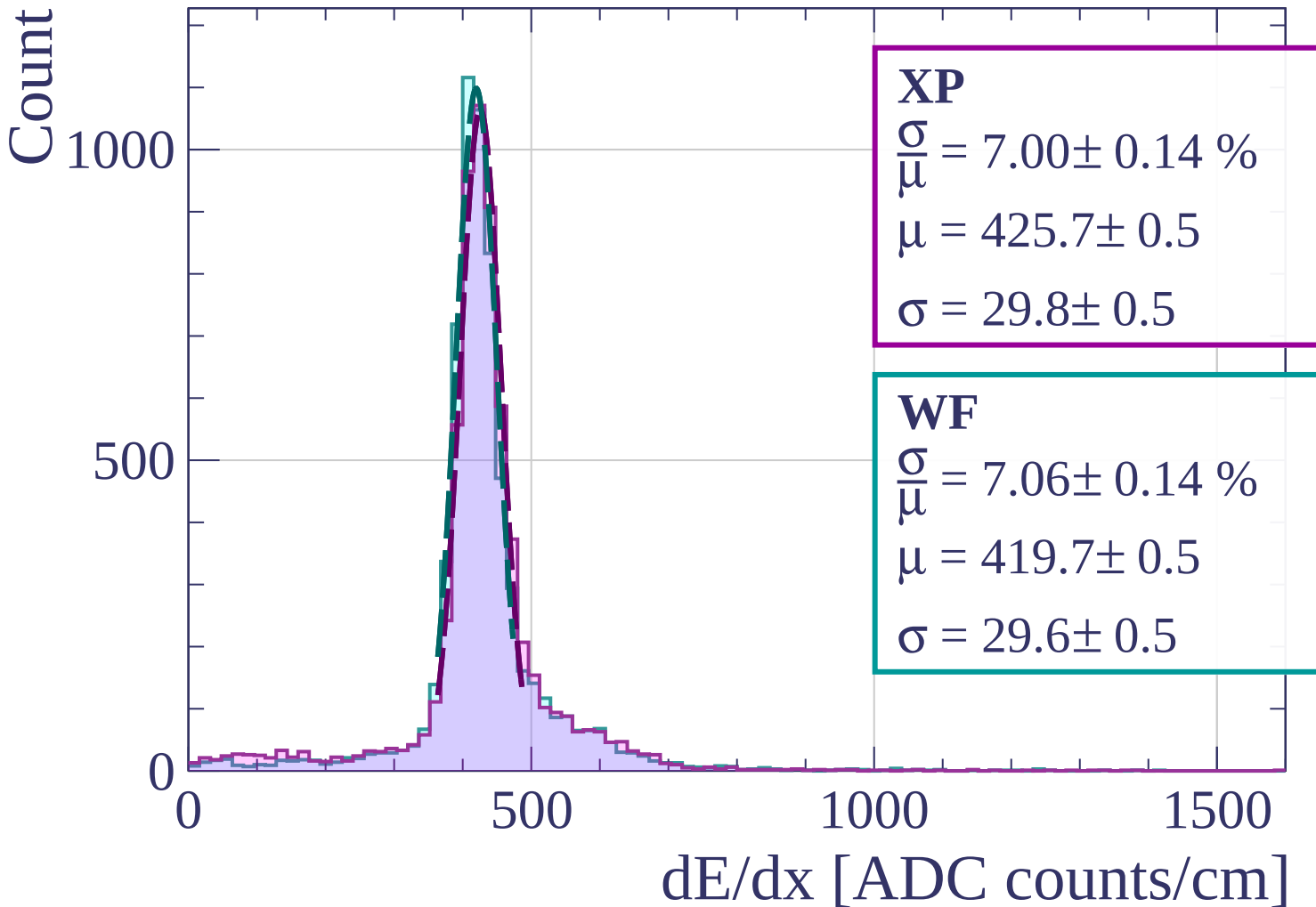
$$\sigma = 27.9 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss |  $21 < \phi < 24$

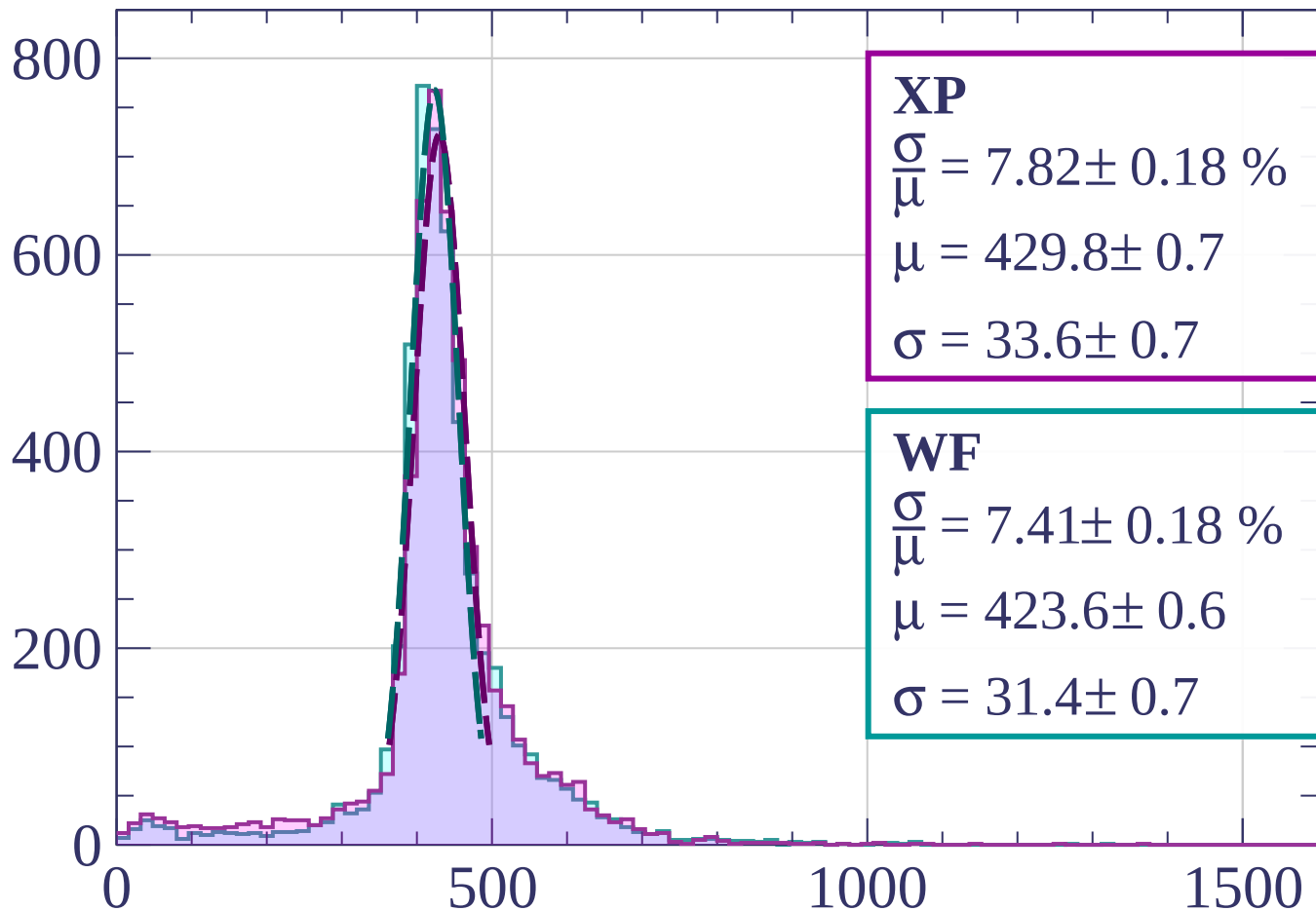


Energy loss |  $24 < \phi < 27$



Energy loss |  $27 < \phi < 30$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.82 \pm 0.18 \%$$

$$\mu = 429.8 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 7.41 \pm 0.18 \%$$

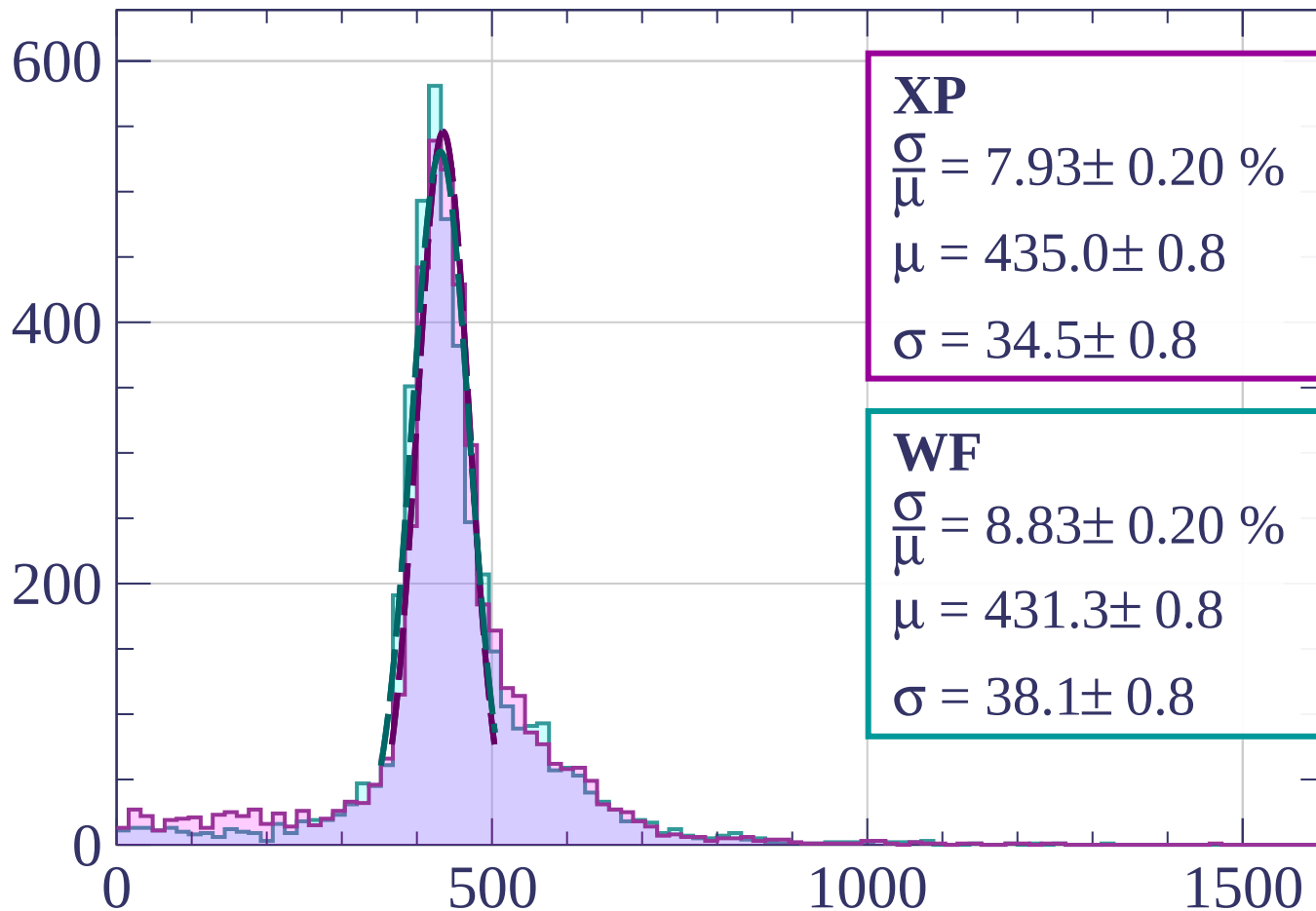
$$\mu = 423.6 \pm 0.6$$

$$\sigma = 31.4 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $30 < \phi < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.93 \pm 0.20 \%$$

$$\mu = 435.0 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.83 \pm 0.20 \%$$

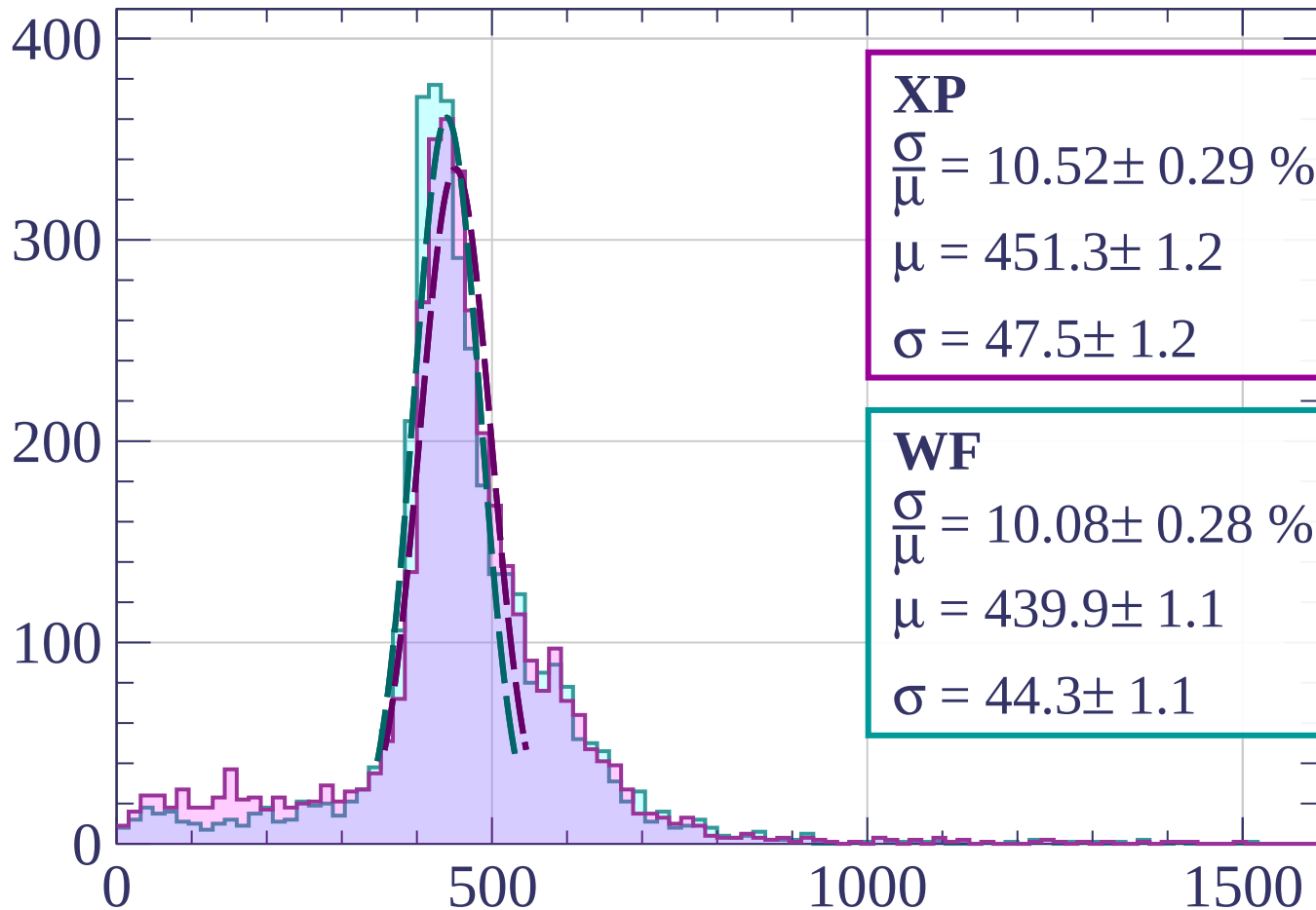
$$\mu = 431.3 \pm 0.8$$

$$\sigma = 38.1 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss |  $33 < \phi < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.52 \pm 0.29 \%$$

$$\mu = 451.3 \pm 1.2$$

$$\sigma = 47.5 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 10.08 \pm 0.28 \%$$

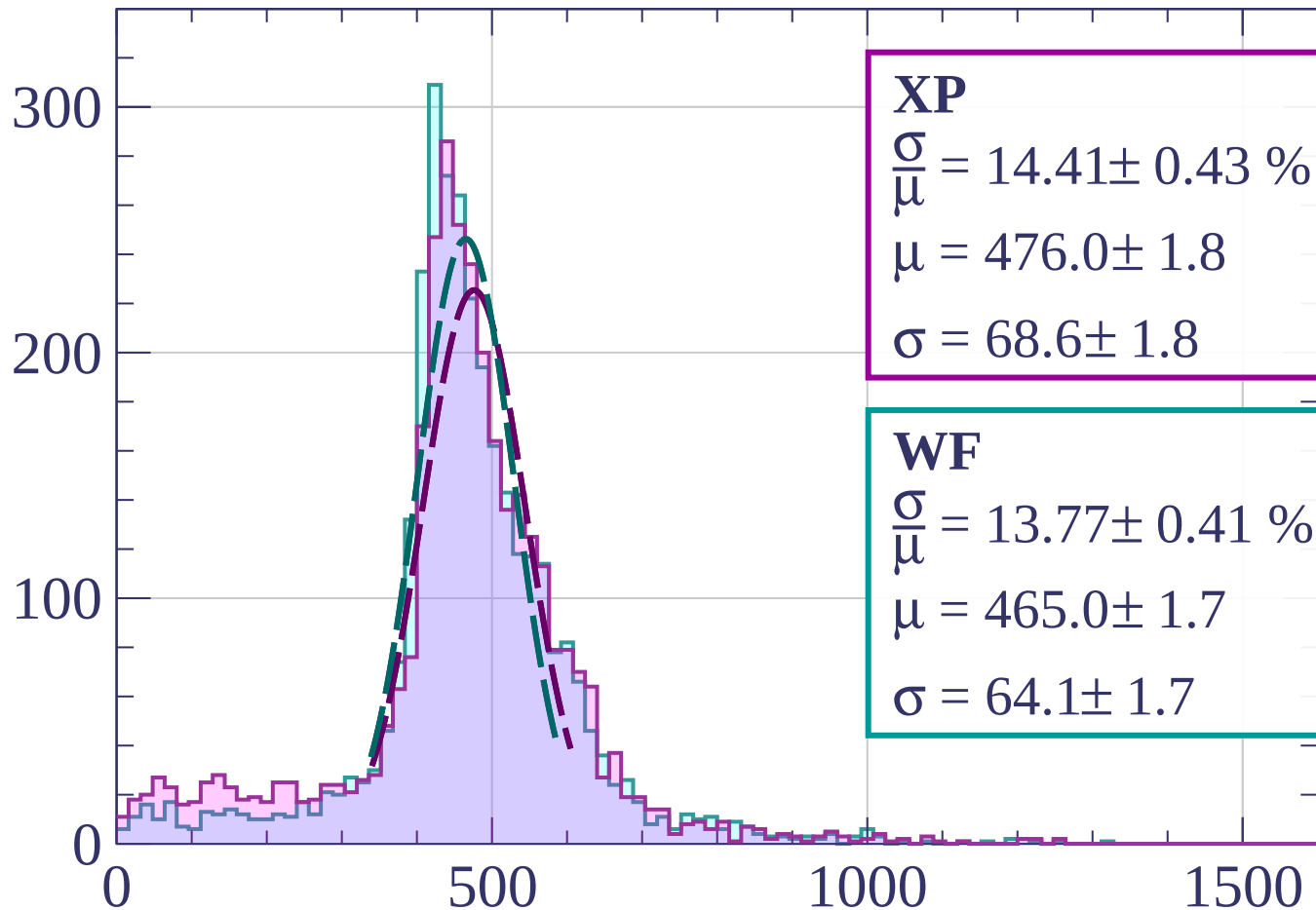
$$\mu = 439.9 \pm 1.1$$

$$\sigma = 44.3 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss |  $36 < \phi < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 14.41 \pm 0.43 \%$$

$$\mu = 476.0 \pm 1.8$$

$$\sigma = 68.6 \pm 1.8$$

**WF**

$$\frac{\sigma}{\mu} = 13.77 \pm 0.41 \%$$

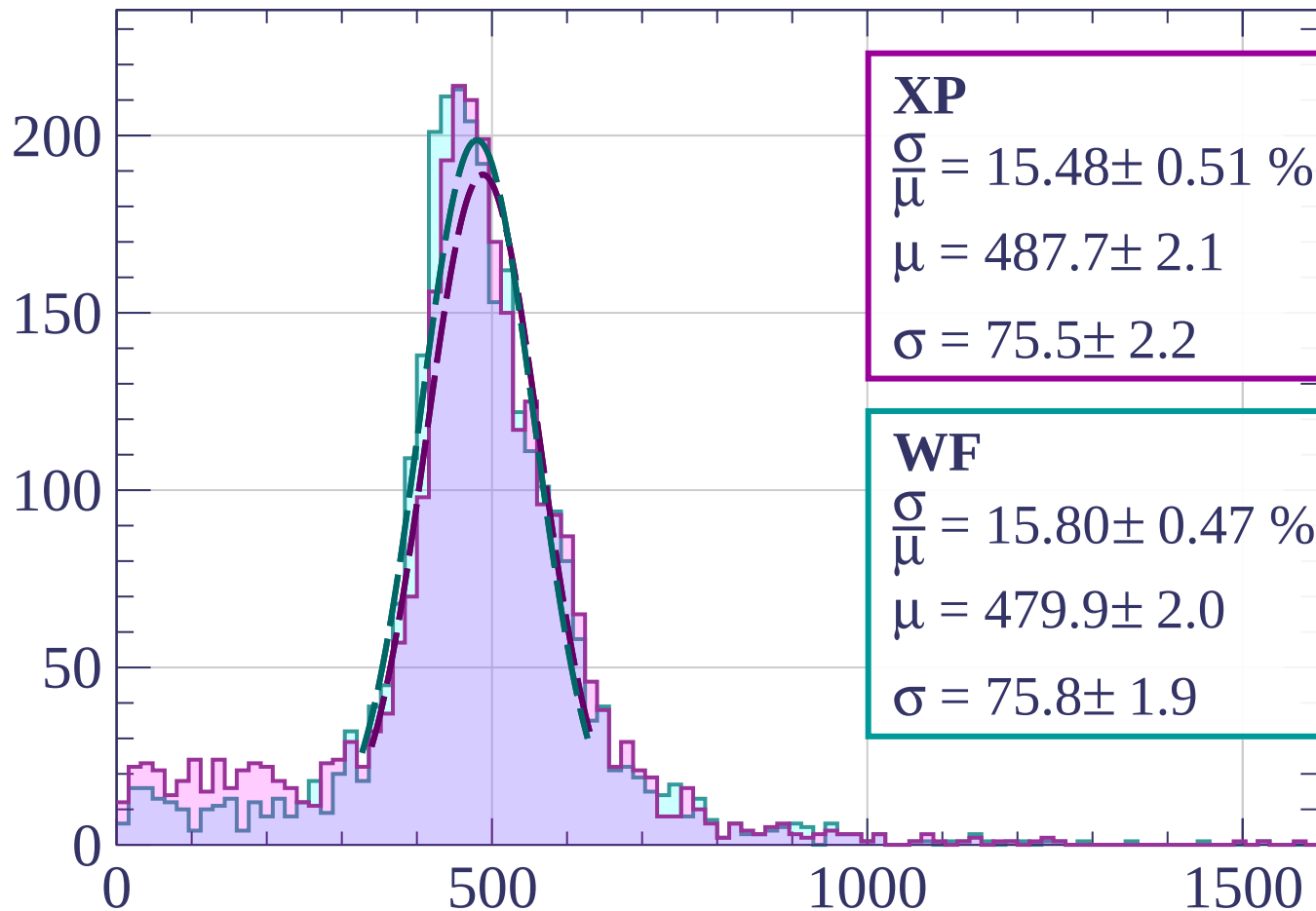
$$\mu = 465.0 \pm 1.7$$

$$\sigma = 64.1 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss |  $39 < \phi < 42$

Count



**XP**

$$\frac{\sigma}{\mu} = 15.48 \pm 0.51 \%$$

$$\mu = 487.7 \pm 2.1$$

$$\sigma = 75.5 \pm 2.2$$

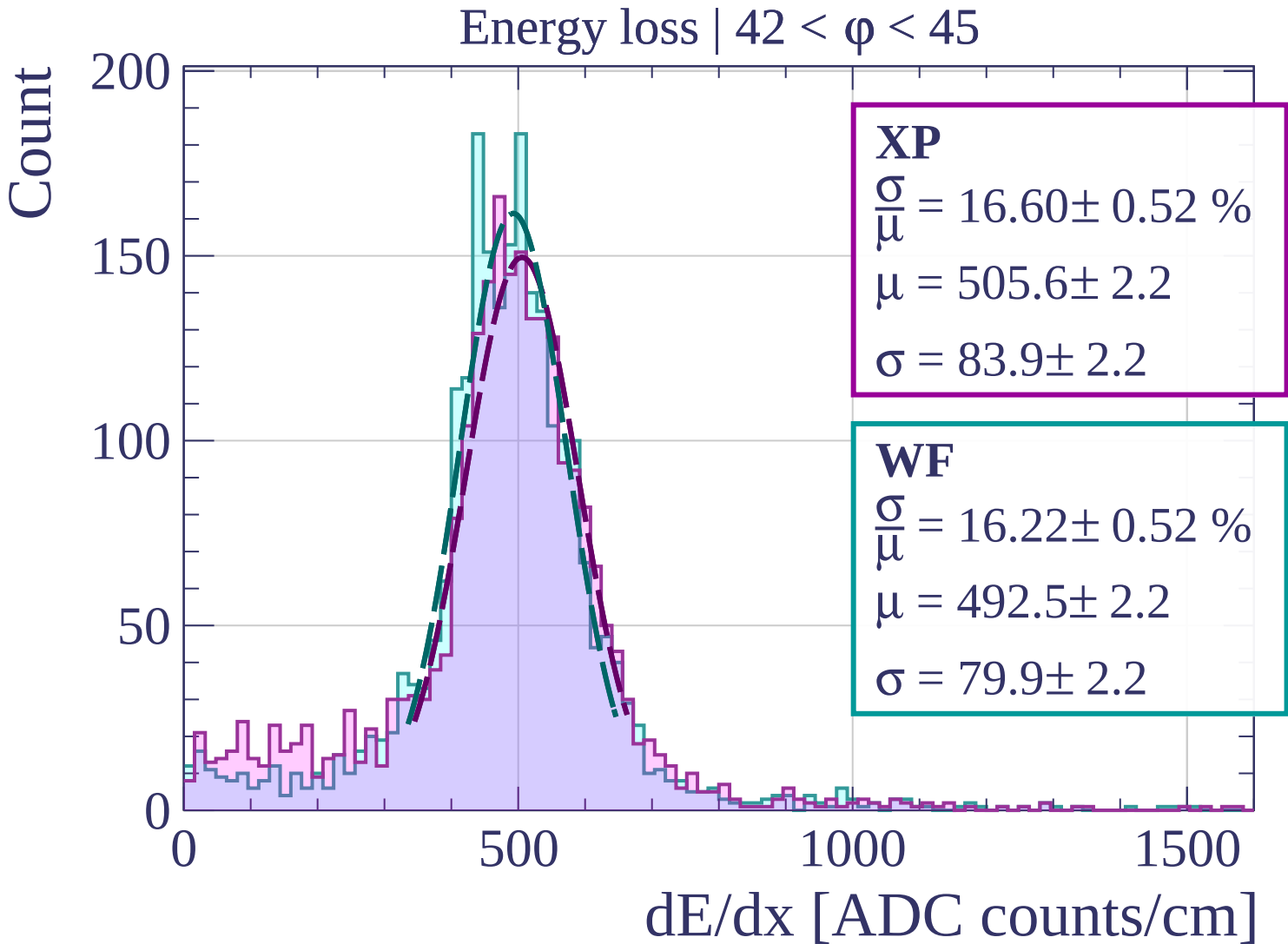
**WF**

$$\frac{\sigma}{\mu} = 15.80 \pm 0.47 \%$$

$$\mu = 479.9 \pm 2.0$$

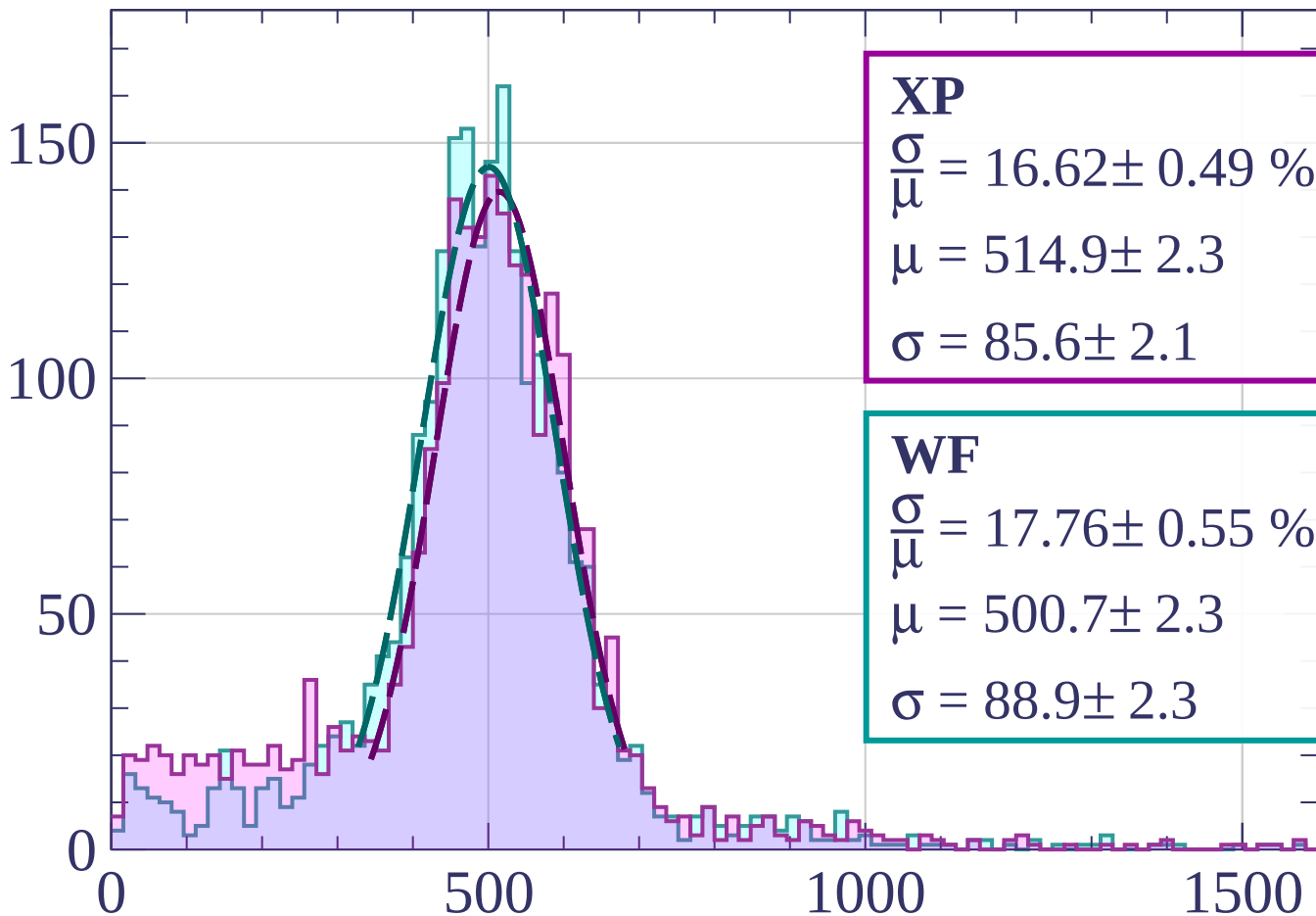
$$\sigma = 75.8 \pm 1.9$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $45 < \phi < 48$

Count



**XP**

$$\frac{\sigma}{\mu} = 16.62 \pm 0.49 \%$$

$$\mu = 514.9 \pm 2.3$$

$$\sigma = 85.6 \pm 2.1$$

**WF**

$$\frac{\sigma}{\mu} = 17.76 \pm 0.55 \%$$

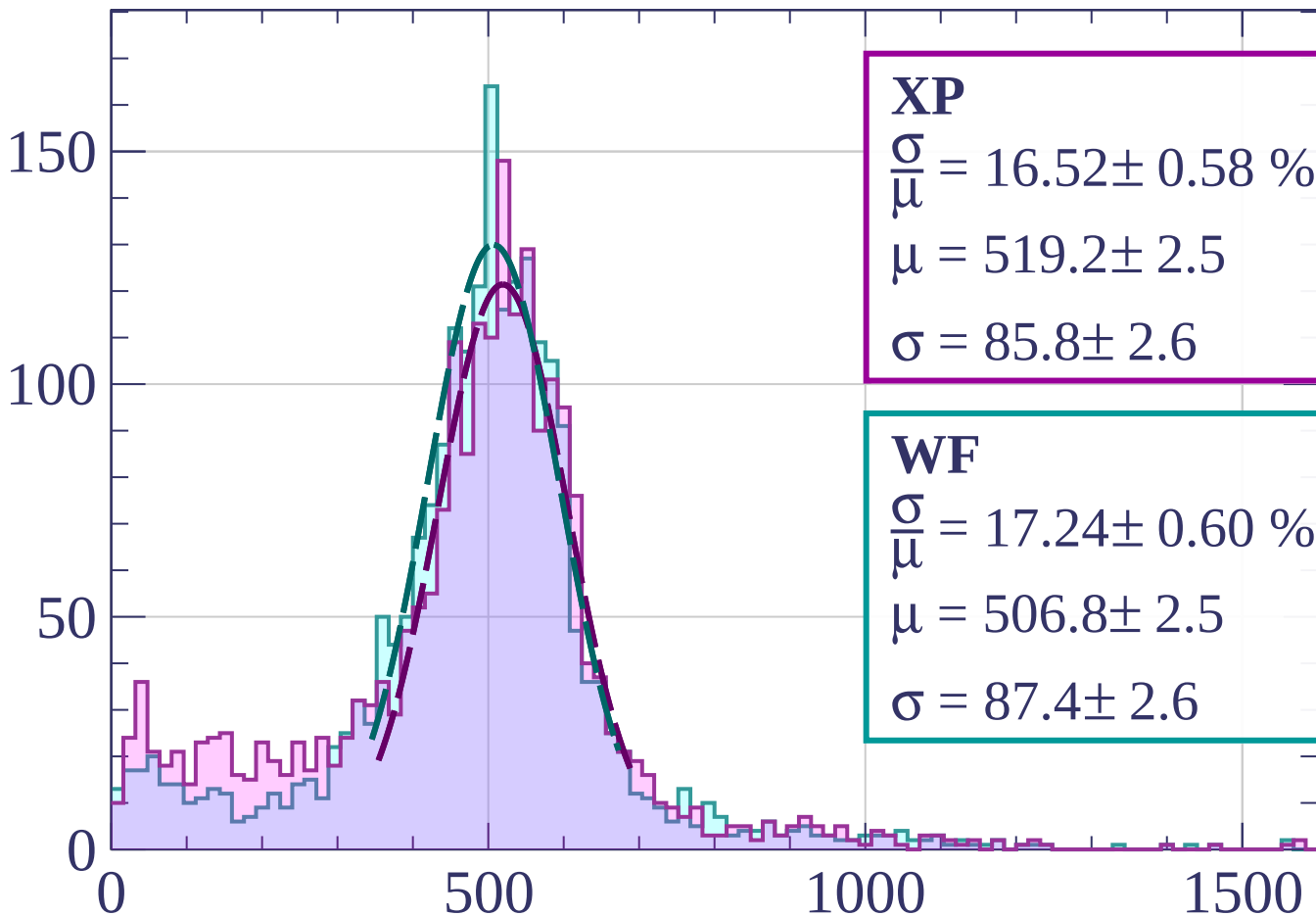
$$\mu = 500.7 \pm 2.3$$

$$\sigma = 88.9 \pm 2.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $48 < \phi < 51$

Count



**XP**

$$\frac{\sigma}{\mu} = 16.52 \pm 0.58 \%$$

$$\mu = 519.2 \pm 2.5$$

$$\sigma = 85.8 \pm 2.6$$

**WF**

$$\frac{\sigma}{\mu} = 17.24 \pm 0.60 \%$$

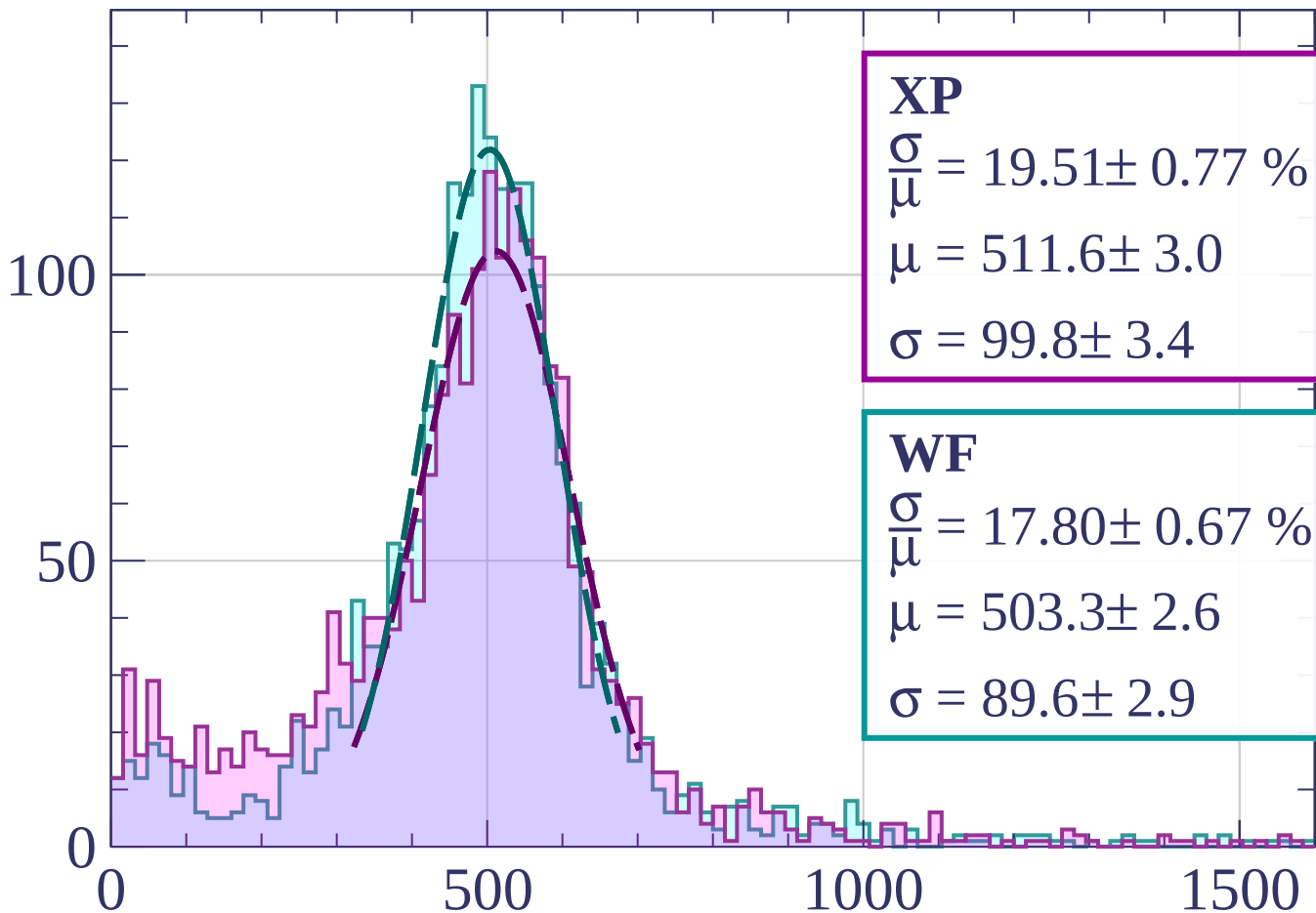
$$\mu = 506.8 \pm 2.5$$

$$\sigma = 87.4 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss |  $51 < \phi < 54$

Count



**XP**

$$\frac{\sigma}{\mu} = 19.51 \pm 0.77 \%$$

$$\mu = 511.6 \pm 3.0$$

$$\sigma = 99.8 \pm 3.4$$

**WF**

$$\frac{\sigma}{\mu} = 17.80 \pm 0.67 \%$$

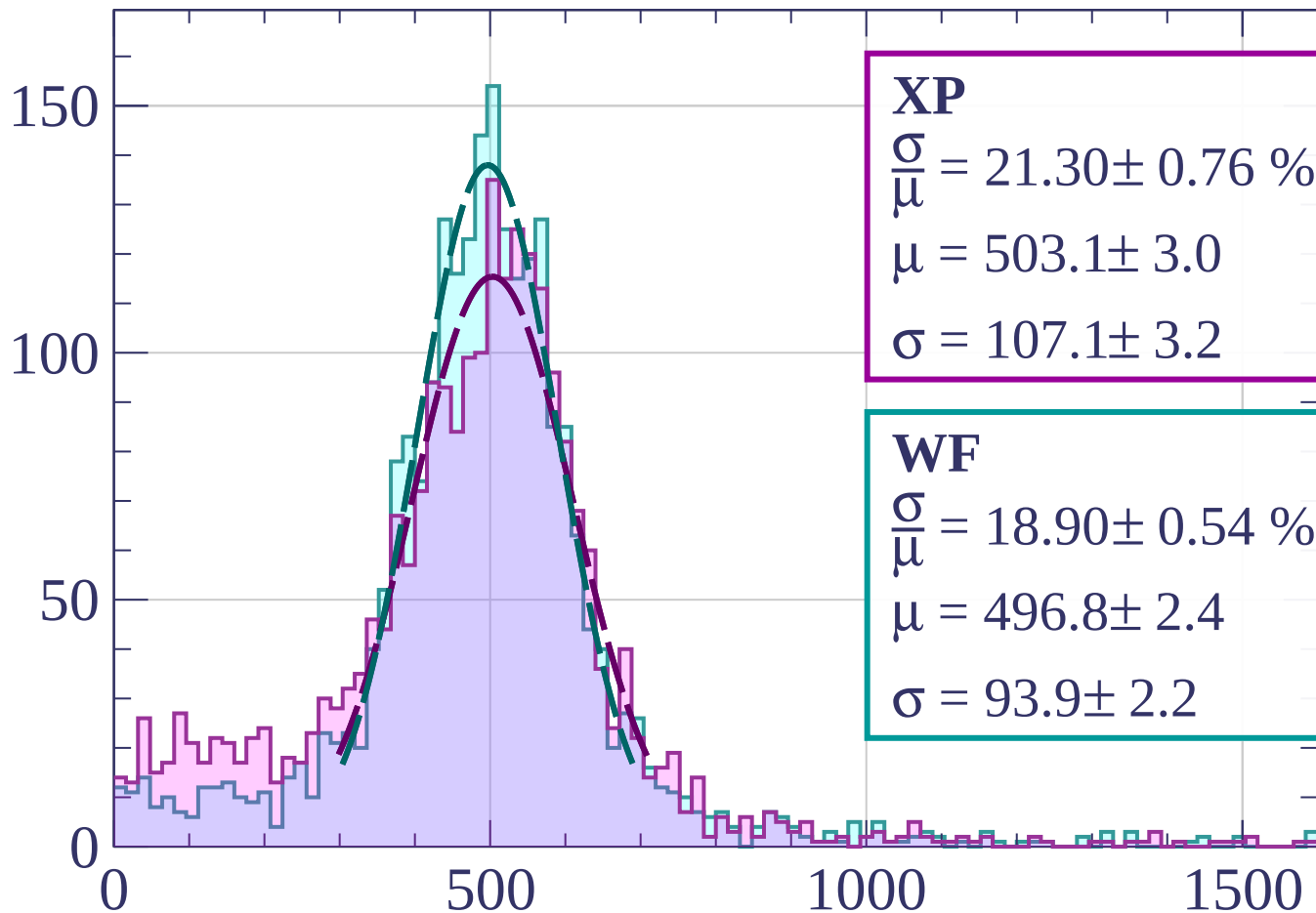
$$\mu = 503.3 \pm 2.6$$

$$\sigma = 89.6 \pm 2.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $54 < \phi < 57$

Count



**XP**

$$\frac{\sigma}{\mu} = 21.30 \pm 0.76 \%$$

$$\mu = 503.1 \pm 3.0$$

$$\sigma = 107.1 \pm 3.2$$

**WF**

$$\frac{\sigma}{\mu} = 18.90 \pm 0.54 \%$$

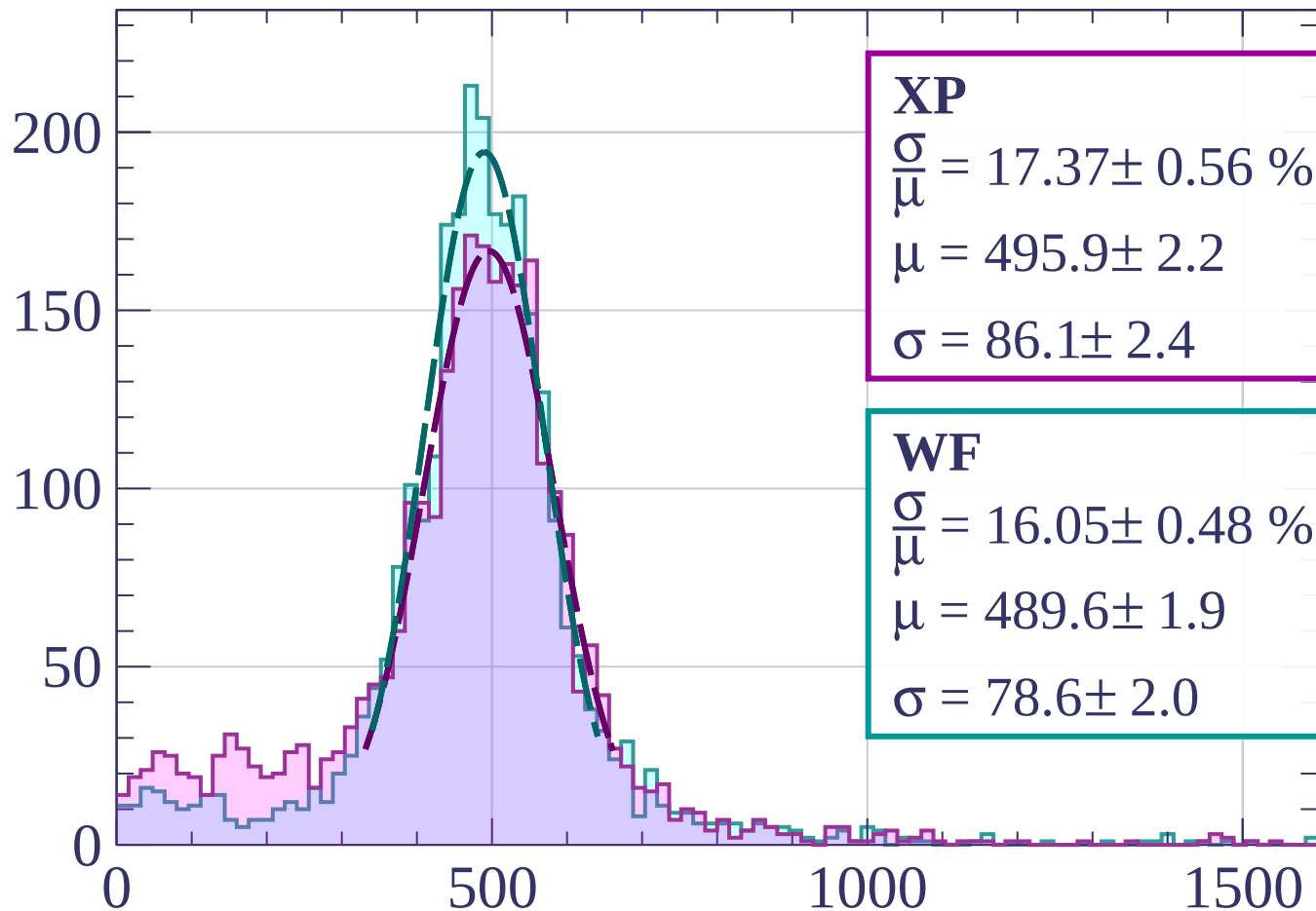
$$\mu = 496.8 \pm 2.4$$

$$\sigma = 93.9 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss |  $57 < \phi < 60$

Count



**XP**

$$\frac{\sigma}{\mu} = 17.37 \pm 0.56 \%$$

$$\mu = 495.9 \pm 2.2$$

$$\sigma = 86.1 \pm 2.4$$

**WF**

$$\frac{\sigma}{\mu} = 16.05 \pm 0.48 \%$$

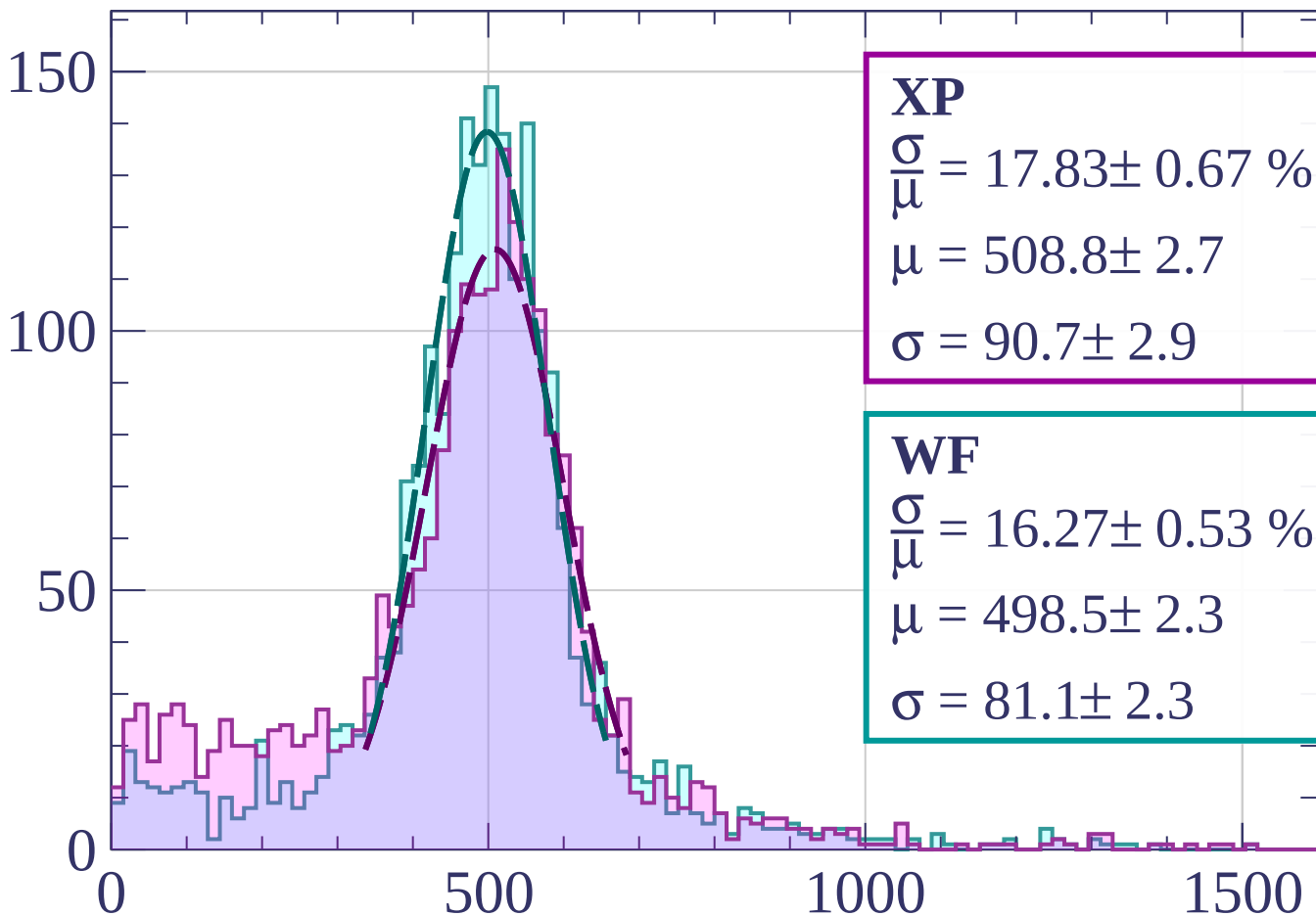
$$\mu = 489.6 \pm 1.9$$

$$\sigma = 78.6 \pm 2.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $60 < \phi < 63$

Count



**XP**

$$\frac{\sigma}{\mu} = 17.83 \pm 0.67 \%$$

$$\mu = 508.8 \pm 2.7$$

$$\sigma = 90.7 \pm 2.9$$

**WF**

$$\frac{\sigma}{\mu} = 16.27 \pm 0.53 \%$$

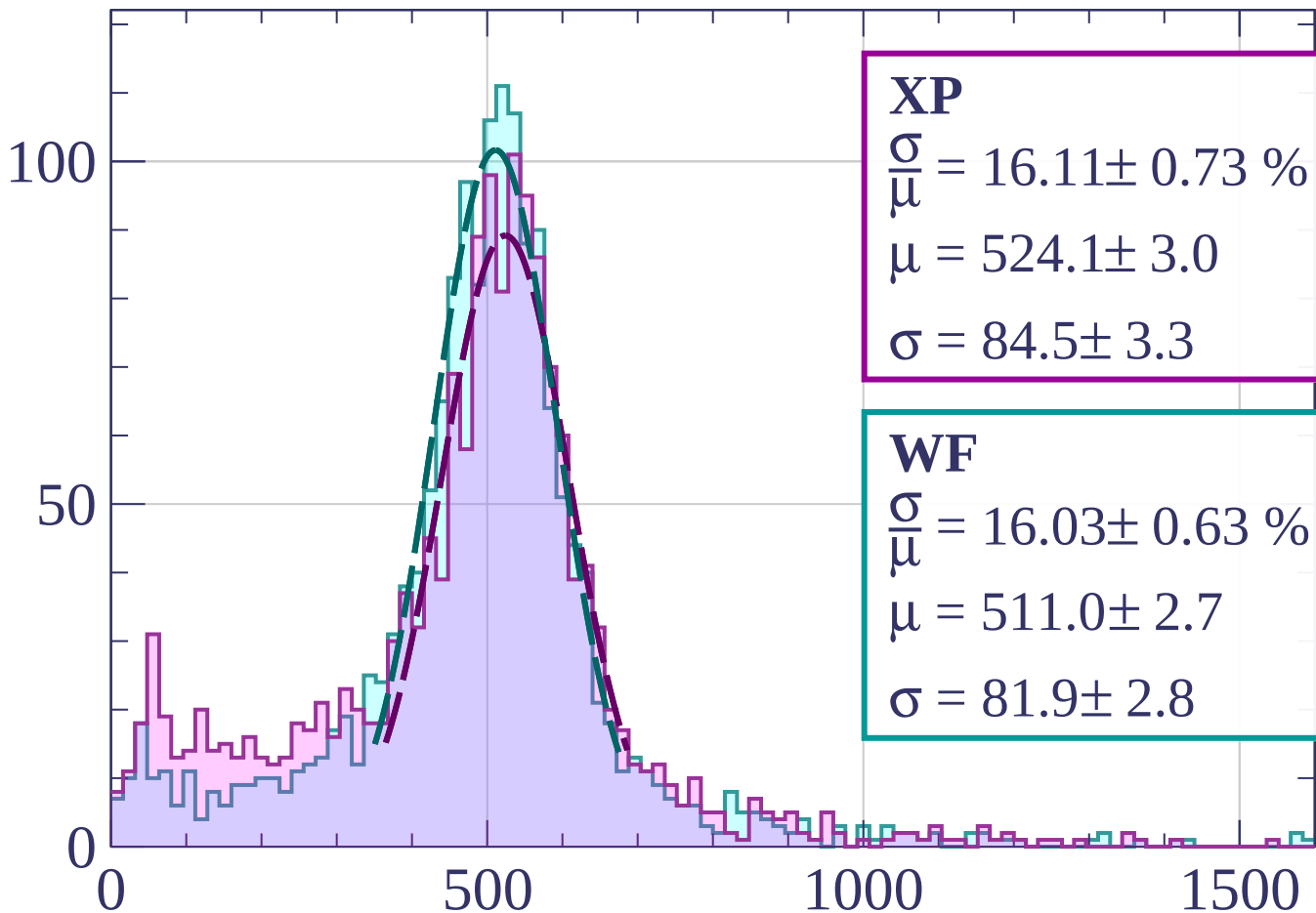
$$\mu = 498.5 \pm 2.3$$

$$\sigma = 81.1 \pm 2.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $63 < \phi < 66$

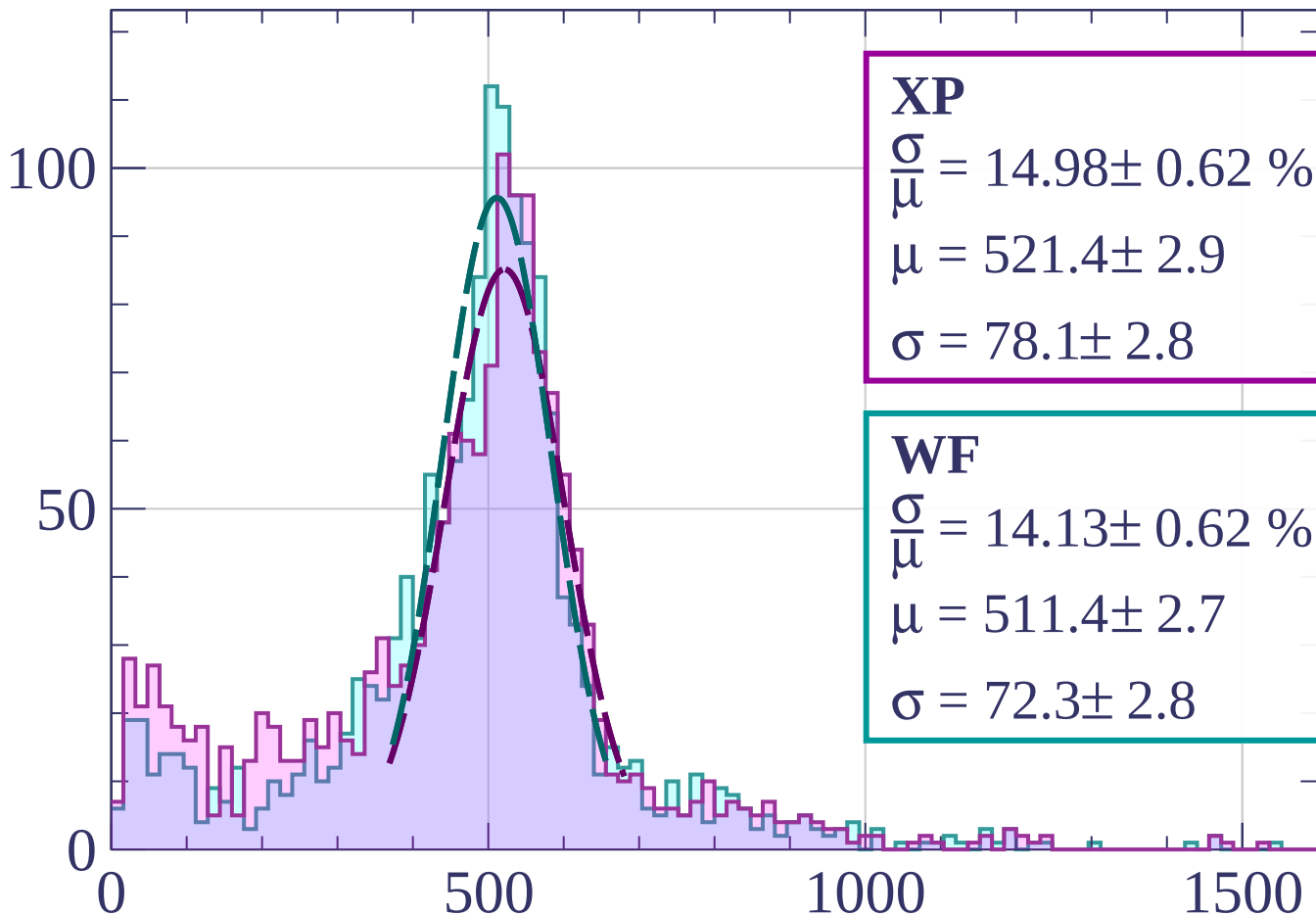
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $66 < \phi < 69$

Count



**XP**

$$\frac{\sigma}{\mu} = 14.98 \pm 0.62 \%$$

$$\mu = 521.4 \pm 2.9$$

$$\sigma = 78.1 \pm 2.8$$

**WF**

$$\frac{\sigma}{\mu} = 14.13 \pm 0.62 \%$$

$$\mu = 511.4 \pm 2.7$$

$$\sigma = 72.3 \pm 2.8$$

$dE/dx$  [ADC counts/cm]

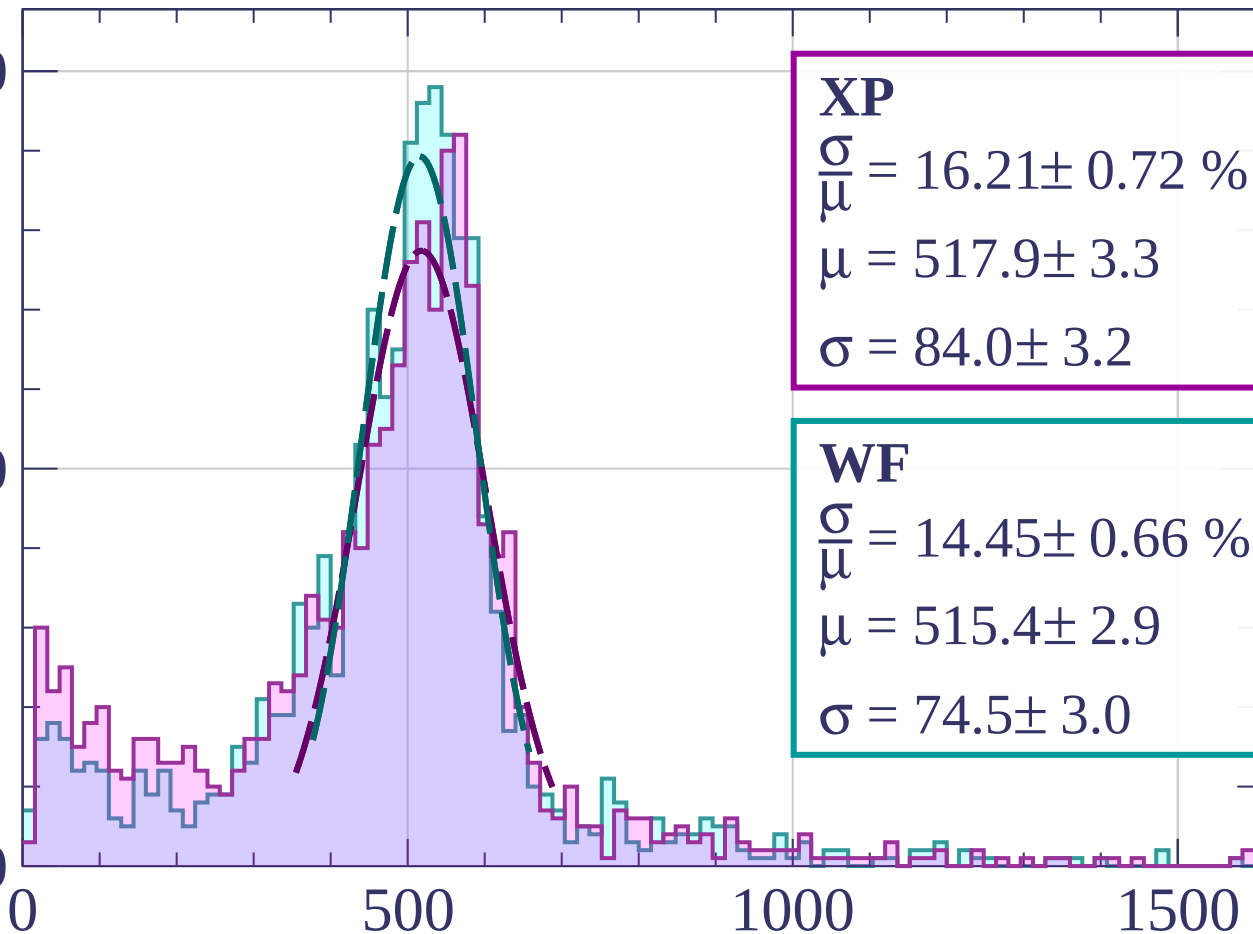
Energy loss |  $69 < \phi < 72$

Count

100

50

0



**XP**

$$\frac{\sigma}{\mu} = 16.21 \pm 0.72 \%$$

$$\mu = 517.9 \pm 3.3$$

$$\sigma = 84.0 \pm 3.2$$

**WF**

$$\frac{\sigma}{\mu} = 14.45 \pm 0.66 \%$$

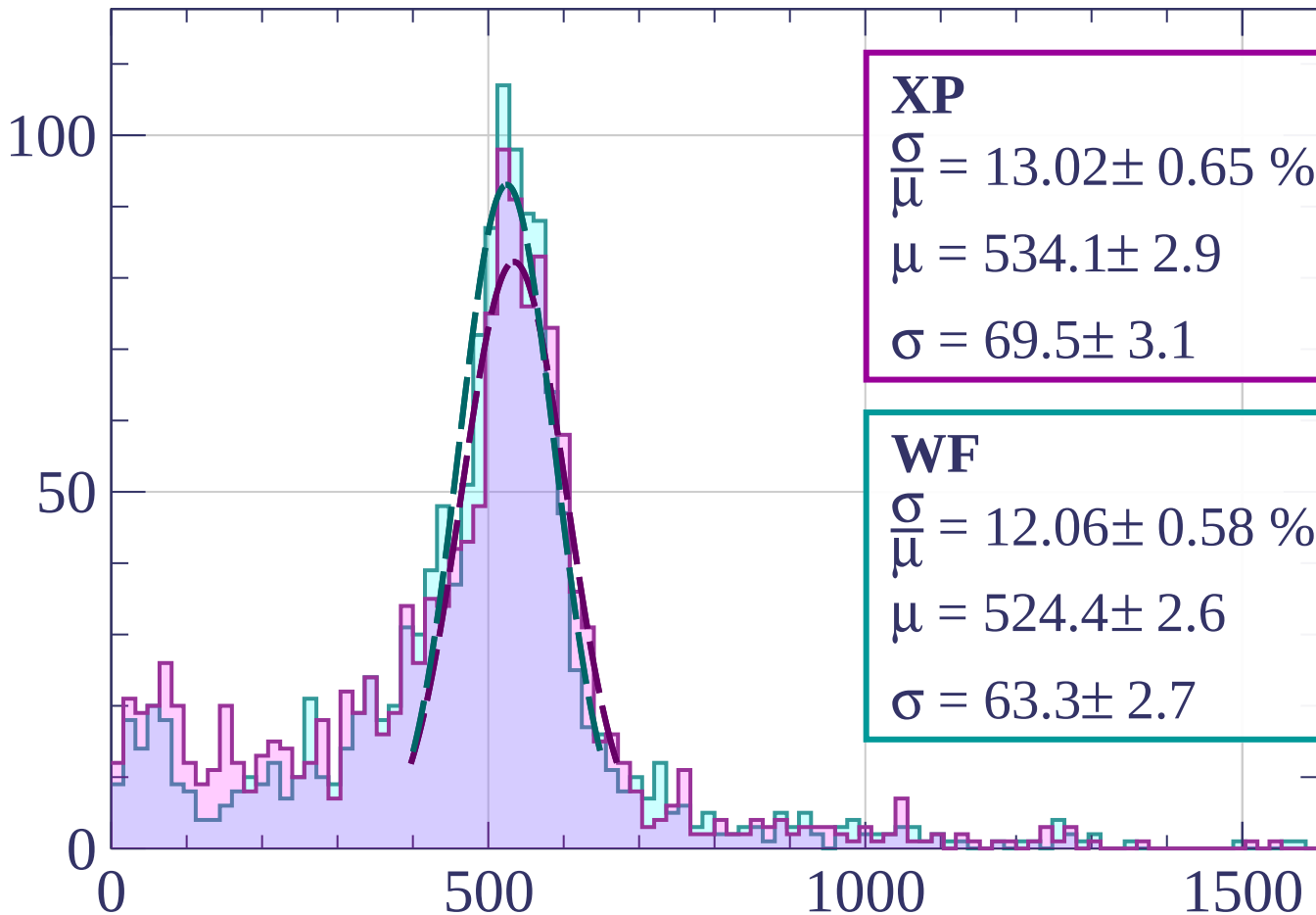
$$\mu = 515.4 \pm 2.9$$

$$\sigma = 74.5 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss |  $72 < \phi < 75$

Count



**XP**

$$\frac{\sigma}{\mu} = 13.02 \pm 0.65 \%$$

$$\mu = 534.1 \pm 2.9$$

$$\sigma = 69.5 \pm 3.1$$

**WF**

$$\frac{\sigma}{\mu} = 12.06 \pm 0.58 \%$$

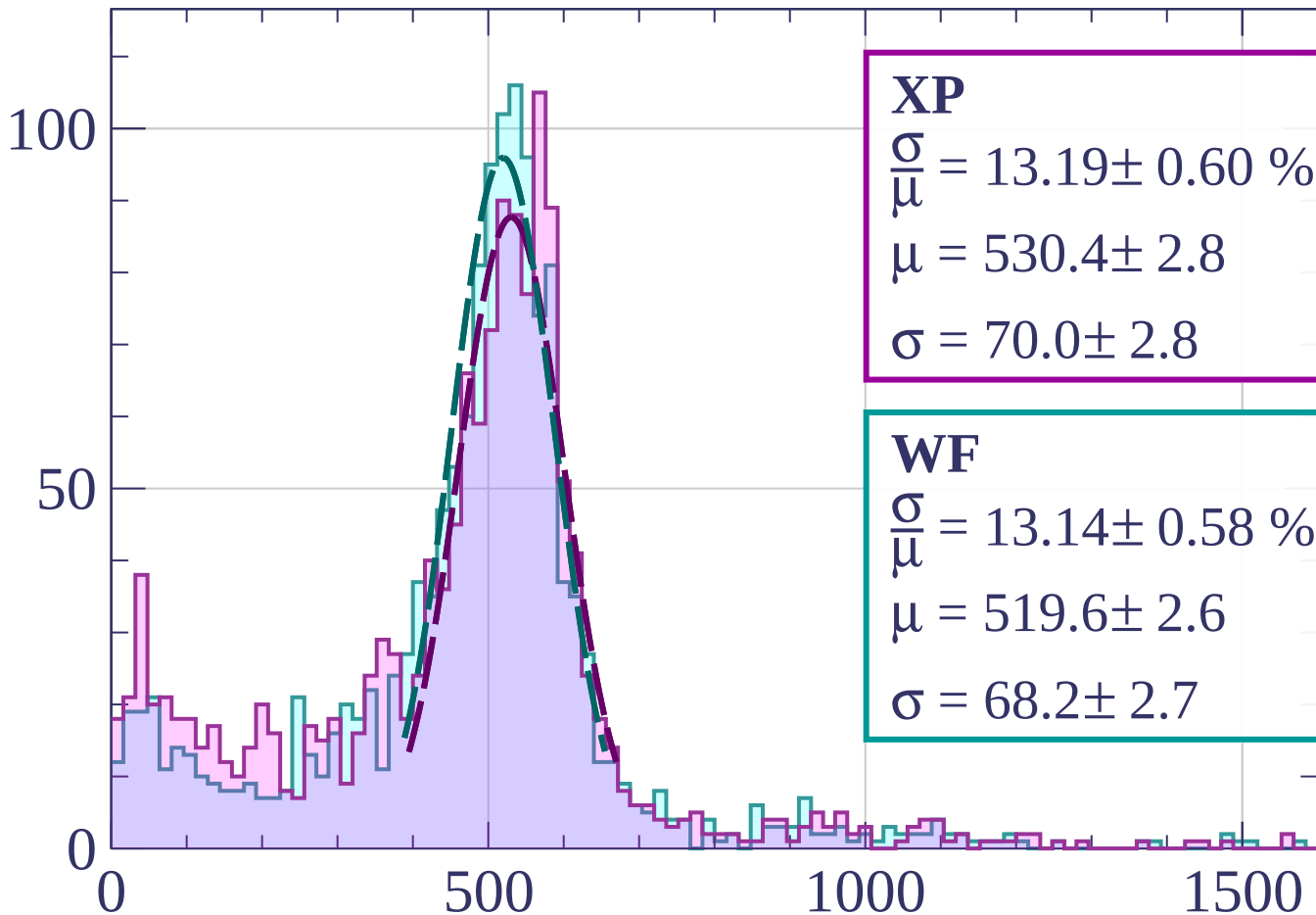
$$\mu = 524.4 \pm 2.6$$

$$\sigma = 63.3 \pm 2.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $75 < \phi < 78$

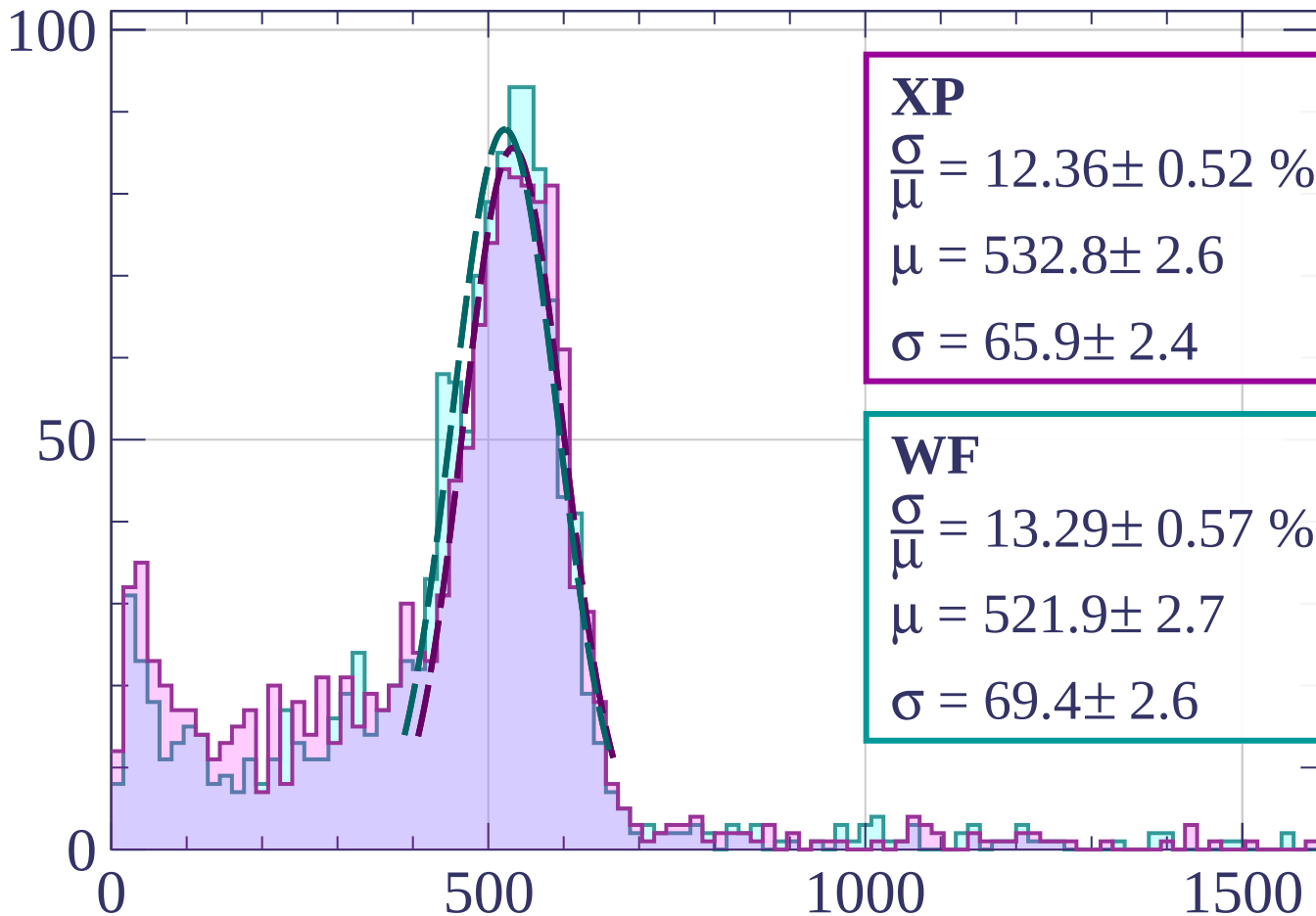
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $78 < \phi < 81$

Count



**XP**

$$\frac{\sigma}{\mu} = 12.36 \pm 0.52 \%$$

$$\mu = 532.8 \pm 2.6$$

$$\sigma = 65.9 \pm 2.4$$

**WF**

$$\frac{\sigma}{\mu} = 13.29 \pm 0.57 \%$$

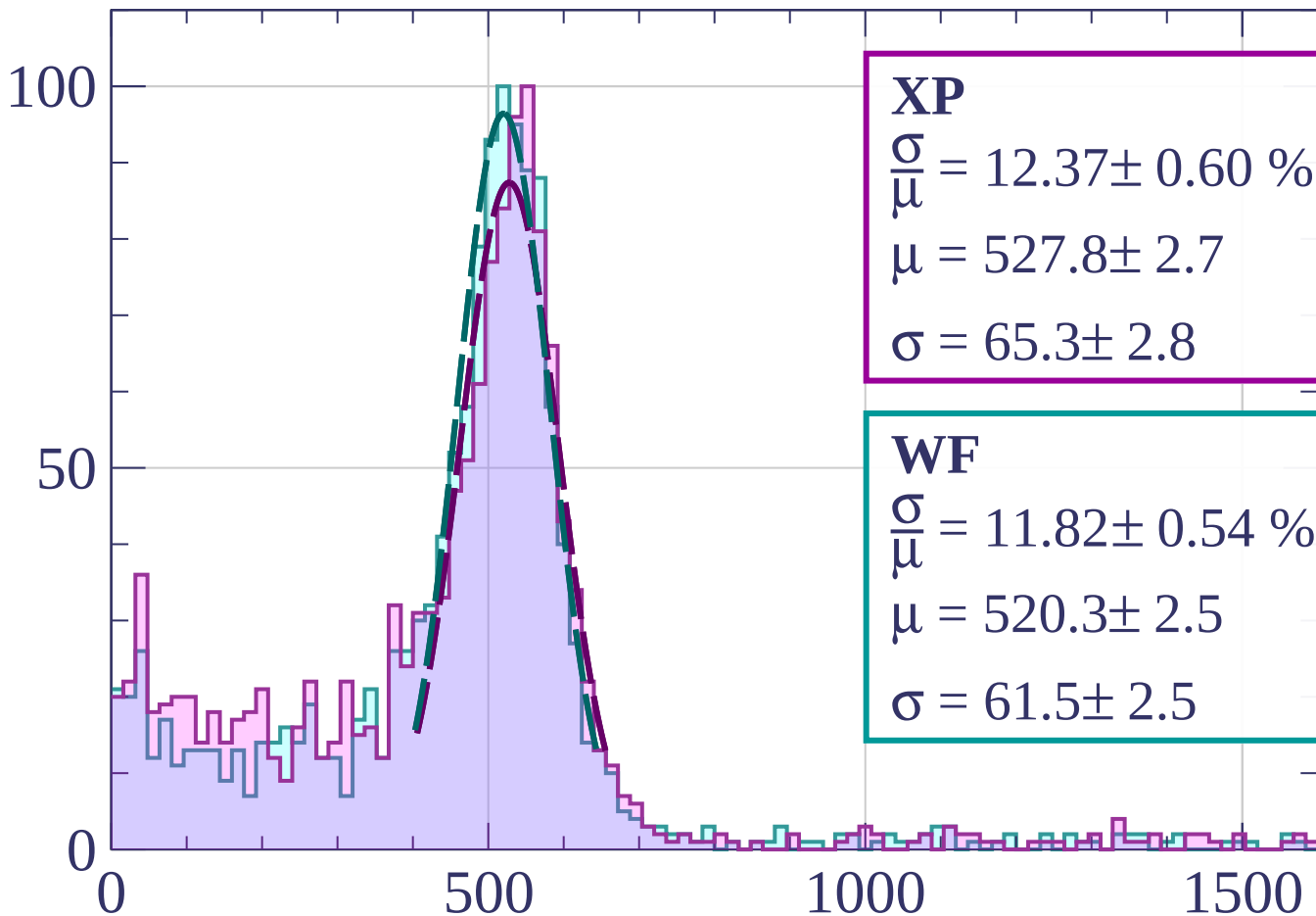
$$\mu = 521.9 \pm 2.7$$

$$\sigma = 69.4 \pm 2.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $81 < \phi < 84$

Count



**XP**

$$\frac{\sigma}{\mu} = 12.37 \pm 0.60 \%$$

$$\mu = 527.8 \pm 2.7$$

$$\sigma = 65.3 \pm 2.8$$

**WF**

$$\frac{\sigma}{\mu} = 11.82 \pm 0.54 \%$$

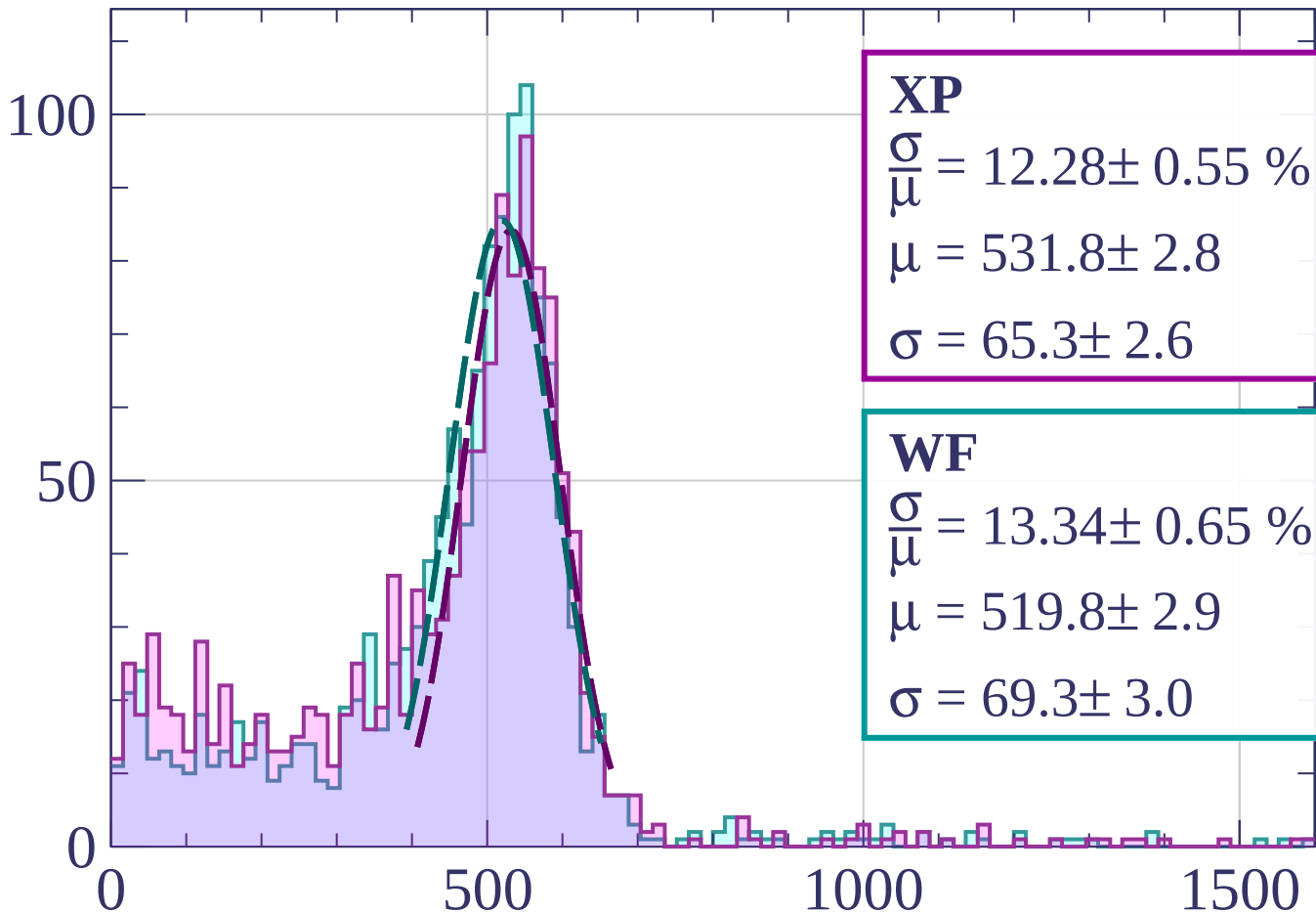
$$\mu = 520.3 \pm 2.5$$

$$\sigma = 61.5 \pm 2.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $84 < \phi < 87$

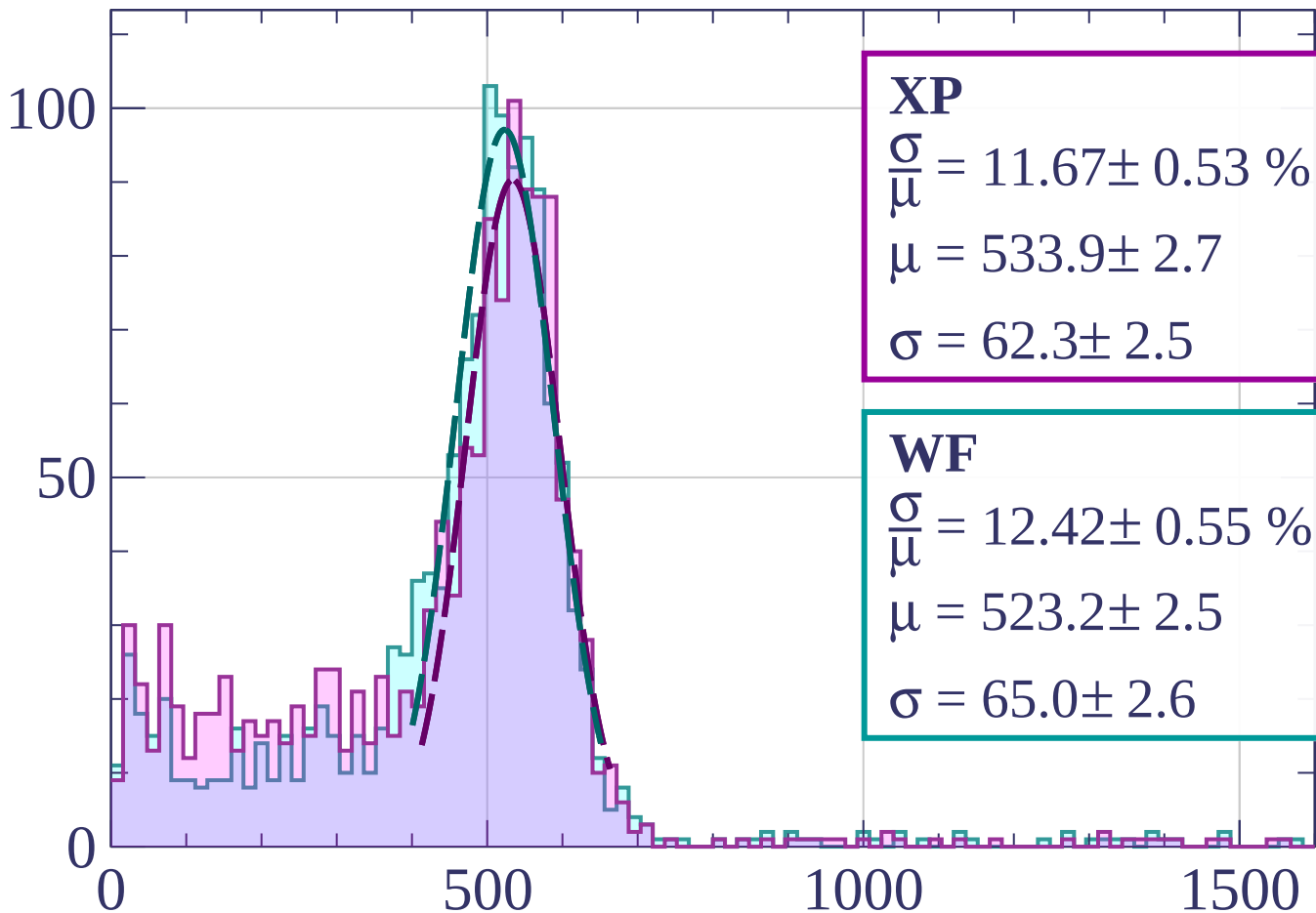
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $87 < \phi < 90$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.67 \pm 0.53 \%$$

$$\mu = 533.9 \pm 2.7$$

$$\sigma = 62.3 \pm 2.5$$

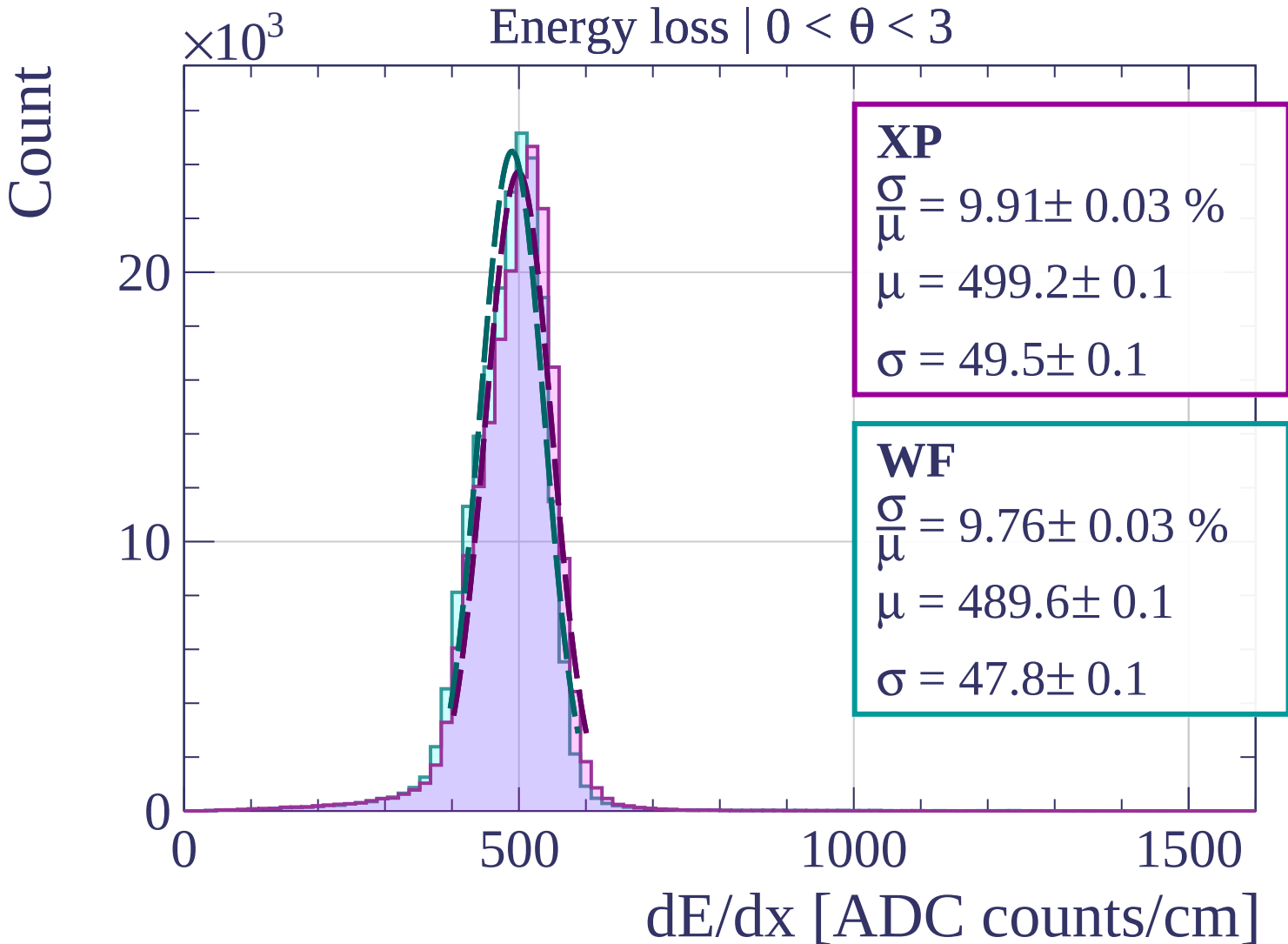
**WF**

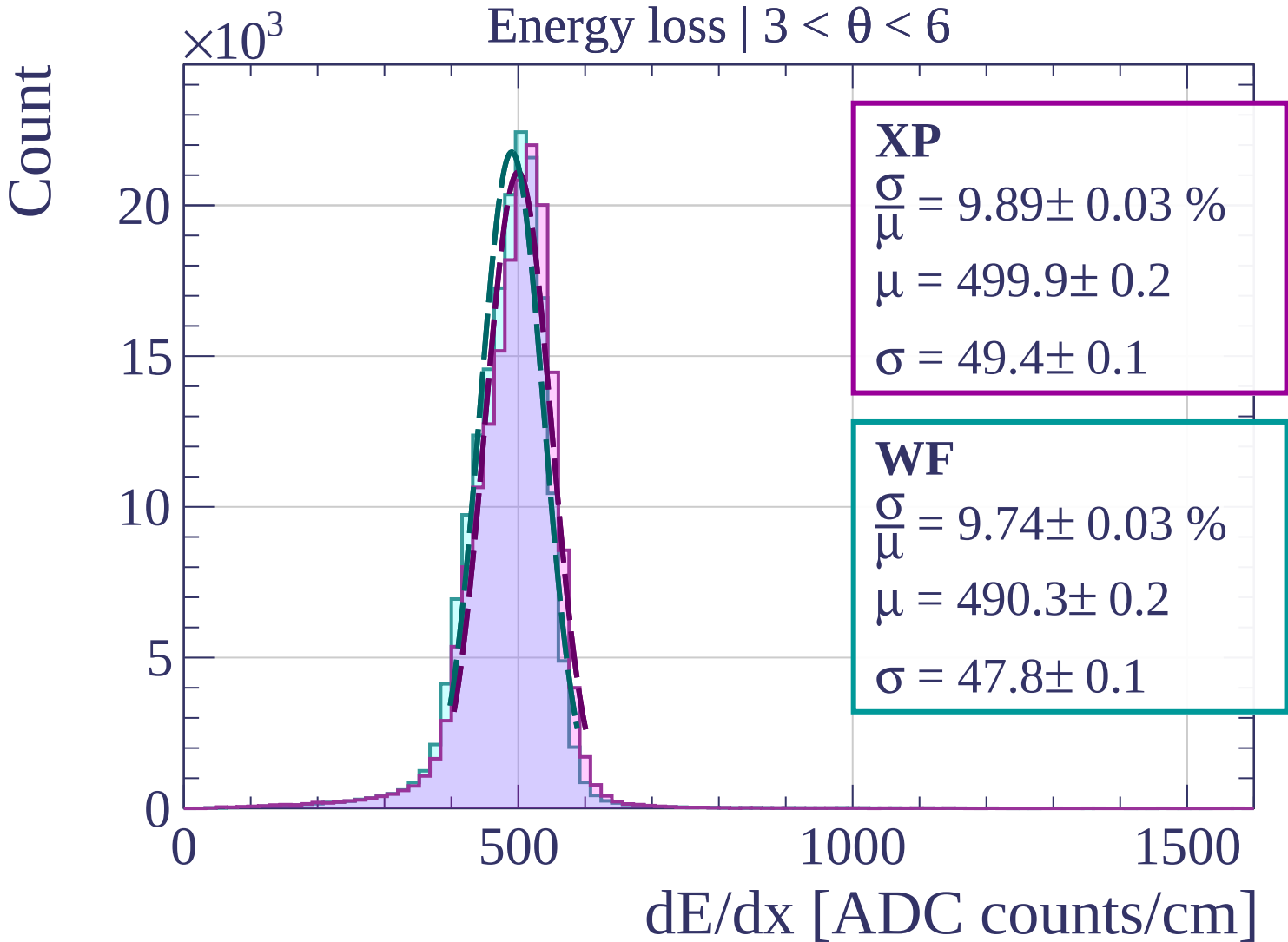
$$\frac{\sigma}{\mu} = 12.42 \pm 0.55 \%$$

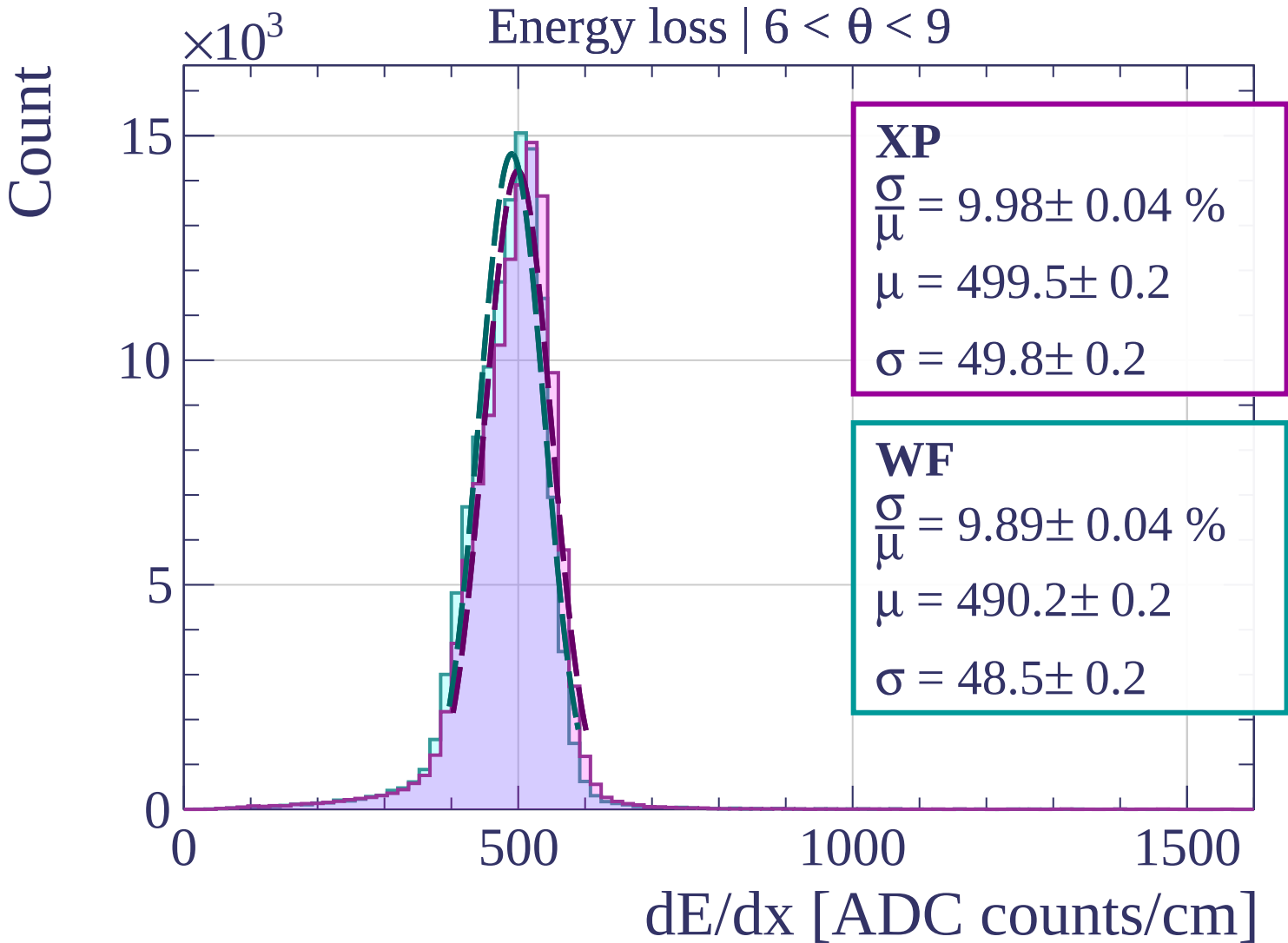
$$\mu = 523.2 \pm 2.5$$

$$\sigma = 65.0 \pm 2.6$$

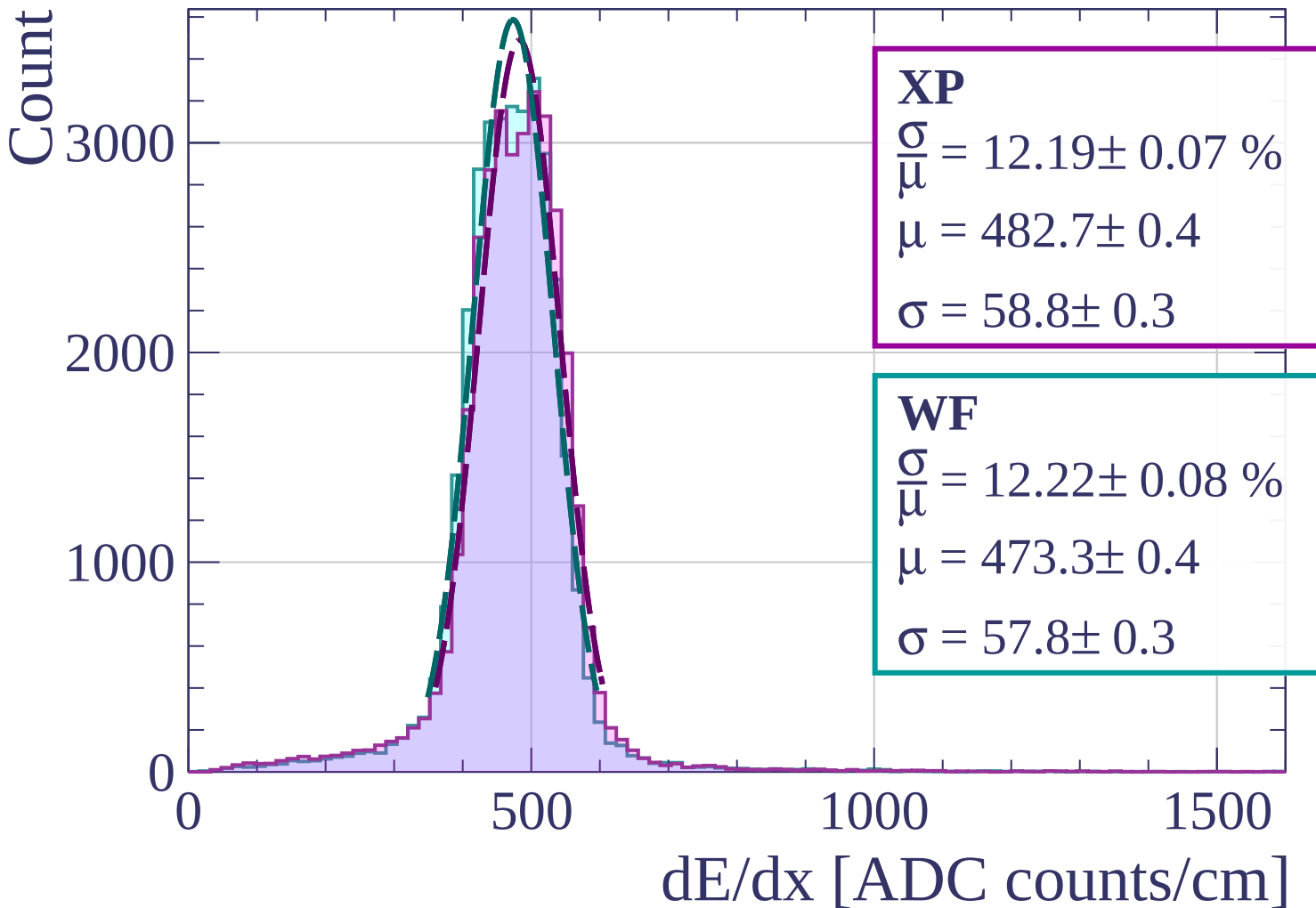
$dE/dx$  [ADC counts/cm]





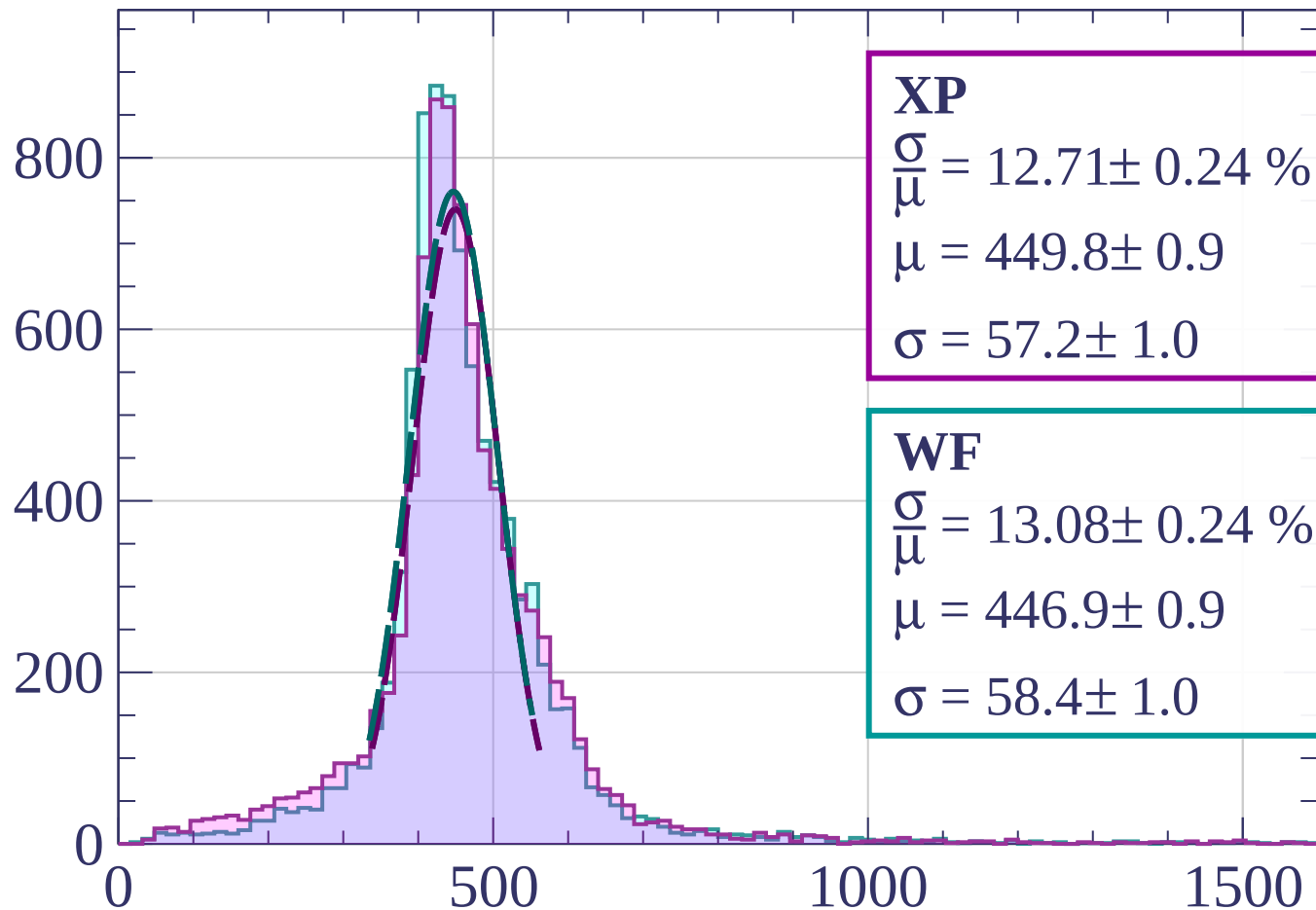


Energy loss |  $9 < \theta < 12$



Energy loss |  $12 < \theta < 15$

Count



**XP**

$$\frac{\sigma}{\mu} = 12.71 \pm 0.24 \%$$

$$\mu = 449.8 \pm 0.9$$

$$\sigma = 57.2 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 13.08 \pm 0.24 \%$$

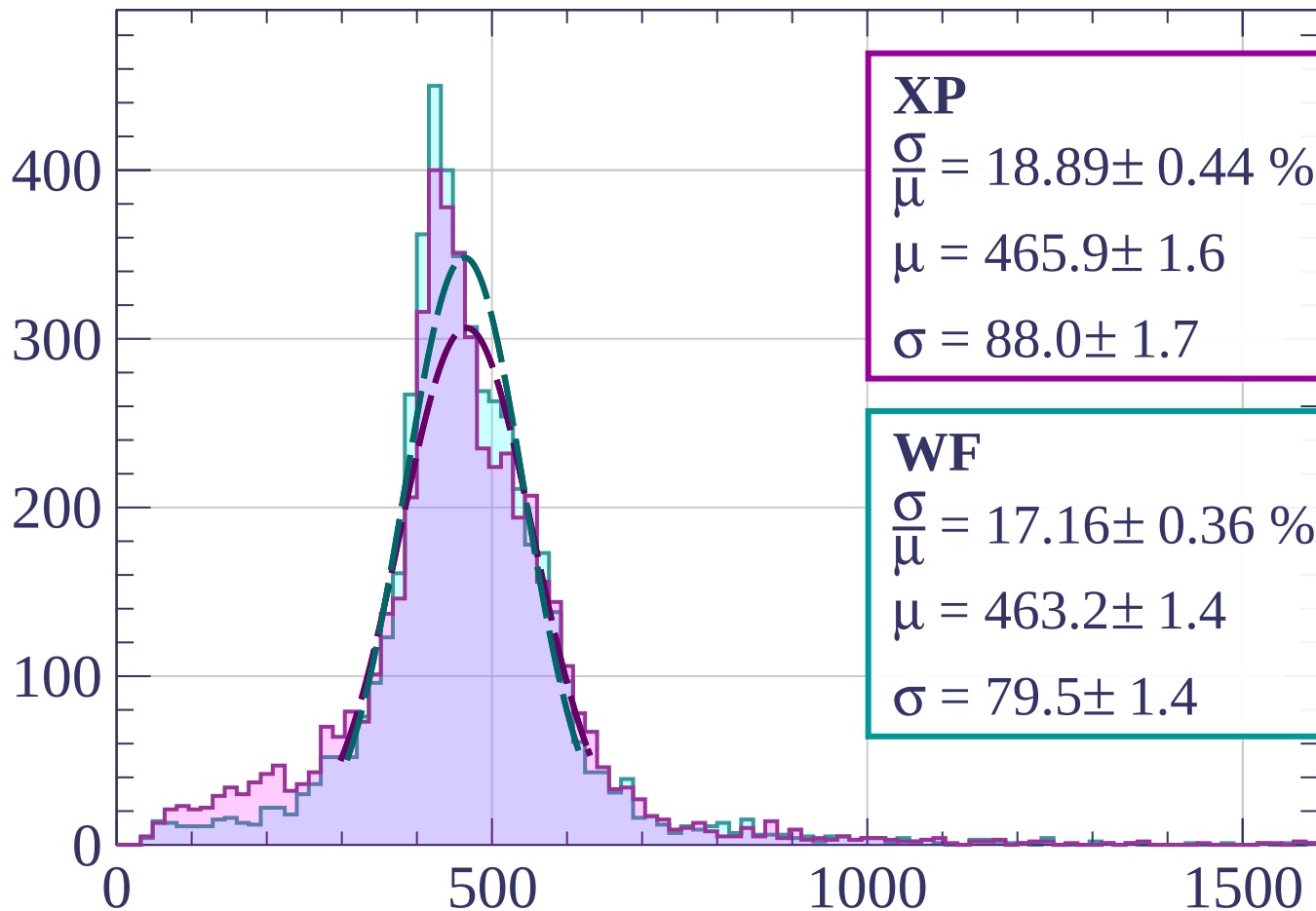
$$\mu = 446.9 \pm 0.9$$

$$\sigma = 58.4 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss |  $15 < \theta < 18$

Count



**XP**

$$\frac{\sigma}{\mu} = 18.89 \pm 0.44 \%$$

$$\mu = 465.9 \pm 1.6$$

$$\sigma = 88.0 \pm 1.7$$

**WF**

$$\frac{\sigma}{\mu} = 17.16 \pm 0.36 \%$$

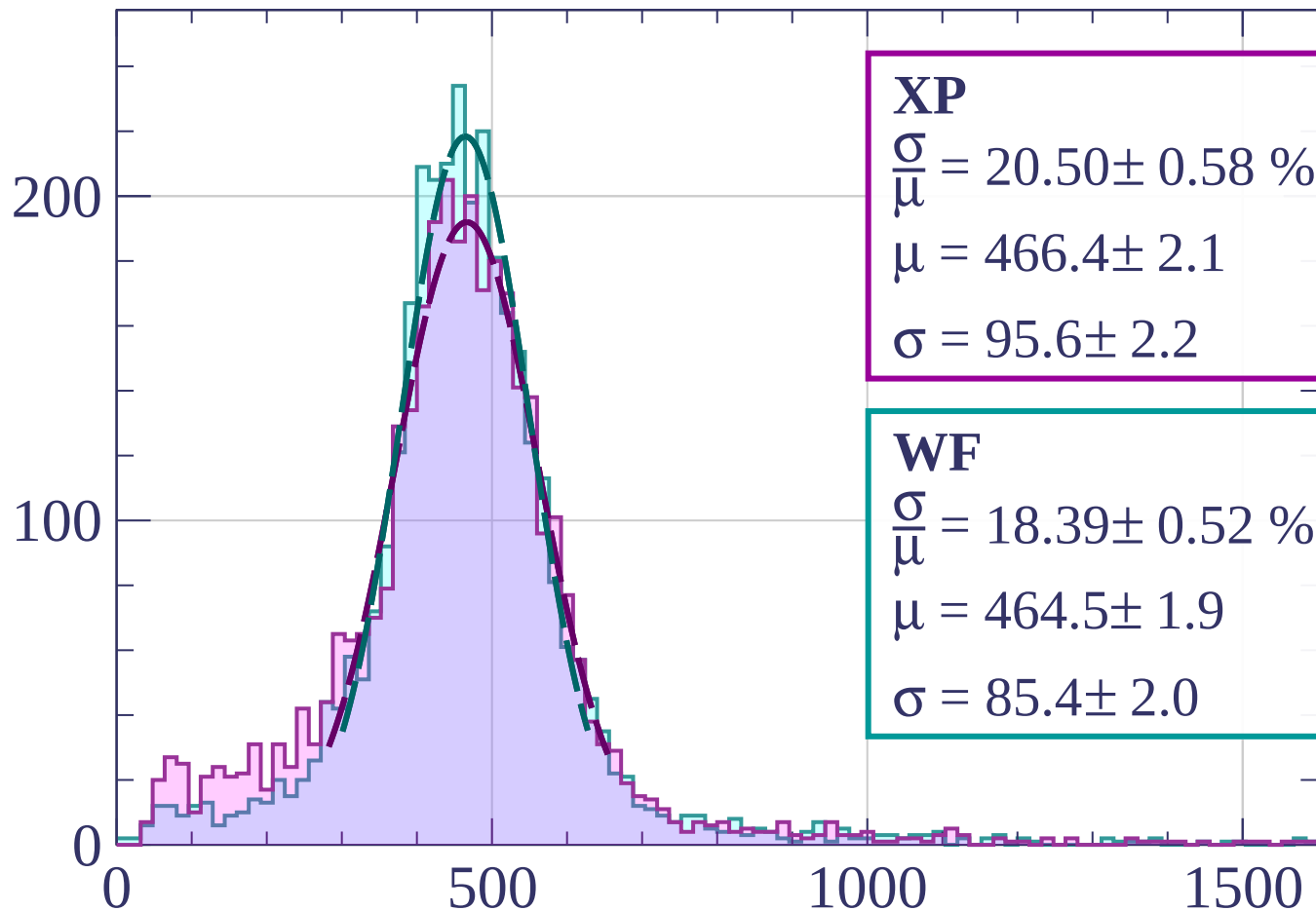
$$\mu = 463.2 \pm 1.4$$

$$\sigma = 79.5 \pm 1.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $18 < \theta < 21$

Count



**XP**

$$\frac{\sigma}{\mu} = 20.50 \pm 0.58 \%$$

$$\mu = 466.4 \pm 2.1$$

$$\sigma = 95.6 \pm 2.2$$

**WF**

$$\frac{\sigma}{\mu} = 18.39 \pm 0.52 \%$$

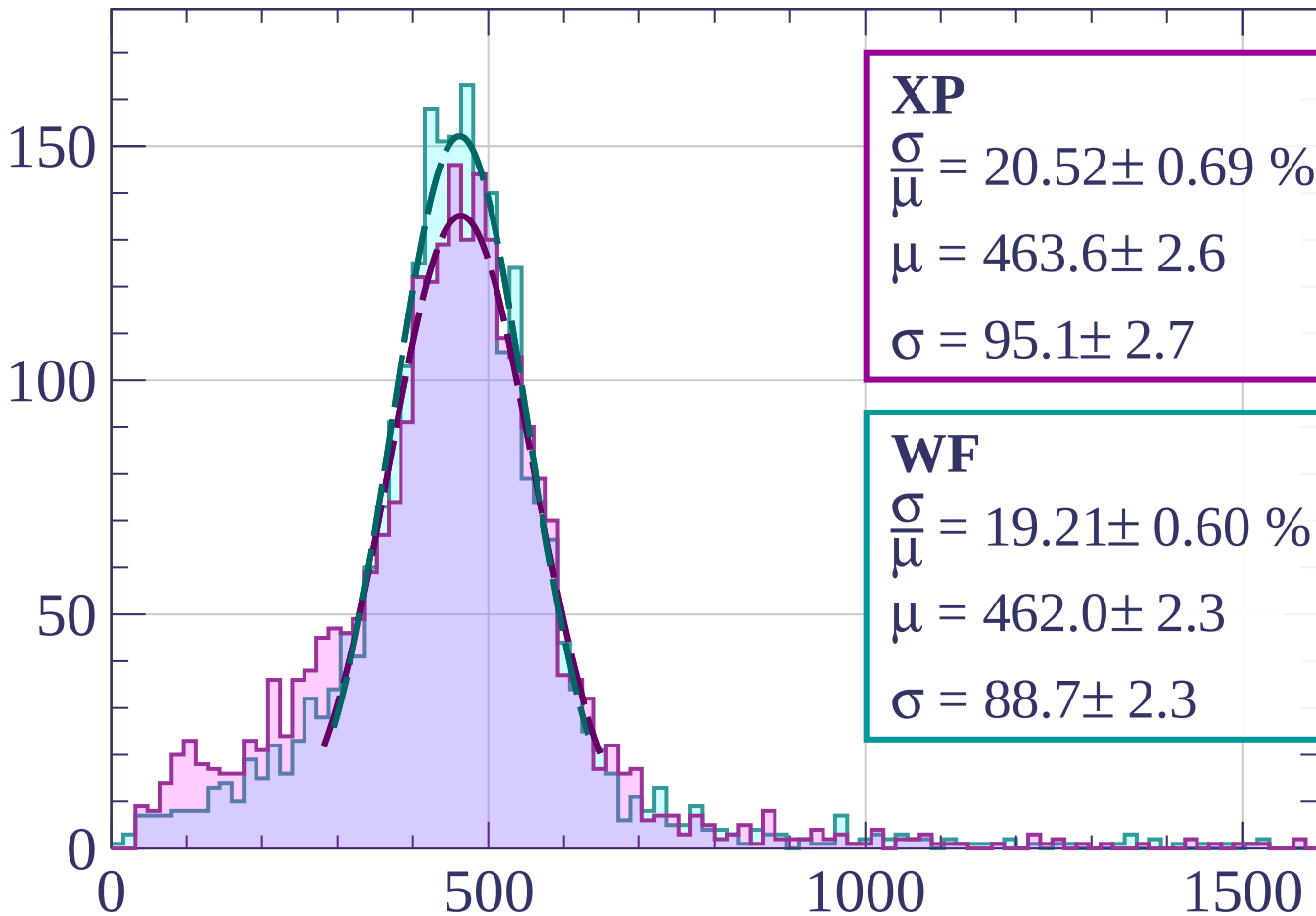
$$\mu = 464.5 \pm 1.9$$

$$\sigma = 85.4 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss |  $21 < \theta < 24$

Count



**XP**

$$\frac{\sigma}{\mu} = 20.52 \pm 0.69 \%$$

$$\mu = 463.6 \pm 2.6$$

$$\sigma = 95.1 \pm 2.7$$

**WF**

$$\frac{\sigma}{\mu} = 19.21 \pm 0.60 \%$$

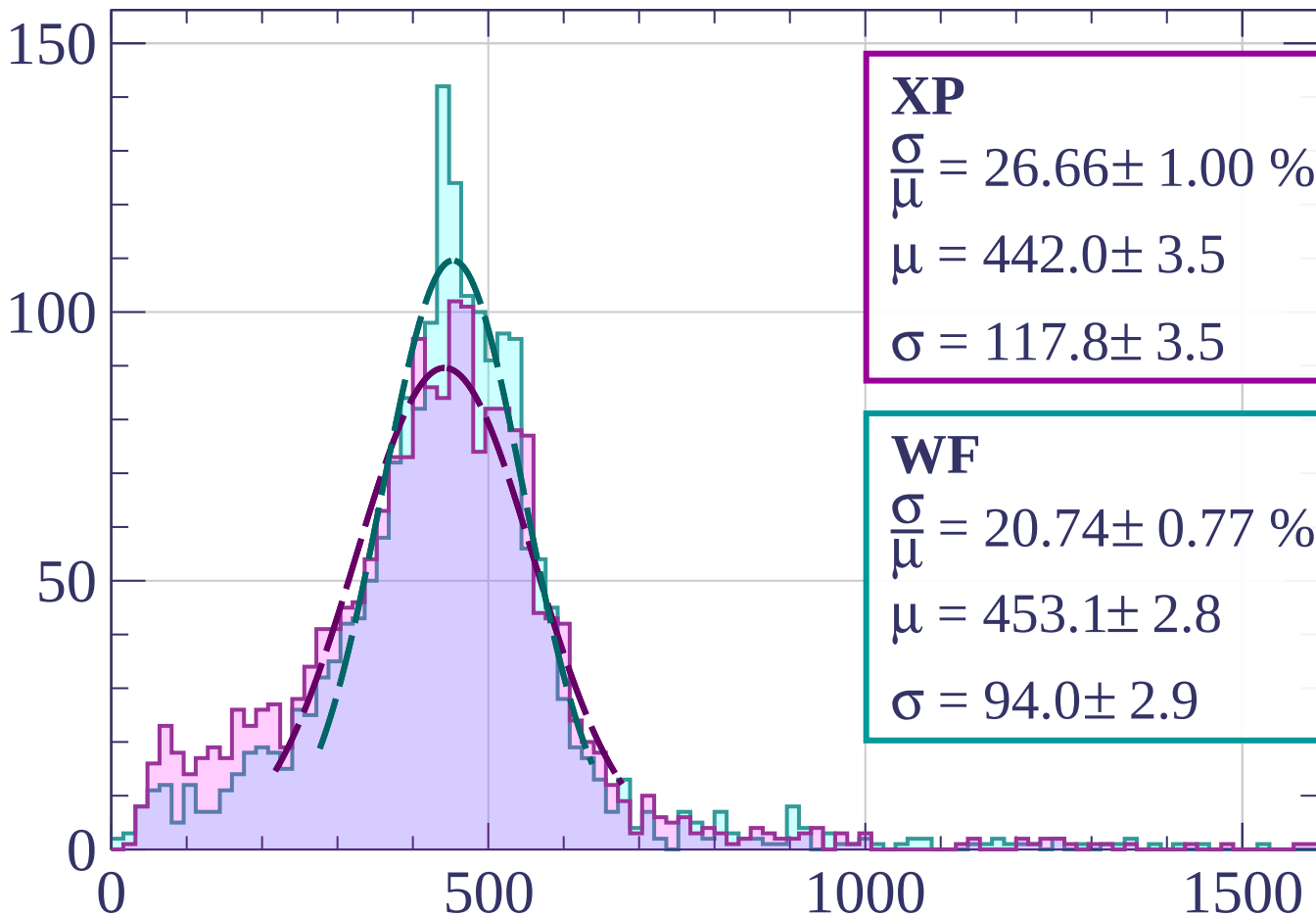
$$\mu = 462.0 \pm 2.3$$

$$\sigma = 88.7 \pm 2.3$$

dE/dx [ADC counts/cm]

Energy loss |  $24 < \theta < 27$

Count



**XP**

$$\frac{\sigma}{\mu} = 26.66 \pm 1.00 \%$$

$$\mu = 442.0 \pm 3.5$$

$$\sigma = 117.8 \pm 3.5$$

**WF**

$$\frac{\sigma}{\mu} = 20.74 \pm 0.77 \%$$

$$\mu = 453.1 \pm 2.8$$

$$\sigma = 94.0 \pm 2.9$$

dE/dx [ADC counts/cm]

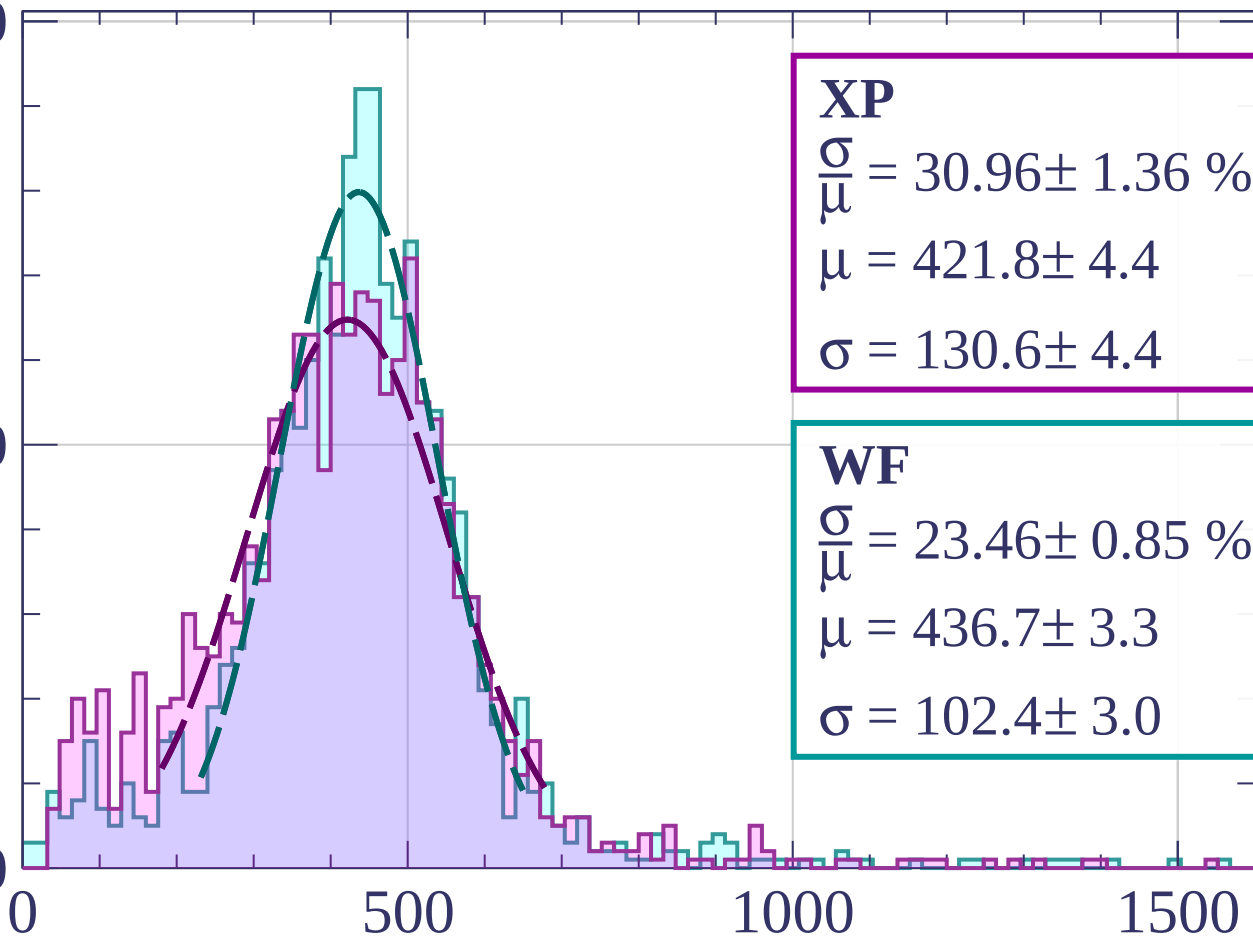
Energy loss |  $27 < \theta < 30$

Count

100

50

0



**XP**

$$\frac{\sigma}{\mu} = 30.96 \pm 1.36 \%$$

$$\mu = 421.8 \pm 4.4$$

$$\sigma = 130.6 \pm 4.4$$

**WF**

$$\frac{\sigma}{\mu} = 23.46 \pm 0.85 \%$$

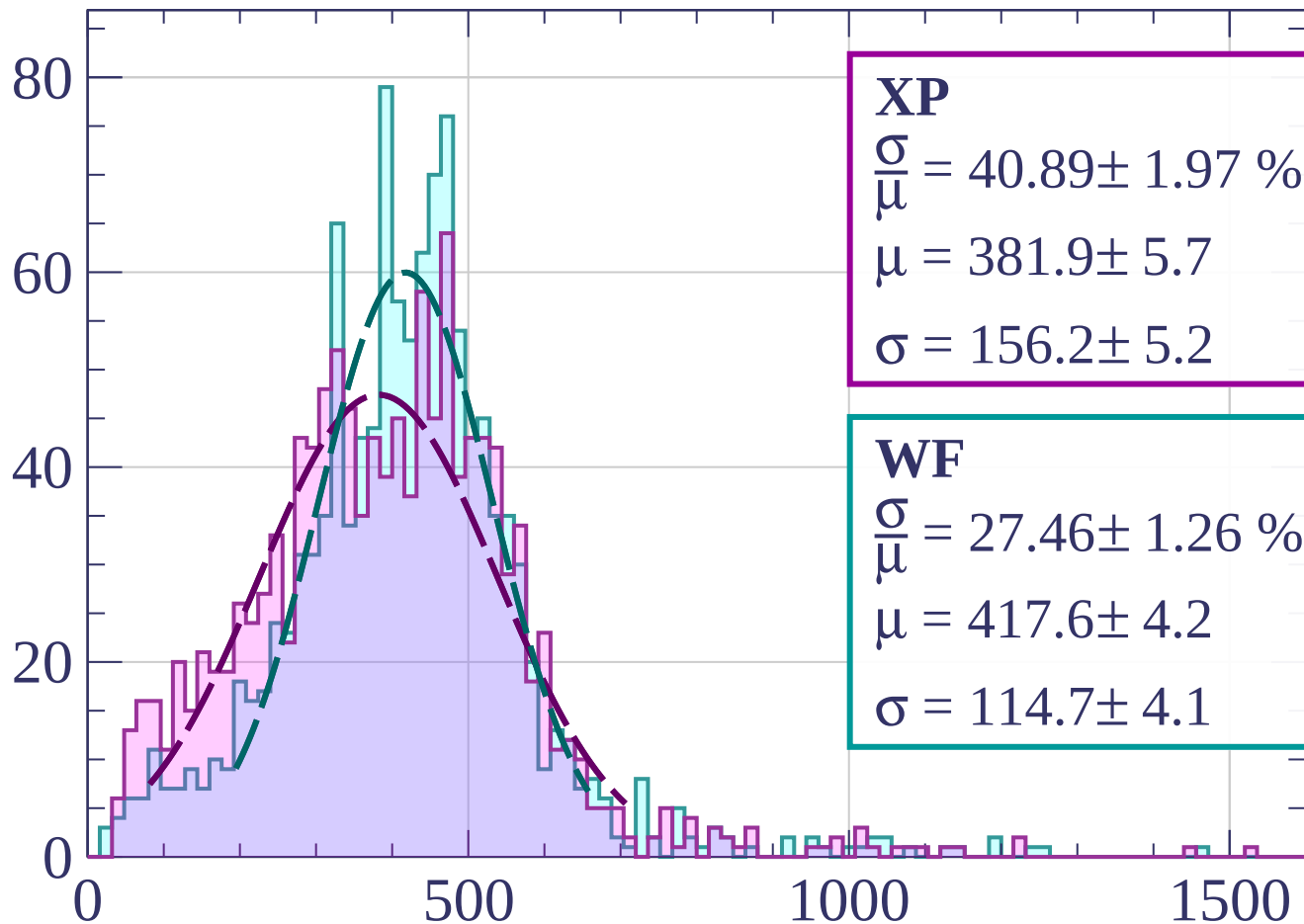
$$\mu = 436.7 \pm 3.3$$

$$\sigma = 102.4 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss |  $30 < \theta < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 40.89 \pm 1.97 \%$$

$$\mu = 381.9 \pm 5.7$$

$$\sigma = 156.2 \pm 5.2$$

**WF**

$$\frac{\sigma}{\mu} = 27.46 \pm 1.26 \%$$

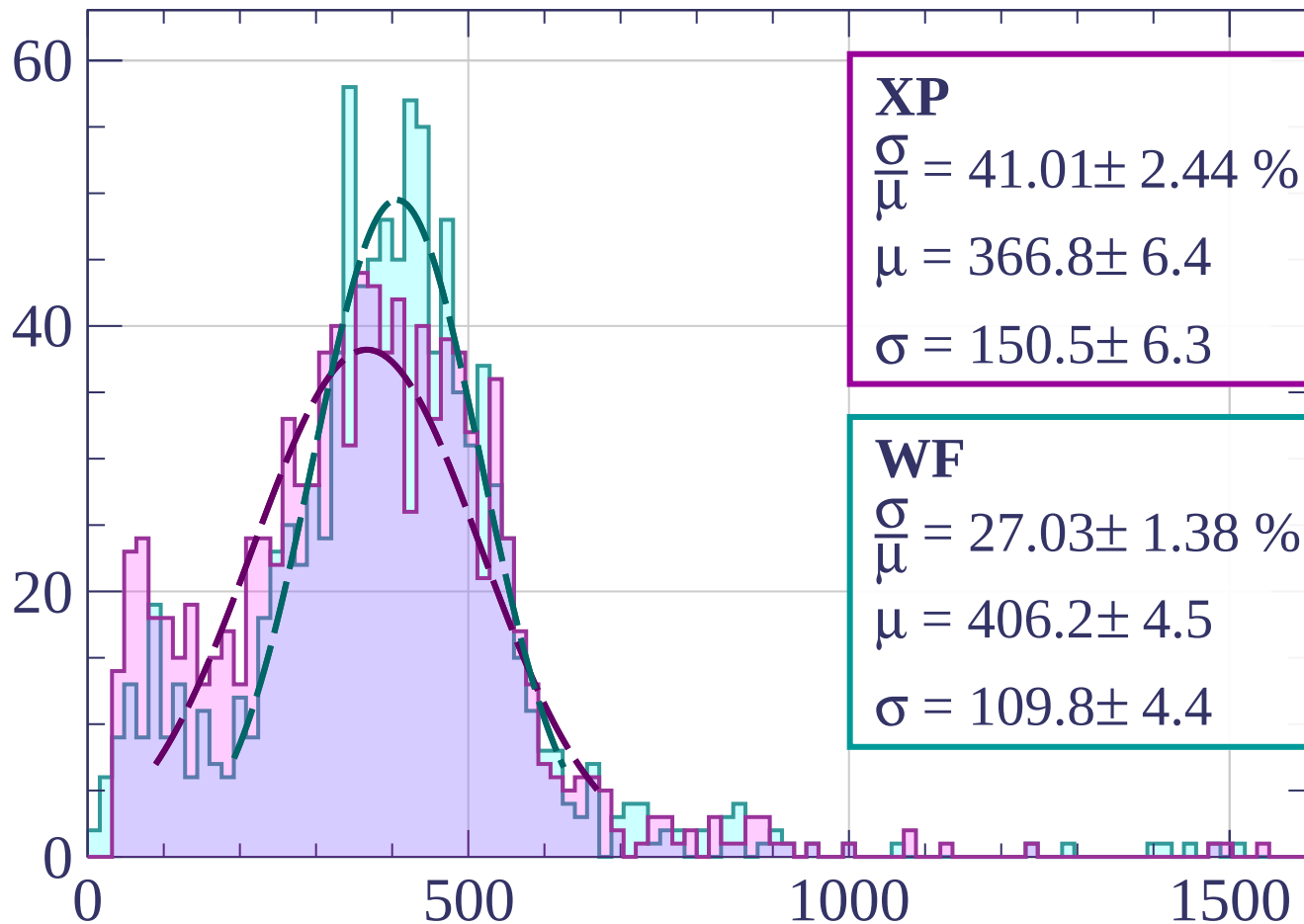
$$\mu = 417.6 \pm 4.2$$

$$\sigma = 114.7 \pm 4.1$$

dE/dx [ADC counts/cm]

Energy loss |  $33 < \theta < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 41.01 \pm 2.44 \%$$

$$\mu = 366.8 \pm 6.4$$

$$\sigma = 150.5 \pm 6.3$$

**WF**

$$\frac{\sigma}{\mu} = 27.03 \pm 1.38 \%$$

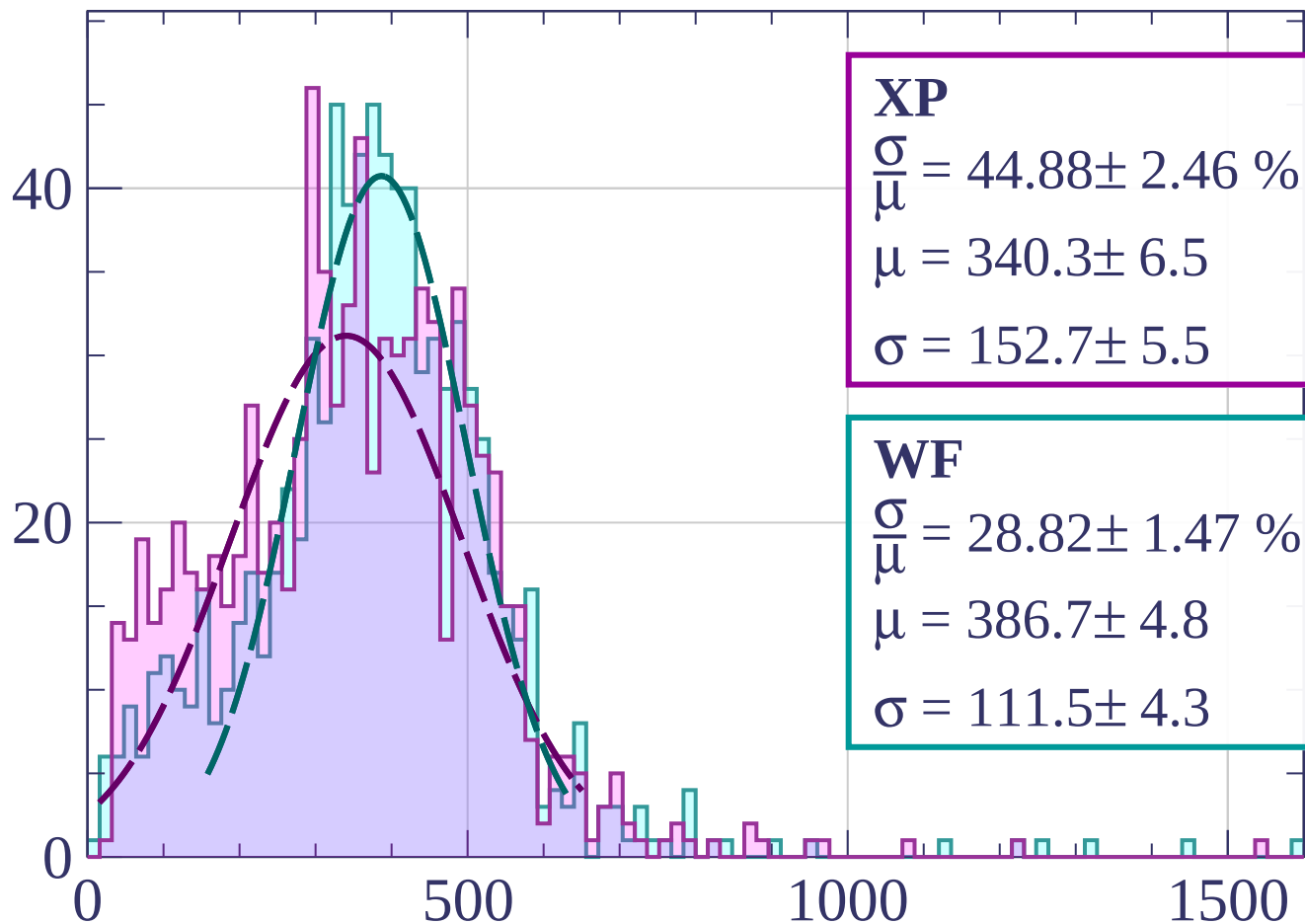
$$\mu = 406.2 \pm 4.5$$

$$\sigma = 109.8 \pm 4.4$$

dE/dx [ADC counts/cm]

Energy loss |  $36 < \theta < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 44.88 \pm 2.46 \%$$

$$\mu = 340.3 \pm 6.5$$

$$\sigma = 152.7 \pm 5.5$$

**WF**

$$\frac{\sigma}{\mu} = 28.82 \pm 1.47 \%$$

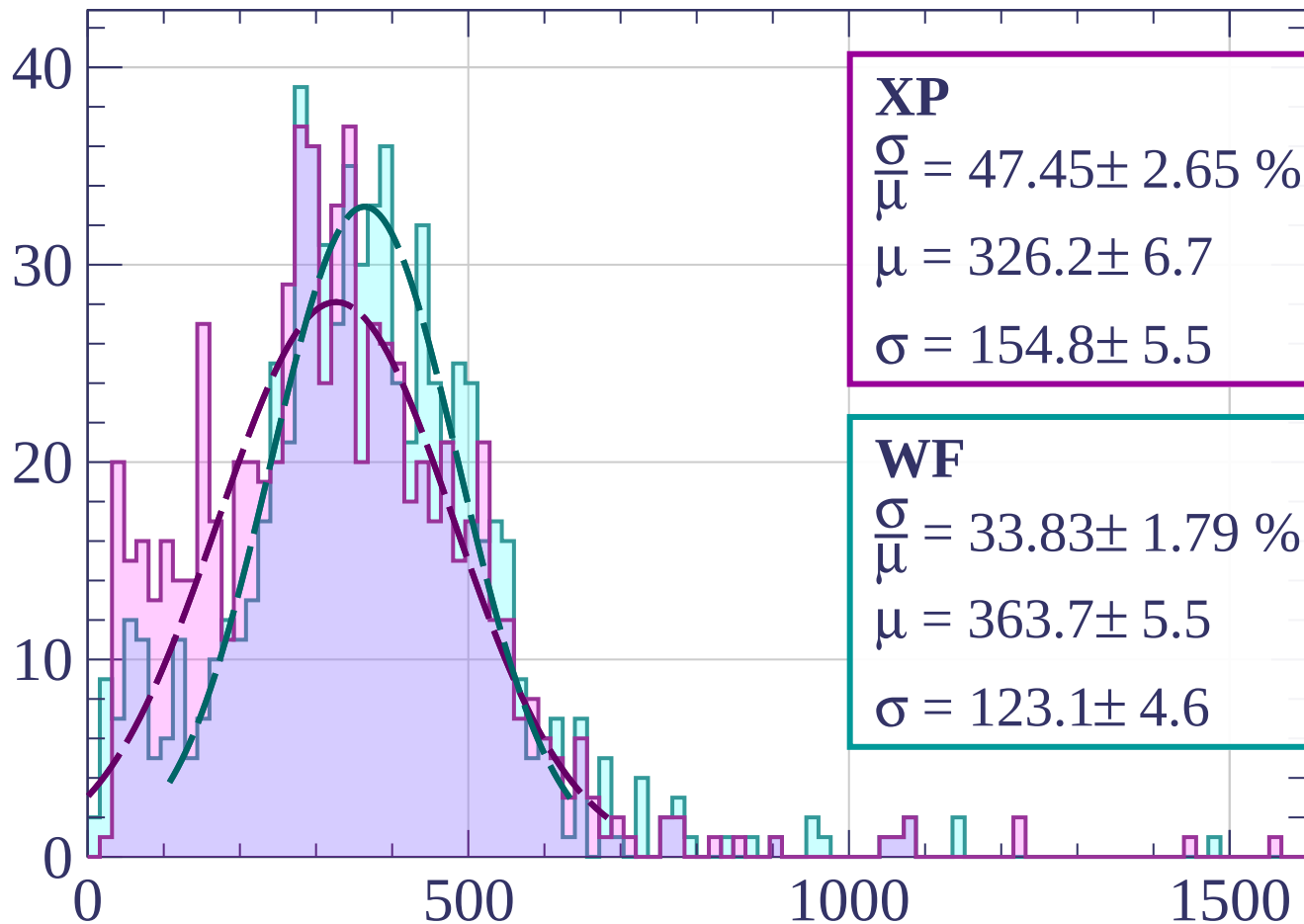
$$\mu = 386.7 \pm 4.8$$

$$\sigma = 111.5 \pm 4.3$$

dE/dx [ADC counts/cm]

Energy loss |  $39 < \theta < 42$

Count



**XP**

$$\frac{\sigma}{\mu} = 47.45 \pm 2.65 \%$$

$$\mu = 326.2 \pm 6.7$$

$$\sigma = 154.8 \pm 5.5$$

**WF**

$$\frac{\sigma}{\mu} = 33.83 \pm 1.79 \%$$

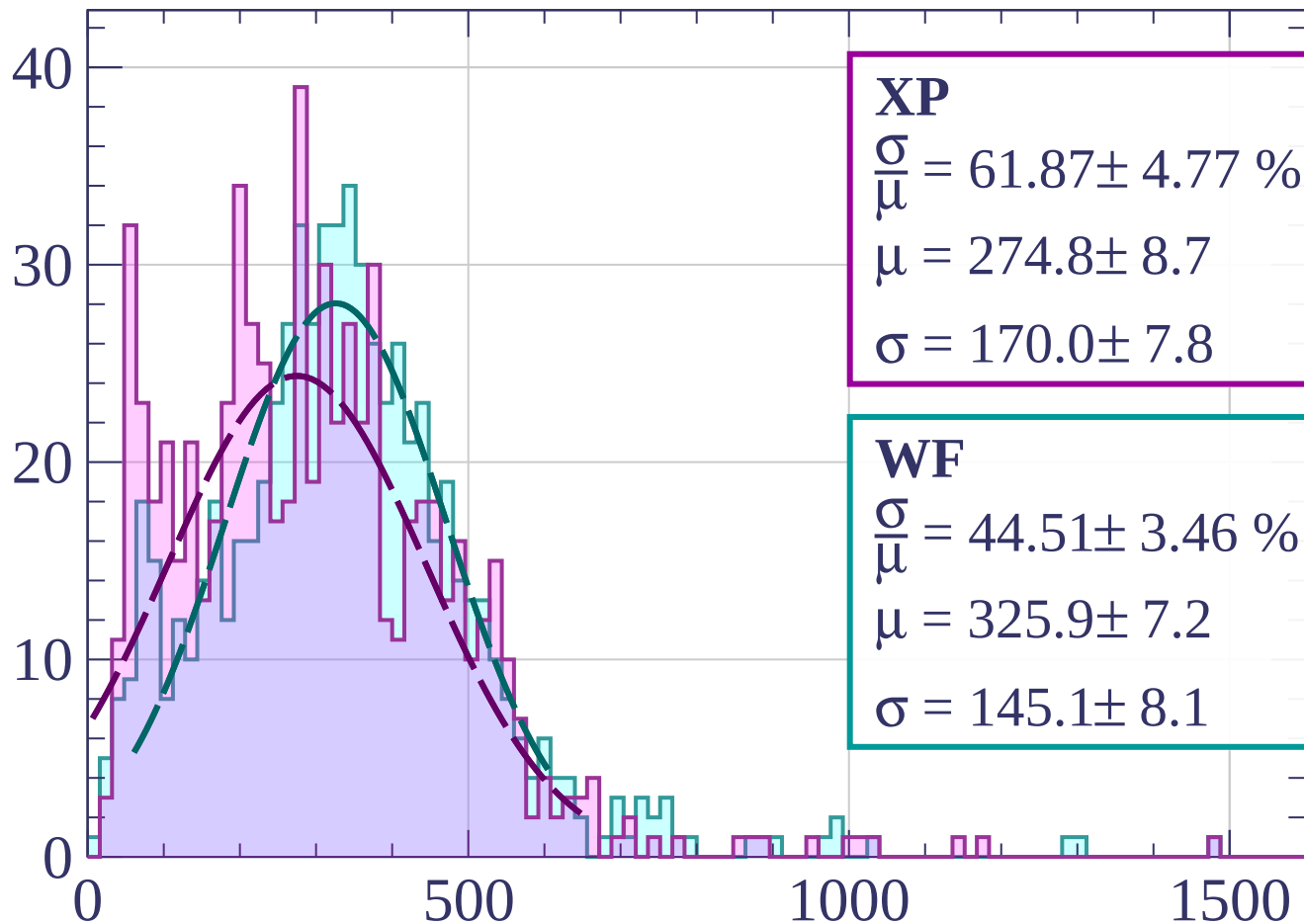
$$\mu = 363.7 \pm 5.5$$

$$\sigma = 123.1 \pm 4.6$$

dE/dx [ADC counts/cm]

Energy loss |  $42 < \theta < 45$

Count



**XP**

$$\frac{\sigma}{\mu} = 61.87 \pm 4.77 \%$$

$$\mu = 274.8 \pm 8.7$$

$$\sigma = 170.0 \pm 7.8$$

**WF**

$$\frac{\sigma}{\mu} = 44.51 \pm 3.46 \%$$

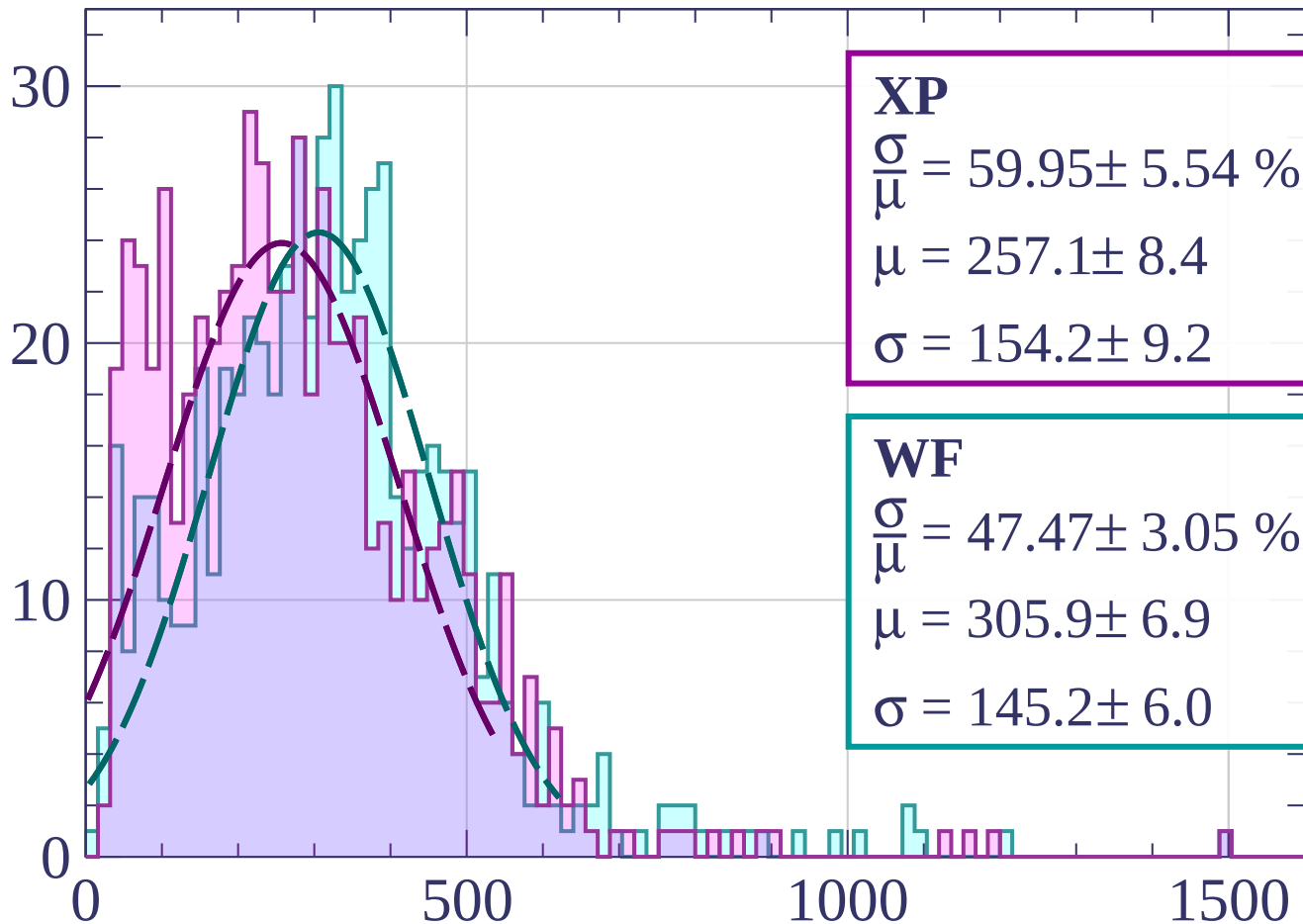
$$\mu = 325.9 \pm 7.2$$

$$\sigma = 145.1 \pm 8.1$$

dE/dx [ADC counts/cm]

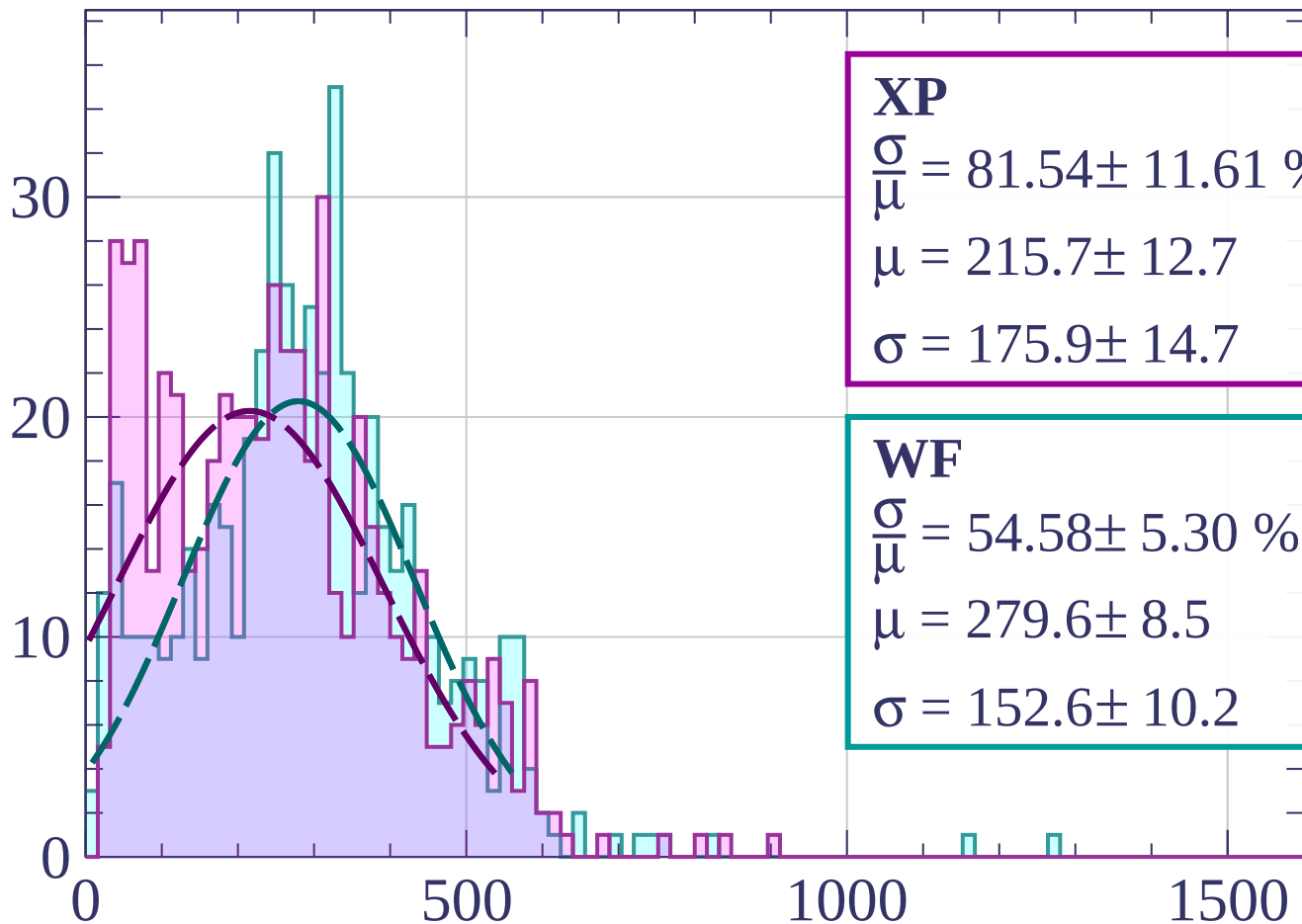
Energy loss |  $45 < \theta < 48$

Count



Energy loss |  $48 < \theta < 51$

Count



**XP**

$$\frac{\sigma}{\mu} = 81.54 \pm 11.61 \%$$

$$\mu = 215.7 \pm 12.7$$

$$\sigma = 175.9 \pm 14.7$$

**WF**

$$\frac{\sigma}{\mu} = 54.58 \pm 5.30 \%$$

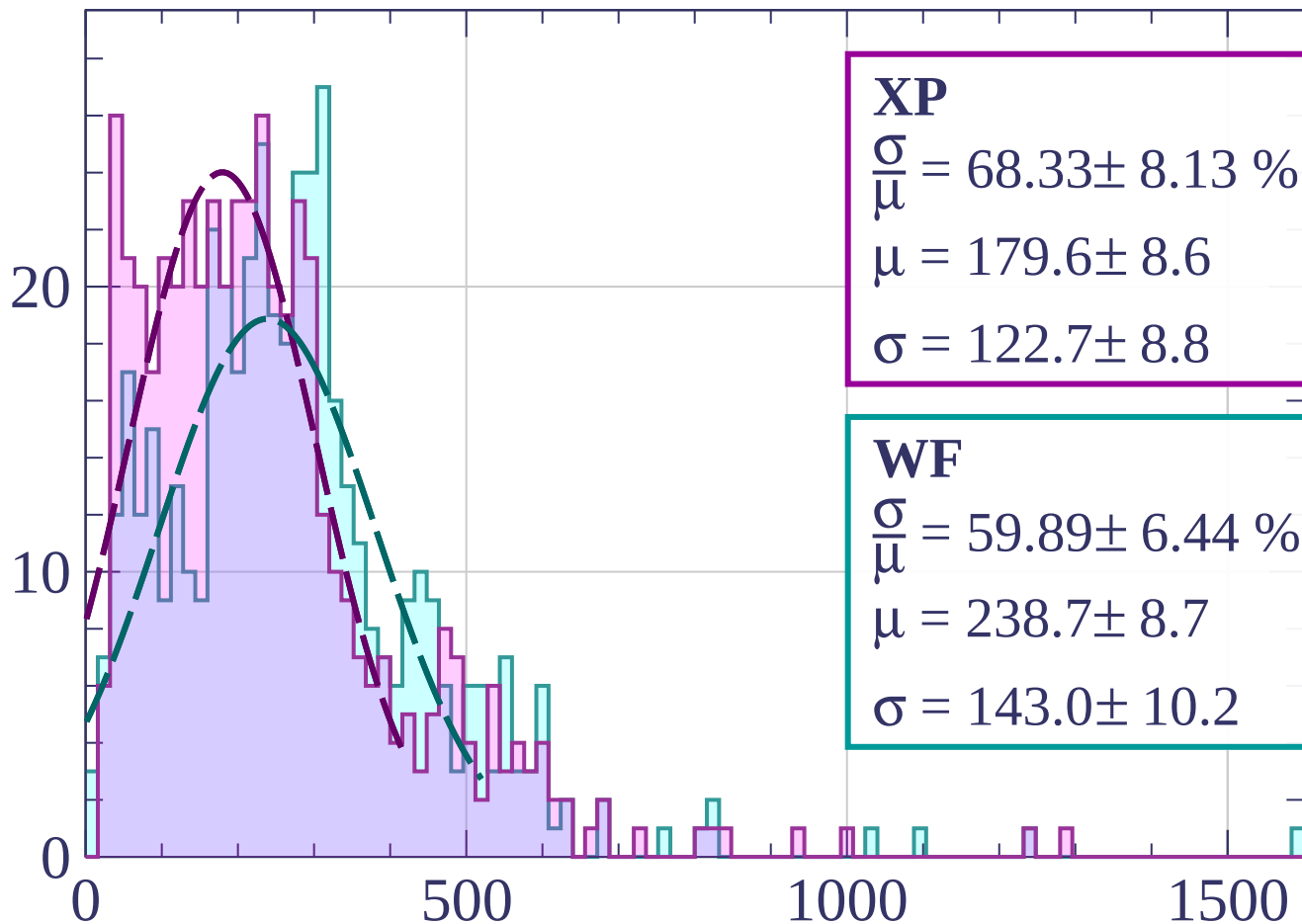
$$\mu = 279.6 \pm 8.5$$

$$\sigma = 152.6 \pm 10.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $51 < \theta < 54$

Count



**XP**

$$\frac{\sigma}{\mu} = 68.33 \pm 8.13 \%$$

$$\mu = 179.6 \pm 8.6$$

$$\sigma = 122.7 \pm 8.8$$

**WF**

$$\frac{\sigma}{\mu} = 59.89 \pm 6.44 \%$$

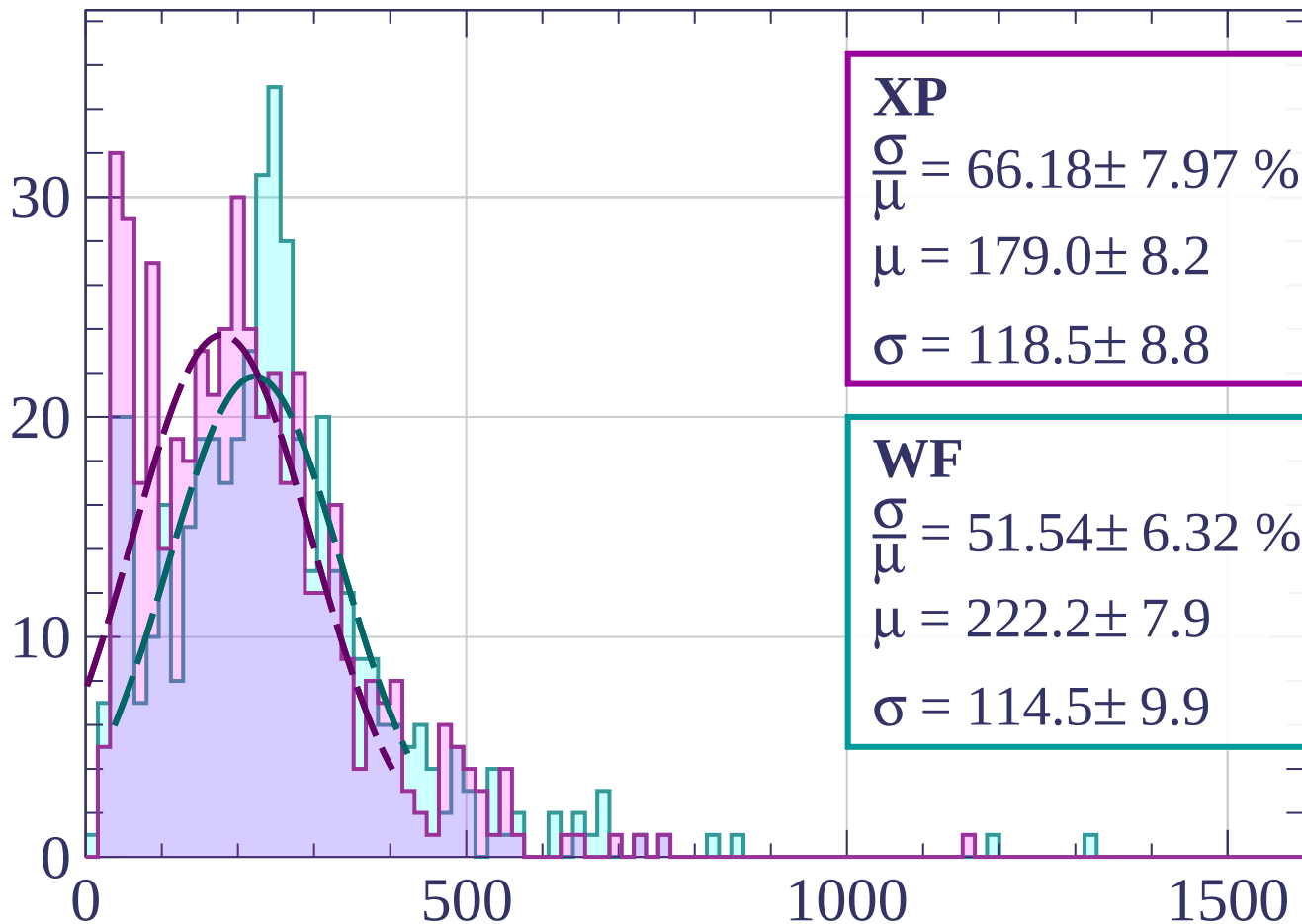
$$\mu = 238.7 \pm 8.7$$

$$\sigma = 143.0 \pm 10.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $54 < \theta < 57$

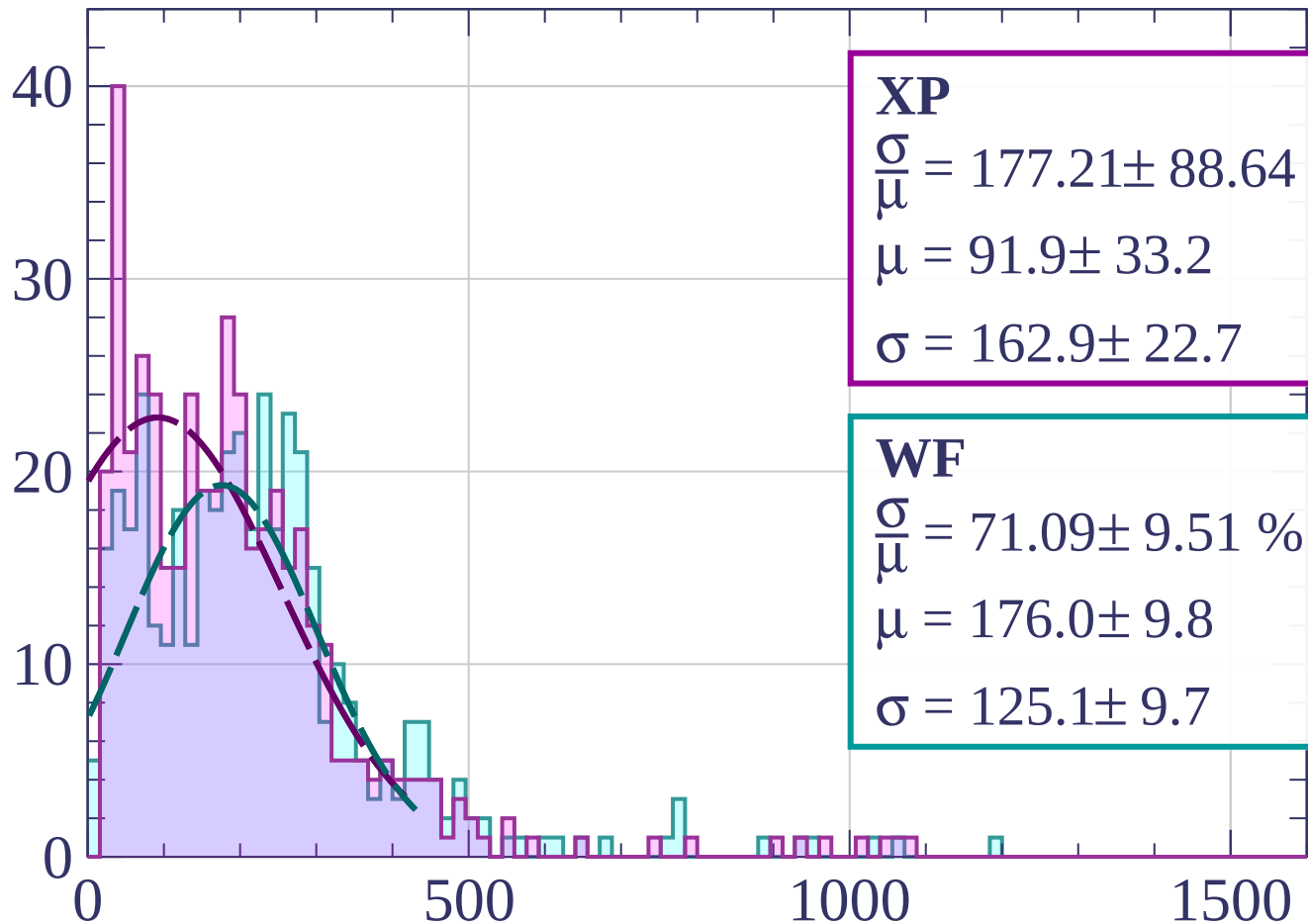
Count



dE/dx [ADC counts/cm]

Energy loss |  $57 < \theta < 60$

Count



**XP**

$$\frac{\sigma}{\mu} = 177.21 \pm 88.64 \%$$

$$\mu = 91.9 \pm 33.2$$

$$\sigma = 162.9 \pm 22.7$$

**WF**

$$\frac{\sigma}{\mu} = 71.09 \pm 9.51 \%$$

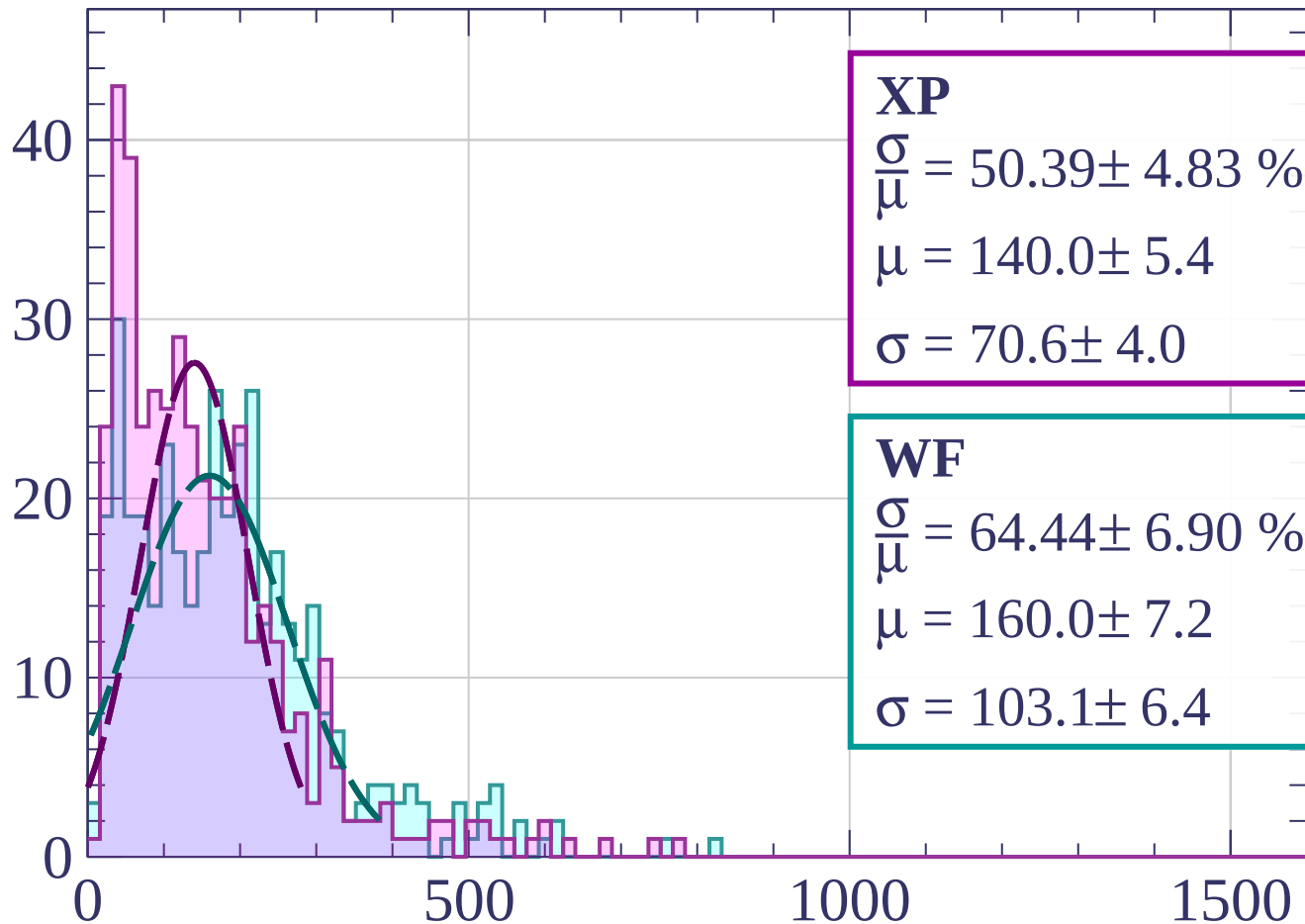
$$\mu = 176.0 \pm 9.8$$

$$\sigma = 125.1 \pm 9.7$$

dE/dx [ADC counts/cm]

Energy loss |  $60 < \theta < 63$

Count



**XP**

$$\frac{\sigma}{\mu} = 50.39 \pm 4.83 \%$$

$$\mu = 140.0 \pm 5.4$$

$$\sigma = 70.6 \pm 4.0$$

**WF**

$$\frac{\sigma}{\mu} = 64.44 \pm 6.90 \%$$

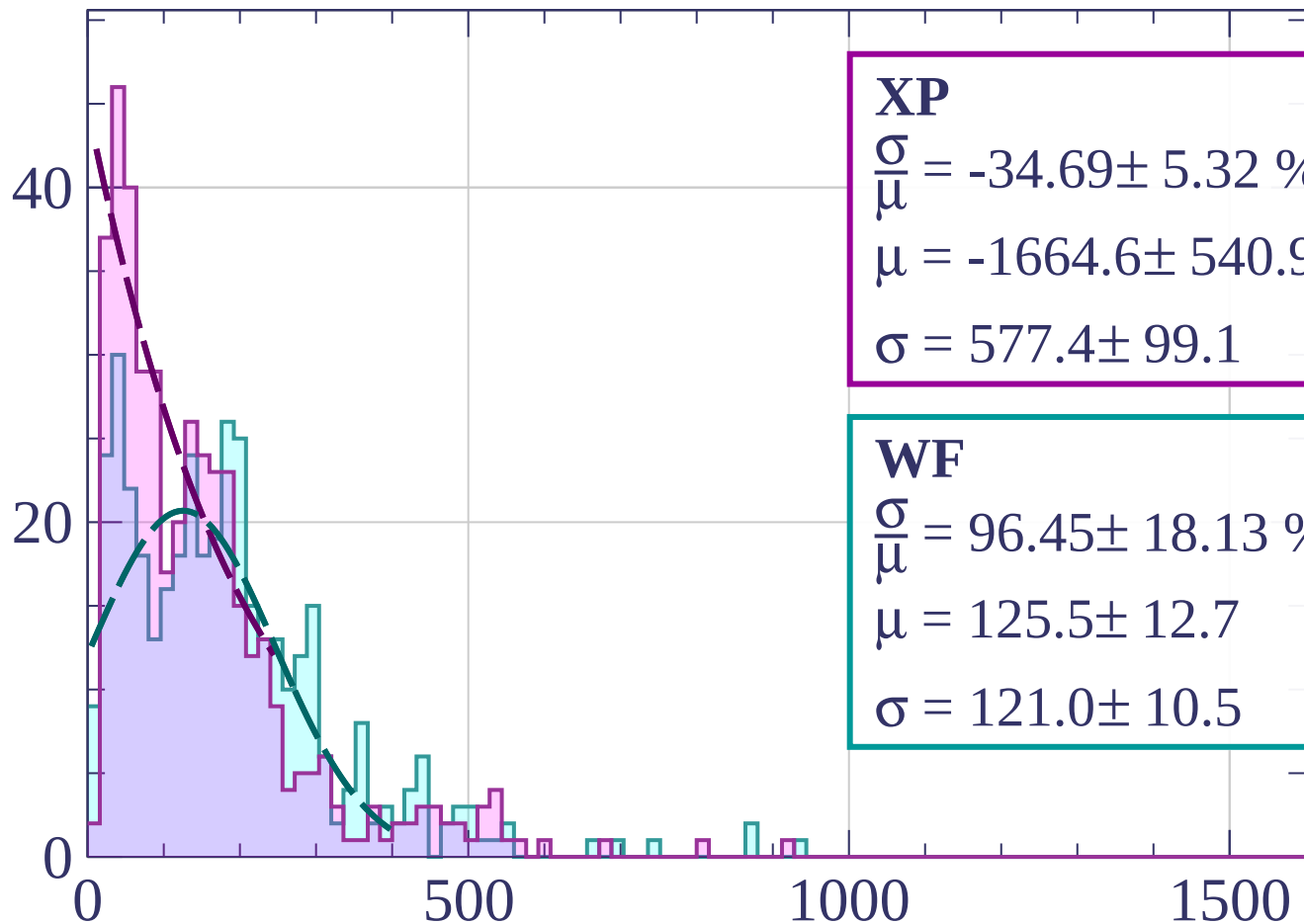
$$\mu = 160.0 \pm 7.2$$

$$\sigma = 103.1 \pm 6.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $63 < \theta < 66$

Count



**XP**

$$\frac{\sigma}{\mu} = -34.69 \pm 5.32 \%$$

$$\mu = -1664.6 \pm 540.9$$

$$\sigma = 577.4 \pm 99.1$$

**WF**

$$\frac{\sigma}{\mu} = 96.45 \pm 18.13 \%$$

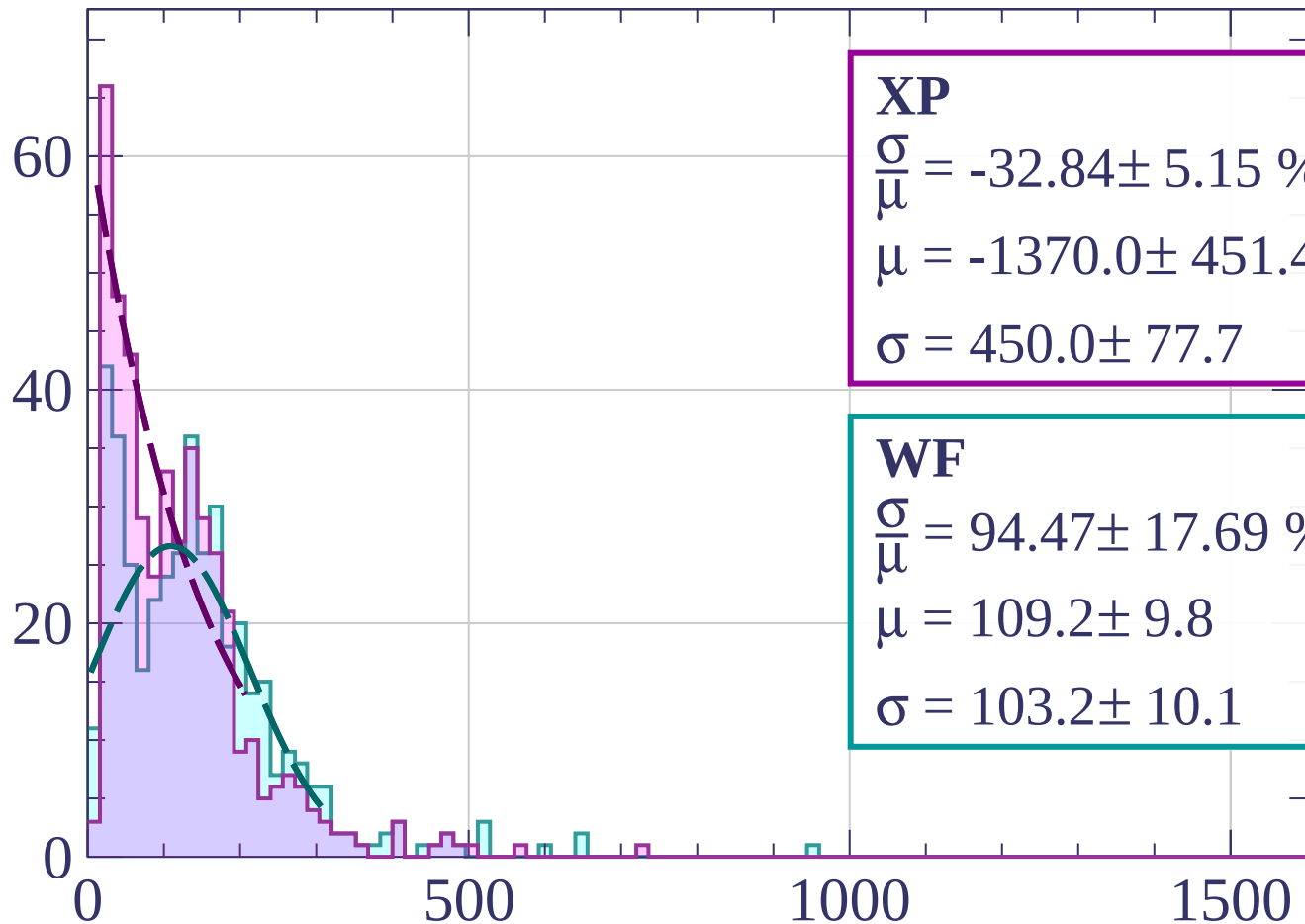
$$\mu = 125.5 \pm 12.7$$

$$\sigma = 121.0 \pm 10.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $66 < \theta < 69$

Count



**XP**

$$\frac{\sigma}{\mu} = -32.84 \pm 5.15 \%$$

$$\mu = -1370.0 \pm 451.4$$

$$\sigma = 450.0 \pm 77.7$$

**WF**

$$\frac{\sigma}{\mu} = 94.47 \pm 17.69 \%$$

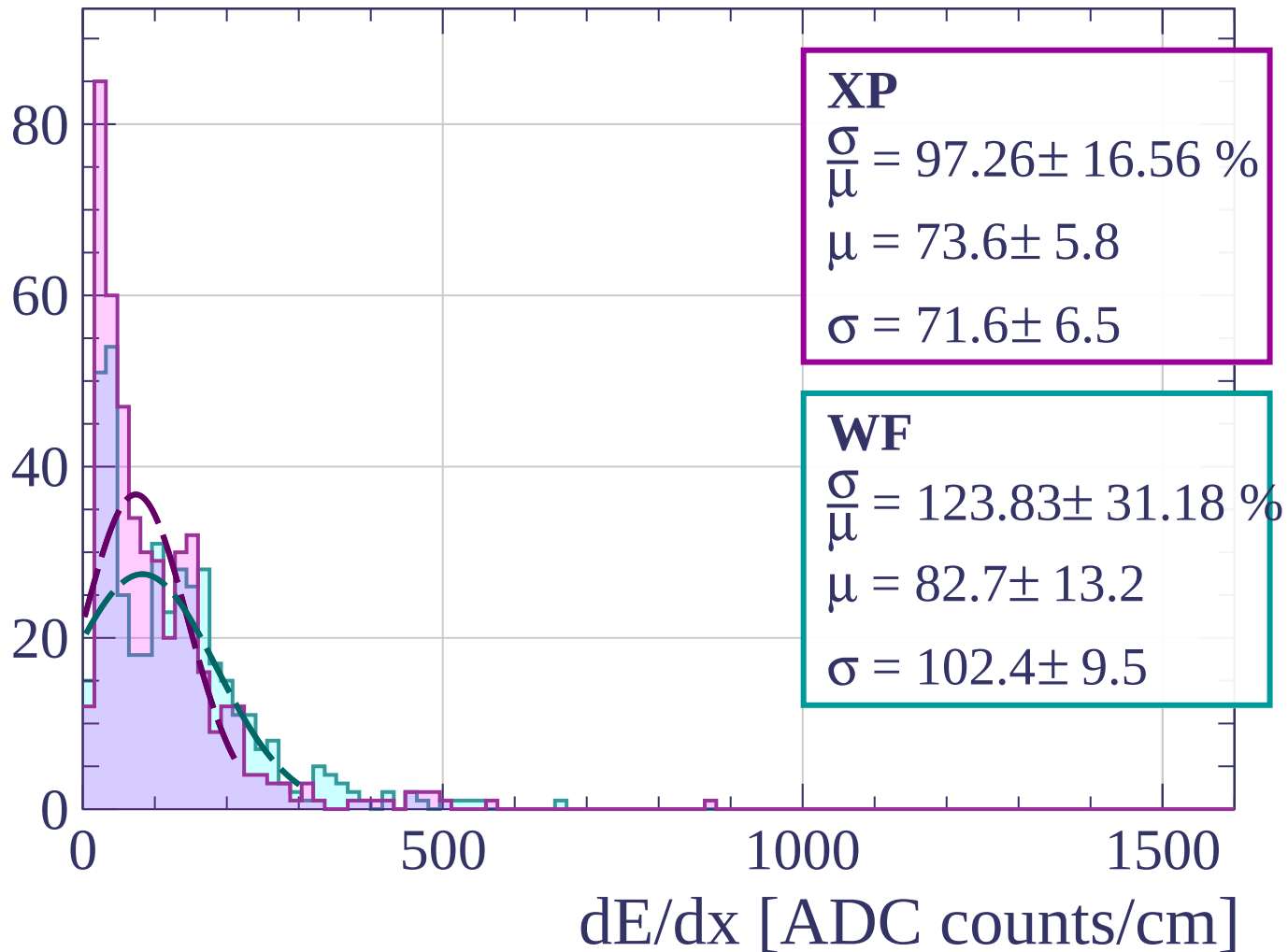
$$\mu = 109.2 \pm 9.8$$

$$\sigma = 103.2 \pm 10.1$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $69 < \theta < 72$

Count



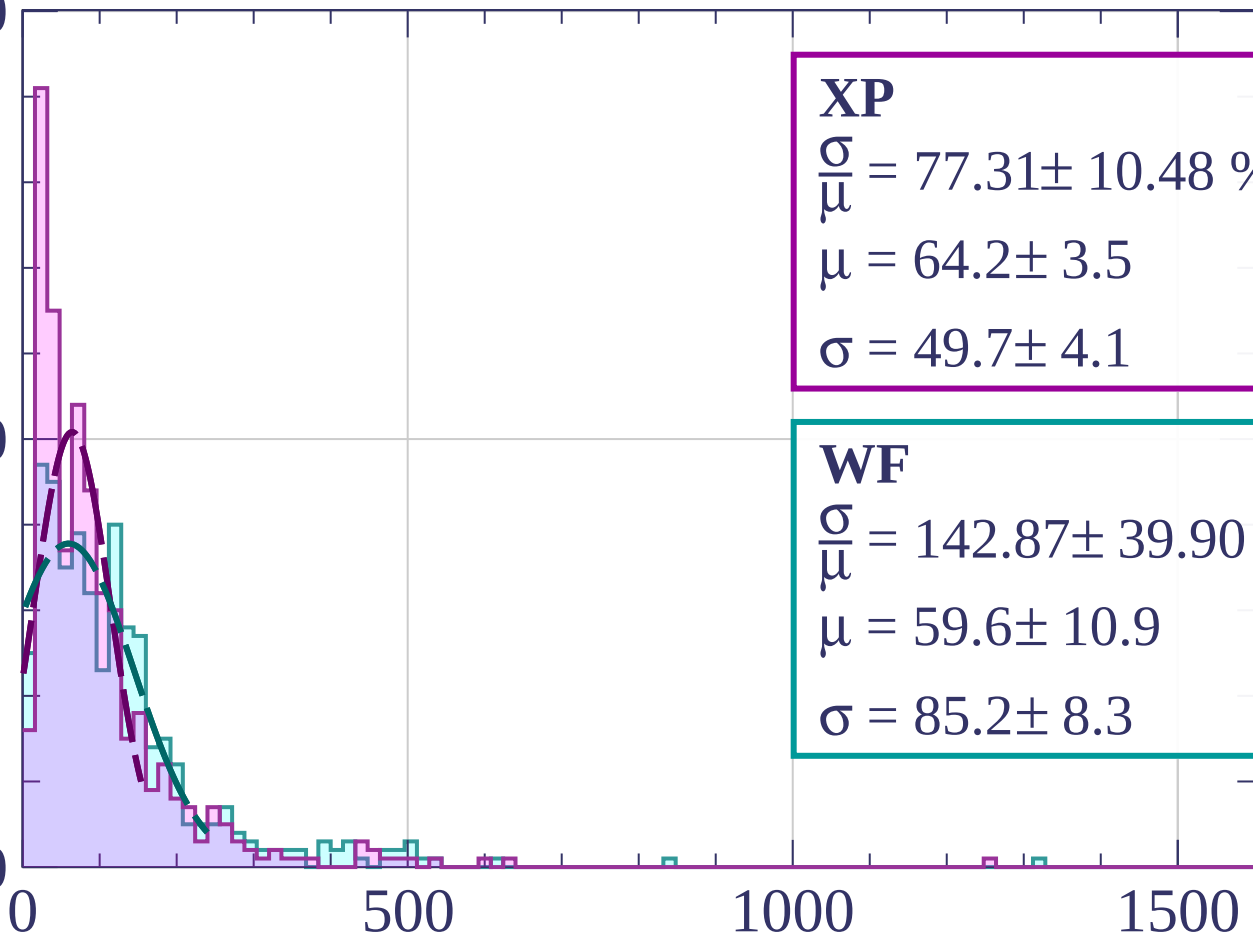
Energy loss |  $72 < \theta < 75$

Count

100

50

0



**XP**

$$\frac{\sigma}{\mu} = 77.31 \pm 10.48 \%$$

$$\mu = 64.2 \pm 3.5$$

$$\sigma = 49.7 \pm 4.1$$

**WF**

$$\frac{\sigma}{\mu} = 142.87 \pm 39.90 \%$$

$$\mu = 59.6 \pm 10.9$$

$$\sigma = 85.2 \pm 8.3$$

$dE/dx$  [ADC counts/cm]

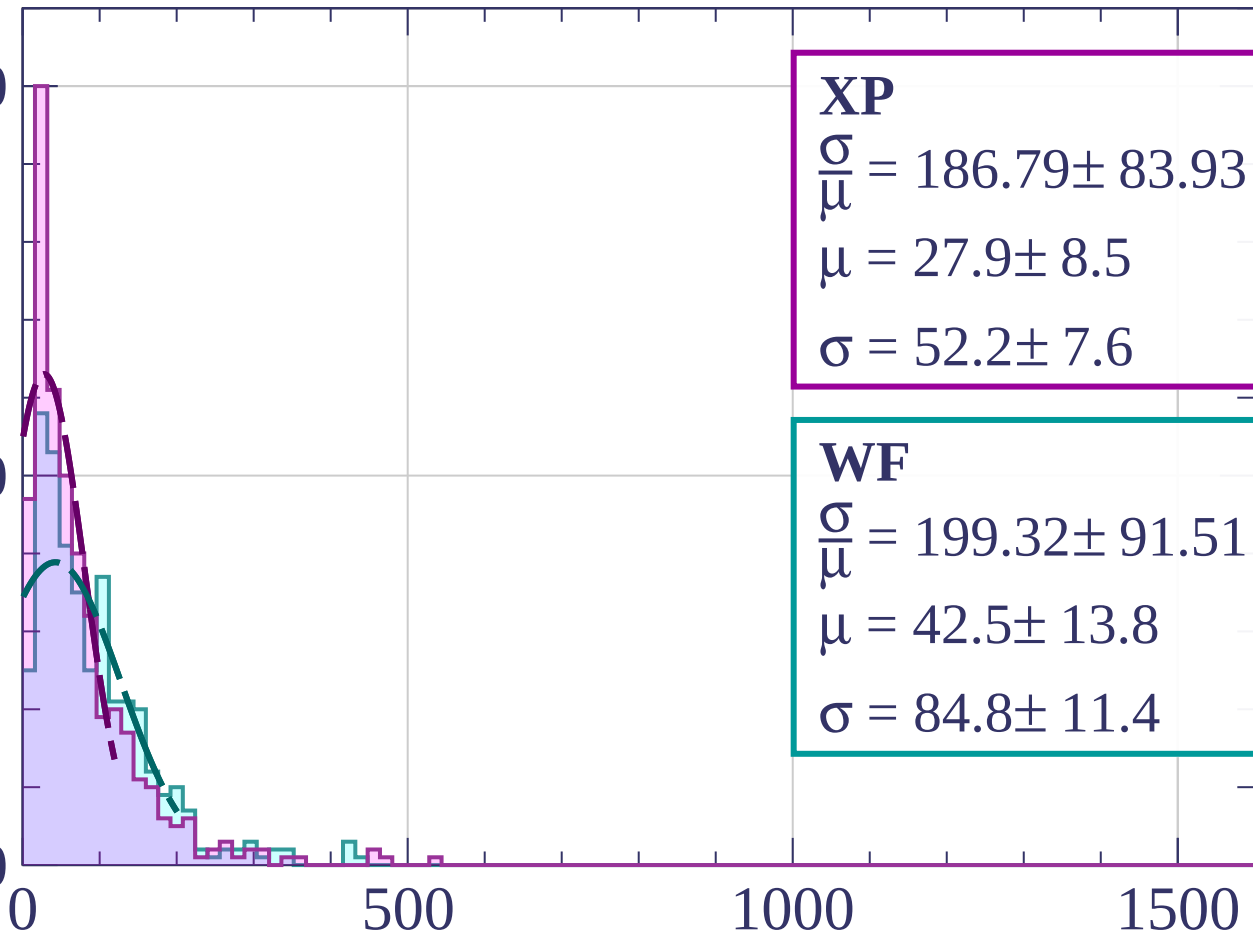
Energy loss |  $75 < \theta < 78$

Count

100

50

0



**XP**

$$\frac{\sigma}{\mu} = 186.79 \pm 83.93 \%$$

$$\mu = 27.9 \pm 8.5$$

$$\sigma = 52.2 \pm 7.6$$

**WF**

$$\frac{\sigma}{\mu} = 199.32 \pm 91.51 \%$$

$$\mu = 42.5 \pm 13.8$$

$$\sigma = 84.8 \pm 11.4$$

dE/dx [ADC counts/cm]

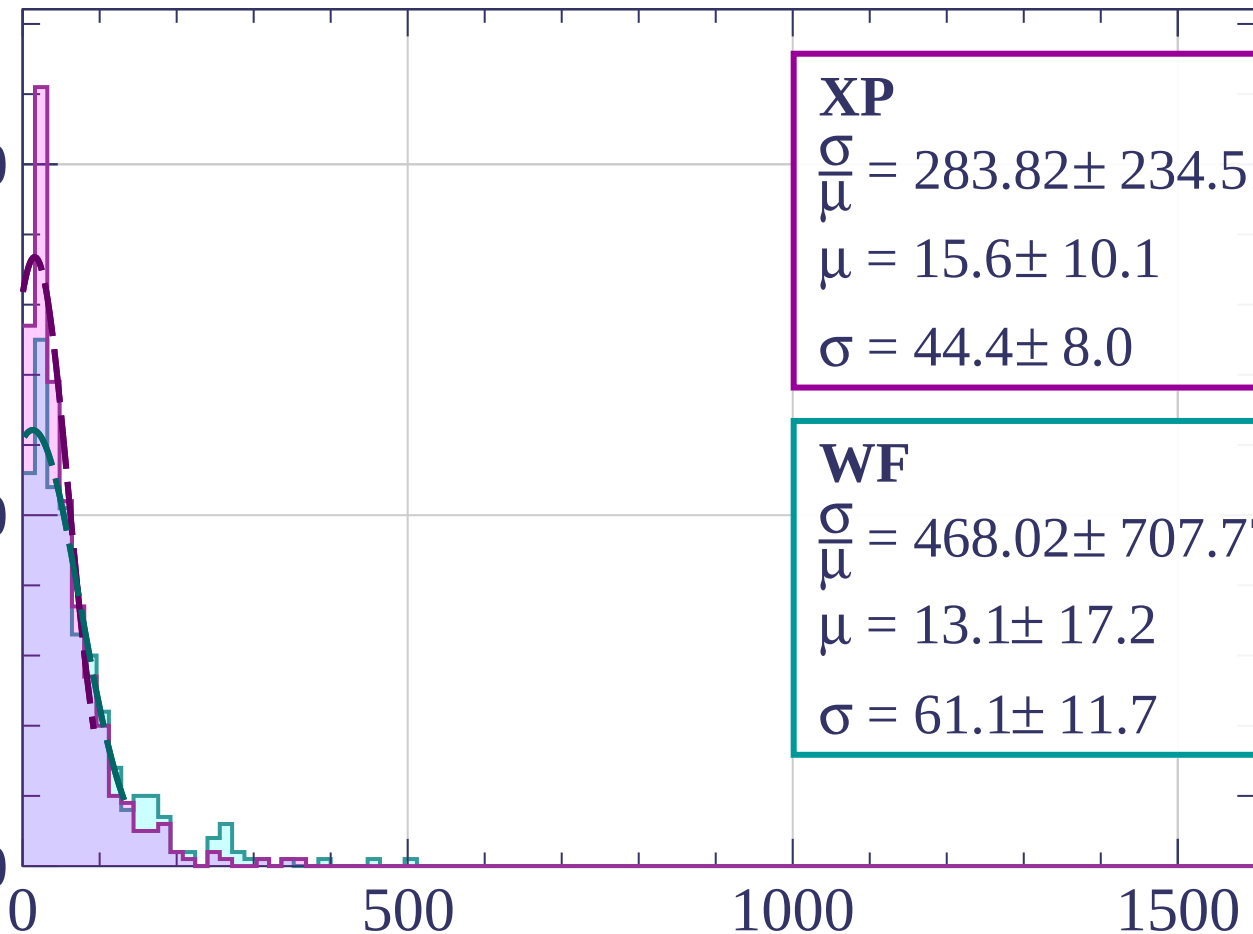
Energy loss |  $78 < \theta < 81$

Count

100

50

0



**XP**

$$\frac{\sigma}{\mu} = 283.82 \pm 234.51 \%$$

$$\mu = 15.6 \pm 10.1$$

$$\sigma = 44.4 \pm 8.0$$

**WF**

$$\frac{\sigma}{\mu} = 468.02 \pm 707.77 \%$$

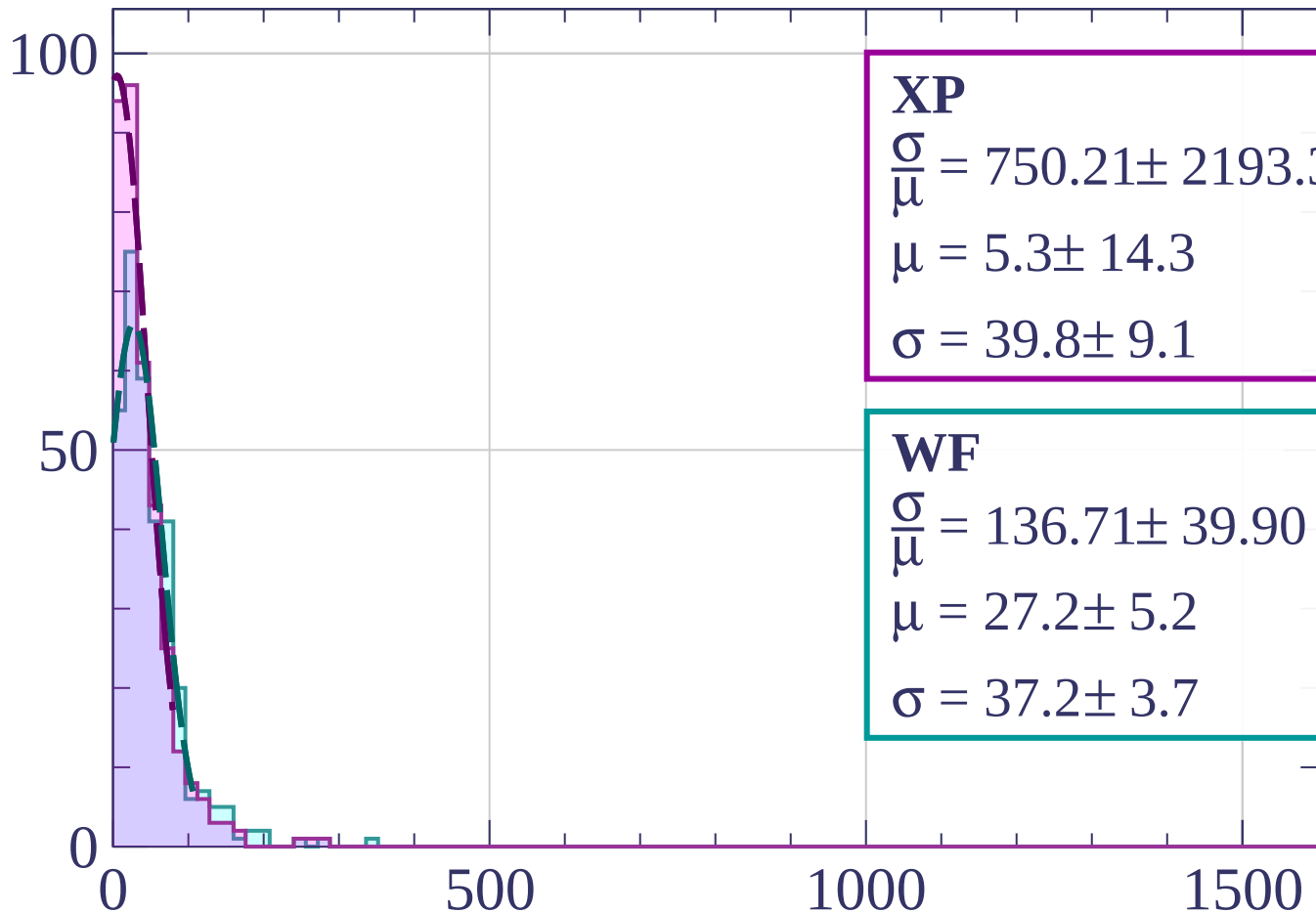
$$\mu = 13.1 \pm 17.2$$

$$\sigma = 61.1 \pm 11.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $81 < \theta < 84$

Count



**XP**

$$\frac{\sigma}{\mu} = 750.21 \pm 2193.37 \%$$

$$\mu = 5.3 \pm 14.3$$

$$\sigma = 39.8 \pm 9.1$$

**WF**

$$\frac{\sigma}{\mu} = 136.71 \pm 39.90 \%$$

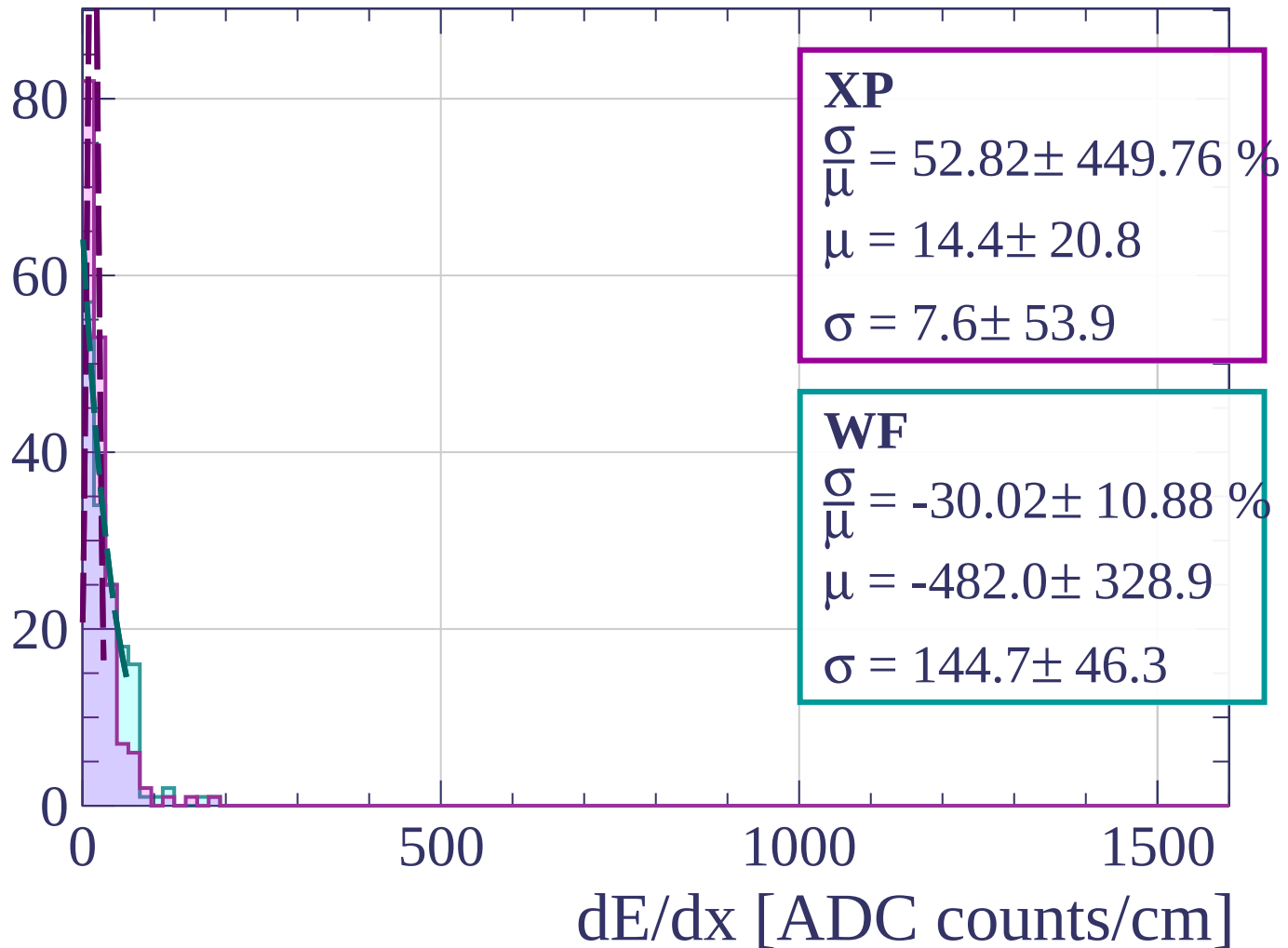
$$\mu = 27.2 \pm 5.2$$

$$\sigma = 37.2 \pm 3.7$$

$dE/dx$  [ADC counts/cm]

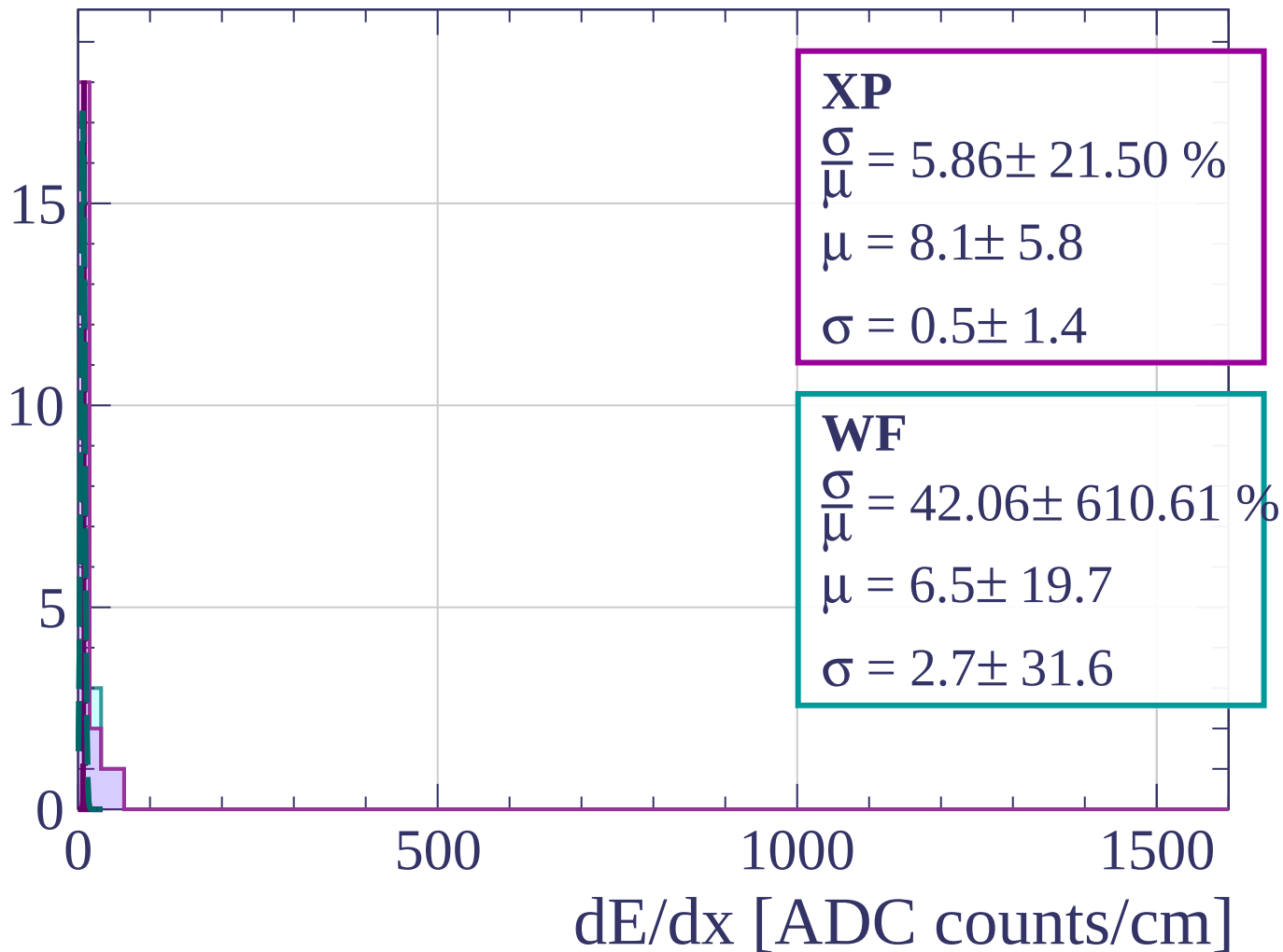
Energy loss |  $84 < \theta < 87$

Count

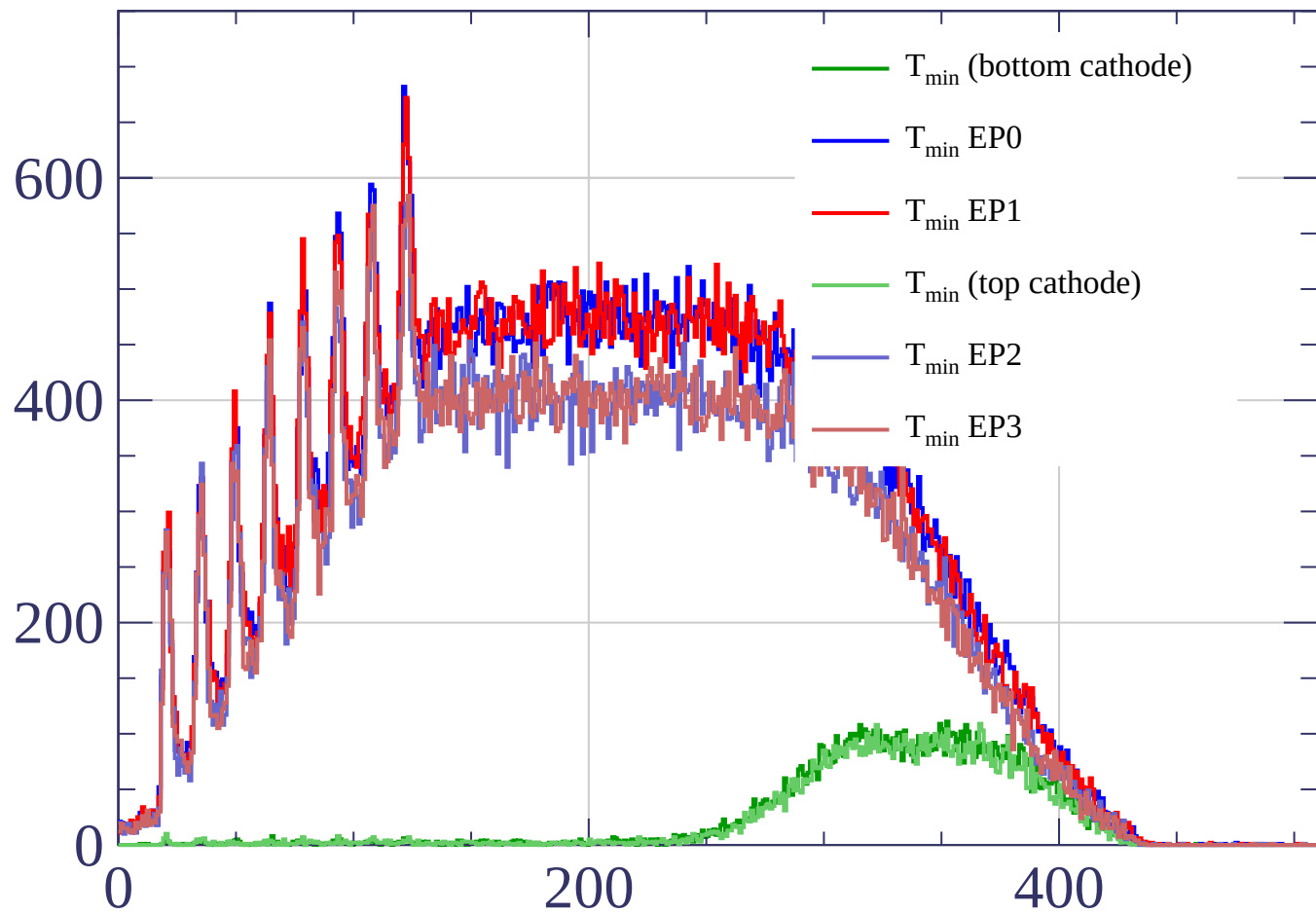


Energy loss |  $87 < \theta < 90$

Count

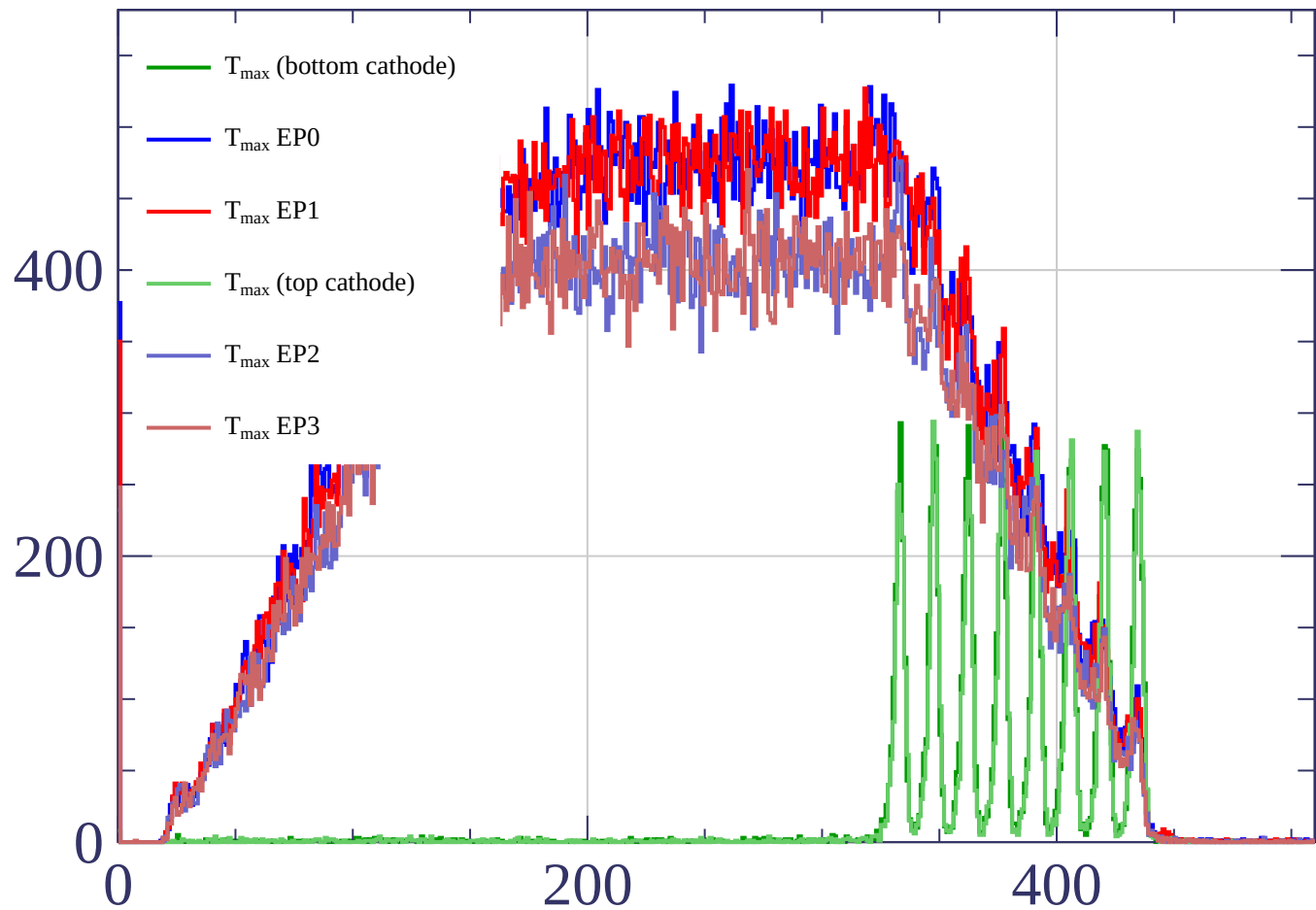


Count



time bin

Count



time bin